Publications of the Minnesota Mathematics Assessment Project (MNMAP)

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August 20, 2015

To: All friends and colleagues of the MNMAP Project.

Ed Andersen	Bill Bush	Arnie (Jan) Cutler
Jon Davis	Danielle Dupuis	Brandon Lebeau
Yukiko Maeda	Amanuel Medhanie	Debra Monson
Mario Moreno	Tom Muchlinski	Bob Reys
Ke Wu		

This bound volume of all MNMAP publications is a token of our appreciation for all of the good work that you performed to enable the completion of these papers.

Mike remarked recently that there was a certain chemistry that we had, one of mutual respect and acceptance. I agree and think this was key to our many accomplishments.

The Project addressed important issues and as far as I know was unique in that it was the only dedicated program of research that continuously addressed, from multiple vantage points, the relationship between high school mathematics curricula (NSF or not) and students' subsequent college level mathematics performance.

The consistency of our findings over ten coordinated papers established two things: first, that the three NSF supported curricula (Core+, IMP and MMOW) prepared students equally well for college mathematics, and, secondly that the anti-reform mathematicians locally and nationally were "shooting from the hip" and had it all wrong. These understandings are important for our field in that they open the door to a wide variety of other successful programs that are more desirable in many ways than traditional textbook approaches to the subject.

All this explanation is by way of saying that what you have accomplished is one of a kind, academically of impressive integrity and quality and important to the field.

Thank you all and well done,

Tom & Mike

August 18, 2015

Publications of the Minnesota Mathematics Assessment Project

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The Minnesota Mathematics Assessment Project (MNMAP) was funded by the National Science Foundation (NSF) from 2006 to 2012. Project writing activity was extended through 2015. MNMAP was a program of coordinated research that published fifteen papers in well-regarded peer reviewed research-oriented journals. Prepared by both faculty and graduate students, ten of the papers examined the relationship between high school mathematics curricula and college level mathematics performance, (paper #'s 2-9,12 &13). These papers analyzed data encompassing four years (or two years in the case of community college students) of college mathematics work.

Five others (#'s 1,10,11,14 &15) were concerned with placement recommendations and their impact on subsequent college level performance, or, pre-college students' performance on standardized tests as a function of their enrollment in high school mathematics curricula – i.e. NSF funded or not.

Thirty-two four-year, and 21 two-year community colleges were involved at one time or another, utilizing a pool of approximately 20,000 students, all from five contiguous Midwest states. Student N's varied with the specific paper. All are listed below in the reverse order in which they were published.

The most important and consistent finding across all of the papers is that the NSF funded integrated high school curricula prepared students equally well for college mathematics when compared with more traditional textbook approaches. The NSF curricula were: (Mathematics in Context - Core+, Interactive Mathematics Project (IMP), and Mathematics Modeling Our World (MMOW). All of the high school integrated students were enrolled for three or more years in one of these three programs. Results indicate that each of these curricula supported students' subsequent college level mathematics course taking patterns (four-year trajectories), achievement, and persistence. This is in direct contrast to the dire prognostications and claims from some members of the post secondary anti-reform mathematics community for the need to get back to more traditional approaches. These ongoing debates became known as the Math Wars.

The no-difference results between traditional and the three NSF funded curricula were consistently demonstrated whether we looked at community college results,

smaller public or private colleges, major university results, STEM majors, students with more intense college mathematics course-taking profiles, one or multiple institutions, partitioned students on the basis of prior mathematics achievement, or examined performance in introductory college statistics coursework. These students performed equally as well as students with a more traditional textbook-oriented high school mathematics background on all but one of the dependent variables that were of interest to project personnel. A higher percentage of integrated students enrolled initially in developmental level coursework in collage, defined here as any course considered a prerequisite for college algebra. After this first developmental course there were no differences in the number or level of subsequent mathematics courses or achievement.

In total, our findings suggest, there are good reasons to consider a more active and intense problem-solving focused approach to high school mathematics as envisioned by the author teams of the three integrated curricula, and, the NCTM Standards related publications. Such an approach is designed to be accessible to a wider range of students and has both psychological and sociological advantages for students. It is also far more consistent with the cognitive perspective of mathematics learning as we now understand it.

We conclude that there is no royal road to success in college mathematics. Our findings suggest that multiple and very different approaches in both content and method are worthy of serious consideration. As of this writing, each of the three NSF funded curricula is available in their second editions.

Hard copies or electronic versions of each of these MNMAP papers are available by contacting your local college librarians or departmental secretary for the photocopy details relevant to your institution.

MNMAP PUBLICATIONS:

1. Harwell, M.R., Moreno, M.A., & Post, T.R. (2015). A study of the relationship between the ACT college mathematics readiness standard and college mathematics achievement. *Journal of Psychoeducational Assessment.* Pre Publication format, pp 1-13. **pp. 5-17.**

2. Dupuis, D., Medhanie, A., Monson, D., LeBeau, B., Harwell, M.R., & Post, T.R. (2015). The relationship between high school mathematics curriculum and mathematics achievement in community college. *MathAMATYC Educator*. May, Vol. 6. No. 3, 41-50. **pp. 18-28.**

3. Harwell M., Post, T., LeBeau, B., Dupuis, D., Medhanie, A. (2014), A multisite study of high school mathematics curricula and the impact of taking a developmental mathematics course in college. *Educational Research Quarterly*. *37 (3),* 22-34*.* **pp. 29-51.**

4. Harwell, M., Dupuis, D., Post, T. R., Medhanie, A., LeBeau, B. (2013). The impact of institutional factors on the relationship between high school mathematics curricula and college mathematics course-taking and achievement. *Educational Research Quarterly*, *36 (3*), 22-46. **pp. 52-77.**

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11. Norman, K. W., Medhanie, A., Harwell, M. R., Andersen, E., & Post, T. R. (2011). High school mathematics curricula, university mathematics placement recommendations, and student university mathematics performance. *Problems Resources and Issues in Mathematics Undergraduate Studies* – PRIMUS 21(5): 434-455. **pp. 184-205.**

12. Post, T. R., Medhanie, A., Harwell, M. R., Norman, K., Dupuis, D., Muchlinski, T., Andersen, E., & Monson, D. (2010). The impact of prior mathematics achievement on the relationship between high school mathematics curricula and postsecondary mathematics performance, course-taking, and persistence. *Journal for Research in Mathematics Education*, 41, 274-308. **pp. 206-240.**

13. Harwell, M., Post, T. R., Cutler, A., Maeda, Y., Anderson, E., Norman, K. W., & Medhanie, A. (2009). The preparation of students from National Science Foundation-funded and commercially developed high school mathematics curricula for their first university mathematics course. *American Educational Research Journal*, 46, 203-231. **pp. 241-269.**

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15. Harwell, M., Post, T. R., Maeda, Y., Davis, J. D., Cutler, A. L., Andersen, E., & Kahan, J. A. (2007). Standards-based mathematics curricula and secondary students' performance on standardized achievement tests. *Journal for Research in Mathematics Education,* 38, 71-101. **299-329.**

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A Study of the Relationship Between the ACT College Mathematics Readiness Standard and College Mathematics Achievement

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Abstract

This study examined the relationship between the American College Testing (ACT) college mathematics readiness standard and college mathematics achievement using a sample of students who met or exceeded the minimum 3 years high school mathematics coursework recommended by ACT. According to ACT, a student who scores 22 or higher on the ACT mathematics test has a relatively high probability of earning a passing grade in his or her first college mathematics course. Data from 11,324 students who attended 1 of 27 four-year postsecondary institutions in the upper Midwest were used to study this relationship. Overall, there was evidence that meeting the mathematics readiness standard was related to college mathematics achievement for students who completed at least 3 years of high school mathematics. The results have important implications for school district staff advising college-intending students on mathematics course-taking, college admissions staff evaluating student readiness for college-level mathematics, and researchers studying college readiness.

Keywords

educational testing, education assessment, mathematics, disciplines and subjects, achievement, education program and leadership, high school, participants

Readiness for college mathematics has drawn a great deal of attention from educators, policy makers, and researchers. College readiness in mathematics was defined by Conley (2008) as the probability a student will succeed in college-level mathematics courses, and by McCormick and Lucas (2011) as the accumulation of knowledge in mathematics that students are required to have to pass their courses in college. The most widely used indicator of college readiness in mathematics appears to be test scores from large-scale mathematics assessments (Conley, 2008; Roderick, Nagaoka, & Coca, 2009; Wiley, Wyatt, & Camara, 2010).

One of the better known efforts around college readiness is the standards developed by the American College Testing (ACT) program to predict a student's readiness to succeed in

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college coursework. Specifically, the standards represent " . . . detailed, research-based descriptions of the skills and knowledge associated with what students are likely to know and to be able to do based on their EXPLORE, PLAN, and/or ACT test scores" (ACT, 2011, p. 2). The EXPLORE and PLAN tests are ACT products used to assess the academic development of 8th/9th-grade and 10th-grade students. The ACT test serves as an admissions test for college-intending students and is typically taken at the end of a student's junior year in high school or early in their senior year. According to ACT, students taking the mathematics portion of the test should ideally have completed at least 3 years (Carnegie units) of core high school mathematics courses consisting of Algebra I, Algebra II, and Geometry because this coursework represents the minimum knowledge and skills needed in this portion of the ACT test (ACT, 2006, 2011).

The ACT College Readiness Standard in Mathematics

Studies of the relationship between ACT scores and college performance began in 1997 and led to the mathematics standard in a 2005 benchmarking study (ACT, 2005) based on a sample of 85 2- and 4-year colleges obtained between 1995 and 2003. Multilevel (students nested within college) logistic regression models with the predictor ACT mathematics score were fitted to student grades for a College Algebra course at each college after grades were dichotomized as a "success" (students earned a grade of at least a B or at least a C) or a "failure" (grade earned was less than a B or a C). The rationale for treating a grade of B or higher as representing "success" in a college mathematics course in the ACT (2005) benchmarking study was largely based on grade inflation, with the authors of that study noting that "Today, it's common for more than 50% of students in a course to earn grades of A or B" (p. 9).

The ACT (2005) benchmarking study used the mathematics score associated with a probability of success of .50 for the fitted model as an institution's benchmark (i.e., a logistic probability curve was fitted to find the score at which 50% of students earned at least a B in their first college mathematics class), with the median of these scores of 22 emerging as the standard. Thus, on average, 50% of the students whose ACT mathematics score was ≥22 earned a B or higher in College Algebra, and on average, 75% of the students scoring ≥22 earned a grade of C or higher. The results from the 2005 study led ACT to characterize students with ACT mathematics scores ≥22 as having met the college readiness standard. In 2011, approximately 55% of the 1.5 million U.S. students who took the ACT failed to meet the standard in mathematics (ACT, 2011). The benchmarks were updated in 2013 (ACT, 2013b), with an ACT mathematics score of 22 again emerging as the mathematics standard.

Precisely how the ACT mathematics standard is used by colleges, high schools, and students and their families in educational planning and decision making has not been rigorously documented. Evidence of the role of test scores in college admission decisions (Clinedinst, Hurley, & Hawkins, 2011) suggests the readiness standards are important, although it is equally important to emphasize that ACT has consistently recommended these standards be combined with other information to support educational planning and decision making (ACT, 2005, 2011, 2013b). It is also likely that many students who meet or fail to meet the mathematics standard have different experiences in the mathematics course(s) they are advised to take their senior year in high school, whether they should take the ACT test a second time in the hope of raising their mathematics score to at least 22, the college mathematics course they should start with, or other forms of academic support that might be appropriate (ACT, 2013b). However the standard is used, its potential impact on the educational trajectory of college-intending students suggests that an extensive literature supporting its use be available.

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Evidence for the ACT Mathematics Standard and College Mathematics Achievement

Recent studies (ACT, 2009, 2012b, 2012c, 2013a) reported that meeting the ACT mathematics standard was positively associated with several college outcomes including earning a grade of B (or C) or higher in mathematics courses taken in the first year of college, continuing to the second year of college at the same institution, achieving a cumulative college grade point average (GPA) of 2.00 or higher, and completing a college degree. An important feature of these studies is the variation in the mathematics preparation of the samples of high school students, which ranged from no high school mathematics through Calculus I (ACT, 2009, 2012c).

Although available evidence indicates that meeting the ACT mathematics standard is generally associated with higher college mathematics achievement, this literature has three important gaps. First, it is unclear how useful the ACT mathematics standard is for students who meet or exceed the minimum 3 years of high school mathematics coursework recommended by ACT (2006, 2011). The ACT mathematics readiness standard does not explicitly take high school mathematics course-taking into account and, as noted above, studies of the standard have sampled students with a wide range of mathematical preparation, rather than students whose mathematics preparation met or exceeded the ACT recommended minimum of 3 years of high school mathematics.

Second, there is a need for studies of the mathematics standard based on data not collected by ACT, which has a vested interest in the standard. Put simply, the potential impact of the ACT college mathematics readiness standard on college-intending students educational trajectory suggests there is a need to provide independent (i.e., non-ACT collected) evidence of the standard's ability to predict students' readiness for college mathematics.

Third, the impact of institutional and student characteristics on the relationship between the ACT mathematics standard and college mathematics performance for students who meet or exceed the high school mathematics coursework recommended by ACT is unclear. For example, institutional selectivity based on average Scholastic Assessment Test (SAT) scores of admitted students and whether a postsecondary institution is publicly or privately funded has produced differential predictive validity of the SAT and first-year college GPA (Kobrin & Patterson, 2011). Selectivity based on ACT scores has been shown to be associated with lower percentages of students completing remedial mathematics coursework (Bettinger & Long, 2009), higher graduation rates, greater graduate school access and success, and higher postgraduation earnings (Carnevale & Rose, 2004), and to predict first-year college GPA (Sawyer, 2013). Student characteristics such as gender and race have also been shown to be related to the likelihood a student will meet the ACT mathematics standard (ACT, 2013a). However, the impact of institutional and student characteristics on the relationship between the ACT mathematics standard and college mathematics performance for students who completed 3 or more years of high school mathematics as recommended by ACT is unclear.

Purpose of the Study and Research Questions

The purpose of the study was to examine the relationship between the mathematics standard and college mathematics achievement for students who met or exceeded the minimum high school mathematics coursework recommended by ACT using data not collected by ACT. Evidence that the standard is related to college mathematics performance supports "business as usual" for how the standard is used in conjunction with other information including high school course-taking and grades, and cognitive, non-cognitive, and socio-demographic factors (ACT, 2010; Allen, Robbins, & Sawyer, 2010). Evidence that the standard is not related to college mathematics achievement suggests the standard is less useful for students who have completed 3 or more years of high school mathematics.

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Our main research questions were as follows:

Research Question 1: Is there a relationship between meeting the ACT mathematics standard and the likelihood that a student will earn at least a B (or at least a C) in his or her first college mathematics course, controlling for gender, the difficulty of college mathematics classes, and race?
Research Question 2: Are the results for Research Question 1 similar across postsecondary institutions with different characteristics?

Method

Subjects and Setting

The target population consisted of students in 4-year postsecondary institutions in the United States who completed at least 3 years of high school mathematics, whereas the sampled population consisted of students who completed at least 3 years of high school mathematics and graduated from a Minnesota high school. The sample of 11,324 students come from a curriculum study that used archival data sets provided by 27 postsecondary 4-year institutions located in one of five upper Midwest states. These students completed (a) 3, 4, or 5 years of high school mathematics and graduated from a Minnesota high school, (b) eight college semesters, and (c) at least one college mathematics course whose difficulty equaled or exceeded that of College Algebra. Because of missing data on different variables, some analyses were based on 11,311 students.

Variables

We define college mathematics as credit bearing courses whose difficulty equals or exceeds that of College Algebra. Hofacker (2006) noted that College Algebra is the most widely enrolled college mathematics course in the United States and is the course that a student who completed 3 years of high school mathematics would likely start with in his or her college mathematics course-taking.

Seventy-five percent of the institutions used a 12-point grading scale (A, A−, B+, . . . , F), 18.75% used a 7-point grading scale (A, AB, B, BC, C, D, F), and 6.25% used a 5-point grading scale (A, B, C, D, F). Throughout this process, we assumed that similar grades earned in similar courses at different institutions reflected similar proficiency. We converted each student's grade in their first college mathematics course to a dichotomous achievement variable such that a student was assigned a 1 if this grade equaled or exceeded a B and 0 otherwise. The same strategy was followed to create a dichotomous achievement variable for earning at least a C in this course. The use of dichotomous grade variables (e.g., ≥B, <B) provided a simple proficient/not proficient categorization of student performance that is less likely to be sensitive to between-institution differences in instruction and grading practices than more refined grading scales (e.g., 12-point scale). Study variables are defined in Table 1.

The difficulty of college mathematics courses was captured in a trichotomous variable following Teitlebaum (2003). Mathematics course descriptions at each institution were reviewed and rated independently by three mathematics educators. Initial levels of disagreement among raters never exceeded 1 point on the difficulty scale and were subsequently resolved by discussion among the raters until a consensus was achieved. A trichotomous institutional profile variable was also created in response to evidence that institutional characteristics affect student outcomes (ACT, 2012a; Berger & Milem, 1999; Sawyer, 2013). The institutional profile variable was treated as possessing a nominal scale of measurement, meaning that an institution with a coded

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Table 1. Study Variables.

Variable	Label	Values
Y	Students earned at least a B in their first college mathematics course	1 = yes; 0 = no
Y	Students earned at least a C in their first college mathematics course	1 = yes; 0 = no
Difficulty1+	Difficulty of a student's first college mathematics course	1 = college mathematics course whose difficulty was consistent with College Algebra, pre-Calculus mathematics, or Finite Mathematics; 0 = course whose difficulty was consistent with Calculus II, Differential Equations, Linear Algebra, or Multivariable Calculus.
Difficulty2+	Difficulty of a student's first college mathematics course	1 = college mathematics course whose difficulty was consistent with Calculus I; 0 = course whose difficulty was consistent with Calculus II, Differential Equations, Linear Algebra, or Multivariable Calculus.
Asian+	Student race (Asian)	1 = Asian; 0 = White
Black+	Student race (Black)	1 = Black; 0 = White
Hispanic+	Student race (Hispanic)	1 = Hispanic; 0 = White
MetStand	Student met the ACT mathematics standard of 22	1 = yes; 0 = no
Profile1+	Institutional characteristics	1 = ≥80% of students enrolled full-time (12 or more semester credits or quarter credits)/ more selective (average ACT/SAT scores of the freshmen class in the upper one fifth of all colleges)/low transfer-in rate (≤10%, proportion of entering students who transferred in from another institution, defined as a student entering an institution for the first time but who previously attended another postsecondary institution as an undergraduate; 0 = mixture of full-time and part-time/selective (average ACT/SAT scores of the freshmen class in the middle two-fifths of all colleges)/high transfer-in rate.
Profile2+	Institutional characteristics	1 = ≥80% of students enrolled full-time/more selective/low transfer-in rate; 0 = mixture of full-time and part-time/selective (average ACT/SAT scores of the freshmen class in the middle two-fifths of all colleges)/high transfer-in rate.

Note. Y = outcome variable, + = dummy-coded variable. ACT = American College Testing; SAT = Scholastic Assessment Test.

value of 1, 2, or 3 was not treated as academically superior or weaker than an institution with a different value but as simply different.

Data Analyses

Two sets of analyses were performed. First, we performed descriptive analyses in which we computed the mean percentage of students who earned at least a B (or at least a C) in their first

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college mathematics course, and the mean model-predicted probability of earning at least a B (or at least a C) in this course. For the latter, we fitted a multilevel logistic model to data indicating whether a student earned at least a B (or at least a C) with a single predictor indicating whether a student met the ACT mathematics standard. The mean percentage of students in an institution who earned least a B (or at least a C) and the mean model-predicted probability of earning at least a B (or at least a C) in an institution were averaged taking differences in the number of students each institution contributed to the sample into account.

Next, we performed inferential analyses examining the relationship between meeting the mathematics standard and college mathematics achievement that fitted multilevel (students within institutions) logistic regression models with control variables to binary outcome data indicating whether a student earned at least a B or at least a C. The inclusion of the variables gender, difficulty of a student's first college mathematics course, and race was intended to control for effects that might otherwise distort estimates of the relationship between meeting the standard and student mathematics achievement. The student model was

$$\eta_{ij} = \ln\left(\frac{\psi_{ij}}{1-\psi_{ij}}\right) = \beta_{0j} + \beta_{1j}\text{Gender}_{ij} + \beta_{2j}\text{Difficulty1}_{ij} + \beta_{3j}\text{Difficulty2}_{ij} + \beta_{4j}\text{Asian}_{ij} + \beta_{5j}\text{Black}_{ij} + \beta_{6j}\text{Hispanic}_{ij} + \beta_{7j}\text{MetStand}_{ij}. \quad (1)$$

In Equation 1, η_{ij} was the logit of the ith student in the jth institution and represented the log odds of a 1 relative to a 0, ψ_{ij} was the probability $Y = 1$ (e.g., earned at least a B in their first college mathematics course) for the ith student in the jth institution given model predictors, β_{0j} was an intercept for the jth institution, β_{1j} estimated the effect of Gender on the outcome, Difficulty1_{ij} and Difficulty2_{ij} were dummy-coded predictors capturing information in the difficulty level variable with the most difficult courses (Calculus II or beyond) serving as the reference category, Asian_{ij}, Black_{ij}, and Hispanic_{ij} were dummy-coded predictors representing race with White students serving as the reference category, and MetStand_{ij} indicated whether a student met the standard (see Table 1).

Preliminary analyses showed that slope variances for the Difficulty1_{ij}, Black_{ij}, Hispanic_{ij}, and MetStand_{ij} predictors were statistically equal to zero, whereas the intercept variance (e.g., average log odds of earning at least a B) and the slope variances for the Gender, Difficulty2_{ij}, and Asian_{ij} student predictors were statistically greater than zero (i.e., varied across institutions), prompting us to fit the following institutional model:

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\text{Profile1}_j + \gamma_{02}\text{Profile2}_j + u_{0j}. \quad (2)$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}\text{Profile1}_j + \gamma_{12}\text{Profile2}_j + u_{1j}.$$

$$\beta_{2j} = \gamma_{20}.$$

$$\beta_{3j} = \gamma_{30} + \gamma_{31}\text{Profile1}_j + \gamma_{32}\text{Profile2}_j + u_{3j}.$$

$$\beta_{4j} = \gamma_{40} + \gamma_{41}\text{Profile1}_j + \gamma_{42}\text{Profile2}_j + u_{4j}.$$

$$\beta_{5j} = \gamma_{50}.$$

$$\beta_{6j} = \gamma_{60}.$$

$$\beta_{7j} = \gamma_{70}.$$

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In Equation 2, γ_{00} was a fixed effect reflecting the weighted average log odds of a student earning, for example, at least a B in his or her first college mathematics course, Profile1_j and Profile2_j were dummy-coded predictors capturing the institutional profile variable (Category 3 served as the reference group), and u_{0j}, u_{1j}, u_{3j}, and u_{4j} were random effects. The multilevel modeling used the unit-average model in which the estimated fixed effects (e.g., $\hat{\gamma}_{01}$) reflected the expected change in the predicted η_{ij} associated with a one-unit increase in a predictor, holding constant all other predictors and random effects (Raudenbush & Bryk, 2002).

How, or whether, to center predictors remains a contentious topic in multilevel modeling (Longford, 1993; Paccagnella, 2006). The results for the fixed-effect slopes (e.g., γ_{11}) for our multilevel model were similar with and without grand mean centering, and the results without centering are reported. We used the HLM 7 software (Bryk, Raudenbush, & Congdon, 2011) to perform the modeling and $\alpha = .05$ for each statistical test. We also report R^2_{pseudo} statistics representing the variance attributable to fixed effects (Nakagawa & Schielzeth, 2013).

Results

The median number of sampled students at an institution was 172, and the mean was 419.4 ($SD = 720.1$) with a range of 3, 393. The student sample was largely White (88.7%) followed by Asian (7.7%), Black (2.2%), and Hispanic (1.4%) with ACT mathematics means of 24.8 (4.29), 23.4 (4.80), 19.9 (4.53), and 22.8 (4.33), respectively. The overall average ACT mathematics score in our sample was 24.6 (4.4), and there were more females (54.4%) although males had a higher ACT mathematics mean (25.3, $SD = 4.55$) than female students (24, $SD = 4.20$). The SD of 4.4 indicates that average variability in ACT mathematics scores about the mean was smaller than the $SD = 5.3$ reported for all ACT mathematics test-takers in 2009 (U.S. Department of Education, 2010), which is likely attributable to the fact that students in our sample completed 3 or more years of high school mathematics.

Twenty percent of the sampled students completed the ACT recommended minimum mathematics coursework (Algebra I and II, Geometry), 50% completed 4 years (Algebra I and II, Geometry, Trigonometry, or Trigonometry replaced with another advanced mathematics course), and 30% completed 5 years that included Calculus I. In comparison, 14% of a national sample of college-intending students completed the ACT recommended minimum high school mathematics coursework of 3 years (Algebra I and II, Geometry; another 5% took 3 years of mathematics, but not all of these were ACT recommended courses), 24% took 4 years of coursework (Algebra I and II, Geometry, and Trigonometry or Trigonometry replaced with another advanced mathematics course; 35% took 4 years of coursework that did not include all of the ACT recommended courses), and 7% completed 5 years including Calculus I with another 9% completing 5 years but replacing Calculus I with other advanced coursework (ACT, 2012a). Thus, a substantial percentage of our student sample met or exceeded the minimum high school mathematics coursework recommended by ACT compared with a national sample of college-intending students.

Table 2 reports the average percentage of students at an institution who earned at least a B (62%) and at least a C (87%) in their first college mathematics course given that they met the ACT mathematics standard (i.e., all values in Table 2 are based on $n = 8{,}460$ students who met the standard). The mean percentage of students at an institution who earned at least a B for the Institutional Profile Categories 1 (≥80% of students enrolled full-time/more selective/low transfer-in rate) and 2 (full-time/selective/high transfer-in rate) were higher (65.1%, 69.3%) than those in Category 3 (mixture of full-time and part-time/selective/high transfer-in rate) institutions (56%). The average percentage of students who earned at least a C in their first college mathematics course for Categories 1 through 3 were 88.4, 89, and 85.3, respectively.

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Table 2. Mean Percentage of Students Who Earned at Least a B or at Least a C in Their First College Mathematics Course Among Those Who Met the ACT Mathematics Standard (i.e., Scored ≥22) by Institutional Profile Category and Difficulty Level.

	M percentage of students earning at least a B in their first mathematics course			*M* percentage of students earning at least a C in their first college mathematics course		
	M	*SD*	*n*	*M*	*SD*	*n*
Met the standard	62	3.3	5,181	87	5.5	7,323
Cat1	65.1	4.1	2,843	88.4	6.2	3,883
Cat2	69.3	4.5	639	89	6	840
Cat3	56	2.4	1,699	85.3	3.8	2,600
Diff1	61	3.0	2,382	85.2	5.6	3,428
Diff2	62.5	3.8	1,994	87.2	5.4	2,823
Diff3	70.2	5.1	805	92.5	7.1	1,072

Note. A total of n = 8,460 students met the ACT mathematics standard, and of these, n = 5,181 earned at least a B and n = 7,323 at least a C in their first college mathematics course. Percentages in the table are weighted by the number of students who met the standard for each school. Cat1 = institution in which ≥80% of students enrolled full-time/more selective/low transfer-in rate (n = 4,465, K = 14 institutions), Cat2 = ≥80% full-time/selective/high transfer-in rate (n = 947, K = 9), Cat3 = mixture of full-time and part-time/selective/high transfer-in rate (n = 3,048, K = 4). Diff1 = courses whose difficulty is consistent with College Algebra (n = 4, 044), Diff2 = courses whose difficulty is consistent with Calculus I (n = 3,246), Diff3 = courses whose difficulty is consistent with Calculus II or beyond (n = 1,170). ACT = American College Testing.

Among students who met the standard, the average percentage at an institution who earned at least a B was highest for courses of Difficulty Level 3 (Calculus II and beyond, 70.2%) and lower for less difficult courses (61%, 62.5%). The average percentage of students earning at least a C in their first college mathematics course followed a similar pattern. The pattern of results for earning at least a B and at least a C was similar for the analyses reported below, and only the former are reported.

We also examined patterns for the n = 2,864 (25.3% of the total sample of N = 11,324) students who did not meet the standard. Among these students, the average percentage at an institution who earned at least a B (not reported in Table 2) was 39.3 (n = 1,033). The average percentages of students who did not meet the standard but earned at least a B for Institutional Profile Categories 1 through 3 were 22.2, 49, and 30.3, respectively; for Difficulty Levels 1 and 2, the averages were 40.1 and 33.5 (there were too few students to compute an average for Difficulty Level 3).

Table 3 reports the average model-predicted probability a student will earn at least a B based on meeting or not meeting the ACT mathematics standard, and by institutional category and course difficulty. These results equal or are generally somewhat higher than the ACT standard of a probability of .50 for earning at least a B. The least selective institutions (Category 3) produced an average probability close to the ACT standard of .50 for earning at least a B (.52), and Difficulty Level 1 courses (e.g., College Algebra) produced an average probability equal to .50.

Multilevel Results for Earning at Least a B in the First College Mathematics Course

We initially fitted a fully unconditional (no predictors) model and found significant differences among institutions in the average log odds of a student earning at least a B relative to earning less than a B. Next, we fitted a student model with a single predictor indicating whether a student met

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Table 3. Mean Predicted Probability That Students Will Earn at Least a B or at Least a C in Their First College Mathematics Course Based on Meeting or not Meeting the ACT Mathematics Standard by Institutional Profile Category and Difficulty Level.

	M probability of earning at least a B in the first mathematics course		*M* probability of earning at least a C in the first college mathematics course	
	M	*SD*	*M*	*SD*
Overall	.60	.11	.82	.07
Cat1	.62	.12	.86	.07
Cat2	.62	.10	.84	.06
Cat3	.52	.07	.83	.05
Diff1	.50	.14	.83	.08
Diff2	.62	.12	.86	.01
Diff3	.68	.02	.91	.02

Note. Values in the table are the mean model-predicted probabilities of earning at least a B and at least a C weighted by the number of sampled students from each institution. Sample size for the overall probabilities was $n = 11{,}131$. Mean percentages are weighted by the number of sampled students for each school. Cat1 = institution in which ≥80% of students enrolled full-time/more selective/low transfer-in rate ($n = 5{,}901$, $K = 14$ institutions), Cat2 = ≥80% full-time/selective/high transfer-in rate ($n = 1{,}386$, $K = 9$), Cat3 = mixture of full-time and part-time/selective/high transfer-in rate ($n = 3{,}844$, $K = 4$). Diff1 = courses whose difficulty is consistent with College Algebra ($n = 6{,}551$, $K = 27$), Diff2 = courses whose difficulty is consistent with Calculus I ($n = 3{,}419$, $K = 27$), Diff3 = courses whose difficulty is consistent with Calculus II or beyond ($n = 1{,}161$, $K = 27$). ACT = ACT = American College Testing.

the standard (institution model was unconditional). Students who met the standard were 2.83 times more likely to earn at least a B in their first college mathematics course compared with students who did not meet the standard.

We then fitted the models in Equations 1 and 2, and report the results in Table 4. The significant fixed effect for whether a student met the ACT mathematics standard ($\hat{\gamma}_{70} = 1.099$) means that, with other predictors held constant and the slope exponentiated, $\exp(1.099) = 3.00$, students who met the standard were 3 times more likely to earn at least a B in their first college mathematics course as students who did not meet the standard. The R^2_{pseudo} for the fixed effects including Met the Standard was .21 (21%), whereas $R^2_{\text{pseudo}} = .14$ for the fixed effects without Met the Standard, suggesting that the Met the Standard predictor accounted for approximately 7% of the variation in logits. Our results showing that meeting the ACT mathematics standard was related to college mathematics performance are consistent with several ACT studies that examined this relationship (ACT, 2005, 2012c, 2013a).

The significant exponentiated effect for the predictor capturing whether a student was Black or White ($\hat{\gamma}_{50} = -.505, \exp(-.505)) = .60$) means Black students were more than 1½ times less likely than White students to earn at least a B in their first college mathematics course. The other significant student predictor in Table 4 was Difficulty1 ($\hat{\gamma}_{20} = -.443, \exp(-.443) = .64)$), so students whose first college mathematics course was Calculus II or a more difficult course were less likely to earn at least a B than students taking a course whose difficulty was consistent with College Algebra.

Two statistically significant institutional effects appeared both linked to the Profile2 predictor. The significant $\hat{\gamma}_{02} = .841$ slope means that students in Category 2 institutions were about $\exp(.841) = 2.32$ times more likely to earn at least a B in their first college mathematics courses compared with students in Category 3 institutions. One significant cross-level interaction also emerged. The significant $\hat{\gamma}_{12} = -.908$, $\exp(-.908) = .40$, effect means that a pattern of males being less likely to earn at least a B in their first college mathematics courses compared with females was exacerbated for Category 3 institutions compared with Category 2 institutions.

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Table 4. Multilevel Results for Students Earning at Least a B in Their First College Mathematics Course.

Fixed effect	Coefficient	*SE*	*t* ratio	Approximate *df*	*p* value
For INTRCPT1, β_0					
INTRCPT2, γ_{00}	−0.211	0.311	−0.678	24	.504
Profile1, γ_{01}	0.496	0.337	1.470	24	.155
Profile2, γ_{02}	0.841	0.358	2.350	24	.027*
For Gender slope, β_1					
INTRCPT2, γ_{10}	−0.209	0.273	−.765	24	.452
Profile1, γ_{11}	−0.109	0.305	−.357	24	.725
Profile2, γ_{12}	−0.908	0.331	−2.743	24	.011*
For Difficulty1 slope, β_2					
INTRCPT2, γ_{20}	−0.443	0.078	−5.678	11,018	.001
For Difficulty2 slope, β_3					
INTRCPT2, γ_{30}	−0.238	0.220	−1.083	24	.290
Profile1, γ_{31}	−0.424	0.237	−1.784	24	.087
Profile2, γ_{32}	−0.081	0.280	−0.289	24	.775
For Asian slope, β_4					
INTRCPT2, γ_{40}	−0.432	0.240	−1.802	24	.084
Profile1, γ_{41}	−0.155	0.300	−0.519	24	.609
Profile2, γ_{42}	0.144	0.404	.356	24	.725
For Black slope, β_5					
INTRCPT2, γ_{50}	−0.505	0.153	−3.306	11,018	.001
For Hispanic slope, β_6					
INTRCPT, γ_{60}	−0.215	0.174	−1.235	11,018	.217
For MetStand slope, β_7					
INTRCPT2, γ_{70}	1.099	0.053	20.606	11,018	.001

Note. * = statistically significant; Profile1 and Profile2 are dummy-coded predictors capturing an institution's profile category with Category 3 serving as the reference group; Gender (1 = male, 0 = female), MetStand (1 = yes, 0 = no), Difficulty1, and Difficulty2 are dummy-coded predictors capturing the difficulty level of a course with Level 3 (Calculus II or beyond) serving as the reference group; Asian (1 = yes, 0 = no), Black (1 = yes, 0 = no), Hispanic (1 = yes, 0 = no) meaning White students served as the reference group. Predictors were not centered, and α = .05 for each statistical test.

As an ancillary analysis, we formally examined whether the mathematics standard was more or less predictive of college mathematics achievement for students who completed 3, 4, or 5 years of high school mathematics by adding this variable as a student-level predictor for model in Equation 1. The multilevel results indicated that the findings reported in Table 4 did not change in any important way.

Discussion

This study examined the relationship between the ACT college readiness mathematics standard and college mathematics achievement using a large extant data set of students who completed 3, 4, or 5 years of high school mathematics and subsequently attended 1 of 27 four-year postsecondary institutions. Our findings help to fill an important gap in the college readiness literature by using data not collected by ACT to examine the relationship between the mathematics standard and college mathematics achievement for students who met or exceeded the minimum high school mathematics coursework recommended by ACT, and the impact of institutional and student characteristics on this relationship.

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Our findings are generally consistent with previous work involving the mathematics standard in which the high school mathematics preparation of samples of students varied from no mathematics to 5 years of coursework (Calculus I; ACT, 2005, 2012b, 2012c, 2013b).

Students in our sample who met the standard were 3 times more likely to earn at least a B in their first college mathematics course compared with those not meeting the standard, and slightly more than 2½ times more likely to earn at least a C. The probability a student will earn at least a B or at least a C in his or her first college mathematics course was generally somewhat higher in our data than the ACT standards of .50 and .75, most likely because of our requirement that sampled students had completed at least 3 years of high school mathematics. In conjunction with previous studies of the relationship between the ACT mathematics standard and college mathematics performance, our results suggests that the standard is a useful predictor of success in the first college mathematic class for students with a range of high school mathematics preparation.

We also found a difference in the likelihood of earning at least a B (or at least a C) among Black and White students, which is consistent with evidence of an achievement gap in college mathematics achievement (ACT, 2013a; Mattern, Patterson, & Kobrin, 2012). These results reinforce the need for continued research to identify strategies to eliminate such differences. However, we found no evidence of overall gender differences, which have sometimes appeared for college mathematics achievement (ACT, 2013a).

Our finding of differences in the probability of earning at least a B (or at least a C) among institutions with different characteristics is consistent with literature showing the impact of these factors on student outcomes (e.g., Kobrin & Patterson, 2011; Sawyer, 2013). Given differences in institutional selectivity, it was not surprising to find that the average log odds of earning at least a B (or at least a C) in a student's first college mathematics course was higher in more selective (Category 2) institutions compared with less selective (Category 3) institutions. The presence of a cross-level interaction in which the relationship between gender and the likelihood of earning at least a B (or at least a C) in a student's first college mathematics course was moderated by institutional selectivity is also consistent with a modest literature in which these effects appear intermittently for several student outcomes (ACT, 2013a; Peters, 2013). However, the mechanism responsible for this pattern is unclear.

Implications

The consistency between our findings and those of previous studies of the impact of the ACT mathematics standard on student performance in their first college mathematics course has at least three important implications. First, the relative homogeneity of the mathematics preparation of our sample, as reflected in 80% of the students completing 4 or 5 years of high school mathematics coursework, suggests that the usefulness of the ACT mathematics standard as a predictor of a student's performance in their first college mathematics class generalizes even when students have a narrow range of high school mathematics preparation.

Second, our results have implications for advising college-intending high school students and newly admitted college students. For example, our results, in conjunction with previous studies, suggest that students with a range of high school mathematics preparation who do not meet the ACT mathematics standard are less likely to earn at least a B in their first college mathematics course compared with meeting the standard. This can inform a decision to advise college-intending students of the importance of continuing to improve their mathematics skills and to take the ACT a second time to learn whether these skills have improved sufficiently to obtain a score of at least 22.

Finally, our results and those of previous studies can inform recommendations about the mathematics course newly admitted college student with a range of high school mathematics preparation should begin with: Meeting the mathematics standard suggests a student has a good chance

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of passing College Algebra. As emphasized by ACT, however, the use of the mathematics standard should be one of several sources of information related to educational planning and decision making.

Limitations

One limitation of our study is that all of the students completed high school in one U.S. state (Minnesota) in which the mathematical achievement of students is on average higher than students in most U.S. states. Minnesota is one of the five highest performing states in mathematics as reflected in the percentage of college-intending students meeting the ACT mathematics standard and average ACT mathematics score (ACT, 2014). Relatedly, all 27 postsecondary institutions were located in one region of the United States meaning that our findings do not necessarily generalize elsewhere.

Another limitation is the potential impact of variation in the quality of teaching within and between institutions and institutional assessment practices (Wayne & Youngs, 2003). Unfortunately, we had no data capturing these factors. Our use of difficulty levels to categorize college mathematics courses and dichotomizing college mathematics grades may help to reduce the impact of such differences, but this remains a limitation.

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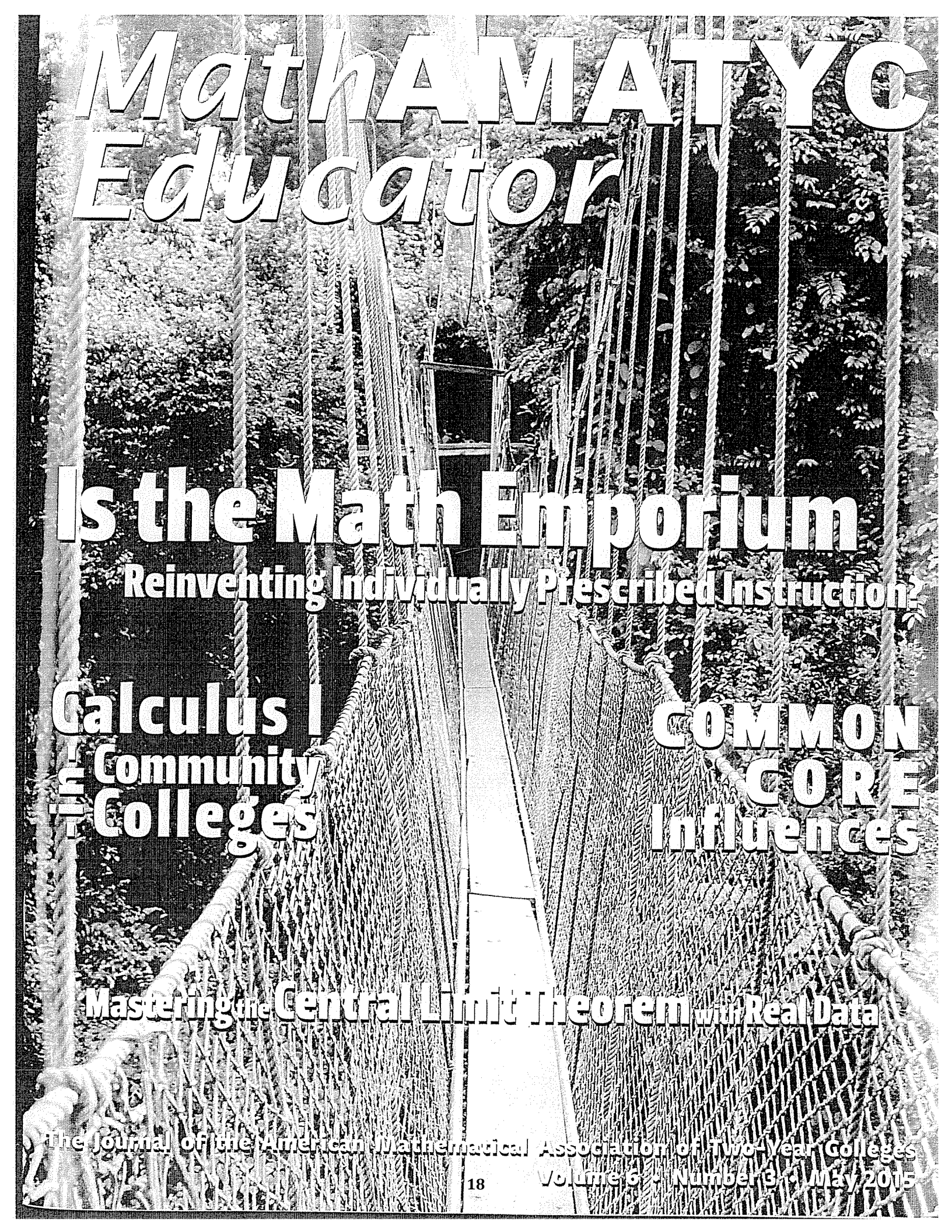

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The Relationship between High School Mathematics Curriculum and Mathematics Course-Taking and Achievement for Students Attending Community College

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Studying the impact of high school mathematics curricula on postsecondary mathematics course-taking and achievement is motivated by two related objectives. First, there is a need to better understand the mathematics preparation of an increasingly diverse group of students planning to enroll in a postsecondary institution (National Center for Education Statistics [NCES], 2014; Programme for International Student Assessment [PISA], 2012). Second is the importance of establishing an empirical research base characterizing the efficacy of high school mathematics curricula in pursuit of the common goal of having more students learn more mathematics more effectively.

Common to meeting these objectives is developing a deeper understanding of the key features of a student's high school mathematics preparation. We focus on the high school mathematics curriculum a student completed, which is widely viewed as an important component of a student's success in postsecondary mathematics (Augustine, 2007; Mathematical Sciences Education Board [MSEB], 2004; National Research Council [NRC], 2002). However, calls for additional research to better understand the impact of high school mathematics curricula on students' college mathematics performance (MSEB, 2004; NRC, 2002; NCES, 2006) means that there is a need to more thoroughly document the preparation of students for postsecondary mathematics.

We focus on understanding the mathematics preparation of students who enroll in a community college for three overlapping reasons. First, approximately 41% of first-time students in postsecondary education enroll in community colleges (NCES, 2014). Moreover, enrollment in community colleges has recently risen while enrollment in four-year institutions has remained steady. As of October 2008, approximately 3.4 million young adults (aged 18–24) in the U.S. were enrolled in community colleges, an increase from 3.1 million the previous year (Pew Research Center, 2009). Thus, community college students represent a large and growing population of students.

Second, community colleges provide an important educational access point to four-year schools, and ensuring that these students have adequate mathematics preparation can be central to students' successfully transferring to a four-year school after satisfactorily completing coursework at a community college. Many students who plan to earn a bachelor's degree opt to begin their postsecondary study in a community college, sometimes because they need additional coursework before being able to enroll at a four-year school, sometimes for financial, family, or job-related reasons, and sometimes because it's simply more convenient (Alfonso, 2006; Bailey & Averianova, 1998). The fact that community colleges have, historically, had proportionally higher enrollments of non-White and low-income students compared to four-year schools also speaks to the access issue (Bahr, 2008). Relatedly, students attending community college typically require more developmental coursework in mathematics than their counterparts in four-year schools (Adelman, 2004; Minnesota State Colleges and Universities, 2011; NCES, 2003), highlighting the importance of understanding their preparation for postsecondary mathematics.

Third, many students earn a terminal degree at a community college and understanding the mathematics preparation of this group of students as a function of the high school mathematics curriculum they completed is important. These students are especially likely to need to complete developmental mathematics courses as part of satisfying institutional mathematics requirements (Alfonso, Bailey, & Scott, 2005). A better understanding of the impact of high school mathematics curriculum may provide important information for institutions seeking to better support the mathematics learning of these students.

Theoretical Framework

High School Mathematics Curricula

There are three general categories of high school mathematics curricula. The most widely used mathematics curricula in the nation's high schools are referred to here as commercially developed (CD) or traditional curricula that stress traditional algorithms and procedures (Schoenfeld, 2004). These curricula tend to be teacher-centered, limit the role of technology, and discourage group work (Roitman, 1999; Schoenfeld, 2004). Available evidence suggests that a theoretical model incorporating research findings on student learning has not guided the development of CD curricula (Willoughby, 2010).

National Science Foundation-funded (NSF-funded) curricula represent a second category of curricula. These curricula were funded through a solicitation of proposals through the National Science Foundation in the early 1990s (RFP NSF 91-100) and were designed to be aligned with the National Council of Teachers of Mathematics' (NCTM) *Curriculum and Evaluation Standards for School Mathematics* (1989). The mathematics curricula funded by NSF included *Contemporary Mathematics in Context* (*CMIC* or Core-Plus) (Coxford, Fey, Hirsch, Schoen, Burrill, Hart, & Watkins, 1998), *Interactive Mathematics Program* (*IMP*) (Fendel, Resnick, Alper, & Fraser, 1998), and *Mathematics: Modeling Our World* (*MMOW* or ARISE) (Garfunkel, Godbold, & Pollak, 1998). A theme of these curricula is their integration of topics in algebra, geometry, probability, statistics and discrete mathematics, their deemphasizing of algorithmic manipulation, and their emphasis on the role of students as active participants in learning that is motivated by real world questions and relies heavily on small groups (Schoenfeld, 2004). Available evidence suggests that the development of NSF-funded curricula has been guided by, but not dictated by, theoretical models incorporating research findings on student learning (Post et al., 2010).

Another category of curriculum is represented by the University of Chicago School Mathematics Project (Usiskin, 1986) curriculum, which was initially developed in 1983 and combines features of NSF-funded and CD curricula (Hirschhorn, 1993; Schoenfeld, 2004; Thompson & Senk, 2001). The UCSMP curriculum has received funding from private sources (AMOCO) as well as NSF support. It is best known through the algebra, geometry, algebra II, functions, statistics, trigonometry, precalculus and discrete mathematics texts that emphasize reading, problem solving, typical or everyday applications of mathematics, and the use of technology including calculators and computers. The ongoing development of the UCSMP curriculum has in large part been informed by a theoretical model that incorporates research findings on student learning.

The Impact of High School Mathematics Curricula on Postsecondary Mathematics Achievement

Research examining the impact of high school mathematics curricula on students' postsecondary mathematics achievement has only recently begun to appear in the literature. The focus on NSF-funded curricula in this research is in part a response to intense criticism of these curricula (Schoenfeld, 2004). The essence of the criticism is that NSF-funded curricula do a poor job of preparing students to succeed in postsecondary mathematics, especially calculus (Klein, 2000; Wu, 1997). However, there is no credible empirical evidence that we could find supporting this criticism for students enrolled in four-year institutions and none for students enrolled in community colleges.

For four-year schools, Schoen and Hirsch (2003) reported that students completing the NSF-funded Core-Plus curriculum had on average higher grades in calculus I and other more advanced mathematics courses compared to students completing a CD curriculum. Hill and Parker (2006) reported that students who completed Core-Plus in high school began their college mathematics coursework with less difficult courses and subsequently completed even less difficult mathematics courses than students who completed a different mathematics curriculum in high school. However, their findings were severely compromised by the fact that often the high school mathematics curriculum students completed could not be determined. Harwell et al. (2009) found no relationship between curriculum and the grade earned in a student's first college mathematics course for a sample of students enrolled in a single large public institution, but did find that students who completed a NSF-funded curriculum tended to enroll in less difficult courses than students who completed a CD curriculum in high school.

For community colleges, Bahr (2007) found evidence of multiple academic deficiencies for freshmen including inadequate mathematics preparation and described their negative impact on postsecondary success. Specifically, Bahr reported that only 28% of community college students who began their college mathematics course-taking with a developmental course went on to successfully complete a nondevelopmental mathematics course. Adelman (2006) reported a strong relationship between completion of four years of high school mathematics culminating in a college algebra course in high school and success in college, a finding also reported for community college students by Bettinger and Long (2005), Fong, Huang, and Goel (2008), and Stern and Pavelchek (2006).

However, there is apparently no research examining the role of specific high school mathematics curricula on students' mathematics performance in community colleges. To better understand this relationship we posed the following questions.

Research questions

1. Do students who completed at least three years of high school mathematics in a NSF-funded curriculum differ from those who had similar exposure to a CD or UCSMP curriculum in the difficulty level of their first college mathematics course or in the grade they earned in that course, when taking into account background factors such as prior mathematics achievement, ethnicity, and sex?
2. Does this relationship vary across a sample of two-year community colleges?

We chose the grade a student earned in their first college mathematics course and the difficulty level of that course as the outcomes of interest because the majority of students (54.2%) in our sample only completed a single course in mathematics in community college (29.9% completed two courses, 15.9% completed three or more courses).

Method

Research Design

A retrospective cohort (quasiexperimental) design was used in which the grade students earned in their first college mathematics course, and the difficulty level of that course, served as outcome variables. The grade data were analyzed separately for students who began community college with a developmental mathematics course and those who began with a nondevelopmental mathematics course (e.g., college algebra). In addition, we analyzed the grade data separately for students who completed two or more courses in mathematics. This approach is consistent with Pedhazur and Schmelkin's (1991) elimination strategy, in which a covariate is converted to a constant and enhances the credibility of inferences. Based on their high school mathematics curriculum students were categorized into one of three curriculum cohorts: NSF-funded, CD, or UCSMP. To identify students' high school mathematics curriculum, we contacted each of the 290 high schools represented in our sample and obtained descriptions of their mathematics programs, including the mathematics courses offered and the mathematics textbooks used during the period the students were in high school (1998–2002). For approximately 25% of students, the high school they attended used more than one curriculum, which required cataloguing the high school mathematics courses that appeared on the transcripts of these students and combining this with collateral information obtained from the high schools to assign these students to a curriculum cohort.

We combined the NSF-funded curricula (Core-Plus, IMP, MMOW) in the analyses for two reasons. First, there are strong conceptual links among these curricula including their adherence to a common set of standards in their construction that features (i) contextually oriented mathematics in which problems emanate from real world situations (voting, environment, change over time etc.) (ii) the use of cooperative group solutions to problems and student-centered investigations (iii) wide-scale use of technology, student-centered investigations and (iv) a focus on algebra, geometry, probability, and statistics each year (NCTM, 1989). Second, we found no important differences among students completing the Core-Plus, IMP, and MMOW curricula on high school variables including ACT mathematics score, high school mathematics GPA, and years of high school mathematics, and, separately, on college variables including first college mathematics course grade, college mathematics GPA, and number of college mathematics courses completed. Thus we did not distinguish among individual NSF-funded curricula in our analyses.

Population and Sample

The sampled populations consist of 3,197 students who enrolled in 21 two-year community colleges in the upper Midwest of the U.S. All 21 institutions are part of a state colleges and universities system in a single state. The institutions sampled here include three state colleges, eight community colleges, eight community and technical colleges, one technical college and one tribal and community college. Fifteen of the institutions are located in rural settings, five in suburban settings, and one in an urban setting. These institutions all have an open enrollment policy.

All students in our sample completed at least three levels (years) of high school mathematics in a CD, NSF-funded, or UCSMP curriculum, and completed at least one postsecondary mathematics course. Also, all students enrolled in a community college in fall 2002 or fall 2003 immediately after graduating from high school. The requirement that students had to have completed at least three years of high school mathematics helped to ensure sufficient exposure to a curriculum for its effects to appear. However, this requirement also makes it likely that our target population of students is on average more academically proficient than would be the case for a broader population of community college students.

Variables

Variables in our student data included the type of high school mathematics curriculum (CD, NSF-funded, UCSMP), which was coded so that the CD curriculum was the reference group, various measures of students' prior mathematics achievement (e.g., high school mathematics GPA, ACT mathematics score), grade earned in the first college mathematics course (0.0–4.0 scale), and demographic information (e.g., ethnicity and sex).

The difficulty of each mathematics course at each institution was categorized into one of four difficulty levels based on course descriptions:

Level 1: Courses that should have been completed at the high school level and generally do not earn college credit. Students entering college without the required or recommended high school background are directed to these courses, which are designed to provide a foundation for students to be successful in future mathematics courses. This is the level we consider to be remedial or developmental.

Level 2: Courses that a student who satisfactorily completed four levels (years) of mathematics in high school would not repeat in a college setting (e.g., college algebra, precalculus mathematics).

Level 3: Beginning calculus.

Level 4: Courses whose difficulty is consistent with that of calculus II or higher (e.g., differential equations, college geometry, calculus II).

These levels are similar to those used in Teitlebaum's analysis of high school mathematics course-taking using the National Educational Longitudinal Study data.

Data Analyses

Initially descriptive analyses were performed to explore patterns in the data. These were followed by two-level hierarchical linear modeling (HLM) (Raudenbush & Bryk, 2002), which treated students as nested within community colleges. To control for compounding of Type I error rates we used an adjusted Type I error rate attributed to Sidak (1967) of the form $\alpha' = 1-(1-\alpha)^{1/k}$, where α is the unadjusted Type I error rate, $k =$ number of statistical tests, and α' is the adjusted Type I error rate. We chose to compute adjusted Type I error rates for each table of statistical test results we report based on an overall $\alpha = 0.10$, which resembles a family-wise error rate strategy often associated with planned comparisons in ANOVA.

Results

Descriptive Analyses

Among the sampled community colleges the average enrollment (full time + part time) was 2,038.31 ($SD = 941.23$), and the average mean ACT mathematics score was 20.64 ($SD = 0.97$). Summaries for number of years of high school mathematics completed, high school mathematics GPA, ACT mathematics score, grade in first college mathematic course, and the difficulty level of that course by high school mathematics curriculum across institutions are reported in Table 1.

The variance in a dependent variable explained by an independent variable was used as an index of effect size for the relationships in Table 1, with larger effects associated with stronger relationships. On the whole, effect sizes comparing the NSF-funded and UCSMP cohorts, and the NSF-funded and CD cohorts, uniformly favored the UCSMP and CD cohorts with most effect sizes in the moderate range (Cohen, 1988). For example, the effect size for ACT mathematics scores as the outcome between the NSF-funded versus UCSMP curriculum cohorts was 0.27, meaning that 27% of the variance in ACT mathematics scores was explained by whether students completed an NSF-funded or UCSMP high school

Table 1. Descriptive Statistics by High School Mathematics Curriculum

	HS Mathematics Curriculum								
	Commercially Developed			UCSMP			NSF-funded		
	M	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*
Yrs. HS mathematics completed	3.55	0.66	1622	3.72	0.63	440	3.55	0.62	422
HS mathematics GPA	2.68	0.75	1618	2.74	0.74	439	2.77	0.75	419
ACT mathematics score	21.05	4.03	662	21.63	3.94	235	20.58	3.87	163
Grade in first college mathematics course	2.64	1.16	1535	2.59	1.18	417	2.6	1.11	393
Difficulty level of first college math course	1.64	0.7	1622	1.69	0.74	440	1.47	0.59	422

Note: Difficulty level 1 = course that should have been completed in high school, 2 = college algebra, 3 = Calculus I, 4 = Calculus II or a more difficult course.

Table 2. Course Difficulty Level by Years of High School Mathematics Completed

		Years of High School Mathematics Completed					
		3 years		4 years		5 years	
		n	%	*n*	%	*n*	%
Difficulty level of first college math course	1	925	55.8	497	40	33	12.4
	2	703	42.4	658	53	138	51.9
	3	20	1.2	67	5.4	66	24.8
	4	10	0.6	19	1.5	29	10.9
Difficulty level of last college math course	1	440	26.5	193	15.6	14	5.3
	2	1151	69.4	887	71.5	130	48.9
	3	37	2.2	84	6.8	41	15.4
	4	30	1.8	77	6.2	81	30.4

Note: Difficulty level 1 = course that should have been completed in high school, 2 = college algebra, 3 = Calculus I, 4 = Calculus II or a more difficult course.

mathematics curriculum (examination of the means showed that NSF-funded students scored on average lower); for NSF-funded versus CD students this effect size was 0.12 favoring CD students.

Descriptive statistics for students by the number of years of high school mathematics completed are reported in Table 2. Of the students sampled, 52.4% (1,658) completed three years of high school mathematics, 39.2% (1,241) completed four years of high school mathematics and the remaining 8.4% (266) completed a fifth year (typically calculus) of high school mathematics. As can be seen in Table 2, the percentage of students who initially enrolled in a developmental mathematics course in college was 55.8%, 40.0%, and 12.4% for students who completed 3, 4, or 5 years of high school mathematics, respectively. The mean (standard deviation) grade earned in students' first college mathematics course was 2.54 (1.14), 2.67 (1.17), and 2.79 (1.16) for students who completed 3, 4, or 5 years of high school mathematics, respectively.

Summary statistics for students for high school mathematics curriculum, gender, and ethnicity by the institution location variable are reported in Table 3. As can be seen, more NSF- funded students (38%) attended urban/suburban institutions than CD students (29%). This is likely the case because the NSF-funded curricula were predominantly implemented in urban and suburban high schools and these students disproportionately went on to attend urban and suburban community colleges. African American, Hispanic, and Asian students predominantly enrolled in urban and suburban institutions whereas nearly 98% of the students attending rural institutions were Caucasian.

Inferential Analyses

Difficulty level of the first mathematics course

First an unconditional hierarchical generalized linear model (HGLM) was fitted to the binary difficulty outcome (1 = started community college with college algebra or a more difficult course, 0 = started community college with a developmental mathematics course). The average (across institutions) log-odds of a student beginning with college algebra or more difficult course as opposed to a developmental mathematics course was 0.24. This means that for a typical institution (i.e. with a random effect of 0), the odds of a student taking a first mathematics course of difficulty level 2 or greater relative to taking a developmental course were $e^{0.24} = 1.27$, which corresponds to a model probability that a randomly selected student would begin with a course of difficulty level 2 or higher of $\frac{e^{0.24}}{1+e^{0.24}} = 0.56$.

The results for the conditional HGLM (see Table 4) revealed that the type of high school mathematics curriculum a student completed was unrelated to the likelihood of students beginning their mathematics coursework at difficulty level 2 or

Table 3. Descriptive Statistics by Community College Location

		Community College Location					
		Rural			Urban/Suburban		
		n	Row %	Column %	*n*	Row %	Column %
Curriculum	Commercially Developed	1147	70.7	67.4	475	29.3	60.8
	UCSMP	295	67	17.3	145	33	18.6
	NSF-Funded	261	61.8	15.3	161	38.2	20.6
Gender	Female	984	70.2	42.5	418	29.8	50.1
	Male	1331	76.2	57.5	416	23.8	49.9
Ethnicity	Asian	22	21.6	1	80	78.4	10.6
	African American	19	43.2	0.9	25	56.8	3.3
	Hispanic	8	40	0.4	12	60	1.6
	Caucasian	2121	76.9	97.7	638	23.1	84.5

higher, which means that the difficulty level of students' first course is unrelated to the high school mathematics curriculum they received. ACT mathematics score, high school mathematics GPA, and years of high school mathematics were the only statistically significant predictors of the likelihood a student would begin their mathematics coursework with a course of difficulty level 2 or higher.

Grade earned in the first mathematics course

To analyze the grade data four models were fitted, one for students who began their community college mathematics coursework with a developmental course (i.e., difficulty level 1) and another for students who began their community college mathematics coursework with college algebra or a more difficult course (i.e., difficulty level ≥ 2). In addition, the same two models described previously were fitted to the subsample of students who completed two or more mathematics courses in community college. An initial analysis showed that there was enough variation in average grades between community colleges to continue modeling ($\hat{\tau}_{00} = 0.05, \hat{\rho}_{\text{int raclass}} = 0.03$). Next, we constructed a predictive model to account for the variation in grades. The results of this analysis for all students who began their community college mathematics coursework with a developmental course are presented in Table 5.

These results indicate that the type of high school mathematics curriculum a student completed was unrelated to the grade earned in the first course. In contrast, both ACT

Table 4. Multilevel Results for Difficulty Level Outcome

Effect	β	$SE(\beta)$	*z*-statistic
Intercept	–6.17	0.641	–9.62*
ACT math score	0.23	0.024	9.51*
HS math GPA	0.30	0.110	2.77*
Years of HS math completed	0.38	0.129	2.95*
African American	1.35	1.419	0.95
Asian	–0.59	0.412	–1.42
Hispanic	–0.97	1.045	–0.93
Gender	–0.16	0.150	–1.06
Location	–1.05	0.668	–1.57
NSF-funded	–0.17	0.226	–0.74
UCSMP	0.04	0.201	0.20

Note: Difficulty level 1 = course that should have been completed in high school, 2 = college algebra, 3 = Calculus I, 4 = Calculus II or a more difficult course. * = statistically significant.

Table 5. Multilevel Results for Grade Outcome for Students who began with a Developmental Course

Effect	β	$SE(\beta)$	t-statistic
Intercept	1.54	0.360	4.27*
ACT math score	0.04	0.016	2.36*
HS math GPA	0.39	0.070	5.52*
Years of HS math completed	−0.05	0.083	−0.62
African American	−0.26	0.988	−0.27
Asian	−0.31	0.213	−1.45
Hispanic	−0.2	0.492	−0.41
Gender	−0.31	0.094	−3.27*
Location	−0.18	0.196	−0.92
NSF-funded	−0.08	0.133	−0.56
UCSMP	−0.25	0.131	−1.88

Note: * = statistically significant.

Table 6. Multilevel Results for Grade Outcome for Students who began with a Non-Developmental Course

Effect	β	$SE(\beta)$	t-statistic
Intercept	0.23	0.311	0.73
ACT math score	0.04	0.013	2.79*
HS math GPA	0.55	0.061	8.93*
Years of HS math completed	0.06	0.073	0.79
African American	−0.39	0.593	−0.65
Asian	0.16	0.303	0.51
Hispanic	−1.61	0.729	−2.21
Gender	−0.12	0.085	−1.34
Location	−0.10	0.146	−0.68
NSF-funded	−0.03	0.136	−0.19
UCSMP	−0.02	0.11	−0.16

Note: * = statistically significant.

mathematics score ($\beta = 0.04$) and high school mathematics GPA ($\beta = 0.39$) were significant predictors of the grades students' earned in their first college mathematics course (in this case a developmental mathematics course). Students with higher ACT mathematics scores and higher high school mathematics GPAs tended to earn higher grades in their first mathematics course compared to students with weaker prior mathematics achievement. Finally, gender ($\beta = -0.31$) was a significant predictor with male students on average earning lower grades than their female peers. The results indicated a similar pattern of results for subsample students who began community college with a development mathematics course but went on to complete two or more courses in college.

Next we examined the sample of students who began community college with a nondevelopmental mathematics course (difficulty level ≥ 2). The results of this analysis for all students who began their community college mathematics coursework with college algebra or a more difficult course are presented in Table 6. These results show that the type of high school mathematics curriculum a student completed was unrelated to the grade earned in the first course, as were the variables capturing a student's demographic characteristics. Both ACT mathematics score ($\beta = 0.04$) and high school mathematics GPA ($\beta = 0.55$) were significant predictors of the grades students' earned in their first community college mathematics course. Students with higher ACT mathematics scores and higher high school mathematics GPAs tended to earn higher grades in their first university mathematics course compared to students with weaker prior mathematics achievement. Again, the results were similar for the subsample of students who began with a nondevelopmental course and completed two or more college mathematics courses.

Discussion

The study provides evidence of the relationship between high school mathematics curriculum and postsecondary mathematics course taking and achievement for students enrolled in community colleges. It also provides useful information for high schools and school districts interested in examining the impact of alternative high school mathematics programs on students' subsequent college level performance. This is important for a growing and increasingly diverse population of postsecondary-bound students, but it is also important for all college-intending students. In addition, this information is helpful to the community colleges that will host these students and help them choose appropriate coursework. The lack of a significant curriculum effect suggests that there are several viable curricular approaches to adequately prepare students for college mathematics. Specifically, NSF funded high school curricula (the Connected Mathematics Project, Mathematics

Modeling Our World and the Interactive Mathematics Project—all currently in their second editions) each developed as a full four-year curriculum option to more traditional approaches that have not adequately served all facets of the high school population. This finding is consistent with previous research conducted by the MNMAP Project that sampled students from 32 four-year postsecondary institutions (Harwell et al. 2009, 2012, 2013; Post et al., 2010). The conclusions from those studies and the present study suggest that NSF students are prepared equally well for college-level mathematics, at the four-year college level and at the community college level.

Using Hierarchical Linear Modeling (HLM), this study found that student high school ACT scores, the number of years of high school mathematics, and students' prior mathematics performance (HS mathematics GPA) were all positive significant predictors of success in community college mathematics coursework. Suggesting, as would be expected, that more advanced mathematics course-taking and higher levels of achievement in high school is important to success in postsecondary mathematics. These findings are consistent with Adelman (2004) suggesting that four years of high school college-intending mathematics is the single most powerful predictor for students earning a college degree. Ethnicity was not found to be a significant predictor in any of the models employed, while gender was, with females achieving below that of males, suggesting the need for colleges to be sensitive to these concomitant variables when counseling students into postsecondary mathematics coursework.

Are there advantages of these NSF-funded curricula when compared to more traditional approaches? To a much greater extent students in these programs have worked in groups developing cooperative social networks that are used to support future learning. Students are expected to communicate mathematically through project reports and ongoing verbal and written assignments, to reason through complex mathematical situations over extended periods of time that are often situated in real-world contexts where students are addressing real world problems. Students are also concurrently exposed to both the procedural and conceptual aspects of mathematics with a decided focus on the latter.

The literature suggests that these generalizable skills are also listed as important for vocational success and to success in other academic content domains including, science, social science or literacy. These approaches are also more compatible with the cognitive psychological perspective as to how students are most likely effectively to learn and understand mathematics. The important point to take away from this research is that an alternate approach to high school mathematics, one that is more student-centered and based on real-world contexts, prepares students equally well for community college mathematics while providing students with opportunities to think, communicate, and reason with others in small and large group settings.

The ubiquitous goal of school mathematics is to have more students learn more mathematics more effectively. This research suggests that alternate approaches as exemplified in the NSF curricula materials, have much to offer in that regard. It also suggests that community college advising and mathematical course placement decisions would be well advised to assimilate this information into their initial student assessment protocols.

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A note regarding this research

This study is situated in the context of an NSF-funded project to provide support for high schools using NSF funded mathematics curricula. In the late1990s, the NSF funded the Minneapolis and St. Paul Merging to Achieve Standards Project (MASP)^2 housed at the University of Minnesota's College of Education and Human Development. Professors Tom Post, Mathematics Education, and Mike Harwell, Quantitative Methods, were PIs and were supported by a variety of school personnel and graduate students. Most of these projects released their beta versions in 1998 or 1999. So, the secondary students all of whom would have completed three or four years of one of these curricula would not have graduated from high school until 2002 or 2003 and a minimum of four years later (2006 or 2007) would have graduated from a four-year institution. Community college students had a similar time trajectory but with a year or two shorter duration until graduation.

The result was a second proposal to the NSF to examine the college mathematics performance of students who had taken three or four years of these curricula while in high school. The proposal was funded circa 2006 and became known as the Minnesota Mathematics Assessment Project (MNMAP) Project. The MNMAP Project was funded by the NSF to compare the mathematical experiences (achievement, difficulty of postsecondary mathematics courses taken over time, etc.) of postsecondary students.

Critical to MNMAP was identifying the high school mathematics curricula completed by each of the students who had graduated from one of 350 Minnesota high schools and subsequently graduated from one of the 56 postsecondary

institutions in 2007 or 2008. This information was obtained by contacting each of the 350 Minnesota high schools with students who had graduated from one of the 56 institutions. This process generated a sample of more than 20,000 students.

We argue that the value of our findings is not undermined by the time it took to obtain and analyze the data from 24 community colleges because the relationship under investigation (i.e., impact of high school mathematics curricula on a student's mathematics achievement) continues to be relevant and important to study because states, school districts, and teachers continue to be faced with choosing a high school mathematics curriculum.

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Thomas Post is a professor of mathematics education and an affiliate of the STEM Education Center at the University of Minnesota. He is a founding member of the Rational Number Project. At 30 years and well over 100 publications, the RNP is the longest lasting federally funded (NSF) cooperative research project in the history of mathematics education. He is currently the co-PI of the Minnesota Mathematics Assessment Project (NSF) that has, to date, published thirteen papers, all investigating the relationship between high school mathematics curricula and subsequent college-level mathematics achievement. This article is the latest in this body of research.

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A Multisite Study of High School Mathematics Curricula and the Impact of Taking a Developmental Mathematics Course in College

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The relationship between high school mathematics curricula and the likelihood of students who enroll in a developmental (non-credit bearing) course in college taking additional mathematics courses was studied. The results showed that high school mathematics curriculum, years of high school mathematics completed, and ACT mathematics scores were related to developmental mathematics course-taking, but curriculum was not related to the subsequent mathematics course-taking of students who began college with developmental mathematics. The results have important implications for educational researchers and policymakers at the college and high school levels.

Students whose preparation for college mathematics is unsatisfactory continue to enroll in U.S. colleges in large numbers, prompting postsecondary institutions to invest significant resources to support mathematics learning (Bettinger & Long, 2009) (We define college as a four-year postsecondary institution offering a bachelor's degree). Among these resources are developmental mathematics courses, which are courses intended to lead to a student attaining the expected mathematics competency of entering college freshmen (Bahr, 2007). A 2003 report by the National Center for Education Statistics (NCES) stated that 16% of freshmen in four-year public institutions in the U.S. complete at least one developmental mathematics course, and 8% in private schools.

The prominence of developmental mathematics coursework in college has led to a rapidly growing research literature whose goals include a deeper understanding of its effectiveness and antecedents. This literature is frequently characterized as a debate between researchers and policy-makers who see developmental course-taking as enhancing college access, and those who see developmental courses as misguided because of the academic deficiencies of students enrolling in these courses (Attewell, Lavin, Domina, & Levey, 2006; Lavin & Weininger, 1998). However, there is agreement that students need to be better prepared for college mathematics (Augustine, 2007; NCES, 2006), thus reducing the need for developmental coursework and, correspondingly, the time, effort, and money invested in this activity by students and colleges.

Evidence of effectiveness has typically examined the impact of completing developmental mathematics courses on subsequent academic attainment especially college graduation. This literature is mixed, with some studies reporting that completing at least one developmental mathematics course reduces the likelihood that a student will stay in college, earn good grades, and graduate (Adelman, 2004; Deil-Amen & Rosenbaum, 2002; Martorell & McFarlin, 2010), whereas other studies report the opposite (Bettinger & Long, 2009; Lavin & Weininger, 1998). The search for antecedents has largely centered on using high school variables (e.g., GPA, percentile rank) as predictors of developmental mathematics course-taking (Attewell et al., 2006; Bettinger & Long, 2009; Hoyt & Sorensen, 2001).

Despite a burgeoning literature in developmental mathematics there are important gaps in our understanding of the antecedents of developmental coursework (Attewell et al., 2006; Tierney & Garcia, 2011). In particular, there is little literature examining the impact of high school mathematics curricula, an important component of the mathematics

preparation of college-bound students, on the likelihood of students enrolling in a developmental mathematics course in college or their subsequent mathematics course-taking. There is also little evidence of the extent to which these patterns generalize across different kinds of postsecondary institutions.

Purpose of the Study

This study examined the relationship between the high school mathematics curriculum a student completed and the likelihood of students who enrolled in a developmental mathematics course in college taking additional mathematics courses for a sample of postsecondary institutions.

Developmental mathematics course-taking in college

How do students wind up in a developmental *mathematics course*?

The most common way to assess the mathematics preparation of newly enrolled students for mathematics involves placement tests. An NCES (2003) report noted that 57% to 61% of U.S. institutions used placement tests to identify students in need of developmental coursework, with some estimates as high as 90% of institutions (Education Week, 1994). Meristosis and Phipps (2000) suggested that institutions claiming not to use placement tests typically use the mathematics portions of the ACT or SAT tests as de facto placement tests. In practice, placement in a developmental mathematics course appears to rely on a combination of evidence including placement test(s) and information reflecting a student's preparation for college mathematics such as number of years of college-intending high school mathematics, GPA in high school mathematics courses, and ACT/SAT mathematics score (Attewell et al., 2006; Mzumara & Shermis, 2001).

What are the characteristics of students in developmental mathematics courses in college?

Students who complete developmental mathematics courses tend to have lower high school GPAs and ACT or SAT scores, are disproportionately non-White, older, female, and more likely to come from a low SES background compared to students not taking such courses (Adelman, 2004; Merisotis & Phipps, 2001; NCES, 2003, 2008). Still, it is important to emphasize that there are many exceptions to these patterns. For example, Attewell et al. (2006) used a sample from the National Educational Longitudinal Survey (NELS) and reported that 24% of the students in the highest SES quartile completed a developmental mathematics course, and Adelman (2004) reported that for the High School and Beyond and NELS datasets a total of 6.8% of the students in the top SES quintile completed at least one developmental mathematics course.

High school mathematics curricula

There are many reasons to believe that the high school mathematics curriculum a student completes plays a key role in their preparation for college mathematics, but there is disagreement over the ability of different curricula to prepare students for college mathematics (Schoenfeld, 2004). High school mathematics curricula in the U.S. can generally be categorized as those that are developed with funding from the National Science Foundation, commercially developed (CD) curricula, or the University of Chicago School Mathematics Project (UCSMP) curriculum.

National Science Foundation-funded curricula were developed with support from NSF in the early 1990's with the expectation that the development of new curricula would be guided by the *Curriculum and Evaluation Standards for School Mathematics* (National Council of Teachers of Mathematics [NCTM], 1989). The current study included students who

completed one of three NSF-funded high school curricula: *Contemporary Mathematics in Context* (CMIC or Core-Plus) (Coxford et al., 1998), *Interactive Mathematics Program* (IMP) (Fendel et al., 1998), and Mathematics: Modeling Our World (MMOW or ARISE) (Garfunkel, Godbold, & Pollack, 1998). These integrated curricula focus on multiple topics each year including algebra, geometry, probability, statistics and topics in discrete mathematics and are intended to be taught in ways that encourage conjecture, exploration, analysis and proof.

The most widely used mathematics curricula in U.S. high schools are non-integrated CD curricula, exemplified by widely-used textbook series such as Houghton-Mifflin, Glencoe-McGraw Hill, and Saxon. Characteristics of these curricula include partitioning the mathematics program beginning in 9th grade into separate years of algebra, geometry, a second year of algebra and for college intending students a senior year college algebra or pre-calculus course. Coursework typically pays significant attention to standard algorithms, repetition for skill development, and a heavy reliance on the teacher for student learning (Schoenfeld, 2004).

Another category is the University of Chicago School Mathematics Project (UCSMP). This curriculum was first developed in the early 1980s (Usiskin, 1986) and combines features of NSF-funded and CD curricula (Schoenfeld, 2004). The UCSMP curricula is perhaps best known through the mathematics textbooks that integrate algebra, geometry, algebra II, functions, statistics, pre-calculus, and discrete mathematics texts and simultaneously emphasize reading, problem-solving, everyday applications of mathematics, and technology.

Evidence of the Relationship between High School Mathematics Preparation and Developmental Mathematics Course-Taking in College

Studies examining high school antecedents of developmental mathematics course-taking have identified several indicators of students' mathematics preparation that predict developmental course-taking. Adelman (2004) reported that students taking developmental mathematics were more likely to have completed high school in an urban area and to be in the lower quintiles of a variable reflecting general learned abilities. Hagedorn et al. (1999) and Bettinger and Long (2009) found that ACT mathematics scores, high school GPA, high school mathematics GPA, and years of high school mathematics predicted the initial mathematics course a student enrolled in including developmental coursework.

The most pertinent finding to the current study is Harwell et al. (2009), who sampled students at a large university and reported that students completing an NSF-funded curriculum were somewhat more likely to take a developmental mathematics course in college than those completing a CD curriculum. The current study extends the work of Harwell et al. (2009) in two important ways. First, our results offer increased generalizability because data from 32 postsecondary institutions are analyzed. Second, we study college mathematics course-taking beyond the developmental course to learn if taking such a course reduces the likelihood of students completing additional mathematics courses. We also explore the data for descriptive evidence of students who successfully completed college level mathematics in high school yet took developmental mathematics in college, and students who took developmental mathematics in college but subsequently completed Calculus I, to provide insight into how students wind up in a developmental course and what happens afterwards.

Research Questions

Our research questions asked

Is there is a relationship between the high school mathematics curriculum a student completed and the (1) likelihood students will complete a developmental mathematics course in college (2) subsequent college mathematics course-taking of students who completed a developmental mathematics course (3) extent to which results for (1) and (2) generalize across different kinds of postsecondary institutions.

Method

Research design

A retrospective cohort (quasi-experimental) cluster design was used in which archival data covering four years of high school and eight college semesters were obtained for students who enrolled in one of 32 post-secondary institutions (clusters/sites) in Fall 2002 or Fall 2003. Students completing the same high school mathematics curriculum were treated as members of the same curriculum cohort. We adopted several of the What Works Clearinghouse Standards (2008) to improve our inferences by taking into account pre-existing differences among cohorts using control variables in the data analyses.

Population and sample

The target population consisted of students enrolled in U.S. colleges who completed at least one mathematics course, whereas the sampled population was students enrolled in 32 colleges in the upper Midwest of the U.S. who completed at least one college mathematics course. The resulting sample of 32 colleges generated 13,188 students who completed at least three years of an NSF-funded, UCSMP, or CD curriculum. Three years was chosen as a cutoff for inclusion in the study because students taking the mathematics portion

of the ACT or SAT test will generally be assumed to have completed at least three years (Carnegie units) of high school mathematics (ACT, 2009; College Board, 2010). Altogether 1, 526 of the 13, 188 students (11.6%) completed at least one developmental mathematics course in college.

Data collection, variables, and data analysis

Student records provided by the 32 institutions were used to extract high school information that included mathematics course titles for grades 9-12, ACT mathematics score, years of high school mathematics, and background information including ethnicity and sex. We also used student records to identify the high school mathematics curriculum a student completed (CD, NSF-funded, UCSMP). We combined the NSF-funded curricula (Core-Plus, MMOW, IMP) in our analyses for two reasons. First, each curriculum was constructed using a common set of standards (NCTM, 1989) and, thus, share a common set of goals and instructional practices designed to support mathematics learning. Second, multivariate analysis of variance found no differences between the NSF-funded curricula for high school (ACT mathematics scores, high school mathematics GPA, years of high school mathematics) or college (grades and difficulty level of mathematics courses) mathematics outcomes. We also coded a student's college major using a modified version of the categories appearing in an NCES (2005) report.

The documented impact of college characteristics on a range of student outcomes including mathematics (Braxton & McClendon, 2002; Ratcliff, Lubinescu, & Gaffney, 2001) prompted us to collect information on each institution's enrollment and educational profile, with the latter primarily a measure of selectivity measured using the ACT score corresponding to the 25th percentile of the freshmen class. This information was obtained from institution websites and the Carnegie Classification Undergraduate Profile website.

Dependent variables

The mathematics courses students completed were available for eight college semesters. To capture course-taking we created a four-point variable using information on the 32 institution websites:

Level 1: This level represents developmental courses that should have been completed in high school.

Level 2: This level includes courses that would be considered pre-Calculus mathematics and include College Algebra, Finite Mathematics, and coursework at a similar level.

Level 3: This would be the typical entry level for well-prepared high school students, who would start their college mathematics coursework with Calculus I.

Level 4: This level is beyond a first course in Calculus. Titles include Calculus II, differential equations, linear algebra, and multivariable Calculus.

The above levels overlap with those used in Teitlebaum's (2003) analysis of NELS data. The difficulty levels of 700 college mathematics courses completed by students at the 32 colleges were assigned by project staff after reviewing course descriptions. To answer our research questions we used the difficulty level variable to create three binary variables: took developmental mathematics (1 = yes, 0 = no); students who began with developmental mathematics and subsequently completed College Algebra/Pre-Calculus (1 = yes, 0 = 0); students who began with developmental mathematics subsequently completed Calculus I (1 = yes, 0 = no).

Data analysis

We fitted two-level (students within institutions) generalized linear models to cross-sectional (binary) data in our inferential analyses (Raudenbush & Bryk, 2002). Where appropriate we constructed between-institution models to predict variation in parameters.

Descriptive analyses

Developmental mathematics course-taking rates varied substantially across the sampled institutions. Sixteen institutions had approximately10% or less of their students in these courses. On the other hand, ten institutions had rates above 30%. The seventeen private colleges in our sample showed a binary pattern in which either few students or significant percentages of students completed a developmental mathematics course, whereas public institutions showed a consistent spread in these rates.

Table 1 reports summary statistics for curriculum cohort, sex, ethnicity, college major, and educational profile by developmental course status. Students in the NSF-funded cohort weremore likely to complete at least one developmental mathematics course (18.6%) as not; for the CD and UCSMP cohorts these percentages were 10% and 8%, respectively. Within ethnic groups African American (36%) and Hispanic (26.1%) students were more likely to complete a developmental mathematics course as not, whereas few STEM students did (1.8%). Among institutions those categorized as more selective, selective, and less selective showed rates of 9.2%, 13%, and 22.2%, respectively.

Table 2 (Page 14) shows the highest college mathematics course difficulty level attained by curriculum cohort and sex for students who started with a developmental mathematics course.

Overall, 54.3% of these students ended their mathematics course-taking at this difficulty level; of these students 19.9% took a second developmental mathematics course. Among the students in Table 2 who took one developmental course 45.7% subsequently completed a course of difficulty level 2 (College Algebra/Pre-Calculus) or a more difficult course. Among the NSF-funded, UCSMP, and CD cohorts the percentages of students who took one

Table 1 Summary Statistics for Curriculum Cohort, Sex, Ethnicity, College Major, and Educational Profile by Developmental Course Status

		Developmental Course Status			
		Yes		No	
		N	Row %	N	Row%
High School Mathematics Curriculum	NSF-funded	355	18.6	1559	81.4
	UCSMP	192	8.0	2213	92.0
	CD	979	10.0	7890	90.0
Sex	Female	1049	13.0	7032	87.0
	Male	656	10.0	5947	90.0
Ethnicity	Asian	187	17.0	916	83.0
	African American	148	36.0	262	64.0
	Hispanic	57	26.1	161	73.9
	Caucasian	1246	10.0	11244	90.0
College Major	STEM	37	1.8	2019	98.2
	Business	229	9.0	2293	91.0
	Humanities	772	15.7	4138	84.3
	Life Science	124	5.4	2158	94.6
	Other	63	11.6	478	88.4
Educational Profile	Full-time/More Selective	636	9.2	6287	90.8
	Full-Time/Selective	925	13.0	6174	87.0
	Full/Part-Time/Less Selective	152	22.2	530	77.8

Note: Developmental course status Yes = completed at least one developmental mathematics course in college, N = sample size, NSF-funded = National Science Foundation-funded, UCSMP = University of Chicago School Mathematics Project, CD = commercially developed curriculum, STEM = Science, Technology, Engineering or Mathematics. For Educational Profile Full-time/More Selective = $\geq$ 80% students attend school on a full-time basis, average ACT score in upper one fifth of all colleges, and low transfer-in rate of $\leq$ 20%; Full/Part-Time/Less Selective = mix of full-time and part-time students, mix of ACT scores, and high transfer-in rate of > 20%.

Table 2 Highest Difficulty Mathematics Course a Student Starting with a Developmental Mathematics Course by the Number of Developmental Mathematics Courses Taken

		Number of Developmental Mathematics Courses Taken							
		1			2			> 2	
	High Diff.	N	Row %	Col %	N	Row %	Col %	N	Row %
Overall	1	659	76.6	54.3	171	19.9	65.2	30	3.5
	2	479	83.6	39.4	83	14.5	31.4	11	1.9
	3	58	89.2	4.8	7	10.8	2.7	0	0.0
	4	18	85.7	1.5	3	14.3	.7	0	0.0
NSF-Funded	1	147	73.1	54.4	48	23.9	65.6	6	3.0
	2	112	81.8	41.5	20	14.6	27.4	5	3.6
	3	9	75.0	3.3	3	25.0	4.1	0	0.0
	4	2	50.0	.8	2	50.0	2.9	0	0.0
UCSMP	1	78	78.0	49.5	19	19.0	70.4	3	3.0
	2	71	86.6	45.1	8	9.8	29.6	3	3.6
	3	8	100.0	5.3	0	0.0	0.0	0	0.0
	4	1	100.0	.1	0	0.0	0.0	0	0.0
CD	1	434	77.6	55.2	104	18.6	63.4	21	3.8
	2	296	83.6	37.7	55	15.5	33.5	3	0.9
	3	41	91.1	5.2	4	8.9	2.4	0	0.0
	4	15	93.8	1.9	1	6.2	.7	0	0.0
Female	1	429	76.3	57.7	114	20.3	67.5	19	3.4
	2	274	82.5	36.9	50	15.1	30.0	8	2.4
	3	28	93.3	3.8	2	6.7	1.2	0	0.0
	4	12	80.0	1.6	3	20.0	1.3	0	0.0
Male	1	230	77.2	48.8	57	19.1	60.0	11	3.7
	2	205	85.1	43.5	33	13.7	34.7	3	1.2
	3	30	85.7	6.4	5	14.3	5.3	0	0.0
	4	6	100.0	1.3	0	0.0	0.0	0	0.0

Note: High Diff. = Highest Difficulty (1 = developmental course, 4 = Calculus II or higher), NSF = National Science Foundation-funded curriculum, UCSMP = University of Chicago School Mathematics Project curriculum, CD = commercially developed curriculum. Maximum sample size =1,526.

developmental course but finished their college mathematics course-taking at difficulty level 2 (College Algebra/Pre-Calculus) were 41.5%, 45.1%, and 37.7% and at difficulty level 3 (Calculus I) were 3.3%, 5.3%, and 5.2%, respectively.

Inferential Analyses

College mathematics course difficulties

We initially fitted a multilevel model to the binary (cross-sectional) data indicating whether a student completed a developmental mathematics course in college (1 = yes, 0 = no). The discontinuous and multimodal nature of college enrollments led us to trichotomize this variable into high, medium, and low enrollments using the data and to dummy-code this variable.

The fixed effects results are presented in Table 3 and show that students completing an NSF-funded curriculum were more likely to take a developmental mathematics course in college than the CD cohort ($\hat{\beta}$ = .278). Exponentiating this result (Raudenbush & Bryk, 2002) shows that students in the NSF-funded cohort were about 1.32 times more likely to take a developmental mathematics course as not, compared to students in the CD cohort. Equivalently, the model-implied probability a student who completed an NSF-funded curriculum will take a developmental mathematics course in college is .57, meaning that for the CD cohort this probability is .43. There were no differences between the UCSMP and CD cohorts in the likelihood of taking a developmental mathematics course. The college major variable had the largest effect, with Humanities majors about 9.6 times more likely to take a developmental mathematics course as not compared to STEM majors. The ACT 25th percentile mathematics score of the freshman class was a significant predictor of intercepts at the institution level, indicating that

Table 3 Fixed Effect Results for the Likelihood of Completing a Developmental Mathematics Course in College (1 = yes, 0 = no)

Between Student Model:	β	SE (β)	t-value
Intercept	-5.205	1.596	-3.261
ACT Mathematics	-0.311	0.015	-20.597
UCSMP	-0.110	0.131	-0.846
NSF	0.278	0.112	2.470
Business	1.231	0.258	4.776
Humanities	2.262	0.249	9.089
Life Sciences	1.153	0.271	4.255
Other	1.562	0.319	4.901
Sex	0.310	0.098	3.149
African American	0.410	0.194	2.109
Asian	-0.154	0.161	-0.957
Hispanic	0.335	0.277	1.210
Years High School Mathematics	-0.629	0.075	-8.342
Between Institution Model:			
Full-time/Selective	-2.818	2.356	-1.196
Full/Part-time/Less Selective	-4.340	3.634	-1.194
ACT 25th percentile	-1.393	0.487	-2.860
Small Enrollment	-2.985	1.480	-2.017
Medium Enrollment	-0.739	1.349	-0.548

Note. **Significant results in bold based on $\alpha' = 1-(1-\alpha)^{1/k} = .009951$ where $\alpha = .15$ is the unadjusted Type I error rate, k = number of statistical tests = 17, and α' is the adjusted error rate. Years high school math = 3,4,5; UCSMP 1 = yes 0 = no, NSF-funded 1 = yes 0 = no so CD served as the reference group; Sex 1 = male, 0 = female; African American 1 = yes, 0 = no; Asian 1 = yes, 0 = no; Hispanic 1 = yes, 0 = no so White students served as the reference group; Business 1 = yes, 0 = no, Humanities 1 = yes, 0 = no, Life Sciences 1= yes, 0 = no, Other 1 = yes 0 = no so STEM served as the reference group; Full-time/Selective 1 = yes, 0 = no, Full/Part-time/Less Selective 1 = yes, 0 = no so Full-time/More Selective served as the reference group; Small Enrollment 1 = < 2110, 0 = > 2110, Medium Enrollment 1 = 2,110 to 3,548, 0 = > 3,548 so Large Enrollment (> 3,548) served as the reference group. Student sample size = 10, 345.**

students in institutions with higher scores were less likely to take a developmental mathematics course.

A second binary difficulty variable reflected whether students who began with adevelopmental mathematics course subsequently took College Algebra/Pre-Calculus. These results (not reported) showed that among students who began college with a developmental mathematics course, high school mathematics curriculum was not related to taking College Algebra/Pre-Calculus. Analysis of a third binary difficulty variable (whether students who began with developmental mathematics subsequently took Calculus I) (not reported) showed that only the college major variable was significant.

Comparing student subgroups who took developmental mathematics in college

We examined two student subgroups in our data with unexpected educational trajectories. Seventy-three students completed five years of high school mathematics (the fifth year is typically Calculus I) yet started college mathematics with a developmental course. The percentage of these students in the CD, NSF-funded, and UCSMP cohorts were 2%, 3.5%, and 1.8%, respectively. Students who completed five years of high school mathematics yet took a developmental course had on average weaker mathematics preparation than students who completed five years of high school mathematics but did not take a developmental course, as suggested by average ACT mathematics scores (21.2 versus 27, respectively). Approximately 58% of the students who completed five years and took a developmental course subsequently completed College Algebra/Pre-Calculus or a more difficult course.

We also found that 77 students began their college mathematics coursework with a developmental course but eventually completed Calculus I. The percentage of these

students for the CD, NSF-funded, and UCSMP cohorts were 1.4%, 0.7%, and 2.7%, respectively.

Discussion

Our results agree with those of Harwell et al. (2009) that students completing an NSF-funded curriculum were more likely to begin college with a developmental mathematics course. One explanation of this result is that NSF-funded curricula simply do not prepare students as well as other curricula, but another explanation is a lack of alignment of the NSF-funded curricula with college mathematics placement tests when compared with the CD and UCSMP curricula (Norman, 2008). These results, as well as our descriptive finding that students who completed five years of high school mathematics (including Calculus I) wound up in a developmental mathematics course, highlights the need for a better understanding of how students are placed in these courses.

Our key finding, which substantially extends the results of Harwell et al. (2009), is that among the approximately 46% of the students who began college with a developmental mathematics course and subsequently completed at least one non-developmental mathematics course, the NSF-funded cohort was just as likely to complete College Algebra or Calculus I as the other cohorts. These results suggest that taking a developmental mathematics course does not permanently limit a student's upwards mathematical mobility.

Implications

For colleges, these results suggest that developmental mathematics coursework can provide a path to success in college mathematics as many of these students go on to take additional courses. For high schools, our results suggest that mathematics curriculum is related to the difficulty level of the

first college mathematics class a student enrolls in but is unrelated to subsequent mathematics course-taking. The latter is important because we think where a student finishes their college mathematics course-taking is more important than where they began. Collectively, we think these results suggest it is premature to abandon or adopt any high school mathematics curricula on the basis that they differentially prepare students for college mathematics, including developmental mathematics course-taking.

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The Impact of Institutional Factors on the Relationship Between High School Mathematics Curricula and College Mathematics Course-Taking and Achievement

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Meta-analytic methods were used to examine the moderating effect of institutional factors on the relationship between high school mathematics curricula and college mathematics course-taking and achievement from a sample of 32 colleges. The findings suggest that the impact of curriculum on college mathematics outcomes is not generally moderated by institutional characteristics such as selectivity and educational profile, providing evidence that the relationships between curriculum and college mathematics outcomes generalize to a range of colleges. The results inform college policies and practices for advising students on mathematics course-taking including enrollment in developmental courses, and high school mathematics curriculum selection.

There is substantial evidence that many U.S. students enter college unprepared to succeed in mathematics (National Center for Educational Statistics [NCES], 2006, 2008, 2009), including the fact that 25%, and perhaps as many as 40%, of all college freshmen complete at least one developmental mathematics class (typically a non-credit bearing class that should have been completed in high school) (Attewell, Lavin, Domina, Levey, 2006; NCES, 2003). The large numbers of students whose preparation for college mathematics is unsatisfactory has prompted postsecondary institutions to invest significant resources in policies and practices that support students' mathematics learning (Bettinger & Long, 2009).

The lack of student preparation has also prompted research on the role of high school mathematics curricula in

preparing students for college mathematics. This literature has generally shown that high school mathematics curriculum is related to the difficulty level of the first college mathematics course a student completes but not to the difficulty of subsequent mathematics courses, whereas evidence of the relationship between curriculum and college mathematics achievement reflected in course grades is mixed (Harwell et al., 2009; Post et al., 2010; Schoen & Hirsch, 2003).

An important next step in increasing our understanding of the relationship between high school mathematics curricula and college mathematics course-taking and achievement is examining the impact of college characteristics on these relationships (We define college as a postsecondary institution offering a bachelor's degree). Research has demonstrated that college characteristics and organizational climates impact a range of student outcomes (Berger & Milem, 1999), and that there is an ongoing need to develop and refine college policies that promote better student outcomes including mathematics outcomes (Bettinger & Long, 2009; Ratcliff, Lubinescu, & Gaffney, 2001). However, none of this work has examined the impact of institutional factors on the relationship between high school mathematics curriculum and college mathematics achievement and course-taking including developmental mathematics.

Rationale for the Study and Research Question

Understanding the impact of institutional factors on the relationship between high school mathematics curriculum and college mathematics outcomes is important for two reasons. First, it informs the use of high school mathematics curricula by college advisors who recommend mathematics coursework to students. Norman (2008) found evidence that high school mathematics curriculum plays an important role in the first

college mathematics course recommended to students by advisors but the basis of this importance was unclear. For example, knowing whether completion of a particular high school mathematics curriculum increases or decreases the likelihood of enrollment in a developmental mathematics course in a more or less selective institution provides guidance to advisors charged with recommending an initial college mathematics course.

Second, the results inform the selection of mathematics curricula by high schools for college bound students. Specifically, the presence or absence of institutional moderators suggests that the relationship between high school mathematics curriculum and college mathematics outcomes does (or does not) generalize to a variety of institutions. The former implies that this relationship generalizes across a range of institutions and suggests that preparation for, and success in, college mathematics does not depend on these characteristics; the latter implies that this relationship is strengthened or weakened by institutional characteristics meaning that preparation for, and success in, college mathematics will be influenced by these characteristics. This information assists high schools in mathematics curriculum selection decisions for college bound students.

High School Mathematics Curricula

The most widely used mathematics curricula in the U.S. are commercially developed (CD) or traditional curricula that stress traditional algorithms and procedures (Education Market Research, 2006). The curricula in the CD category are typically organized as separate years of algebra, geometry, and advanced algebra, and within each year the texts are divided into chapters which focus on algorithmic development with problem solving that concentrates on solving word problems. Student communication is not a central focus of

these texts and mathematical content is, in general, explained by the teacher.

A second category are National Science Foundation (NSF)-funded curricula that are designed to be aligned with the National Council of Teachers of Mathematics' (NCTM) *Curriculum and Evaluation Standards for School Mathematics (CESSM)* (NCTM, 1989, 2000). The mathematics curricula funded by NSF included: *Contemporary Mathematics in Context* (*CMIC* or Core-Plus) (Coxford et al., 1998), *Interactive Mathematics Program* (*IMP*) (Fendel, Resnick, Alper & Fraser, 1997), and *Mathematics: Modeling Our World* (*MMOW* or ARISE) (Garfunkel, Godbold, & Pollak, 1998). These curricula focus on algebra, geometry, probability, statistics and topics in discrete mathematics yearly, de-emphasize algorithmic manipulation, and emphasize the role of students as active participants in the learning process and the value of small groups (Schoenfeld, 2004). A third curriculum is the University of Chicago School Mathematics Project (UCSMP), which is often described as having attributes of both the CD and NSF-funded curricula (Schoenfeld, 2004).

Studies of the relationship between high school mathematics curricula and college mathematics performance have typically compared the impact of NSF-funded and CD curricula. The focus on NSF-funded curricula is in large part because of the intense criticism of these curricula as detailed in Schoenfeld (2004), and in part because of the unsatisfactory mathematics performance of U.S. high school students (NCES, 2006, 2008) most of whom completed a CD curriculum.

The Impact of College Characteristics and Organizational Climates

There is considerable empirical evidence of the impact of college characteristics and organizational climates on a range of student outcomes (Berger & Milem, 1999), and of the need

to develop college policies that promote better student outcomes including mathematics outcomes (Bettinger & Long, 2009; Ratcliff et al., 2001). There is also empirical evidence of the impact of institutional factors on developmental mathematics outcomes (e.g., Attewell et al., 2006; Fong, Huang, & Goel, 2008), but there are no theoretical models of this process.

The importance of understanding the role of postsecondary institutional factors on preparation for college mathematics (reflected in high school mathematics curricula) and college mathematics outcomes led us to pose the following research question:

> What institutional factors moderate the relationship between high school mathematics curriculum and college mathematics course-taking and achievement as reflected in the likelihood of completing a (i) developmental mathematics course and the grade earned (ii) college algebra course and the grade earned (iii) calculus I course and the grade earned?

We limit our examination to the impact of CD and NSF-funded curricula because of the intense scrutiny and criticism the latter have received (Klein, 2001). Given our research question, institutional (aggregate) data rather than disaggregated (student) data were analyzed, a practice similar to coarsening data in epidemiological research to identify overall patterns (see, e.g., Iacus, King, & Porro, 2008). In this process data are rescaled to a more coarse form (e.g., student level to institutional level), which implies a change in inferences to aggregated units of analysis. Using institutional level data provided compelling and succinct representations of the impact of institutional moderators on the relationship between high school mathematics curricula and college mathematics outcomes. Of course, it is important to avoid

the ecological fallacy discussed by Robinson (1950) in which inferences about disaggregated data (students) are made based on aggregated data (institutions). While such inferences are permissible under quite restrictive conditions (Schuessler, 1999) we limit our inferences to institutions.

We focused on the likelihood of enrolling in a college developmental mathematics course because of evidence that high school mathematics curriculum plays a key role in the first college mathematics course a student is advised to take (Norman, 2008), and the possibility that this relationship could be moderated by institutional factors. College algebra was included as an outcome due to the large numbers of students who enroll in these courses (Hofacker, 2006), and calculus I because it serves as a gateway to higher level mathematics and completion of a major in science, technology, engineering, or mathematics (STEM) (Tyson, 2011).

Methods

Research Design

A retrospective cohort cluster design was used in which postsecondary institutions (colleges) served as clusters. College mathematics outcomes for CD and NSF-funded curricula were compared for each institution, producing a quasi-experimental design.

Population and Sample

The target population consisted of colleges in the U.S. whereas the sampled population consisted of colleges in the upper Midwest of the U.S. that varied in several ways (e.g., educational profile). Following the purposive sampling arguments of Shadish, Cook, and Campbell (2002, pp. 354-372), the sample allows us to generalize our results at least approximately to similar institutions. Altogether 32 four-year institutions were sampled.

Students enrolling in college will generally be expected to have completed at least three years (Carnegie units) of high school mathematics, as reflected in the mathematics portions of the ACT and SAT tests (ACT, 2009; College Board, 2009). To be included in our study students enrolled at an institution must have completed at least three years of high school mathematics in a CD or NSF-funded curriculum and completed at least one college mathematics course.

Data Sources

Data were obtained from two sources. The first, college records, included the number of years of high school mathematics a student completed and titles of their high school mathematics courses which were used to identify a student's high school mathematics curriculum. College records also yielded the titles and grades of student's mathematics coursework across eight college semesters, demographic information, and a student's college major which was captured using a trichotomous variable: STEM, Humanities/Social Sciences, or Other.

The second data source reflected information about post-secondary institutions. Variables taken from institution websites and a State Department of Education website reflected important college characteristics and organizational climates: institutional size (captured by enrollment = number of full-time + part-time students), selectivity (ACT mathematics score of the freshmen class corresponding to the 25th percentile as reported on institution websites), percentages of African American students and STEM majors, institutional profile (publicly or privately supported), and educational profile. The latter variable combined selectivity and the nature of an institution's student body and was generated using information on the Carnegie Foundation website http://classifications.carnegiefoundation.org/

lookup_listings/standard.php?key=783. For our sample the educational profile variable had three categories: (i) 1 = full-time = ≥ 80% students attend school on a full-time basis, selective (more selective = average ACT score in upper one-fifth of all colleges, selective = average ACT score in middle two-fifths of all colleges), and transfer rate (high transfer-in = ≥ 20%, low transfer-in rate = < 20%) (ii) 2 = full-time/more selective/high transfer-in rate (iii) 3 = mixture of full-time and part-time students/mixture of selective and less selective/high transfer-in rate.

The dependent variables were used to compute effect sizes corresponding to our research question. Dependent variables reflecting the difficulty level of a college mathematics course were created by examining the description of each mathematics course at each sampled institution. This process led to college mathematics courses being categorized as representing one of three difficulty levels: Level 1 reflected course that should have been completed in high school (developmental), Level 2 a course that a student who satisfactorily completed four levels (years) of mathematics in high school would not repeat in a college setting (e.g., college algebra, pre-calculus mathematics), and Level 3 calculus I. Seventy-five percent of the sampled institutions used a 12-point grading scale (A, A-, B+, ..., F), 18.75% used a seven-point scale (A, AB, B, BC, C, D, F), and 6.25% used a five-point point grade scale (A, B, C, D, F). To reduce the impact of between-institution differences related to grades we rescaled all grades to a five-point scale (A = 5, B = 4, ...).

Data Analyses

Meta-analysis was used to study patterns in effect sizes to provide insight into the moderating effect of institutional factors on the high school mathematics curricula and college mathematics course-taking and achievement relationships.

Both graphical and inferential analyses were used to explore patterns in the effect sizes and the impact of institutional factors. Data for each dependent variable were analyzed separately.

Course-taking

We focused our examination on the relationship between high school mathematics curriculum and college mathematics course-taking by creating three binary dependent variables corresponding to our research question: Completed a developmental mathematics course in college 1 = yes (difficulty level = 1), 0 = no; Completed college algebra in college 1 = yes, (difficulty level = 2), 0 = no; Completed calculus I in college 1 = yes (difficulty level = 3), 0 = no. The proportion of 1's reflected enrollment in a curriculum cohort (e.g., CD with proportion $\hat{p}_{CD_k}$) among students who subsequently enrolled in a developmental mathematics course at the kth institution. A similar quantity was generated for students in the NSF-funded cohort at the kth institution (e.g., $\hat{p}_{NSF_k}$). We then estimated an effect size for an institution ($\hat{\delta}_k$) as:

$$\hat{\delta}_k = \frac{(\hat{p}_{CD_k} - \hat{p}_{NSF_k})}{\hat{\sigma}_{\hat{p}_{CD_k} - \hat{p}_{NSF_k}}} \sim \text{Normal}(\delta_k, \sigma^2_{\delta_k}),$$

$$\hat{\sigma}^2_{\hat{p}_{CD_k} - \hat{p}_{NSF_k}} = \frac{\hat{p}_{CD_k}\hat{q}_{CD_k}}{N_{CD_k}} + \frac{\hat{p}_{NSF_k}\hat{q}_{NSF_k}}{N_{NSF_k}} \quad (1)$$

In equation (1) $\hat{\delta}_k$ is a standardized effect size similar to Cohen's d (Cohen, 1988), $\hat{\sigma}^2_{\hat{p}_{CD_k} - \hat{p}_{NSF_k}}$ is the variance of the difference in the proportions, N is a sample size, and $\hat{q} = 1 - \hat{p}$. A similar process was followed for the outcomes

reflecting the likelihood of completing a college algebra course and calculus I, respectively.

Grades

The process to compute $\hat{\delta}_k$ for grades followed Konstantopoulos and Hedges (2009). For all outcomes we examined plots of effect sizes and used two-level hierarchical linear modeling to explore variation among effect sizes and to construct explanatory models.

Results

Descriptive Analyses

The average enrollment at the 32 institutions was 5,100 and ranged from 440 to 37,383 with first, second, and third quartile values of 1897, 2672, and 6655, respectively. The average institutional ACT mathematics score corresponding to the 25th percentile was 21 and ranged from 15 to 29 in our sample (scores corresponding to the 50th percentile were unavailable). The distribution of institutions in the three categories of the educational profile variable (full-time/more selective/high transfer-in rate, full-time/selective/high transfer-in rate, mixture of full-time and part-time students/mixture of selective and less selective/high transfer-in rate) was 28.1%, 59.4, and 12.5, respectively. The percentage of African American students enrolled at an institution was on average quite small (2.4%) and ranged from almost 0 to 12.73%, whereas the average percentage of STEM majors at an institution was 7.8% and ranged from almost 0 to 21.1%.

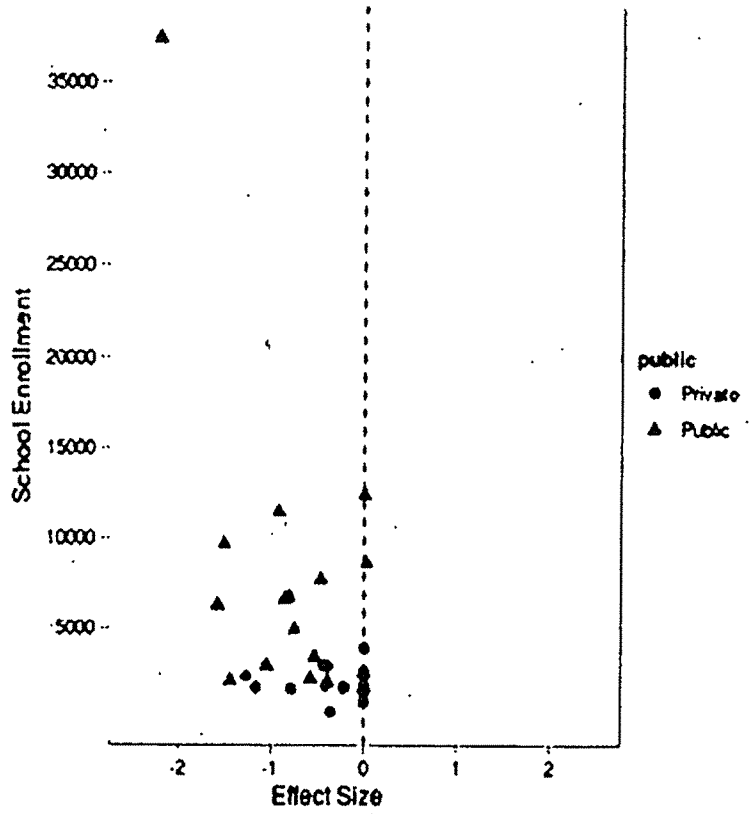
The institutional proportions for completing a developmental mathematics course in college for the CD and NSF-funded cohorts were .12 (N = 1,211) and .22 (468), respectively; for students completing three, four, or five years of high school mathematics the proportions completing a

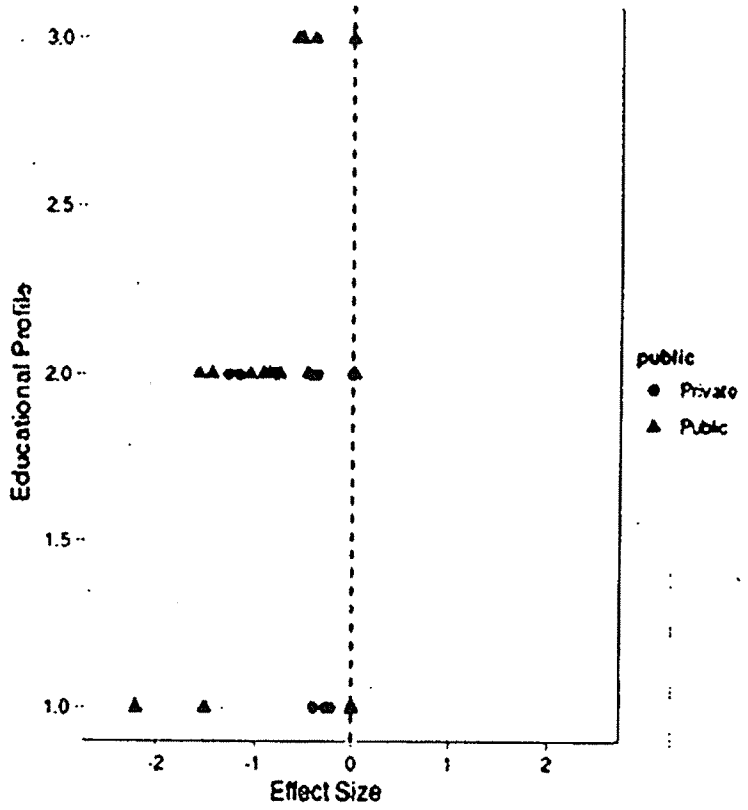
developmental mathematics course for the CD and NSF-funded cohorts were .12 (702) and .22 (265), .10 (453) and .19 (184), and .02 (56) and .04 (19), respectively. Institutional proportions for completing a college algebra course for the CD and NSF-funded cohorts were .53 (5,091) and .57 (1,216), respectively.

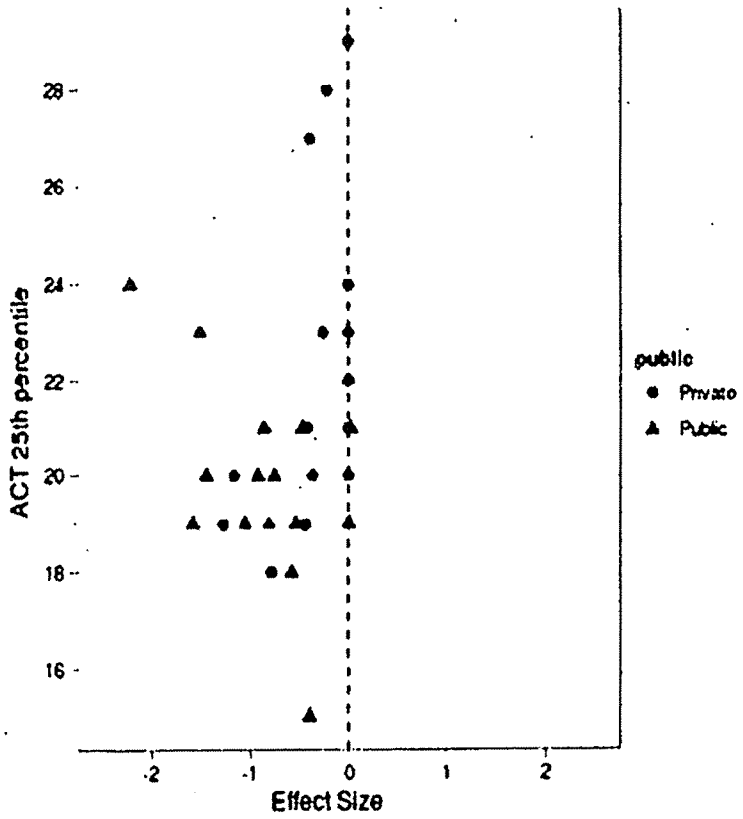
Plots of effect sizes ($\hat{\delta}_k$) computed using equation (1) are given for the likelihood of completing a developmental (Figures 1-5) and calculus I course (Figures 6-10). Patterns for college algebra were similar and are not reported. Figures 1-5 reflect the difference between the CD and NSF-funded cohorts at an institution in the likelihood of completing a developmental mathematics course by an institution's size (enrollment), educational profile, ACT mathematics 25th percentile, percentage of STEM majors, and percentage of African Americans as a function of financial profile (publicly or privately supported). The overwhelming majority of effect sizes are negative meaning that at most institutions the NSF-funded cohort was on average more likely to initially enroll in a developmental mathematics course compared to the CD cohort. An effect size of zero suggests that at that institution the CD and NSF-funded cohorts are equally likely to enroll in a developmental course.

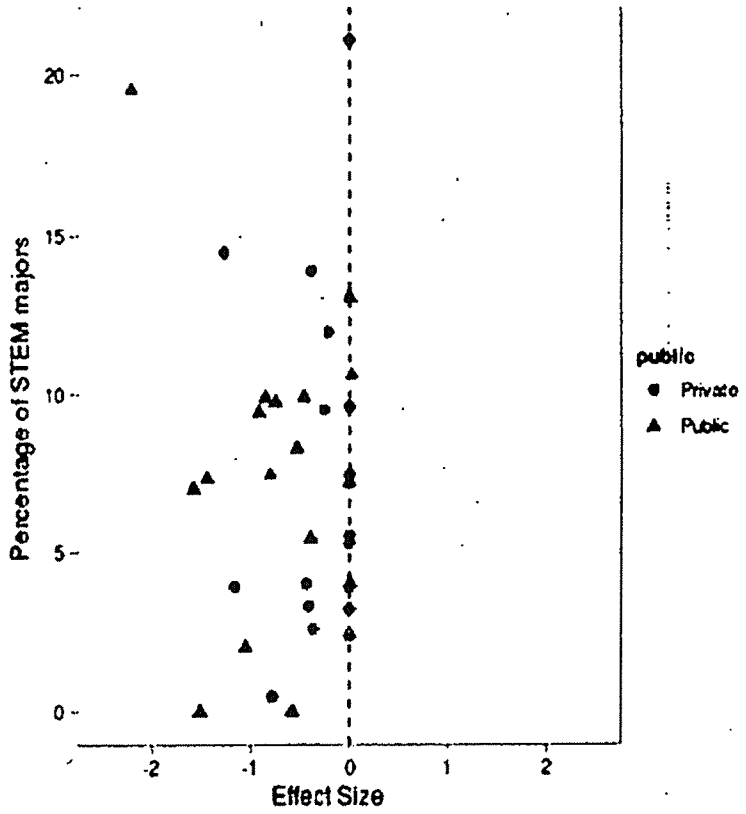
Educational profile, ACT 25th percentile score, and percentages of African American and percentage of STEM majors appeared to weakly moderate the relationship between curriculum and likelihood of taking a developmental mathematics course appearing in Figures 1-5.

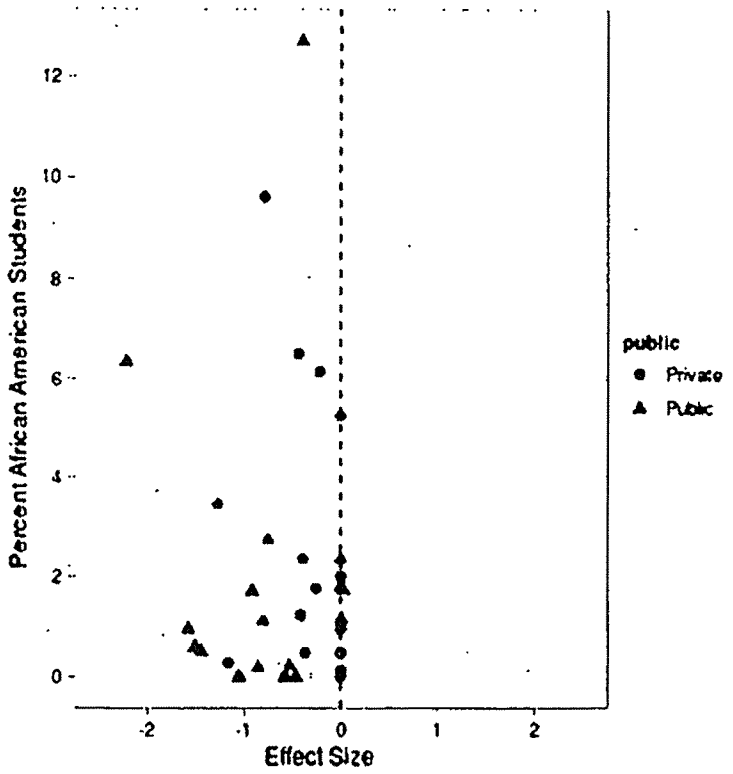
Although enrollment did not appear to moderate this relationship, the institution with the largest enrollment also had the largest effect size, which could be due to factors such as the alignment between mathematics placement tests and high school mathematics curricula or the way that these curricula are used by advisors in recommending a student's first college mathematics course. Effect sizes appeared to


School Enrollment
35000
30000
25000
20000
15000
10000
5000
-2
-1
0
1
2
Effect Size
public
Private
Public


Educational Profile
3.0
2.5
2.0
1.5
1.0
-2
-1
0
1
2
Effect Size
public
Private
Public

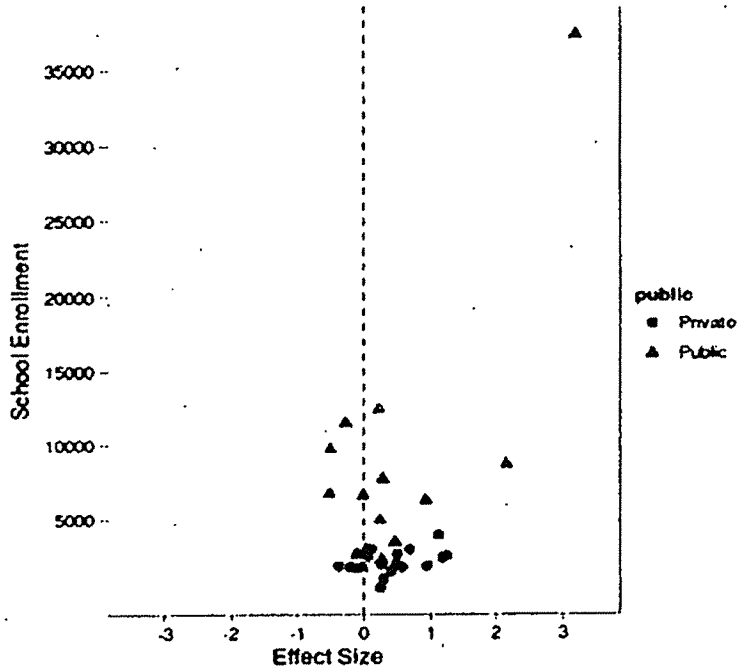

ACT 25th percentile
28
26
24
22
20
18
16
-2
-1
0
1
2
Effect Size
public
Private
Public

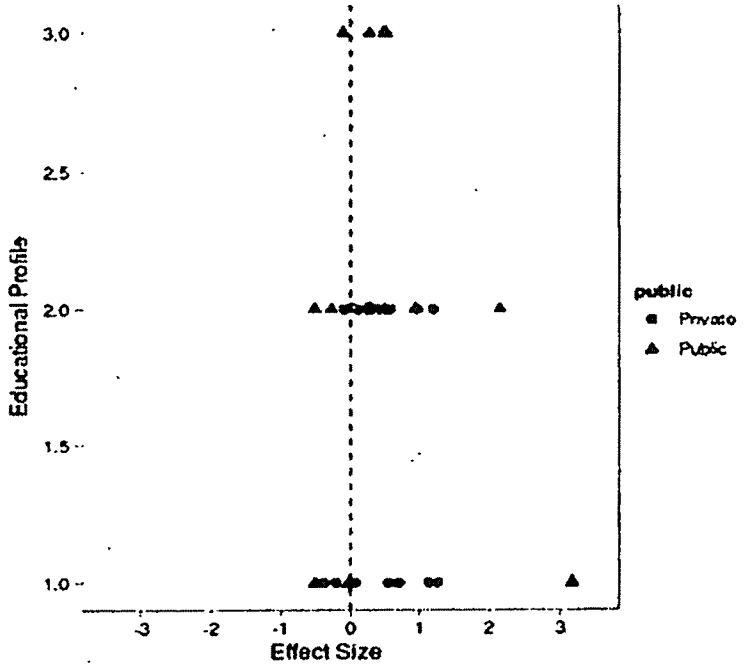

Percentage of STEM majors
20
15
10
5
0
-2
-1
0
1
2
Effect Size
public
Private
Public



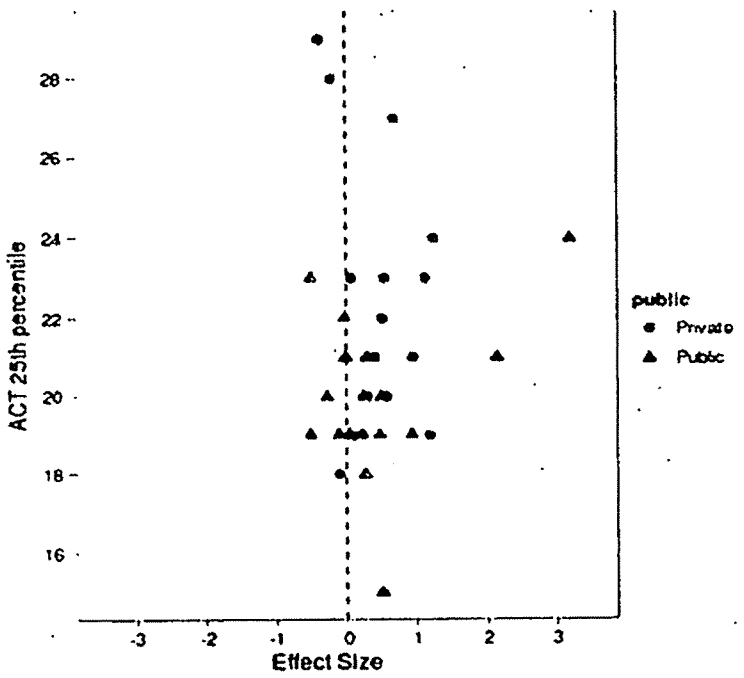
Figures 1-5. Plots of effect sizes reflecting the difference between the CD and NSF-funded cohorts in the likelihood of taking a developmental mathematics course.

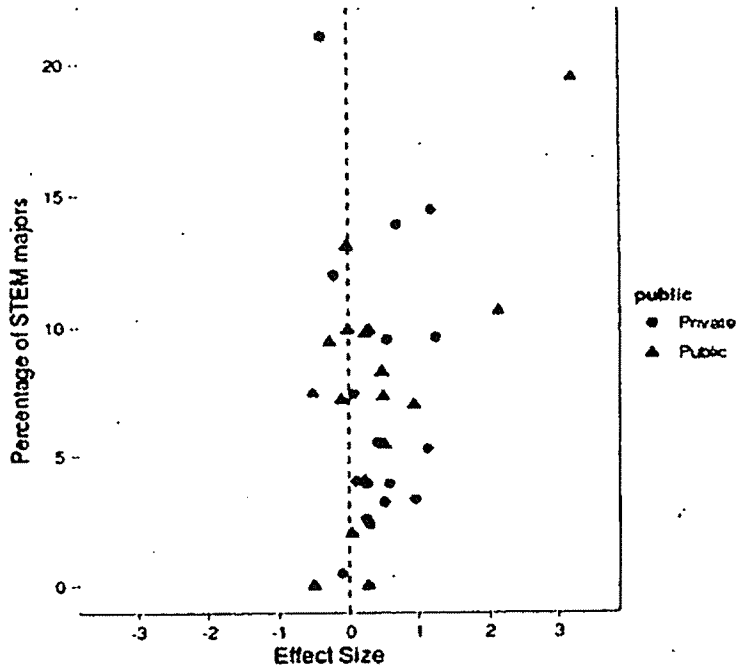
Figures 6-10. Plots of effect sizes reflecting the difference between the CD and NSF-funded cohorts in the likelihood of taking calculus I.

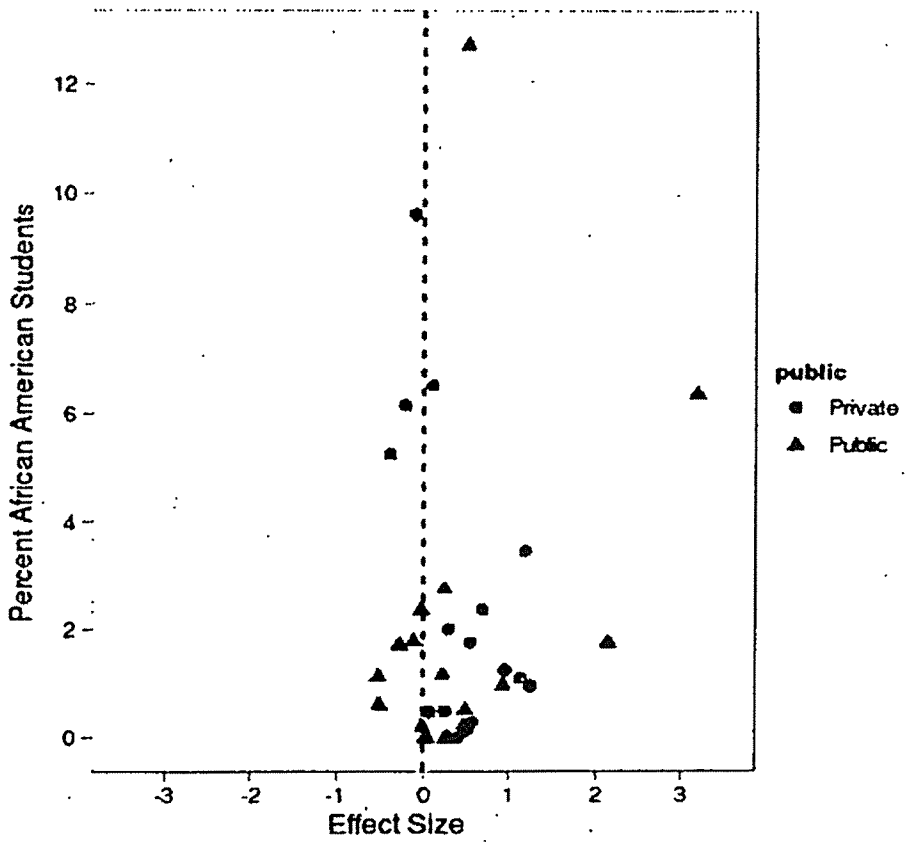




99


ACT 25th percentile
28
26
24
22
20
18
16
-3
-2
-1
0
1
2
3
Effect Size
public
Private
Public


Percentage of STEM majors
20
15
10
5
0
-3
-2
-1
0
1
2
3
Effect Size
public
Private
Public


Percent African American Students
12
10
8
6
4
2
0
-3
-2
-1
0
1
2
3
Effect Size
public
Private
Public

increase somewhat for more selective/selective institutions (coded 1) compared to selective/less selective institutions. For higher percentages of STEM majors the effect sizes seemed to move toward zero, which might be expected because the mathematics requirements of many STEM majors results in a sample of mathematically able students for whom high school mathematics curriculum differences on college mathematics outcomes are negligible (Post et al., 2010).

Figures 6-10 reflect the likelihood of completing calculus I in college for the various institutional variables. On average the CD cohort was more likely to complete calculus I especially in institutions with higher concentrations of STEM majors. In general, educational profile seemed to mildly moderate the relationship between curriculum and likelihood of completing calculus I with positive effect sizes for more selective institutions (especially for a few public institutions) and near zero for less selective institutions.

Effect sizes showed an especially interesting pattern for the ACT 25th percentile moderator, with values of 18 -24 generally associated with positive effect sizes favoring the CD cohort. However, for ACT 25th percentile values above 26 the effect sizes are close to zero and for two institutions the NSF-funded cohorts are more likely to complete calculus I. This is likely attributable to increasingly small differences among high school mathematics curricula for college mathematics outcomes for more mathematically proficient students. For quite small percentages of African American students at an institution there was no distinct pattern among effect sizes but for higher percentages the effect sizes centered on zero, with the exception of one public institution with approximately 6% African American students which had a large effect size favoring the NSF-funded cohort.

Similar figures for the average grade in a developmental, college algebra, and calculus I course were

constructed. These figures (not reported) suggested that the CD cohort was somewhat more likely to earn higher grades than the NSF-funded cohort and that this pattern was not clearly moderated by any of the institutional factors.

Inferential Analyses

Hierarchical linear models in which the level one model (within institutions) did not have predictors and level two (between institutions) contained predictors reflecting institutional factors were fitted to effect sizes (see (Raudenbush & Bryk, 2002). The statistical shortcomings of proportions led us to transform each proportion using the arcsine transformation and to use the transformed values to compute effect sizes (Cox, 1970). We opted to first fit models to the proportions and then to the transformed values. Because the two sets of findings were very similar we report results for proportions.

Preliminary analyses with no predictors showed significant variation among effect sizes for each outcome. Further examination of the variance of the effect sizes for each outcome indicated medium variation among effect sizes (Hedges & Pigott, 2004), prompting us to construct explanatory models evaluating the presence of moderating effects. However, only a few statistically significant results emerged. One was the percentage of %STEM majors (rescaled to quartiles of the original variable to avoid implausible estimates) whose slope of .07 indicated that a one unit increase (i.e., from the first to the second quartile) was associated with a small expected increase in the likelihood that the NSF-funded cohort would enroll in a developmental mathematics course. Another significant meta-analytic finding was that the CD cohorts were more likely to complete college algebra relative to the NSF-funded cohorts.

Conclusions and Implications

This study sought to identify moderating effects of institutional factors such as educational profile and selectivity on the relationship between high school mathematics curricula and college mathematics course-taking and achievement. Two important findings emerged. First, there was significant variation among effect sizes in the likelihood that the NSF-funded and CD cohorts enroll in developmental mathematics, college algebra, or calculus I courses in college. In our view the result of greatest concern is for the likelihood of enrolling in a developmental course since it suggests differences between the NSF-funded and CD cohorts at the college entry point.

Second, the general lack of a significant relationship between effect sizes and institutional factors for most outcomes suggests that the NSF-funded and CD cohort differences (or lack thereof) generalize to a variety of post-secondary institutions (e.g., more or less selective institutions). More generally, our results suggests that variables reflecting organizational characteristics and climate had on average little or no impact on the relationship between high school mathematics curricula and college mathematics course-taking and achievement.

Implications

We think our findings have two important implications. One is that the absence of significant moderators suggests that the development of policies and practices that promote greater student mathematics learning regardless of the high school mathematics curriculum a student completed will not be informed by institutional selectivity, educational mission, enrollment, etc. For example, the similar enrollment rates in calculus I in college regardless of high school mathematics curriculum suggests that this factor be given less weight in recommending an initial college mathematics course. Put

another way, the absence of moderating effects suggests that this issue is not affected by various characteristics of colleges and thus is applicable to a range of institutions. Second, our results suggesting that the relationship between mathematics curricula and college mathematics course-taking and achievement generalizes to a variety of colleges informs curriculum adoption decisions at the high school level for college bound students.

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Mixed Effects?

A Multi-Institutional Study of High School Mathematics Curricula and College Mathematics Achievement and Course Taking

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This study examined the relationship between high school mathematics curricula and student achievement and course-taking patterns over 4 years of college course taking for a sample of over 10,000 students from 32 postsecondary 4-year institutions. Three types of curricula were studied: National Science Foundation (NSF) funded curricula, the University of Chicago School Mathematics Project curriculum, and commercially developed curricula. The major result was that high school mathematics curricula were unrelated to college mathematics achievement or students' course-taking patterns for students who began college with precalculus (college algebra) or a more difficult course. However, students of the NSF-funded curricula were statistically more likely to begin their college mathematics at the developmental level. Implications of these results for research and practice are discussed.

Keywords: Achievement; College mathematics; Integrated curricula; Large-scale studies; Longitudinal studies; Mathematics education

Surprisingly little is known about the relationship between high school mathematics curricula and student preparation for college mathematics, prompting calls for studies that clarify this relationship (Augustine, 2007; Mathematical Sciences Education Board [MSEB], 2004). Such studies are important for three related reasons. First, it is important to establish an empirical research base characterizing the likelihood that various curricula differentially prepare students for college mathematics (MSEB, 2004; Planty et al., 2008), particularly in light of evidence that many students are not prepared to succeed at this level (Planty et al., 2008; Snyder, Dillow, & Hoffman, 2009).[1] Second, it is important to examine the ability of high school mathematics curricula to prepare an increasingly diverse group of students for college mathematics in postsecondary institutions that vary in enrollment, selectivity, and educational and financial profiles. Third, there is significant interest in identifying predictors of developmental mathematics course enrollment. Developmental mathematics courses address content that is ideally covered in high school and do not usually count toward college degree completion (Adelman, 2004).

[1] We define *college* as a 4-year postsecondary institution offering a bachelor's degree.

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Approximately 25% of college freshmen complete at least one developmental mathematics course (Parsad & Lewis, 2003), and many states are considering postsecondary policies related to mathematics remediation (Bettinger & Long, 2009). Current literature is largely limited to using student outcomes in high school, such as GPA and percentile rank, as predictors of developmental mathematics course taking in college (Attewell, Lavin, Domina, & Levey, 2006; Bettinger & Long, 2009); however, none have studied the role of high school mathematics curriculum on developmental course taking.

The purpose of this study is to deepen our understanding of the relationship between high school mathematics curricula and college mathematics achievement and course taking for a diverse group of students and postsecondary institutions. We begin by briefly describing the high school mathematics curricula we studied, and then we summarize available literature on the relationship between these curricula and student achievement and course taking in college mathematics, including developmental courses. After posing research questions reflecting the study's purpose, we describe our methodology and findings. We conclude by discussing the implications of our results and suggestions for additional research.

High School Mathematics Curricula

The development of almost all high school mathematics curricula currently used in the United States can be traced to one of three sources: the National Science Foundation, for-profit companies (usually publishers), or a combination of the two. High school mathematics curricula linked to these sources share several features, but they also differ in important ways in their approach to mathematics content, instructional practices, student assessment, and in the populations of students the curricula were designed for (e.g., students who will take advanced mathematics courses and those who will not). These differences have led to concerns about the ability of different curricula to prepare students for college mathematics, especially calculus (Schoenfeld, 2004).

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National Science Foundation Funded Curricula

National Science Foundation (NSF) funded curricula represent those funded through a proposal solicitation in the early 1990s (NSF RFP 91-100). The expectation was that the development of new curricula would be guided by the *Curriculum and Evaluation Standards for School Mathematics* (National Council of Teachers of Mathematics [NCTM], 1989). In many ways this was a theory-based, best practices document in its incorporation of findings from cognitively oriented research on human learning and instructional practices (Bruner, 1964; Shulman & Sparks, 1992). The Standards also focused on engaging students in a learning environment whose features include group work and problem solving motivated by real-world problems and data (Schoenfeld, 2004). The current study included three NSF-funded first edition high school curricula: (a) Contemporary Mathematics in Context (CMIC or Core-Plus), (b) Interactive Mathematics Program (IMP), and (c) Mathematics: Modeling Our World (MMOW). These curricula are similar in their focus on algebra, geometry, probability, statistics, and topics in discrete mathematics every school year in a way that emphasizes problem solving and deemphasizes algorithmic manipulation. These curricula also share a heavy reliance on contextually oriented mathematical activity with real-world problems selected because of their high potential for the development of important mathematical topics.

Commercially Developed Curricula

In the late 20th and early 21st centuries, commercially developed curriculum was the most widely used curriculum type, reaching approximately 75% of the nation's students (Education Market Research, 2006). Commercially developed (commercial) curricula are sometimes referred to as *traditional curricula* because of their focus on longstanding conventions about which mathematics topics students should learn and how this material should be learned. These curricula partitioned the high school mathematics program into algebra (Year 1), geometry (Year 2), and a second course in algebra (Year 3). They tended to be teacher centered and depended more heavily on algorithmic development and repetition (Schoenfeld, 2002; Schoen, Fey, Hirsch, & Coxford, 1999). For college-intending students, these curricula typically included a senior year precalculus (college algebra) course or, in many cases, a first year Advanced Placement Calculus course. Examples of publishers whose programs are included in this category are Addison Wesley, Saxon, Glencoe/McGraw Hill, and Scott Foresman. These curricula represent a distinct alternative to NSF-funded curricula approaches.

University of Chicago School Mathematics Project Curriculum

Another category of curriculum is represented by the University of Chicago School Mathematics Project (Usiskin, 1986). This program combines features of both NSF-funded and commercial curricula (Hirschhorn, 1993; Schoenfeld, 2004; Thompson & Senk, 2001). In our estimation, UCSMP leans more heavily in the direction of the NSF approaches.

mathematics achievement plays in developmental mathematics course taking in college (e.g., Adelman, 2004; Bettinger & Long, 2009), but only two studies are currently available that address the relationship between specific high school mathematics curricula and developmental course taking in college mathematics.

Harwell et al. (2009) used data from a single postsecondary institution and found that students who completed an NSF-funded curriculum were statistically more likely to initially enroll in a developmental mathematics course in college compared to students who completed a commercial curriculum; among students who began with a nondevelopmental (higher difficulty) course, no differences among the curriculum cohorts emerged. Also, when students' ACT mathematics scores were partitioned into thirds, there were no differences in course-taking patterns among the curriculum cohorts for students in the middle and high ACT levels (Post et al., 2010).

A related area of research has focused on the relationship between student ethnicity and mathematics achievement and course taking. It is well known that a mathematics achievement gap exists between African American and white students in high school (Braun, Wang, Jenkins, & Weinbaum, 2006; Riegle-Crumb & Grodsky, 2010; Snyder et al., 2009) and college (Adelman, 2006; Spenner, Buchmann, & Landerman, 2004; Vars & Bowen, 1998). Studying the impact of high school mathematics curricula is important because of arguments that the achievement gap can in part be attributed to instructional practices within schools that can be modified (Gamoran & Berends, 1987; Snyder, 2007). Furthermore, one common goal of NSF-funded, commercial, and UCSMP curricula is to make mathematics accessible to all students (Thompson & Senk, 2001), which would be expected to affect the achievement gap.

In summary, a few studies have examined the impact of high school mathematics curricula on students' preparation for college mathematics and have provided insight into this important relationship. These studies have relied on sampling students from a single postsecondary institution and—with the exception of the Schoen and Hirsch (2003) study that used random assignment of curriculum to high schools but with a small sample—employed various statistical methods to try to strengthen causal inferences. Moreover, none of these studies examined students' achievement and the difficulty level of their mathematics courses over their 4-year college career, including their performance in their last college mathematics course. The latter is an important benchmark in the sense that where (and how) students finish their college mathematics is arguably more important than where they began.

To better understand the relationship between high school mathematics curricula and college mathematics achievement and course taking for a diverse group of students and postsecondary institutions, we posed the following research questions:

1. What is the nature and magnitude of the relationship between the high school mathematics curriculum completed and a student's (a) grade in college Calculus I, (b) grade in last college mathematics course, and (c) mathematics course grades across eight college semesters?

2. What is the nature and magnitude of the relationship between the high school mathematics curriculum students complete and the (a) likelihood that the difficulty level of their last college mathematics course equaled or exceeded that of Calculus I, (b) pattern of difficulty levels of their college mathematics course(s) when difficulty equaled or exceeded college Calculus I across eight semesters, and (c) likelihood that they completed a developmental mathematics course in college?
3. Do the relationships in the research questions above differ across ethnic groups and postsecondary institutions varying in enrollment, selectivity, and educational and financial profiles?

We included a student's college Calculus I class grade and the likelihood that the difficulty level of a student's last college mathematics course equaled or exceeded that of Calculus I as outcomes to provide an empirical test of the ability of the three high school mathematics programs to prepare students for advanced college mathematics courses. The likelihood that a student completed a developmental mathematics course was included because of the high percentage of students in the United States who completed at least one developmental mathematics course, 16% in public colleges and 8% in private colleges (Parsad & Lewis, 2003). Also, a difference among curricula would imply a difference in student preparation for college mathematics. Question 3 was included because of literature documenting the impact of student ethnicity on college mathematics performance (Adelman, 2006; Spenner et al., 2004). We note that SES was not included because good measures of a high school student's SES (e.g., parental occupation or education) were not available.

Method

Research Design

A retrospective cohort (nonequivalent, posttest only, quasi-experimental) cluster design was used in which archival data, including college mathematics achievement and course-taking data covering eight semesters, were obtained for students who enrolled at one of 32 postsecondary institutions (clusters) in Fall 2002 or Fall 2003. Students who completed the same high school mathematics curriculum were treated as members of the same curriculum cohort. Following the What Works Clearinghouse (2008) standards, preexisting differences among cohorts were taken into account using control variables in the data analyses. This helps to ensure, but does not guarantee, credible inferences. We also acknowledge that our use of a quasi-experimental design implies that our study does not meet all of the What Works Clearinghouse standards for estimating an intervention (curriculum) effect, but we believe that our design and analyses produce credible answers to our research questions.

Sampling

The target population consisted of colleges in the United States, whereas the sampled population consisted of colleges in the upper Midwest that differed in

several ways. The 46 colleges we invited to participate varied in enrollment (full-time and part-time students), selectivity (ACT score of the freshman class corresponding to the 25th percentile), and financial profile (publicly or privately supported), based on a classification developed by the Carnegie Foundation (available on the Carnegie Foundation website; see http://classifications.carnegiefoundation.org/lookup_listings/standard.php?key=783) and an educational profile variable used by the Carnegie Foundation to categorize colleges.

The educational profile variable used by the Carnegie Foundation is based on three components (variables). The first is the proportion of undergraduates who attend college part- or full time, where a full-time student is one who enrolled for 12 or more semester credits or quarter credits, or has 24 hours of instructional contact each week of a term (United States Department of Education, 2013). A second reflects the achievement characteristics of first-year, first-time students, defined using the ACT/SAT scores of the freshman class expressed in quintiles, for example the average ACT score in the upper one fifth of all colleges. The third characteristic represents the proportion of entering students who transfer in from another institution. A transfer-in student is defined as a student entering an institution for the first time but who previously attended another postsecondary institution as an undergraduate (United States Department of Education, 2013).

Although average ACT scores expressed in quintiles are used in constructing the educational profile variable, we also used a version of this variable as a standalone measure of selectivity (ACT score of the freshman class corresponding to the 25th percentile) because of its documented predictive power and because it showed a relatively small correlation with educational profile. Focusing on institutional characteristics like selectivity (average ACT score of the freshman class corresponding to the 25th percentile) and a college's educational profile, and ensuring that variation in these characteristics in our sample is consistent with that appearing on the Carnegie Foundation website, enhances the generalizability of our results.

Thirty-two of the 46 invited colleges (76%) agreed to participate and provided data for this study. Comparisons of colleges that did or did not participate provided no evidence of differences on the above institutional variables. For example, a two-sample *t*-test comparing average ACT mathematics scores corresponding to the 25th percentile for the 32 participating colleges and the 14 colleges not participating was not significant ($p > .05$). Similarly, a comparison of the two groups of colleges on the educational profile variable using a Kruskal-Wallis test for ranks was not significant ($p > .05$). The analyses reported below were based on 32 colleges in five states (Minnesota, Wisconsin, Iowa, North Dakota, and South Dakota).

The 32 colleges represent a purposive sample in the sense that they varied in ways we think enhanced the viability of generalizing to the target population. Shadish, Cook, and Campbell (2002) pointed out that purposive sampling does not ensure generalizability, but we think that the similarity of the 32 institutions to colleges in the United States on key indicators such as college preparation (numbers of students in the United States with ACT scores greater than or equal to 26 and SAT scores greater than or equal to 1200 per 1,000 high school graduates) and

college accessibility (average tuition for public and private institutions in the United States) suggests generalizability of our results to a broader population.

The initial sample consisted of 13,337 students who completed at least 3 years of an NSF-funded, commercial, or UCSMP curriculum in a high school in Minnesota and subsequently enrolled in one of the 32 colleges in five contiguous states (Minnesota, Wisconsin, Iowa, North Dakota, and South Dakota). Participation was limited to students who completed at least 3 years of high school mathematics because students enrolling in college are generally assumed to have completed at least 3 years (Carnegie units) of high school mathematics, as reflected in the mathematics portions of the ACT and SAT (ACT, 2013; College Board, 2013).

All students had completed at least one mathematics course while they were enrolled in college. Fall 2002 was chosen as the first term to study college mathematics because it allowed for significant numbers of sampled students who had completed three or four years of the first published editions of NSF-funded curricula. Sampling students who enrolled in Fall 2003 essentially doubled the sample size. No evidence emerged that students enrolling in Fall 2002 differed systematically from those enrolling in Fall 2003, and students were pooled for the various analyses.

Data

Archival data were obtained from college records and reflected both high school and college information.

High school information. High school information was obtained from college records provided by the sampled postsecondary institutions and included titles of the high school mathematics courses a student had completed and the associated course grades for Grades 9–12, ACT mathematics score, years of high school mathematics completed, and demographic information. We adopted a two-step process to link each student to a curriculum cohort.

First, we contacted each of the 290 high schools (all in Minnesota) represented in our sample of students and obtained descriptions of their mathematics programs, including the mathematics courses offered and the mathematics textbooks used during the period the students were in high school (1998–2002). Thus, the response rate of our sample of high schools was 100%. Approximately 75% of the sampled students graduated from a high school in which a single curriculum (NSF-funded, commercial, or UCSMP) was offered, making it straightforward to assign these students to a curriculum cohort.

The remaining students (more than 3,000) graduated from a high school that offered two mathematics curricula (NSF-funded/commercial, NSF-funded/UCSMP, or commercial/UCSMP). Identifying the high school mathematics curriculum these students completed required a second step: cataloging the high school mathematics courses that appeared on the transcripts of these students and combining this with collateral information obtained from the high schools to assign a student to a curriculum cohort. Three project staff members completed Step 2.

Two of these staff members collectively possessed more than 60 years of teaching experience with high school mathematics. In all cases, a student was assigned to a curriculum cohort after the three staff members agreed on the assignment.

College information. An archival data file containing available records for students who enrolled in college in Fall 2002 or Fall 2003 was obtained from the sampled postsecondary institutions. Longitudinal data for students who began college in Fall 2002 or 2003 were obtained through the Spring 2006 or 2007 college semesters, respectively. These records contained student college mathematics grades and courses completed across eight semesters and a student's declared college major. Information about institutions was obtained from each institution's website and from the Carnegie Classification of Institutions of Higher Education.

Variables

Predictors. The high school mathematics curriculum that a student completed was the key predictor. We opted to combine the NSF-funded curricula (Core-Plus, IMP, MMOW) in the analyses for two reasons. First, there are strong conceptual links among these curricula, including their adherence to a common set of standards in their construction. Second, we used MANOVA to look for differences among students who completed the Core-Plus, IMP, and MMOW curricula on high school variables including ACT mathematics score and high school mathematics GPA and, separately, on college variables including first college mathematics course grade and college mathematics GPA. The MANOVAs were statistically significant at $p < .05$ but produced squared canonical correlations of about .01, meaning that the particular NSF-funded curriculum accounted for little variation in the outcomes. Thus, we did not distinguish among individual NSF-funded curricula in our analyses.

Dummy-coded predictors were used to capture curriculum cohort—NSF-funded (1 = yes, 0 = no) and UCSMP (1 = yes, 0 = no)—and commercial served as the reference group, meaning, for example, that the slope for the NSF-funded predictor represented the difference on the outcome between the NSF-funded and commercial cohorts. Other student predictors were ACT mathematics score, female (1 = yes, 0 = no), and college major, which was coded using a modified version of the categories appearing in a United States Department of Education report (Chen, 2005): Life Sciences/Health Sciences and Services (1 = yes, 0 = no), Humanities/Fine and Performing Arts/Social Sciences/Education (1 = yes, 0 = no), Business (1 = yes, 0 = no), and Other (1 = yes, 0 = no); Engineering/Mathematics/Computer Science/Physical Sciences (STEM) served as the reference group. Number of years of high school mathematics completed (3, 4, or 5) also served as a student-level predictor. In constructing this variable, it was apparent that a small number of students completed more than one mathematics course in a single year. For these students, we coded the higher level course taken. The high school mathematics GPA variable was omitted from our final models because results with and without this predictor were quite similar.

We coded ethnicity as a student predictor—African American (1 = yes, 0 = no), Asian American (1 = yes, 0 = no), and Hispanic (1 = yes, 0 = no)—so that white students served as the reference group. This pattern of coding ethnicity is common in regression analyses of achievement data (e.g., Attewell, Heil, & Reisel, 2011; Riegle-Crumb & Grodsky, 2010; Spenner et al., 2004). It is important to emphasize that the statistical results are identical regardless of which ethnic group serves as the reference group. Student predictors representing the interaction of ethnicity and curriculum cohort with respect to college mathematics outcomes also were constructed. These predictors allowed us to examine whether the relationship between college mathematics outcomes and ethnicity was related to curriculum (e.g., was one curriculum associated with a strengthening or weakening of this relationship).

Predictors at the institution level were enrollment, selectivity, financial profile, and educational profile. After examining institutional data for the 32 colleges, we found that three categories of educational profile characterized this feature of the institutions: (a) full-time/more selective/low transfer-in rate, (b) full-time/selective/high transfer-in rate, and (c) mixture of full-time and part-time/less selective/high transfer-in rate. Each institution's financial profile (public or private) also was coded.

Mathematics course taking was captured by creating a variable reflecting course difficulty level that matches the three-level scale used in Teitlebaum's (2003) analysis of the National Educational Longitudinal Study data but adds a fourth level:

- *Level 1*: This level includes developmental courses that should have been completed in high school. Students entering postsecondary study without the required or recommended high school mathematics background are typically directed to these courses, which are designed to provide a foundation for students to be successful in future mathematics courses and generally do not confer credits towards graduation.
- *Level 2*: This level includes courses that would be considered precalculus mathematics, including college algebra, precalculus, and finite mathematics.
- *Level 3*: This is the typical entry level for well-prepared high school students, who would start their college mathematics coursework with Calculus I.
- *Level 4*: This level is beyond a first course in calculus. Titles include Calculus II, differential equations, linear algebra, and multivariable calculus.

The difficulty-level variable was constructed by examining mathematics course descriptions at the 32 colleges and rating each course using the 4-point scale discussed above. The median number of mathematics courses offered by the colleges was approximately 23 and ranged from 3 to 47. Each course description was reviewed and rated independently by the three project staff members who examined high school mathematics courses. Initial levels of disagreement among

raters never exceeded 1 point on the difficulty scale and constituted approximately 2% of the courses rated. The disagreements that did occur were usually confined to one or two courses at a few institutions (typically for difficulty level 2) and were subsequently resolved through discussion among the raters. Thus, the consistency of ratings used in our analyses was 100%.

Dependent variables. College mathematics course grades, which were based on different grading scales across institutions, served as dependent variables in several analyses. Seventy-five percent of the sampled institutions used a 12-point grading scale (A, A–, B+, . . . , F), 18.75% used a 7-point grading scale (A, AB, B, BC, C, D, F), and 6.25% used a 5-point grading scale (A, B, C, D, F). We wished to use data from all institutions in each analysis, and we examined the impact of differences in grading scale on our statistical modeling results to assess the credibility of doing so. Specifically, we performed sensitivity analyses using three versions of the grade variable (transforming all grades to a 5-point scale, transforming all grades to a 7-point scale, using the original grading scales) in our major analyses and compared the results. To create a 7-point scale based on a 12-point scale, we transformed grades such that A = A, A–/B+ = AB, B = B, B–/C+ = BC, C = C, D = D, F = F. To transform grades based on a 5-point scale to a 7-point scale, we assumed that the grades of A through F were earned on a 7-point scale (A = 7, A–/B+ = 6, . . . , F = 0) even though this was not always the case, meaning that students at institutions using a 5-point scale had no opportunity to earn an AB or BC grade. Throughout this process, we assumed that similar grades earned in similar courses at different institutions reflected similar proficiency. The statistical modeling results for the three versions of the grade variable were similar, and we report findings using the 7-point scale.

Two other dependent variables captured the likelihood that a student completed a college mathematics course with a difficulty equaling or exceeding that of Calculus I or that was developmental in nature. Each dependent variable was analyzed separately.

Important Methodological Issues

Selection bias. One potential source of bias appeared in high schools offering both NSF-funded and commercial curricula. The selection bias consisted of less mathematically proficient students being channeled into NSF-funded curricula at the beginning of high school (Harwell et al., 2009), creating preexisting differences among the cohorts. A selection bias of this nature would likely favor the commercial cohort as they moved into college mathematics coursework and disadvantage students who completed an NSF-funded curriculum. There may be a similar bias in high schools offering NSF-funded and UCSMP curricula or a commercial and UCSMP curriculum, but we have no evidence of this. In our study, this potential bias applies to 2,005 or 15% of the sampled students who graduated from one of 27 high schools in which NSF-funded and commercial curricula were offered.

To respond to the possibility of selection bias, we included statistical control

variables in the analyses for which the curriculum cohorts differed significantly, including ACT mathematics scores. We note that previous results have suggested that mathematics proficiency is a fairly stable trait over time (e.g., Ding & Davison, 2005), a finding supported by Post et al. (2010), who found that ACT mathematics scores were essentially unaffected by high school mathematics curriculum. We also performed analyses to assess the sensitivity of our findings to such bias.

Bias due to missing data. An important data-based issue was the presence of missing college grade and difficulty level data. To respond to missing data and the associated possibility of biased inferences we turned to Little and Rubin's (2002) characterization of incomplete data through ignorable and non-ignorable response mechanisms. The data consisted of an $N \times P$ grid (where N represents the number of students and P represents the number of courses) in which 13,337 students had grades and difficulty levels for between 1 and 14 college mathematics courses. Because 87.3% of the students in our sample completed three or fewer college mathematics courses and 93.8% completed four or fewer courses, most of the cell entries in this grid were empty (incomplete or missing).

In the $N \times P$ data grid, most of the missing grade and difficulty values can be explained by the fact that college majors generally have requirements that specify the minimum number of mathematics courses students must successfully complete as part of their degree work and that students generally complete only required mathematics courses (Harwell et al., 2009). The inclusion of college major in the statistical modeling should minimize parameter estimation bias relative to ignoring missing grade and difficulty data (Little & Rubin, 2002).

We also examined our data for evidence of the impact of missing values for variables such as ACT mathematics score and years of high school mathematics. Data were missing relatively infrequently for most student variables (e.g., ACT mathematics scores were missing for 2.3% of students), although college major was missing more often (17% of the time). Missing college major values were overwhelmingly linked to students with Other majors, which included technologies, technicians, trades, military, and culinary arts. A consequence of missing data is that different analyses are based on different subsets of students. Extensive analyses comparing students for whom data were or were not available (e.g., ACT mathematics scores, college major) did not provide evidence of important differences in the pattern of significant results or the estimated magnitude of a statistically significant effect.

Other potential sources of bias. The fidelity with which high school mathematics curricula are implemented is an important methodological issue (O'Donnell, 2008) and applies equally to the implementation of NSF-funded, commercial, and UCSMP curricula. Although recent evidence suggests that fidelity is unrelated to student achievement (Krupa, 2011), we think that assessing fidelity of implementation continues to be both important and difficult, especially in a retrospective study like ours. One factor promoting fidelity for the NSF-funded curricula is that

an earlier NSF-funded study (Minneapolis and St. Paul Merging to Achieve Standards Project [MASP]) provided more than 400 high school mathematics teachers from the Twin Cities area with an average of 120 hours of targeted professional development using NSF-funded curricula (see Harwell et al., 2007) between 1996 and 2001.

After completing the targeted professional development in NSF-funded curricula through MASP, the 400 high school mathematics teachers (representing almost all those teaching NSF curricula in the Twin Cities) collectively taught about 25,000 high school students during 1996–2002. Among the students in our sample who completed an NSF-funded curriculum, almost all graduated from a Twin Cities high school, meaning that they were very likely to have been taught by a MASP teacher. This suggests at least moderate fidelity of implementation of NSF curricula, although significant variation is likely even among MASP teachers (Tarr et al., 2008).

The alignment (or lack thereof) between high school mathematics curricula and college mathematics placement tests was also a potential source of bias in its impact on college placement test scores and advising students on which college mathematics course to enroll in initially. Available evidence suggests that the strongest alignment is between a commercial curriculum and material likely to appear on college mathematics placement tests and, thus, is most likely to benefit students who completed a commercial curriculum (Norman, Medhanie, Harwell, Anderson, & Post, 2011). There does not appear to be any available evidence of this effect for the UCSMP curriculum.

Another important issue in estimating the relationship between high school mathematics curricula and college mathematics achievement and course taking is variation in the quality of teaching within and between institutions and institutional assessment practices (Wayne & Youngs, 2003). However, we had no data capturing these factors.

Cross-curricular contamination in a high school offering two curricula could also be a source of bias. For example, teachers teaching a commercial curriculum and those teaching an NSF-funded curriculum in a high school could, as a result of professional or social interactions, actively incorporate features of the other curriculum that they thought useful into their teaching and bias the results. Student transience (students transferring to high schools that use a different curriculum) and variation in high school and class size, which have been shown to affect student achievement (Nye, Hedges, & Konstantopoulos, 1999), also represented potential sources of bias. Unfortunately, we did not have data to empirically test for these biases.

In summary, the potentially damaging effects of different sources of bias on our inferences represent important methodological challenges. Although we were unable to respond to all threats, we think that our response to several threats, using control variables and sensitivity analyses, is consistent with recommended practice (What Works Clearinghouse, 2008), and, combined with evidence that key threats were unlikely to seriously distort our findings, allows us to credibly use our results to answer our research questions.

Data Analysis

We used descriptive analyses—two-level (students within institutions) and three-level (longitudinal grade and difficulty level data within students within institutions) multilevel models—to capture the hierarchical nature of the data and the associated dependency in outcomes within students and institutions (see Raudenbush & Bryk, 2002). Dependencies in college mathematics outcomes are likely among students with similar pre-college backgrounds (e.g., high school experiences, academic background) who enroll in the same institution (Spenner et al., 2004; Whitt, Pascarella, Nesheim, & Marth, 2003).

We did not include high school as a level in our analyses because we did not have sufficient data to support a model that also included student- and institution-level predictors. To assess the possible impact of not including high school in our models, we fitted an unconditional (no predictors) cross-classified random effects model with high schools and colleges as factors (Raudenbush & Bryk, 2002) to our cross-sectional outcomes. There was no evidence of important differences between high schools for these outcomes. For example, the cross-classified random effects model fitted to Calculus I grades showed that the intraclass correlation for high schools (i.e., correlation among Calculus I grades of two students from the same high school who attended different colleges) was .01.

We reformulated the semester variable (1 through 8) to capture mathematics courses completed (first mathematics course completed, second mathematics course completed, and so on) to facilitate interpretation. Most students completed 3 or fewer college mathematics courses, but a few completed up to 14 courses, raising concerns that these latter values might reflect outliers in the data analyses and disproportionately impact the findings (Langford & Lewis, 1998). To respond to this possibility, we limited the number of mathematics courses considered in our analyses to seven. For the few students who completed more than seven mathematics courses, the course beyond the seventh with the highest difficulty level was used as the seventh course and the remainder of the course data was excluded.

To construct a three-level model, we generally followed the steps recommended by Bliese and Ployhart (2002) to identify the explanatory model that provided the best fit to the data. This led us to divide the analyses into two phases for each outcome variable that involved (a) identifying a between- or within-student model with predictors and an error structure consistent with the data and determining if slopes of predictors should be fixed (not allowed to vary across students/institutions) or random (allowed to vary across students/institutions) effects and (b) performing statistical tests of the fixed effects.

To control for compounding of Type I error rates and ensure that the overall error rate equaled that specified, we used an adjusted error rate attributed to Sidak (1967). The adjusted error rate was computed as $\alpha' = 1-(1-\alpha)^{1/m}$ where α is the unadjusted Type I error rate, m represents the number of statistical tests within an analysis phase for a given outcome, and α' is the adjusted Type I error rate. We chose an overall $\alpha = .15$ for the statistical tests of fixed effects associated with each outcome variable. This practice is consistent with the frequently recommended

familywise error rate strategy for controlling Type I error rates without severely dampening statistical power (Keselman, Cribbie, & Holland, 2002). Preliminary analyses showed that the financial profile variable had little explanatory power, so it was removed from the model.

Grades. For cross-sectional analyses of the Calculus I grade data (Research Question 1a) we fitted a two-level model (students within institutions). Research Questions 1b and 1c were answered by fitting a three-level model (longitudinal grades within students within institutions). A review of the literature for college mathematics grades did not suggest a theoretical model for the expected pattern of student grades over time, and an empirical examination of the grade and difficulty level trajectories led us to use models in which the patterns over courses were linear. For longitudinal analyses of grades, we assumed that within-student error terms followed an autoregressive structure such that the error associated with the grade of the ith student in his or her pth college mathematics course depended on the error associated with the grade of his or her $(p-1)$th mathematics course (Raudenbush & Bryk, 2002).

Following the advice of Biesanz, Deeb-Sossa, Papadakis, Bollen, and Curran (2004), we rescaled the predictors in the within-student model to answer questions of interest. Specifically, mathematics courses completed was rescaled so that 0 represented the last college mathematics course completed, 1 represented the next to last college mathematics course completed, and so on. Similarly, the 4-point difficulty level variable was rescaled so that 0 represented the most difficult college mathematics course a student completed, which was almost always a student's last mathematics course. The rescaling allowed us to interpret the intercept estimated for each student in the three-level model as the expected grade in a student's last and most difficult college mathematics course and to use multilevel analyses involving student intercepts to answer key research questions.

Because the difficulty level variable had an ordinal scale of measurement, the models for grade were also fitted using three dummy-coded predictors to capture the information in the difficulty level variable. The multilevel findings using difficulty level as a predictor in its original form and those obtained using dummy-coded predictors to represent difficulty level produced the same pattern of findings, and the results for difficulty level in its original (4-point) form are reported.

Predictors for the student and institution models were grand-mean-centered so that interpretations of slopes are in terms of a student or institution whose values on these variables are average. An average student is one whose value for these predictors equals the grand mean of these variables. For example, the estimated grand mean for ACT mathematics scores was 24, and an average student is one whose ACT mathematics score equals this value. Frequently, there are no actual cases with values for the predictors exactly equal to the estimated grand means of these predictors, and the phrase *average student* is for descriptive purposes.

Difficulty levels. To answer our research questions related to the difficulty level outcome, the original 4-point difficulty level variable was dichotomized to create

two separate dependent variables. One dependent variable for difficulty level was defined such that 1 indicated that a student completed Calculus I or beyond and 0 otherwise. Research Questions 2a and 2b were answered by fitting generalized hierarchical proportional odds models to these data (see Hosmer & Lemeshow, 2000). The predictors in these models were the same as those for the grade data except that difficulty level was not a predictor in the within-student model and an autoregressive error structure for difficulty levels was assumed.

A second dependent variable for difficulty was defined such that 1 indicated that a student completed a developmental mathematics course and 0 otherwise. Research Question 2c was answered by fitting generalized hierarchical proportional odds models to this cross-sectional outcome. The predictors in this model were the same as those used for analyses of grades.

Model checking revealed some evidence of departures from underlying assumptions, primarily for a subsample of two to three institutions whose residuals did not appear to be consistent with the assumption of multivariate normality. Refitting our models without this subsample (whose composition varied over dependent variables) produced the same general pattern of findings as those using all of the institutions, and the results reported are based on the full sample of institutions. Also, no evidence of collinearity emerged in preliminary analyses. Multilevel analyses were performed with the R 2.10.1 lme4 package (R Development Core Team, 2012), and full maximum likelihood (Raudenbush & Bryk, 2002) was used to estimate parameters.

Results

Next, we present results for a series of descriptive and multilevel analyses of the college mathematics grade and difficulty level data. The descriptive analyses reflect patterns in the data, and the results of the multilevel analyses provide evidence about the research questions.

Descriptive Analyses

Among postsecondary institutions, the average enrollment based on information provided on each institution's website was 4,953 (SD = 6,577), but there was substantial variability across institutions as summarized using quartiles (Q1 = 1,795, Q2 = 2,449, Q3 = 4,201). Samples from each institution also varied, with seven colleges providing less than 100 students and seven providing more than 500. Table 1 reports various summary statistics for students by educational profile. These statistics suggest that the distribution of students across the curriculum cohorts was approximately equal within each of the three categories of the educational profile variable. For example, the percentage of full-time students in more selective institutions who completed the NSF, UCSMP, or commercial curricula in high school were 55.8%, 57.3%, and 56.7%, respectively.

Table 1 also shows that far fewer (N = 301) but proportionally more African American students (73.1%) were in more selective institutions compared to white

Table 1
Summary Statistics for Curriculum Cohort, Female, Ethnicity, and College Major by Educational Profile of the Sample Postsecondary Institutions

		Educational profile					
		Full-time/more selective/ low transfer-in rate[a]		Full-time/ selective/ high transfer-in rate[b]		Mixture of full-time and part-time, less selective/ high transfer-in rate[c]	
		N	%	*N*	%	*N*	%
High school mathematics curriculum	NSF-funded	971	55.8	754	43.3	16	0.9
	UCSMP	1,292	57.3	950	42.1	12	0.5
	Commercial	4,598	56.7	3,486	43.0	22	0.3
Female	Yes	3,689	50.2	3,633	49.5	22	0.3
	No	3,691	61.7	2,254	37.7	37	0.6
Ethnicity	Asian	843	78.6	229	21.4	0	0.0
	African American	301	73.1	111	26.9	0	0.0
	Hispanic	135	63.1	79	36.9	0	0.0
	White	5,993	52.6	5,345	46.9	57	0.5
College major	Life Sciences	1,144	54.7	949	45.3	0	0.0
	Humanities	2,757	60.2	1,819	39.8	0	0.0
	Business	849	38.3	1,368	61.7	0	0.0
	Other	67	21.3	247	78.7	0	0.0
	STEM	1,247	66.6	3,486	43.0	0	0.0

[a] Full-time/more selective means at least 80% of students attend school on a full-time basis and average ACT score is in upper one fifth of all colleges. Low transfer-in rate means less than 20%. [b]Full-time/selective/high transfer-in rate means at least 80% of students attend school on a full-time basis, average ACT score is in middle two fifths of all colleges, and transfer-in rate of at least 20%. [c]Less selective means average ACT score is in bottom two fifths of all colleges.

Table 2
Summary Statistics by Educational Profile

		Educational profile								
		Full-time/more selective/ low transfer-in rate			Full-time/ selective/ high transfer-in rate			Mixture of full-time and part-time, less selective/ high transfer-in rate		
		M	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*
ACT Math 25th %[a]		25.16	4.62	7,239	22.50	4.22	5,728	22.05	4.07	57
Grade first college math course		2.82	1.02	7,225	2.61	1.15	5,693	2.55	1.19	51
Grade last college math course		2.77	0.89	6,915	2.63	1.06	5,566	2.66	1.24	59
Average grade by difficulty level[b]	1 (Dev)	2.31	1.19	1,513	2.32	1.19	1,286	2.06	1.05	32
	2 (Coll Alg)	2.49	1.12	6,944	2.53	1.17	6,429	2.46	1.10	76
	3 (Calc I)	2.67	1.06	8,035	2.71	1.08	4,142	3.00	0.88	52
	4 (> Calc I)	2.89	1.01	8,029	2.92	0.97	2,976	3.21	0.75	28
		%		*N*	%		*N*	%		*N*
First college math course (proportion in each difficulty level)	1 (Dev)	9.2		678	12.8		747	23.7		14
	2 (Coll Alg)	42.8		3,145	63.9		3,713	50.8		30
	3 (Calc I)	33.7		2,477	19.9		1,158	25.4		15
	4 (> Calc I)	14.2		2,045	3.4		197	0.0		0
Last college math course (proportion in each difficulty level)	1 (Dev)	4.6		320	8.7		488	6.8		4
	2 (Coll Alg)	38.4		2,689	60.6		3,398	59.3		35
	3 (Calc I)	25.2		1,766	17.0		951	23.7		14
	4 (> Calc I)	31.8		2,225	13.7		768	10.2		6

Note. Dev = developmental (difficulty level 1, courses that should have been taken in high school), Coll Alg = college algebra/precalculus (difficulty level 2), Calc I = Calculus I (difficulty level 3), > Calc I = beyond Calculus I (difficulty level 4).
[a]Scores correspond to the 25th percentile and were taken from institution websites. [b]The sample sizes represent the total number of students who completed a course at that level; students who took courses at different difficulty levels were counted in the sample size for each level.

students (5,993 and 52.6%, respectively). The number of students in our sample who were full-time or part-time students at less selective institutions with high transfer-in rates was quite small, and inferences based on the reported percentages should be made cautiously.

Table 2 reports summary statistics for ACT mathematics scores corresponding to the 25th percentile, average grade earned in each difficulty level, and the proportion of students in each difficulty level for the first and last college mathematics courses by educational profile. These results suggest that students in more selective institutions tended to be more proficient in mathematics and to enroll in higher level mathematics courses, compared to students in less selective institutions. For example, 33.7% of full-time students in more selective institutions with low transfer-in rates began their mathematics coursework with Calculus I, whereas 19.9% of full-time students in less selective institutions with high transfer-in rates did so.

Table 3 reports summaries for students' ACT mathematics score, first and last college mathematics course grades, and the difficulty level of students' first and last college mathematics courses by curriculum cohort across institutions. There was little difference in effect sizes between the UCSMP and commercial curriculum cohorts.

Bivariate correlations among student variables were also computed. The largest correlation was between grade earned in a student's first and last college mathematics courses (.76), followed by the correlation between ACT mathematics score

Table 3
Summary Statistics for Student ACT Mathematics Scores, Grades, and Difficulty Level of First and Last College Mathematics Courses by Curriculum Cohort

	High school mathematics curriculum								
	NSF-funded			UCSMP			Commercial		
	M	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*
ACT Math	22.30	4.50	1,705	24.60	4.30	2,190	24.20	4.70	7,933
Grade first college math course	2.61	1.12	1,690	2.74	1.08	2,203	2.74	1.09	7,895
Grade last college math course	2.62	0.99	1,637	2.73	0.97	2,122	2.71	0.97	7,619
Difficulty level first college math course	2.11	0.80	1,728	2.48	0.87	2,243	2.41	0.87	8,037
Difficulty level last college math course	2.50	1.10	1,655	2.86	1.14	2,143	2.80	1.16	7,696

and number of years of high school mathematics (.46); all other correlations were less than .30 and thus reflect moderately weak to weak relationships.

Multilevel Analyses

Multilevel models were used to study the relationship between curriculum cohort and college mathematics achievement and course taking. These models reflect data obtained at multiple levels of a data hierarchy and include both fixed effects and random effects. The grade and difficulty level variables were analyzed separately, and the findings presented below are organized by research question. Predictors representing the interaction of ethnicity and curriculum with respect to college mathematics outcomes were initially included in the models but were not statistically significant, meaning that the relationship between college mathematics outcomes and ethnicity was unrelated to the high school mathematics curriculum a student completed.

Research Question 1a: College Calculus I grades. Two-level (students within institutions) cross-sectional fixed effects results for Calculus I grades are reported in Table 4. These analyses estimated the effect of each predictor on the outcome via a slope (B) and tested each slope using a t-test. Table 4 also reports standardized slopes (*Stand B*) for statistically significant effects to enhance interpretation.

The largest effects in Table 4 were for college major, with an average student in STEM (i.e., a student whose values for the predictors in the model, such as ACT math, equaled the grand mean of those variables) earning a grade that was 0.324 units (on a 7-point scale) higher than an average student whose major was categorized as Other. Equivalently, an average STEM major earned a grade that was 0.334 standard deviations higher than students whose college major was Other. Average male students earned a Calculus I grade that was 0.12 units (0.13 *SD*) higher than that earned by average female students. ACT mathematics scores and years of high school mathematics were also significant predictors of Calculus I grades. However, ethnicity was not a significant predictor, meaning that African American and white students had, with the other predictors held constant, similar Calculus I grades. The predictors representing Asian American and Hispanic students, and the institutional predictors, were not significant.

We performed a sensitivity analysis for these findings to assess the impact of specific features of the data and the model on the results. Findings that do not appear to be overly influenced by these specific features provide evidence that our results are not seriously distorted by threats to inferences such as selection bias.

One sensitivity analysis was performed in response to concerns about the impact of less mathematically proficient students being channeled into NSF-funded curricula at the beginning of high school (Harwell et al., 2009). We conditioned the analysis of college Calculus I grades on the difficulty level of a student's first college mathematics course (determined by project staff as described earlier) by refitting our model for students who began their college mathematics course taking

Table 4
Multilevel Cross-sectional Fixed Effects Results for College Calculus I Grades

Between-student model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Intercept	2.588	0.042	61.72	—
NSF-funded	−0.004	0.048	n.s.	—
UCSMP	− 0.02	0.039	n.s.	—
Female	− 0.120	0.031	− 3.83	−0.130
African American	0.061	0.105	n.s.	—
Asian	− 0.029	0.057	n.s.	—
Hispanic	0.122	0.141	n.s.	—
Life Sciences	− 0.007	0.042	n.s.	—
Humanities	− 0.217	0.043	− 5.03	−0.224
Business	− 0.050	0.051	n.s.	—
Other	− 0.324	0.081	− 3.99	−0.334
ACT Math	0.074	0.004	16.56	0.077
Years HS Math	0.169	0.026	6.58	0.175
Between-institution model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Enrollment	0.00	0.00	n.s.	—
Selectivity[a]	0.023	0.023	n.s.	—
Educational Profile	−0.099	0.112	n.s.	—

Note. Analyses were based on data from 30 postsecondary institutions and 3,649 students. *Stand B* = standardized effect sizes. n.s. = not significant at $\alpha' = 1 - (1 - \alpha)^{1/m} = .00101$, where $\alpha = .15$ is the unadjusted Type I error rate, m = number of statistical tests (16 fixed effects), and α' is the adjusted error rate.

[a]ACT score of freshman class corresponding to the 25th percentile.

with a course of similar difficulty. This strategy helped to minimize the possible effects of channeling. This resulted in two sets of analyses that corresponded to the second and third difficulty levels, College Algebra and Calculus I. Students who began their college mathematics course taking with a course of difficulty level 1 were omitted from these sensitivity analyses because there were a small number of cases, and those who began with a course of difficulty level 4 (Calculus II or beyond) never took Calculus I in college. There was agreement between the above findings and those produced by the sensitivity analyses for the effect of curriculum cohort on Calculus I grades. Specifically, the same pattern of significant and nonsignificant effects emerged, providing evidence that the main results were not overly influenced by selection bias attributable to channeling.

We also refitted the Calculus I grade model using a sample of students known to have graduated from a high school offering a single mathematics curriculum, and, separately, to a sample of students who graduated from a high school offering two different mathematics curricula. The pattern of significant and nonsignificant results was the same and provided evidence that our results were not sensitive to whether high schools offered one or two curricula.

Research Questions 1b and 1c: Grades across college courses. Three-level analyses of college mathematics course grades are reported in Table 5. Curriculum cohort was not a significant predictor of grade earned in a student's last college mathematics course (NSF-funded slope = 0.022, n.s.; UCSMP slope = –0.038, n.s.), but other student variables were. For example, average African American and Asian American students earned a grade in their last college mathematics course that was 0.28 units (0.37 *SD*) and 0.18 units (0.23 *SD*), respectively, below that earned by average white students. Average female students tended to earn a grade that was 0.19 units (0.25 *SD*) lower than males in their last mathematics course. Overall, the between-student intercept model reported in Table 5 for last college mathematics grade accounted for 12% of the variance compared to the unconditional model.

The within-student model in Table 5 shows that the more mathematics courses a student completed, the more difficult these courses tended to be (B = 0.403 or 0.52 *SD*). Student grades increased slightly per mathematics course completed, even though courses typically increased in difficulty (B = –0.079 or –0.102 *SD*). Mathematics courses completed was coded so that the last course completed was coded 0, the next to last course was coded 1, and so on. This pattern is likely attributable to the fact that students who completed more college mathematics courses were more mathematically proficient than students who completed fewer mathematics courses before entering college. For example, the average ACT mathematics score for students who completed one nondevelopmental course (almost always college algebra) was 23.60 (SD = 4.28); for students who completed two or more nondevelopmental mathematics courses (usually College Algebra and Calculus I), the average ACT mathematics score was 25.16 (SD = 4.64). This difference in means is statistically significant ($p < .05$) and suggests that students who completed more college mathematics courses were more mathematically proficient than their peers.

Table 5
Multilevel Results for Longitudinal Grade Data

Within-student model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Intercept	2.682	0.040	67.21	—
Linear slope	−0.079	0.014	−5.61	−0.102
Difficulty slope	0.403	0.016	25.51	0.521
Between-student model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Intercepts				
NSF-funded	0.022	0.029	n.s.	—
UCSMP	−0.038	0.026	n.s.	—
Female	−0.193	0.021	−9.33	−0.253
African American	−0.284	0.065	−4.39	−0.373
Asian	−0.176	0.039	−4.57	−0.231
Hispanic	−0.072	0.083	n.s.	—
Life Sciences	0.046	0.034	n.s.	—
Humanities	0.150	0.030	4.95	0.197
Business	0.137	0.034	4.00	0.180
Other	−0.154	0.059	−2.61	− 0.202
ACT Math	0.065	0.003	23.78	0.085
Years HS Math	0.062	0.016	3.77	0.081
Linear Slopes				
NSF-funded	−0.032	0.017	n.s.	—
UCSMP	0.00	0.014	n.s.	—
Female	0.021	0.012	n.s.	—
African American	0.085	0.031	2.71	0.111

Asian	0.041	0.019	n.s.	—
Hispanic	0.109	0.052	n.s.	—
Life Sciences	−0.054	0.019	−2.88	−0.071
Humanities	−0.022	0.015	n.s.	—
Business	−0.009	0.022	n.s.	—
Other	−0.042	0.031	n.s.	—
ACT Math	0.002	0.001	n.s.	—
Years HS Math	0.013	0.009	n.s.	—

Between-institution model

Effect	*B*	*SE*(*B*)	*t*-value	
Intercepts				
Enrollment	0.00	0.00	n.s.	—
Selectivity	−0.001	0.023	n.s.	—
Educational profile	−0.052	0.113	n.s.	—
Linear Slope				
Enrollment	0.00	0.00	n.s.	—
Selectivitya	0.012	0.007	n.s.	—
Educational profile	−0.003	0.035	n.s.	—

Note. Analyses were based on data from 32 postsecondary institutions and 10,288 students. *Stand B* = standardized effect sizes. n.s. = not significant at $\alpha' = 1-(1-\alpha)^{1/m} = .00101$, where $\alpha = .15$ is the unadjusted Type I error rate, m = number of statistical tests (33 fixed effects), and α' is the adjusted error rate. Reported coefficients were statistically significant.
[a]ACT score of freshman class corresponding to the 25th percentile.

Similarly, one half of STEM majors (about 29% of the sample) completed Calculus II or a more difficult course and earned an average grade of 2.77 in these classes whereas for non-STEM majors, one half completed college algebra or a more difficult course and earned an average grade of 2.71 in these classes. Thus, taking more mathematics was associated with more difficult courses in which students earned slightly higher grades. Overall, 5.5% of the variance within students in the model in Table 5 was accounted for, a relatively modest effect.

Curriculum cohort was not a significant predictor of students' linear grade trajectories (pattern of grades over mathematics courses). Table 5 shows that the largest

effect in the slope models involved African American students (0.085 units, 0.11 *SD*), meaning that an average African American student's linear grade trajectory increased at a slower rate than that of an average white student. None of the institution predictors were significant. Difficulty level slopes varied significantly, meaning that the difficulty of mathematics courses completed varied across students but none of the predictors in this model were significant.

We performed a sensitivity analysis for Table 5 results by conditioning the longitudinal analysis of college mathematics grades on the difficulty level of a student's first college mathematics course. Once again, we refitted our multilevel models to the grade data for students who began their college mathematics course taking with a course of similar difficulty, and there was strong agreement between the main findings and those produced by the sensitivity analyses, including no significant impact of curriculum on grades.

We also refitted the longitudinal grade models to subsamples of students known to have graduated from a single or a dual curriculum high school. The pattern of the results was similar to those reported in Table 5 with one exception: For students from high schools offering both NSF-funded and commercial curricula, the NSF-funded curriculum (1 = yes, 0 = no) by mathematics course completed slope was significant (–0.06), meaning that the linear grade trajectories of the NSF-funded cohort increased at a slightly faster rate than that of the commercial curriculum cohort. To explore this pattern further we refitted the main longitudinal grade model after adding a predictor indicating whether a student graduated from a single or a dual curriculum high school and found that the pattern of significant and nonsignificant results was the same as that reported in Table 5.

Research Questions 2a and 2b: Course difficulty levels. The results of fitting a three-level model to a binary outcome variable, in which 1 indicated a course difficulty equal to or greater than Calculus I and 0 otherwise, showed that none of the predictors were statistically significant. Thus, curriculum cohort, ethnicity, and gender were not related to the difficulty level of a student's last college mathematics course or to whether they completed Calculus I or a higher level college mathematics course.

Research Question 2c: Developmental course taking. Our data indicated that overall, 22%, 12%, and 10% of the sampled students who completed an NSF-funded, commercial, or UCSMP curriculum, respectively, began their college mathematics with a developmental course. To explore this further, we fitted a two-level (students within institutions) model to a binary outcome in which 1 indicated a student completed a developmental mathematics course in college and 0 otherwise. Fitting this model with the predictors appearing in Table 5 showed that curriculum cohort was a significant predictor of the likelihood that a student completed a developmental mathematics course in college. An average student in the NSF-funded cohort was about 1 1/3 times as likely to initially take a developmental mathematics course in college than to take a more difficult course, compared

to students in the commercial curriculum cohort. Equivalently, the model-implied probability (Raudenbush & Bryk, 2002) that a student who completed an NSF-funded curriculum will take a developmental mathematics course was .58.

Discussion

Advances in learning theory and the acknowledgement of students' interests and abilities increasingly play significant roles in the conceptualization of the mathematics curricula designed for school-age students. The NSF-funded and UCSMP programs studied here were guided by the 1989 NCTM Standards and can be thought of as reflecting the best practices in the field of teaching and learning of school mathematics at the time (NCTM, 1991). However, there have been relatively few studies comparing the impact of different curricula on students' mathematics proficiency, especially among college students.

This study contributes to the empirical research base by characterizing the extent to which high school mathematics curricula prepare college-bound students for college mathematics. Our findings can be organized into three complementary categories. First, our results suggest that NSF-funded, commercial, and UCSMP curriculum prepared students who began with a nondevelopmental course in college (e.g., College Algebra or Calculus I) equally well in terms of longitudinal mathematics achievement and course-taking patterns across a range of colleges, including both public and private institutions. Collectively, these results suggest that the largely anecdotally based criticisms that NSF-funded curricula leave students unprepared to succeed in college calculus and more difficult courses are unsubstantiated. The results also extend earlier findings by Schoen and Hirsch (2003), Harwell et al. (2009), and Post et al. (2010), who examined similar data from a single postsecondary institution.

In examining our findings, the absence of statistical significance for the high school mathematics curriculum variable raises the question of whether there was sufficient statistical power to detect this effect. Prior to conducting our study, a power/sample size analysis that focused on testing the curriculum effect for cross-sectional and longitudinal data was done using the Optimal Design software (Raudenbush, Liu, Spybrook, Martinez, & Congdon, 2006). Specifically, we projected that a sample size of 35 postsecondary institutions and 5,000 students would allow us to detect a standardized curriculum effect of .35 (a medium effect following Cohen, 1988) with a power exceeding .80 in most analyses. Although the final number of sampled students generally exceeded 10,000, the number of institutions in our sample was 32. Juxtaposed against the slightly smaller than anticipated number of postsecondary institutions is that p-values for tests of the estimated curriculum effect for the various outcomes typically exceeded .30, suggesting that the lack of statistically significant curriculum effects is unlikely to be attributable to inadequate power.

Second, our analyses indicate that students who completed an NSF-funded curriculum were more likely to enroll initially in a developmental mathematics

course in college, compared to students who completed a commercial or UCSMP curriculum. This result may of course be due to differences in the ability of these curricula to prepare students for courses situated at difficulty level 2 (college algebra), but other explanations are also plausible. For example, these results may reflect a bias in which less proficient students tend to be channeled into an NSF-funded curriculum in high schools that also offer a commercial curriculum, the differential advising NSF-funded curriculum students may receive as to which college mathematics course to enroll in initially, a relative lack of alignment between NSF-funded curricula and college mathematics placement tests, or some combination of these (Norman et al., 2011). We also found that students from different curricula were equally likely to finish their college mathematics course taking with a course of similar difficulty, which we think is important.

The study by Norman, Medhanie, Harwell, Anderson, and Post (2011) highlights the importance of both high schools and colleges participating in serious efforts at realignment. For example, high school curricula should ideally include content appearing in college placement tests, and colleges need to be aware of the content differences of the various high school curricula and ensure that placement tests reflect the content and approaches of these newer curricula. However, the locally developed nature of many college mathematics placement tests, and the absence of evidence that these tests predict subsequent student mathematics achievement (Medhanie, Dupuis, LeBeau, Harwell, & Post, 2012), significantly complicates this task. Still, the introduction of the Common Core State Standards for Mathematics (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010) offers directions for the changes that both high schools and postsecondary institutions must make to reach alignment with the standards. The Common Core State Standards for Mathematics have parallel tracks, one labeled traditional and the other designated as integrated, with the latter seemingly designed to accommodate alternative approaches to high school mathematics.

Third, the presence of a grade trajectory for African American students that did not increase as fast as that of white students, and differences between African American and white students in the grade earned in their last college mathematics course, are consistent with literature indicating that the achievement gap in mathematics in high school continues into college. Spenner, Buchmann, and Landerman (2004) argued that much of the gap in mathematics achievement between African American and white students in their first college semester can be explained by differences in educational experiences beginning in early childhood, as well as family backgrounds of students prior to college enrollment (see also Treisman, 1992).

Spenner et al. (2004) also suggested that institutional climate, reflecting the extent to which campus environments (e.g., classroom, dormitory) are perceived as welcoming and inclusive or hostile and exclusive, may impact the mathematics achievement gap between African American and white students. Spenner et al. did not have data to test this assertion empirically, and, unfortunately, neither did we. However, our finding of no interaction between student ethnicity and curriculum and our outcomes provide evidence that enrollment in new and different high

school mathematics curricula, in and of itself, is unlikely to significantly impact the mathematics achievement gap.

In summary, our results offer a clear take-home message: There is no single road to success in college mathematics. Alternate conceptualizations of the content, varied instructional approaches, and varied assessment practices resulted in similar success in mathematics for thousands of students who began college mathematics with college algebra or a higher level mathematics course. These results can be seen as a contribution to understanding the efficacy of these best practices for the mathematics classroom. The NSF-funded curriculum students who began their college mathematics with college algebra or a more difficult course showed college achievement patterns and course difficulty levels that equaled that of students who completed other curricula in high school. These results provide support for the instructional approaches and organization of the content domains reflected in NSF-funded curricula.

For postsecondary institutions, our findings suggest that the mathematics curriculum a student completed in high school is unlikely to predict differential success in college mathematics. This should inform the development and modification of college mathematics placement tests, initial course enrollment recommendations, and perhaps even the design of college courses.

Limitations

There are important limitations of our study that should be considered in examining our results. One is evidence of a selection bias connected to student enrollment in an NSF-funded curriculum (less proficient students tended to be channeled into an NSF-funded curriculum in high schools also offering a commercial curriculum). The use of control variables in our main analyses and the associated sensitivity analyses provide evidence that this bias does not overly influence our findings, but it is impossible to be sure. It is also possible that a school-level bias exists in which schools with lower levels of student achievement are more likely to experiment with the adoption of NSF-funded curricula.

A second limitation is fidelity of implementation. A portion of the high school mathematics teachers were involved in a targeted NSF-funded professional development program of 120 hours or more over a 2-year period. Our evidence suggested that they implemented the NSF-funded curricula with a reasonable amount of fidelity (Harwell et al., 2009; Post et al., 2010). This, however, does not eliminate concerns about the potentially important biasing effects of this factor on a wider sample of teachers (O'Donnell, 2008) and applies to the implementation of all curricula, commercial and UCSMP included. We recognize that fidelity of curricula implementation is a key factor in attempts to assess the viability of a new mathematics curriculum and that fidelity is a complex concept that can be measured in more than one way (e.g., content coverage, pedagogical approach, interview, classroom observation).

We also acknowledge that the degree of departure from more traditional curricular and pedagogical approaches may be significantly larger and more complex for

teachers of NSF-funded curricula than for teachers of the other two curricula. In our experience, implementing NSF-funded curricula in ways that accurately reflect the authors' intent requires a particularly intense effort by teachers that requires substantial time and resources. Whether this leads to implementing NSF-funded curricula with more or less fidelity is at this point unclear.

Other important limitations include purposive sampling of colleges, which does not ensure that inferences can be generalized beyond our sample, and a quasi-experimental design that places limits on the causal inferences that can be drawn. The impact of missing data and student mobility and the possibility of important differences within and between institutions in the quality of mathematics instruction and variation in assessment practices should also be kept in mind in examining our results.

A more general limitation is that the analyses conducted here were based on NSF-funded first edition materials, meaning that our study would be difficult to replicate using second edition materials, as these have only recently been released after renewed funding by NSF. Assuming that a replication would require 4 years of high school performance data followed by 4 years of college mathematics achievement and course-taking data, an additional 8 years would be needed to allow the kind of curriculum comparisons reported in this study. Of course, there is no guarantee that the new editions released in 2008–2009 are more (or less) effective than the first editions, but it is reasonable to posit that the authors of the second edition materials have learned a great deal after a decade of implementation of their first edition materials.

What Now?

The general finding that the high school mathematics curriculum students complete has no significant relationship with their college mathematics achievement in college algebra or a more difficult course raises the question of what should happen next in this line of research. Our findings suggest a need for studies identifying the factors associated with the slower rate of increase in mathematics grades across courses taken by African American students and the nature of the differences between African American and white students' grades in their last college mathematics course. Information needed to address this important topic was not available to us in this study. Moreover, our sense is that future work might be well served by employing a mixed methods research design that draws on both qualitative and quantitative methodologies. There is also a need to explore the disproportional representation of college students who completed an NSF-funded curriculum in developmental mathematics courses.

Still, it is essential to consider the implications of the general finding that the various curricula are preparing students equally well for college mathematics. Is this license to continue current instructional practices, which are largely defined by a commercial curriculum with less than stellar student achievement and retention results over the past several decades? Should the research focus shift to other facets related to how students who completed various high school mathematics

curricula perform in the mathematics classroom, such as tenacity, motivation, teacher reactions and insights, problem-solving ability, and fidelity of implementation? Should the research community continue to examine more closely differential achievement patterns for specific mathematical topics in the various high school and college curricula, and should the focus be on employing longitudinal studies rather than cross-sectional studies? Should we try to answer all the questions above?

Is there an argument for using NSF-funded curricula based on their links to the NCTM-endorsed Standards and the national goal of having more students complete 4 years of high school mathematics in preparation for college? As Adelman (2004) suggested, 4 years of such mathematics at the high school level is the best indicator of a student's likelihood of graduating from college.

As a community, it is important to continue to engage in research dedicated to a deep understanding of the nature and source of student differences in mathematics, and the identification of appropriate responses to the conditions thought to underlie those differences. The objective should be to document instructional materials and practices that promote the goal of more students learning more mathematics more effectively.

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Integrated Curricula and Preparation for College Mathematics

In the early 1990s, after a long series of disappointing results on national and international mathematics achievement tests—for example, TIMSS (1998) and NAEP (Campbell, Hombo, and Mazzeo 2000)—the National Science Foundation (NSF) funded the development of thirteen complete mathematics programs at the elementary school, middle school, and secondary school levels.

By NSF edict, each reform curriculum had to align with the basic process principles and content specifications articulated in NCTM's *Curriculum and Evaluation Standards for School Mathematics* (1989). At the secondary school level, these NSF-funded curricula, released between 1997 and 1999, extended the mathematical domains addressed each year by including attention to algebra, geometry, probability, statistics, and topics in discrete mathematics. The pedagogical guidelines were also more expansive and quite different from the methodology commonly used at the time. As a result, implementing these materials posed a different set of challenges for teachers.

Many teachers were quite happy with the new curricula and impressed with the levels of student involvement and accomplishment. However, some parents and mathematicians were opposed to them, referring to the new curricula as "fuzzy mathematics." In response, in 2007 NSF funded the Minnesota Mathematics Achievement Project (MNMAP) to explore the empirical relationship between high school mathematics curricula and college-level mathematics performance.

MNMAP is a longitudinal program of research. Various subsets of its database of more than 16,000 students from 366 high schools who attended thirty-two four-year regional colleges and universities in the upper Midwest were involved in the MNMAP studies. All students graduated from a Minnesota high school and completed three or more years of college-intending high school mathematics. This group presented a unique opportunity to examine the mathematics achievement of students who had recently completed one of two distinctively different types of high school curricula—one NSF-funded and one commercially developed. These students have now graduated from regional four-year colleges and universities.

We will summarize five of these MNMAP studies, each of which used quantitative methodology. All studies were concerned with the efficacy of three first-edition secondary school curricula—Contemporary Mathematics in Context (Core Plus), Interactive Mathematics Project (IMP), and Mathematics Modeling Our World (MMOW)—as measured by student performance in college mathematics. Each program received an additional grant from NSF to prepare second-edition materials.

TWO VIEWS OF HIGH SCHOOL MATHEMATICS CURRICULA

The three NSF-funded curricula considered here offer students opportunities to communicate and reason mathematically in ways not found in traditional textbooks, referred to here as commercially developed curricula. In the NSF-funded curricula,

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problems are introduced within some mathematically rich, real-world context. Rarely are raw numbers or symbols manipulated simply for practice.

Two criticisms of NSF-funded curricula were that they did not pay enough attention to algorithmic development and that more extensive practice needed to be included in the lessons. The cognitive perspective underlying NSF-funded curricula expands the nature and extent of mathematical and instructional goals. This means that algorithmic development and maintenance is one of several important goals but, because the content has been expanded, a lesser portion of classroom time is available for the development and maintenance of skills and procedures. Critics believed that this approach would leave students unprepared for college-level mathematics. This hypothesis is unsupported by empirical evidence.

Method

In the past, the school mathematics curriculum has not received the research attention that it deserves. Rarely have two curricula been contrasted to see which has been more effective over the long run. The NSF-funded curricula are probably the most researched curricula in our nation's history. Previous studies assessed topic-specific achievement, calculus 1, problem-solving ability, attitude, and tenacity, but we identified no studies that used large databases from multiple postsecondary institutions to follow students through four full years of college coursework. The MNMAP studies discussed here are an attempt to fill this void and provide insight into the relationship between high school mathematics curricula and college-level performance.

In educational settings, units of analysis—students, schools, institutions—are often embedded, or nested, within larger units. For example, students are nested within classrooms that are nested within schools. Units within a larger unit tend to be more similar to one another and are therefore not independent (e.g., students in a classroom tend to be more similar to one another than they are to students in a different classroom). Traditional analyses, such as multiple regression, does not take this dependence into account, but hierarchical linear modeling, the primary analytic tool used in the MNMAP studies, does, thus significantly improving our ability to understand relationships among the many meaningful and relevant variables that typically are present in educational studies.

The first four MNMAP studies discussed here compared the achievement and course-taking patterns of students who had completed three, four, or five years of an NSF-funded curriculum with students who had used a commercially developed program or the University of Chicago School Mathematics Project. The fifth study relates to the likelihood of students completing a college STEM major—engineering or mathematics, in particular—as a function of the type of their high school mathematics curriculum.

New college students in 2002 and 2003 were some of the first students to have completed three or four years of one of the NSF-funded curricula. Some also completed a fifth year of secondary school mathematics (usually calculus), neither a non-NSF-funded nor a noncommercially developed course. Students in the latter cohort used materials published by one of approximately twenty programs, including those of Laidlaw, Holt, Addison Wesley, and Saxon. Students in the NSF-funded cohort were enrolled in courses using either Core Plus, IMP, or MMOW. The MNMAP website (www.cehd.umn.edu/ci/mnmap/) contains additional information about each study discussed here. In the three longitudinal studies, MNMAP examined all students' mathematics coursework taken over eight semesters of college work.

The MNMAP studies discussed here are an attempt to provide insight into the relationship between high school mathematics curricula and college-level performance.

Independent Variables

Student level variables (independent variables or covariates) were (1) high school curricula compared — Variable of major interest in each of these studies: (1a) NSF-funded vs. commercially developed curricula; (1b) Core Plus vs. commercially developed curricula, (1c) NSF-funded vs. commercially developed and UCSMP; (2) ACT scores; (3) number of years of high school mathematics (three, four, or five); (4) high school mathematics GPA; (5) high school percentile rank; (6) college major; (7) gender; (8) ethnicity; and (9) basic skills state test scores.

Dependent Variables

The dependent variables considered included these: (a) the difficulty level of the first college mathematics course taken (developmental, college algebra, calculus, etc.); (b) the difficulty level of the last college mathematics course taken (developmental through upper-division mathematics courses); (c) the grades or patterns of grades that students earned in mathematics courses (achievement); (d) the number of mathematics courses completed (persistence); (e) the pattern of difficulty levels of all mathematics courses completed over multiple semesters; (f) the likelihood of completing a STEM

Table 1 Summary of the Design and Participants of the Five MNMAP Studies

Study	Reference	Participants	Independent (I) and Dependent (D) Variables	Design
1	Harwell, Post, et al. (2009)	$N = 1{,}667$; 85 high schools	I: (1a), (2), (3), (4), (5), (8), (9) D: (a), (c)	Retrospective quasi-experimental, cross-sectional student data from a single institution at one point in time
2	Harwell, Medhanie, et al. (2011)	$N = 1{,}588$; 209 high schools	I: (1b), (2), (3), (4), (6), (7), (8) D: (a), (b), (c), (e), (g)	Retrospective, longitudinal analysis (four years of college coursework); data from a single institution
3	Post et al. (2010)	$N = 4{,}144$; 266 high schools	I: (1c), (2), (3), (4), (6), (7), (8) D: (a), (b), (c), (d), (e)	Retrospective, longitudinal analysis; data from a single institution
4	Harwell, Post, et al. (in press)	$N = 10{,}190$; 266 high schools; 32 four-year postsecondary institutions	I: (1c), (2), (3), (4), (6), (7), (8) D: (a), (b), (c), (e), (g)	Retrospective, longitudinal analysis; data from multiple institutions
5	LeBeau et al. (2012)	$N = 3{,}459$; 1,166 STEM majors; 537 completed an engineering or mathematics degree	I: (1c), (2), (3), (5), (7), (8) D: (f), (h)	Retrospective, cross-sectional analysis; data from a single institution

major in college; (g) the student's grade in a first calculus course; and (h) within STEM majors, the likelihood of completing an engineering or a mathematics major. This research was conducted initially with students from one large university and later with students from thirty-two four-year regional institutions varying in size, selectivity, mission, and type (public or private). The key findings are consistent over the five studies and across the thirty-two institutions.

See **table 1** for a summary of the design and participants for each study. Note that the references to independent and dependent variables refer to the lists above. **Table 2** provides a summary of the results across the first four studies.

SUMMARY OF RESULTS

Study 1: Harwell, Post, et al. (2009)

When we examined the entire NSF-funded cohort, we found differences in the difficulty level of initial course enrollment favoring students in commercially developed programs. These enrollments, from both types of curricula, included developmental (non-credit-bearing) and nondevelopmental courses (college algebra, calculus 1 and 2, and higher-level courses). There were no curricular-related differences in the grades attained in these courses.

Study 2: Harwell, Medhanie, et al. (2011)

When we examined only students who took two or more college mathematics courses, the curricular-related differences mentioned in study 1 disappeared. That is, there were no differences in the difficulty level or achievement (grades) of the initial course enrollment. Further, there were no differences in the longitudinal difficulty levels of coursework completed or in the grades attained in those courses examined over eight semesters of coursework. We also found no curricular-related differences in the number of mathematics courses completed.

Study 3: Post et al. (2010)

When we partitioned students according to their previous achievement scores (forming three groups), the middle group (those with ACT scores of 22 to 26) and the high group (those with ACT scores higher than 26) showed no significant differences in any of the dependent variables (the difficulty levels of the initial and final courses, achievement, and persistence) that we examined. For the group of students with the lowest ACT scores, we found a difference in the difficulty level of initial course enrollment (primarily developmental) favoring the cohort using commercially developed materials, with no curricular differences in grades earned.

Table 2 MNMAP Summary of Results for Studies 1–4

	Number of Students	Difficulty Level of Initial College Mathematics Course	Grade of Initial College Mathematics Course	Persistence (Number of College Mathematics Courses)	Longitudinal Grade Trajectories	Longitudinal Difficulty Trajectories
Study 1	1,667	CD > NSFF	NSD			
Study 2	1,704	NSD	NSD	NSD	NSD	NSD
Study 3 (ACT strata)	4,144	Low 1/3: CD > NSFF Middle and Top 1/3: NSD	NSD	NSD	NSD	NSD
Study 4 (Multi-institutional)	10,300	CD > NSFF	NSD (last course)		NSD	NSD

Note: NSFF = National Science Foundation funded; CD = Commercially developed curricula; NSD = No significant difference

Study 4: Harwell, Post, et al. (in press)

When we expanded the sample to thirty-two four-year regional institutions and more than 10,000 students, the results were consistent. When these students were compared with students in the cohort using commercially developed materials, the only difference was in relation to the difficulty level of the first college mathematics course. That is, students in the NSF-funded cohort, regardless of institutional parameters, were as well prepared for college mathematics as their counterparts who used commercially developed materials, suggesting that earlier results related to preparing students equally well for college mathematics are generalizable. Again, students in the NSF-funded cohort enrolled initially in lower-level coursework but achieved equally well in that course and in subsequent courses. We found no differences at any of the difficulty levels of the last college mathematics course completed.

Study 5: LeBeau et al. (2012)

When we restricted our sample to only those students who had completed a STEM major and then to students who had completed an engineering or mathematics major, the results were again consistent. That is, we found no relationship between students' high school curriculum and their likelihood of completing an engineering or a mathematics major in college.

EXPLANATION OF RESULTS

Teachers will notice that we used the terms "equally well prepared," "no differences," and "no curricular relationship" in our summary. What exactly are the implications of these phrases, and why are these implications important for classroom teachers? These results suggest that multiple, very different approaches can prepare high school students equally well for college mathematics.

To qualify for funding, all NSF-funded curricula must have adhered to NCTM's *Curriculum and Evaluation Standards for School Mathematics* (1989). It follows that each of these curricula reflect in various ways the "best practices" in our field. Although NCTM's 1989 publication and its *Principles and Standards for School Mathematics* (NCTM 2000) are not exclusively research based, they do reflect many research implications that have accumulated over the past several decades relating to student learning and conceptual development, effective teaching, the sociology of the classroom, uses of technology, and appropriate assessment practices. Classrooms based on these principles portray a very different learning environment—one that is student centered. In such classrooms, the teacher's role changes from primarily that of a lecturer to that of a coach who promotes both student independence and interdependence. This new role routinely highlights the applications of mathematics in the real world and preempts the ever-present student question, "Why do I have to learn this?"

Does such a student-oriented environment become a disadvantage for students later on? Our results suggest that it does not. In fact, these studies suggest that, aside from curricular-related differences in NSF-funded and commercially developed curricula in the difficulty level of the first college mathematics course, primarily for students enrolling in lower-level courses, we found no other differences in any of the dependent variables.

As we attempted to explain these results further, we focused on the characteristics of NSF-funded curricula. Each of the three curricula—Core Plus, IMP, and MMOW—was designed to appeal to a wide spectrum of the high school population. Key elements include the integration of five mathematical domains every year, widespread use of contextually laden problem situations, group-oriented problem solving, and use of a wide variety of

technology, including spreadsheets, graphing calculators, computer algebra systems (CAS), and applications such as The Geometer's Sketchpad® and Fathom Dynamic Data™ software.

These elements engage students in the conceptually oriented aspects of mathematics and its applications while necessarily limiting the time devoted to routine paper-and-pencil manipulations. This approach represents a substantial departure from the status quo and consequently has contributed to the acrimonious interchanges alluded to earlier.

The NSF-funded curricula included in these studies do prepare students for college mathematics equally as well as commercially developed approaches.

The problem-solving aspects of NSF-funded curricula engage students in problems whose solutions often consume several days of class time. This more accurately depicts the interdisciplinary nature of problem solving as it occurs in the real world. The tenacity acquired from these elaborate problem-solving experiences apparently served students well as they pursued college-level mathematics.

These characteristics do not appear only in the NSF-funded curricula. However, they are found there in a more coordinated fashion and in more abundance than in more traditional approaches.

A LOOK AHEAD

The Common Core State Standards for Mathematics (CCSSI 2010), designed to establish national standards for school mathematics, has been endorsed by more than forty states and the District of Columbia. The intent is "the specification of content that all students must study in order to be college and career ready" (CCSSI 2010, p. 4).

CCSSM has developed two major model course pathways—a traditional one and and an integrated one. Specific mathematics content by grade or course level is suggested within each of the specified pathways, including a transpathway perspective of an expanded view of the various mathematical content domains to be addressed each year. The NSF-funded curricular approaches, by virtue of addressing five mathematical domains each year and selecting problems from a wide variety of disciplines, has had an integrated perspective from the beginning.

CCSSM also endorses a series of mathematical practices that relate to NCTM's original five Process Standards: Problem Solving, Reasoning and Proof, Communication, Connections, and Representation. The Common Core Standards for Mathematical Practice are (1) make sense of problems and persevere in solving them; (2) reason abstractly and quantitatively; (3) construct viable arguments and critique the reasoning of others; (4) model with mathematics; (5) use appropriate tools strategically; (6) attend to precision; (7) look for and make use of structure; and (8) look for and express regularity in repeated reasoning (CCSSI 2010, pp. 6–8).

These practices are subtle yet profound, and implementation in the manner intended will represent a substantial departure in both content and method from the status quo. These core practices are not unrelated to more traditional approaches to high school mathematics; rather, these practices are more ably reflected in the NSF-funded approaches. This should not be surprising—these curricula have reflected alternative types of classroom mathematical activities for more than a decade.

SUMMARY AND IMPLICATIONS

Newer approaches to school curricula are always adapted reluctantly, especially so in mathematics. After all, everyone is interested in providing the best possible experiences for his or her own students. In one local suburban district, NSF-funded curricula were implemented for grades K–12 for all students more than a decade ago. Even though this district was high achieving before the change, it is now one of the top two or three highest-performing districts in the state regarding graduation rates, standarized test scores, the number and percentage of students in AP calculus and AP statistics courses, the percentage of students who take four full years of college-intending mathematics (through Core Plus IV), and the percentage of students enrolled in four-year post-secondary institutions (Monson 2011).

Earlier concerns about NSF-funded curricula—primarily that they will not prepare students for college-level mathematics in general and calculus in particular—have not been validated by these studies. The NSF-funded curricula included in these studies do prepare students for college mathematics equally as well as commercially developed approaches. Further, there may also be legitimate precollege pedagogical and sociological reasons for adopting NSF-funded curricula.

For schools and teachers interested in materials that have a decidedly different perspective, that appeal to a wider audience, and that promote a higher degree of student engagement, the NSF-funded curricula are a viable option.

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Student and high-school characteristics related to completing a science, technology, engineering or mathematics (STEM) major in college

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Background: The importance of increasing the number of US college students completing degrees in science, technology, engineering or mathematics (STEM) has prompted calls for research to provide a better understanding of factors related to student participation in these majors, including the impact of a student's high-school mathematics curriculum.
Purpose: This study examines the relationship between various student and high-school characteristics and completion of a STEM major in college. Of specific interest is the influence of a student's high-school mathematics curriculum on the completion of a STEM major in college.
Sample: The sample consisted of approximately 3500 students from 229 high schools. Students were predominantly Caucasian (80%), with slightly more males than females (52% vs 48%).
Design and method: A quasi-experimental design with archival data was used for students who enrolled in, and graduated from, a post-secondary institution in the upper Midwest. To be included in the sample, students needed to have completed at least three years of high-school mathematics. A generalized linear mixed model was used with students nested within high schools. The data were cross-sectional.
Results: High-school predictors were not found to have a significant impact on the completion of a STEM major. Significant student-level predictors included ACT mathematics score, gender and high-school mathematics GPA.
Conclusions: The results provide evidence that on average students are equally prepared for the rigorous mathematics coursework regardless of the high-school mathematics curriculum they completed.

Keywords: STEM; high-school mathematics curricula; college majors; engineering/mathematics

Introduction

Over the past several decades the USA has been graduating fewer students with science, technology, engineering and mathematics (STEM) majors in college. Currently, the USA ranks 17th among developed nations in producing science and engineering students, a decline from third place three decades ago, and is 26th in

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producing mathematics majors. This decline has prompted significant concern over the ability of the USA to produce enough STEM majors to meet the country's technological needs. Current estimates are that the USA produces 200,000 fewer STEM graduates per year than are needed (Committee on Prospering in the Global Economy of the 21st Century 2007). The shortage of engineering and mathematics majors is especially acute. The need to increase the number of students in STEM majors has led to a focus on pre-college factors related to student preparation and STEM literacy [National Academy of Engineers (NAE) and National Research Council (NRC) 2009]. STEM majors in the current study are defined as follows: life and physical sciences majors (e.g. biology, chemistry, physics), technology majors (e.g. computer science), engineering majors, and mathematics majors. A clear finding from this work is that successful completion of many STEM majors (especially engineering and mathematics majors) generally requires significant mathematics coursework in college, typically beginning with calculus I or a more difficult course (Adelman 2006). This in turn requires adequate mathematics preparation prior to college enrollment, but available evidence of mathematics preparation tends to be general; for example, students who take more (and more difficult) mathematics courses in high school are more likely to earn a college degree (Adelman 1999, 2006). Among the factors believed to affect a student's preparation for rigorous college mathematics coursework, the high-school mathematics curriculum a student completes is expected to play a key role (Post et al. 2010; Schoenfeld 2004).

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High-school mathematics curriculum

There are three major categories of high-school mathematics curricula in the USA (Post et al. 2010). The most common are commercially developed (CD) curricula, which generally include a year of geometry sandwiched between two years of algebra and followed by a year of pre-calculus. Most students who graduate from USA high schools, perhaps as many as 75%, complete a CD curriculum (Education Market Research 2006). However, evidence supporting the effectiveness of CD curricula is limited (Harwell et al. 2007; Post et al. 2008).

A second category of high-school mathematics curricula was developed with support from the National Science Foundation (NSF) between 1991 and 1998. These curricula were based on the Curriculum and Evaluation Standards for School Mathematics (NCTM 1989), and represented an effort to bring best practices from the mathematics education literature into the classroom (Senk and Thompson 2003). These NSF-funded curricula represented a sharp departure from CD curricula in their focus on integrating algebra, geometry, probability, statistics and topics in discrete mathematics every school year in a way that emphasized contextually laden problems, use of technology and group work, and de-emphasized algorithmic manipulation.

A third curriculum category is the University of Chicago School Mathematics Project (UCSMP), which began in the early 1980s. This curriculum combines features of the CD and NSF-funded curricula and new editions appear regularly.

Attempts to adopt standards-based curricula (i.e. NSF-funded and UCSMP) in USA schools have frequently faced a variety of obstacles (Schoenfeld 2004). One such obstacle has been the absence of a compelling empirical literature documenting the effectiveness of these curricula relative to CD curricula; another is criticism that these curricula (especially NSF-funded curricula) do not adequately prepare students for rigorous college mathematics coursework (Klein 2000; Wu 1997),

although empirical evidence supporting this criticism is lacking. Often the adoption of a standards-based curriculum is used in tandem with a CD curriculum. There is anecdotal evidence that the standards-based curriculum (specifically NSF-funded) are disproportionately used for low-achieving students (Norman 2008). In addition, many college or university mathematics courses tend to follow a CD curriculum framework (i.e. lecture based with repetition using fictional data) which may favor students who completed a CD curriculum in high school.

The relationship between the high-school curriculum a student completes and their performance in college mathematics is not well understood, and available evidence of this relationship is mixed. For example, some studies have concluded that the high-school mathematics curriculum a student completes has no impact on grades earned in college mathematics courses, whereas other studies have concluded the opposite (Post et al. 2010; Harwell et al. 2009; O'Neill 2002; Schoen and Hirsch 2003). However, these studies have only examined a general population of students; none have focused on students who complete a STEM major in college.

It is also likely that understanding the relationship between completing a particular high-school mathematics curriculum and completing a STEM major in college has implications for other efforts to increase student interest in STEM majors. For example, in 2003 the Engineering in Elementary (EiE) project was formed to introduce an engineering curriculum in K-12 school districts in the USA (Cunningham 2009). The goal of this curriculum is to educate and expand awareness about the field of engineering and technology. As of 2009, only two states in the USA have developed educational standards at the elementary level to include engineering: Massachusetts and Minnesota (Cunningham 2009). Even though engineering-related curricula in elementary and middle schools have not yet been widely adopted, many of the standards in this curriculum are found within the standards-based mathematics curricula described above.

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Predicting completion of a STEM major

Owing to the decades-long drop in the number of STEM majors the USA has produced, and the growing need for college graduates with these skills, researchers have been focusing on identifying variables that help to predict the likelihood of a student enrolling in, and completing, a STEM major. Some of the factors commonly reported are high-school mathematics and science course-taking, standardized test scores (e.g. ACT, SAT) and student socio-economic status (Allen et al. 2008; Maple and Stage 1991; Tan 2002; Tyson et al. 2007). Not surprisingly, these studies have found that students who take more mathematics and science courses in high school are more likely to graduate from college with a STEM major.

There has also been a good deal of interest in the gender and ethnicity of students who complete a STEM major. Over the years there has been a gender gap, with more males starting and graduating with a STEM major (Tyson et al. 2007). Historically, there has also been an ethnic gap, with white and Asian students more likely to graduate with a STEM major compared to black and Hispanic students, although there is some evidence of recent changes in this pattern (Tyson et al. 2007). Gender and ethnicity gaps have been documented for students graduating with an engineering major (NSF 2011). The percentage of female undergraduate engineering majors has been decreasing, from 19.7% in 1998 to 17.5% in 2008 (NSF 2011). A recent report by the NAE and NRC summarized the results of a two-year study exploring the scope and

nature of efforts in primary school to teach engineering to students. One of the committee's recommendations included funding research that attempts to close gender and ethnicity gaps alluded to above (NAE and NRC 2009).

One other important feature of the STEM literature is the difference between the number of students initially declaring a STEM major and the number who have actually completed a STEM degree. Changing majors or dropping out of college altogether has been shown to be as high as 59% for students who initially declared a STEM major (Daempfle 2003; Scott, Tolson, and Huang 2009; Tan 2002). An engineering persistence study by Burtner (2005) found that expectations and perceptions of the profession (specifically job outlook) and indicators of a student's self-confidence were the strongest predictors of status in engineering in years 1 and 3. On the whole, though, relatively little is known about the factors that prompt students to complete (or not complete) a STEM major.

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Research questions

Given the gap in our understanding of student and high-school factors related to completion of a college STEM major and the growing problem of the USA failing to produce sufficient numbers of engineering and mathematics graduates, the following research questions were asked: What is the nature and magnitude of the relationship between student and high-school characteristics (including the high-school mathematics curriculum a student completed) and the likelihood that a student (1) completes an engineering or mathematics major, (2) completes a STEM major; and (3) can the relationships in (1) and (2) be generalized across high schools of varying sizes, percentages of college-bound students, socio-economic status and location in the USA?

Methods

Research design

A quasi-experimental design for cross-sectional data (Pedhazur and Schmelkin 1991) with archival data was used for students who enrolled in, and graduated from, a single university in the upper Midwest in the USA. To be included in the study students had to have completed at least three years of a high-school mathematics curriculum, a condition which helped to ensure that students had sufficient exposure to a mathematics curriculum for its effects to appear. A total of 3459 students who graduated from one of 229 high schools constituted the sample. Of these, 537 completed an engineering or mathematics major and an additional 629 completed a STEM major other than engineering or mathematics. Thus, 1166 students in the sample graduated with a STEM major and 2293 students did not graduate with a STEM major. Categorization of college majors was guided by the Comprehensive Instruction Program (National Center for Educational Statistics 2002).

Data

Data reflecting a student's high-school mathematics course-taking, high-school mathematics curriculum, college mathematics course-taking, college major and college course grades across eight semesters were obtained from students who started college in fall 2002 or fall 2003. These years were chosen for two reasons: (1) they

included the first cohort of high-school students to have been a part of the NSF-funded curricula for at least three years; and (2) many student records were in electronic form, which made data collection easier. Data were collected from three sources: the state, the university and the high school. High-school transcript data were obtained from university records which contained variables such as ACT mathematics score, high-school mathematics GPA, percentile rank, gender, number of years of high-school mathematics completed (three, four or five) and ethnicity. The high schools verified transcript indications of the mathematics curriculum that a student had completed.

Although the university maintained electronic records of high-school course titles for all applicants, discrepancies were found between the coding of these courses in the university records and the correct titles. As a result, every high school was contacted to ascertain exactly which high-school curricula were used during years the students in the sample were in high school. A variable reflecting students' prior mathematics ability as measured in eighth grade by a state-mandated Basic Skills Test for mathematics was obtained from the Department of Education in a Midwestern state in the USA.

The student-level variables included in the analysis were: ACT mathematics score, years of high-school mathematics completed (3, 4, 5), high-school mathematics GPA, high-school percentile rank, gender, African American/Hispanic status (1 = yes, 0 = no) (these subgroups were combined because of small numbers) and Asian status (1 = yes, 0 = no). White students served as the reference group. High-school-level variables included the percentage of African American/Hispanic students and Asian students, aggregated ACT mathematics score, percentage of males, high-school location (urban, suburban, rural) and high-school mathematics curriculum (CD, NSF-funded, UCSMP), the key independent variable. With the exception of location and high-school mathematics curriculum, the high-school level variables were aggregated from the student-level data. A variable reflecting whether a high school had a mixed curriculum (used at least two curricula) was initially included in the models but was not significant and the results are presented without the mixed curriculum variable.

The primary dependent variables in this study were completion of a STEM major (1 = STEM, 0 = otherwise) and completion of an engineering or mathematics major (1 = engineering/mathematics, 0 = otherwise). The latter dependent variable responds to calls for a better understanding of the antecedents of enrollment in these majors and the strong interest in developing more engineering and mathematics majors in the USA (NAE and NCR 2009). Engineering and mathematics majors tend to need significantly more mathematics knowledge than other STEM majors, and results capturing the effect of high-school mathematics curriculum may differ from other STEM majors. These dependent variables were analyzed separately.

Analyses

Descriptive and inferential analyses were used. For inferential analyses, a generalized linear mixed model (GLMM) for binary data (Fitzmaurice, Laird, and Ware 2004) was used to examine the relationship between student and high-school characteristics and the likelihood that a student completed (1) a STEM major or (2) an engineering or mathematics major. The GLMM for binary variables models the odds of being a STEM major within each high school. In this two-level model students were treated as nested within high schools and the data were cross-

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sectional. This model is flexible to allow variables at both levels to be modeled (i.e. student and high-school variables can be included in the same model).

Preliminary analyses were completed to determine the number of random effects needed and revealed that only a random intercept was needed; therefore, only the intercepts were modeled with high-school-level variables. This was accomplished by fitting a model with and without the random slope and performing a likelihood ratio test to see whether there was significant variation in high-school slopes for variables such as average ACT score. As a result, the models took the form:

$$log(\boldsymbol{Y}_i) = \boldsymbol{X}_i\beta + \boldsymbol{b}_i \quad (1)$$

In Equation (1), $\mathbf{Y}_i$ is a vector of outcomes on a binary dependent variable (e.g. completed a STEM major 1 = yes, 0 = no) for the ith high school, $\mathbf{X}_i$ is the design matrix for the fixed-effects, β are the fixed effects, and $\mathbf{b}_i$ represents random intercepts for each high school. To help control for compounding type I error rates, the per test α value was 0.001.

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Limitations

The quasi-experimental design does not allow for strong causal inferences; however, many statistical control variables were included to increase the strength of the inferences. Owing to data limitations, other potentially important high-school variables, such as number of AP/IB courses offered; biology, chemistry, and physics high-school course data, teacher experience or percentage of students in extracurricular activities, were not available. Substantial care needs to be taken when generalizing results to countries outside the USA, and such generalizations would depend on countries having similar high-school and college characteristics to those described here. Students in the sample were academically proficient as they had to graduate from college to be in the sample. Therefore, no information on dropout or transfer could be obtained from the data.

Results

Descriptive and inferential statistics are provided.

Descriptive statistics

Table 1 reports descriptive statistics to show the characteristics of students in the sample. Most students were white, with slightly more than 50% of the students identified as males. The average number of years of high-school mathematics completed was 3.92, suggesting, on average, the completion of a college algebra or pre-calculus course.

Table 2 shows the number of students who graduated with an engineering/mathematics major or a STEM major by the high-school mathematics curriculum. As can be seen from the table, many students who graduated with an engineering and mathematics major took a CD mathematics curriculum in high school. A smaller number of students who graduated with an engineering or mathematics major took an NSF-funded mathematics curriculum in high school. Table 2 also shows that the proportion of students (shown in square brackets) within a curriculum who graduated with an engineering or mathematics major was similar for students who

Table 1. Proportion or mean for ethnicity, gender and years of high-school mathematics for students in the sample.

Variable	Proportion/mean
African American	0.05
Asian	0.13
Caucasian	0.79
Hispanic	0.02
Native American	0.001
Male	0.52
Years of high-school mathematics completed	3.92
Sample size = 3459	

Table 2. Number of students graduating in each college major group by high-school mathematics curriculum.

	CD	UCSMP	NSF
Engineering/mathematics majors	389 (0.72) [0.16]	97 (0.18) [0.17]	51 (0.09) [0.11]
All STEM majors	804 (0.69) [0.33]	231 (0.20) [0.40]	131 (0.11) [0.27]

Notes: Total number of engineering/mathematics majors and STEM majors = 537 and 1166, respectively. Proportions in parentheses are computed within rows. Proportions in brackets reflect the proportion of students within a curriculum to graduate with an engineering or mathematics major or a STEM major.
CD: commercially developed; UCSMP: University of Chicago School Mathematics Project; NSF: National Science Foundation.

took the CD or UCSMP curricula, but was lower for students who took the NSF curriculum. Similar trends can be seen with the STEM majors.

Inferential statistics

An unconditional GLMM for the engineering and mathematics majors data was fitted to gain an idea of the average log-odds of the likelihood of a student graduating with an engineering or mathematics major across high schools relative to not graduating with an engineering or mathematics major. For a 'typical' high school (i.e. with a random effect of 0), the expected log-odds for engineering/mathematics graduation is –1.817, corresponding to an odds of $\exp(-1.817) = 0.163$, and a probability of $1/(1 + \exp\{1.817\}) = 0.139$. Thus, among all students in the sample the likelihood of a student graduating with an engineering or mathematics major is modest (about 0.14).

The results of the conditional GLMM for engineering and mathematics majors (see Table 3) revealed at the student level that ACT mathematics score, years of high-school mathematics, high-school mathematics GPA and gender were all significant predictors of the likelihood that a student graduated with an engineering or mathematics major for a 'typical' high school. No predictors at the high-school level were significant, which includes the key independent variable high-school mathematics curriculum. This indicates that none of the high-school-level variables affected the baseline (intercept) likelihood of students graduating with an engineering/mathematics major.

Of the significant predictors at the student level, ACT mathematics score, gender and high-school mathematics GPA had the largest impact on the likelihood that a student would graduate with an engineering or mathematics major. For example, for

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Table 3. Results of the generalized linear mixed model for engineering and mathematics majors.

	B	SE	*Z*-value
Student level			
Intercept	−14.719	1.254	−11.736*
ACT mathematics	0.223	0.017	12.791*
High-school mathematics level	0.254	0.088	2.882
High-school mathematics GPA	1.251	0.162	7.715*
High-school percentile rank	−0.001	0.002	−0.375
Male	1.884	0.150	12.533*
African American/Hispanic	0.084	0.318	0.264
Asian	0.435	0.162	2.674
High-school level (intercepts)			
% African American/Hispanic	0.316	1.048	0.302
% Asian	1.496	0.687	2.179
ACT mathematics	0.012	0.047	0.265
% Male	−0.515	0.581	−0.887
UCSMP	−0.294	0.250	−1.176
CD	−0.196	0.250	−0.892
Rural	−0.054	0.309	−0.175
Suburban	0.170	0.274	0.620

Notes: *A significant *Z*-value in this case is 3.09 in absolute value. $N = 3459$.
UCSMP: University of Chicago School Mathematics Project; CD: commercially developed.

this sample of students, males were associated with an expected increase in the log-odds for the likelihood that a student graduates with an engineering or mathematics major of 1.884, for a 'typical' high school, holding the other predictors constant. Converting from the log-odds scale to the odds scale produces exp(1.884) = 6.580. Thus, males are about six and one-half times more likely than females to graduate with an engineering or mathematics major (relative to not graduating with an engineering or mathematics major). Similar interpretations can be made for other significant predictors.

Figure 1 shows the model predicted probabilities as a function of ACT mathematics score. From the plot it can be seen that there is a sharp increase in the likelihood of a student graduating with an engineering or mathematics major for an ACT mathematics score of 25. This steep increase continues throughout the rest of the ACT mathematics scale and produces a probability of 0.5 for an ACT mathematics score of 33. Thus, ACT mathematics score appears to be a good predictor of the likelihood of graduating with an engineering or mathematics major, but other variables also matter at the student level.

As in the analysis above, an unconditional GLMM for STEM majors was fitted to gain an idea of the average log-odds of the likelihood of a student graduating with a STEM major across high schools. For a 'typical' high school (i.e. with a random effect of 0), the expected log-odds for STEM graduation is –0.712, corresponding to an odds of exp(–0.712) = 0.491, and a probability of $1/(1 + \exp\{0.712\}) = 0.329$. Thus, among graduating students the likelihood that a student graduated with a STEM major is moderate in size (about 0.33).

Results for the GLMM for STEM majors revealed that ACT mathematics, high-school mathematics GPA, gender and Asian status at the student level were significant (Table 4). Again, the key independent variable, high-school mathematics curriculum, and all other high-school variables were not significant. The same inter-

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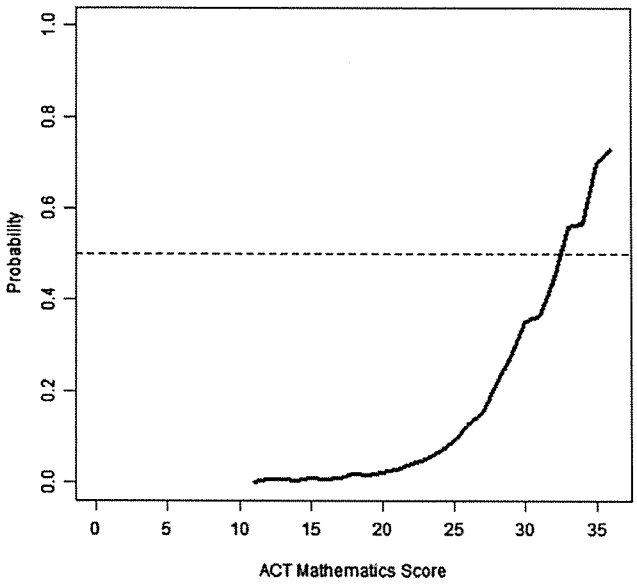

Figure 1. Model-predicted probabilities for engineering and mathematics majors by ACT mathematics scores.

pretation from the previous model can be used for this model as well. For example, Asian status results in an expected increase in the log-odds for the likelihood that a student graduates with a STEM major of 0.383 compared to white students, for a 'typical' high school, holding the other predictors constant. Converting from the log-odds scale to the odds scale, exp(0.383) = 1.467. Thus, Asian students are about one-and-a-half times more likely than white students to graduate with a STEM major (relative to not graduating with a STEM major). Similar interpretations can be made for other significant predictors.

Table 4. Results of the generalized linear mixed model for STEM majors.

	B	SE	*Z*-value
Student level			
Intercept	−6.308	0.847	−7.447*
ACT mathematics	0.117	0.012	10.039*
High-school mathematics level	0.181	0.062	2.938
High-school mathematics GPA	0.717	0.097	7.397*
High-school percentile rank	0.001	0.002	0.598
Male	0.484	0.086	5.612*
African American/Hispanic	0.358	0.180	1.990
Asian	0.383	0.117	3.256*
High-school level (intercepts)			
% African American/Hispanic	−0.277	0.659	−0.420
% Asian	−0.403	0.502	−0.802
ACT mathematics	−0.030	0.035	−0.852
% Male	−0.496	0.422	−1.176
UCSMP	−0.021	0.187	−0.110
CD	−0.259	0.164	−1.575
Rural	0.150	0.219	0.685
Suburban	0.173	0.198	0.876

Note: *A significant Z-value in this case is 3.09 in absolute value. N = 3459.

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Figure 2 shows the model-predicted probabilities for STEM majors by ACT mathematics scores. Unlike Figure 1, there is a steady increase in the likelihood of graduating with a STEM major for an ACT mathematics score of 20 and above. Figure 2 also reaches a probability of 0.5 more quickly than Figure 1, which is expected as a larger variety of majors is incorporated into the general STEM majors. Some of these majors (e.g. biology) require significantly less mathematics than engineering/mathematics majors.

Discussion

This study examined the relationship between various student and high-school characteristics and completion of a STEM major in college for a sample of 3459 students from 229 high schools in the USA. Two findings were of particular interest. First, the high-school mathematics curriculum a student completed was unrelated to completing a STEM major in general or more specifically to completing a major in engineering or mathematics. Thus, on average, high-school students from each of the types of curriculum considered here are equally prepared for the kind of rigorous mathematics work associated with completing a STEM major and in particular an engineering or mathematics major. This empirical evidence contradicts assertions that standards-based curricula do not adequately prepare students for rigorous college mathematics coursework (Klein 2000; Wu 1997). Importantly, this finding helps to inform high-school and college policies related to mathematics course-taking and advising high-school students interested in a STEM major in college. For example, the results provide no evidence that high-school students interested in a STEM major should be steered towards a particular mathematics curriculum (e.g. NSF-funded or UCSMP) in high schools offering more than a single curriculum option. Even though the standards-based mathematics curricula may have similar aspects to those being promoted by the EiE, no evidence was found that these standards-based mathematics curricula influence the likelihood that a student would graduate with an engineering or

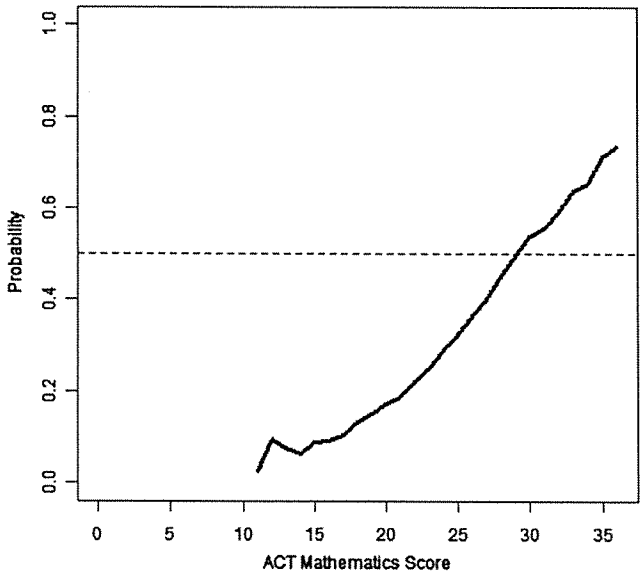

Figure 2. Model-predicted probabilities for STEM majors by ACT mathematics scores.

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mathematics major. However, other skills identified by the EiE as necessary to a successful career as an engineer, in addition to a strong science and mathematical background, such as the ability to function as an effective group member or solving contextually laden real-world problems, may be better developed using a standards-based mathematics curriculum. Studying the relationship between high-school mathematics curriculum and STEM literacy would be helpful in assessing whether these curricula help to raise STEM awareness and interest.

Second, and relatedly, the lack of significant findings for any of the models at the high-school level implies that the central finding of no relationship between high-school mathematics curriculum and likelihood of completing a STEM major (including engineering and mathematics majors) is not dependent on particular high-school characteristics such as location or whether a school offers more than a single mathematics option to its students. This enhances the generalizability of the central finding beyond high schools in this sample.

Still, several predictors were significant at the student level, including a student's ACT mathematics score, high-school mathematics GPA, gender and Asian status for STEM majors. In considering the student and high-school results collectively, an important conclusion is that the likelihood of completing a STEM major (including engineering and mathematics majors) appears to depend largely on a student's mathematics proficiency and certain student characteristics (e.g. gender), and not on high-school characteristics. In some ways these are comforting findings because they imply that students receive similar high-school mathematics preparation, and that high-school characteristics are unrelated to opportunities to complete a STEM major but are related to students' mathematics proficiency. On the other hand, the findings provide evidence supporting recommendations for additional research to explore ways to close gender and ethnicity gaps (NAE and NRC 2009).

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Preparation of Students Completing a Core-Plus or Commercially Developed High School Mathematics Curriculum for Intense College Mathematics Coursework

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The purpose of this study was to examine the college mathematics achievement and course-taking of students at a large public research university who completed a commercially developed or standards-based (Core-Plus) high school mathematics curriculum, and who subsequently completed at least 2 college mathematics courses of difficulty level at or beyond precalculus mathematics. Mathematics course-taking and achievement data across 8 college semesters were analyzed for a sample of 1,588 students. Findings indicated that students (including science, technology, engineering, and mathematics majors) were equally prepared for intense college mathematics coursework regardless of which high school mathematics curriculum they completed. These findings inform high school mathematics curriculum adoption decisions for college-bound students, and college policies and practices for advising students enrolling in mathematics courses.

Keywords *college mathematics, curriculum, high school, longitudinal*

THERE IS WIDESPREAD AGREEMENT of the importance of documenting the ability of high school mathematics curricula to prepare college-bound students in the United States for college mathematics (Augustine, 2007; Mathematical Sciences Education Board, 2004; National Center for Education Statistics [NCES], 2006; National Research Council, 2002) for at least two reasons. First, there is substantial evidence that the mathematics preparation of many college-bound high school students is inadequate. Perhaps the most compelling evidence of inadequate preparation is that approximately one quarter of all college freshmen enroll in at least one developmental mathematics class, typically a non–credit-bearing class that should have been completed in high

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school (NCES, 2003). This percentage would be even higher except that many institutions do not allow students to take a developmental class or discourage them from doing so (Conley, 2007).

Second, questions about the preparation of students for college mathematics coincide with growing concern over the inability of the United States to produce sufficient numbers of students majoring in science, technology, engineering, and mathematics (STEM) to meet the country's technological, academic, and intellectual needs. The United States currently ranks 17th among developed nations in the proportion of college students receiving degrees in science or engineering, a fall from third place three decades ago, and ranks 26th in the proportion receiving undergraduate degrees in mathematics. Recent estimates are that the United States will soon graduate 200,000 fewer STEM majors per year than are needed (Committee on Prospering in the Global Economy of the 21st Century, 2007; Congressional Research Service, 2008). Because many STEM and STEM-related disciplines require substantial mathematics coursework, college-bound students interested in these majors need to be adequately prepared for college mathematics when they graduate from high school.

Empirical studies involving high school mathematics curricula typically include traditional curricula as well as curricula developed with support from the National Science Foundation (NSF) in the late 1990s. However, the introduction of NSF-funded curricula was quickly followed by harsh criticism of these curricula (Schoenfeld, 2004). A precise measure of the effect of the criticism is unavailable, but there is anecdotal evidence that the criticism has reduced the number of school districts using these curricula in many states (Post, Monson, Dupuis, Medhanie, & LeBeau, 2010b).

Criticism of NSF-funded curricula has several facets but a common thread is that the goal of these curricula, to teach mathematics to all students, fails to adequately support the learning of college-bound students. Specific criticisms of these curricula are that they lack mathematical rigor and encourage misconceptions by the careless manner in which logical arguments are treated, rely too much on realistic problems, and use imprecise language (Wu, 1997; see also the Mathematically Correct website). Critics have also claimed that the use of group work encouraged by these curricula compromises the learning of more proficient students (Wu, 1997). These characteristics are argued to lead to poor mathematics performance with potentially dire consequences, such as poor performance on the mathematics portions of standardized tests taken by college-bound students and an increase in the likelihood that college-bound students will not be adequately prepared for rigorous college mathematics coursework such as Calculus I and II (Hill & Parker, 2006; Klein, 2000a, 2000b). Schoenfeld (2004) provided additional discussion of these criticisms.

The present study examined the preparation of students who completed a traditional (commercially developed) or NSF-funded high school mathematics curriculum for intense college mathematics coursework, including those who completed a STEM major. We build on the definition of academic/curricular intensity used by Adelman (1999, 2006), in which intense high school mathematics coursework was characterized as completing an average of almost four Carnegie units (one unit is typically defined to be the content equivalent of an academic year course taught 1 hr per day; Boyer, 1983), including trigonometry, precalculus mathematics (college algebra, finite mathematics, and coursework at a similar level), or calculus. Adelman's (1999, 2006) index combined indicators of high school mathematics with information from other domains such as science and English. Students who completed more (and more advanced) coursework in high school showed greater academic intensity in mathematics and were more likely to earn a college degree.

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Academic/curricular intensity has been defined in several research contexts as the completion of advanced coursework. For example, Attewell and Domina (2008) studied inequality in access to an advanced curriculum in high school and its effect on test scores and college entry, and defined an intensive curriculum as one in which a high school student takes many advanced courses in different content areas. Goldrick-Rab (2006) cited Adelman's (1999) work in defining academic/curricular intensity as the rigor of a high school student's curriculum across several areas of study as reflected in the difficulty and quality of courses. Other researchers have defined academic/curricular intensity in a similar way (e.g., Attewell, Lavin, Domina, & Levey, 2006; Doyle, 2009; Hargrove, Grodin, & Dodd, 2008). Similarly, we defined academically intense college mathematics coursework as having completed a minimum of two college mathematics courses of difficulty level at or beyond precalculus mathematics.

High School Mathematics Curricula

Commercially developed curricula

There are three general categories of high school mathematics curricula in the United States. The most widely used are commercially developed curricula, with approximately 75% of U.S. students completing one of these curricula in high school (Education Market Research, 2006). General characteristics of these curricula include an emphasis on algebra as the fundamental underpinning of high school mathematics, attention to algorithms and repetition, and an expectation that learning takes place in relatively small increments and is heavily dependent on the actions of the teacher (Roitman, 1999; Schoenfeld, 2004). Examples of publishers of commercially developed curricula used for the sample of students in our study include Houghton-Mifflin, Glencoe-McGraw Hill, and Saxon. We did not distinguish among the various publishers of commercially developed curricula in our analyses.

NSF-funded curricula

NSF-funded curricula were aligned with the National Council of Teachers of Mathematics' (NCTM, 1989) Curriculum and Evaluation Standards for School Mathematics. The NCTM Standards and the follow-up Principles and Standards for School Mathematics published in 2000 (NCTM, 2000) reflect a framework for mathematics learning that includes students, schools, curricula, assessment, technology, parents, and communities.

NSF-funded curricula reflect a significant expansion of the scope of the mathematical content considered each year, along with dramatic changes in the conceptions of school mathematical content from a fixed set of facts and procedures to a dynamic body of knowledge that is continually enriched through conjecture, exploration, analysis, and proof (Smith, 1996). In addition, these curricula are associated with substantial changes in models of teaching and learning along with a reconceptualized role for the student.

Advocates of NSF-funded curricula claim that the potential benefit is deeper and broader student understanding compared with commercially developed curricula (Riordan & Noyce, 2001). These benefits result from orienting these curricula around (a) problems that actually occur in, or are derived from, real-world experiences, often involving multiple strategies and content domains in their solution, which can enhance student engagement; (b) problems whose solution

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requires more than a few minutes and more than simply applying previously learned procedures and thus encourages student explanation and investigation; (c) providing opportunities for group discussion and problem solving, thus emphasizing the role of students as active participants in the learning process; (d) using various forms of technology where appropriate. These curricula are also consistent with the cognitive perspective on the nature of human learning (Post et al., 2010a).

The present study focuses on one of the NSF-funded curricula: contemporary mathematics in context (or Core-Plus; Coxford et al., 1998). The Core-Plus curriculum focuses on algebra, geometry, probability, statistics and topics in discrete mathematics every school year in a way that emphasizes problem solving and deemphasizes algorithmic manipulation. As a result, this curriculum represents a significant departure from the traditional Algebra 1, geometry, Algebra 2, precalculus mathematics sequence of high school mathematics.

High School Mathematics Curricula and Student Preparation for College Mathematics

It is surprising that literature documenting the ability of high school mathematics curricula to prepare students for college mathematics is sparse and somewhat contradictory. For example, Schoen and Hirsch (2003) reported that students who completed a Core-Plus curriculum had higher overall grade point averages in a Calculus I course in college and in subsequent mathematics courses compared with students who completed a commercially developed curriculum. However, Harwell et al. (2009) and Post et al. (2010a) found no relation between high school mathematics curricula (including Core-Plus) and college mathematics course grades.

Findings of the effect of high school mathematics curricula on the difficulty level of the college mathematics courses a student completes have also varied. Hill and Parker (2006) reported that students completing Core-Plus in high school initially enrolled in less difficult college mathematics courses, and subsequently completed even less difficult college mathematics courses, than students who did not complete Core-Plus. A key deficiency of this study was that the high school mathematics curriculum completed by many students was unknown. Harwell et al. (2009) found a relation between a student's high school mathematics curriculum and the difficulty of a students' first college mathematics course in which students completing an NSF-funded curriculum were more likely to take a less difficult course initially compared with those who completed a commercially developed curriculum. However, Post et al. (2010a) reported that high school curriculum was unrelated to the pattern of course difficulties over multiple college semesters.

Other benefits of completing an NSF-funded curriculum have also been documented albeit for middle school and high school students. Webb (2003) found that students studying from an NSF-funded curriculum completed more mathematics courses than students studying from a commercially developed curriculum. The results of Cichon and Ellis (2003) indicated that about two thirds of the students completing an NSF-funded curriculum in an inner-city school took additional mathematics courses as high school seniors compared with students not completing one of these curricula. There is also evidence that NSF-funded curricula promote more positive student attitudes towards mathematics (Cichon & Ellis, 2003; Schoen & Pritchett, 1998).

In sum, there is mixed evidence of the preparation of students completing an NSF-funded curriculum in high school for college mathematics coursework. Moreover, no studies have examined the effect of these curricula on student preparation for intense college mathematics coursework.

Research Questions

Our research questions focused on the relation between high school mathematics curricula (commercially developed, Core-Plus) and course grades and difficulty levels for students completing two or more college mathematics courses whose difficulty level equaled or exceeded that of precalculus mathematics courses:

What is the nature and magnitude of the relation between the high school mathematics curriculum a student completed (commercially developed, Core-Plus) and the (a) difficulty level and grade of a student's first college mathematics course, (b) pattern of mathematics grades earned over multiple college semesters, and (c) pattern of difficulty levels of mathematics courses completed over multiple college semesters, taking into account a student's STEM status and student background factors?

We defined academically intense college mathematics coursework as having completed a minimum of two college mathematics courses of difficulty level at or beyond precalculus mathematics.

METHOD

Research Design

A retrospective cohort (quasi-experimental) design was used in which college mathematics course grades and difficulty levels were examined longitudinally. Based on their high school mathematics curriculum students were categorized into one of two curriculum cohorts: commercially developed or Core-Plus. Because of small numbers students who completed other NSF-funded curricula such as the Interactive Mathematics Program (Fendel, Resek, Alper, & Fraser, 1998) were omitted from our sample. Mathematics course grades and difficulty level data were collected for five or seven semesters depending on whether a student began college in Fall 2003 or Fall 2002, respectively.

Population and Sample

The sampled population consisted of college students in a Midwestern state in the United States who graduated from a public high school, completed at least three years of high school mathematics using a commercially developed or Core-Plus curriculum, and completed a minimum of two college mathematics courses of difficulty level at or beyond precalculus mathematics. The requirement that students had to have completed at least three years of high school mathematics is consistent with the material tested on the American College Testing (ACT)/Scholastic Assessment Test (SAT) (ACT, 2009; College Board, 2009).

The resulting sample of 1,588 students represented approximately 29% of the total number of students enrolled at a large public university in the upper Midwest. The sampled students completed 2–14 college mathematics courses of difficulty level at or beyond that associated with

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precalculus mathematics, with 45 (21.5%) of the 209 high schools represented in the sample offering both curricula (commercially developed, Core-Plus).

Our sample was not random in the classic sense (Cochran, 1977). However, following the purposive sampling arguments of Shadish, Cook, and Campbell (2002, pp. 354–372), the sample allowed us to generalize our results at least approximately to students attending other large public research institutions.

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Data Collection

Our data were obtained from college records and reflected a number of variables at the high school and college levels.

The most important independent variable was the high school mathematics curriculum a student completed (commercially developed, Core-Plus). Students were assigned to a curriculum cohort (commercially developed, Core-Plus) after carefully collecting information on the titles of the high school mathematics courses a student had completed and contacting high schools via phone calls, emails, and/or in person to identify the mathematics textbooks used when the sampled students were in attendance there. Details of this process appear in Harwell et al. (2009).

Another important independent variable was a student's college major. This information was needed because examining student mathematics course-taking required that we take into account the mathematics requirements of each campus admitting unit represented in our sample. An examination of the mathematics requirements of the majors in our student sample led us to generate a trichotomous variable to represent college major: students who completed a STEM major (e.g., biology, engineering; 47%), those whose majors were in humanities (e.g., political science, education; 32%), and other (e.g., trades, military, culinary; 21%).

Other independent variables were as follows: ACT mathematics score, high school mathematics grade point average (GPA), number of years of high school mathematics (3, 4, 5), sex, ethnicity (African American, Hispanic, Asian, White), and the date a student enrolled in college (Fall 2002 or Fall 2003). Fall 2002 was chosen as the first term to study college mathematics performance because it allowed for significant numbers of students to be sampled who had completed at least three years of the Core-Plus curriculum. Sampling students who enrolled in Fall 2003 essentially doubled our sample size, and there was no evidence in our analyses that these students differed systematically from those enrolling in Fall 2002. Students who enrolled in Fall 2002 provided seven semesters of data and those who enrolled in Fall 2003 provided five semesters of data.

One dependent variable reflected student achievement captured through mathematics course grades. This dependent variable was examined to explore the possibility that the cohorts tended to achieve at different levels in their college mathematics courses. The grade scale was A (scale value of 4.0), A– (3.70), B+ (3.30), B (3.00), B– (2.70), C+ (2.30), C (2.0), D (1.0), and F (0.0) and was treated as showing an interval scale of measurement.

We also created a variable reflecting the difficulty level of the college mathematics courses students completed. In some analyses, difficulty level served as a dependent variable; in others, it served as a covariate. This variable was constructed after examining mathematics course descriptions and produced a Likert-type scale:

Level 1: This level includes courses that should have been completed at the high school level (i.e., developmental).

Level 2: This level includes courses that would be considered precalculus mathematics and include college algebra, finite mathematics, and coursework at a similar level.

Level 3: This would be the typical entry level for mathematically well-prepared high school students, who would start their college mathematics coursework with beginning calculus.

Level 4: Beyond Calculus I. This would be the typical entry level for very well-prepared high school students who have successfully completed a high school AP Calculus course, normally with a score of 4 or 5.

We examined college mathematics achievement (grades) and course-taking (difficulty level) using longitudinal data. Only students completing college mathematics courses of difficulty level 2 or higher were included in the analyses.

Data Analyses

We used descriptive analyses to describe patterns in the data and two-level multilevel modeling (Raudenbush & Bryk, 2002) with longitudinal grade and difficulty level data to address the research questions. We began by reformulating the semester variable (1–5 or 1–7) to mathematics courses completed (first mathematics course completed, second mathematics course completed, and so on) to facilitate interpretation. Following the advice of Biesanz, Deeb-Sossa, Papadakis, Bollen, and Curran (2004), we recoded these values so that the first mathematics course completed was coded 0, the second course completed was coded 1, and so on, which allowed the grade and difficulty level of a student's first college mathematics course to be studied in the intercept model in the multilevel analyses.

Few students completed more than seven mathematics courses (1.7%), and those beyond the seventh were excluded. This was done because larger values of number of mathematics courses completed (e.g., 12), although sparse in our data, would disproportionately influence the findings (Neter, Kutner, Nachtsheim, & Wasserman, 1996).

For the multilevel analyses of grade data we fitted a within-student model of the form:

$$Y_{p_i} = \pi_{0_i} + \pi_{1_i} a_{p_i} + \pi_{2_i} b_{p_i} + e_{p_i}, \tag{1}$$

where Y_{p_i} is the mathematics grade of the ith ($i = 1, 2, \ldots, N$) student in the pth ($p = 1,2, \ldots, P$) college mathematics course completed, a_{p_i} is a covariate indicating the mathematics course completed (e.g., first, second), b_{p_i} is a covariate reflecting the difficulty level of a mathematics course that served as a control variable, π_{1_i} is the slope for the ith student reflecting the linear trajectory of grades over mathematics courses completed adjusted for difficulty level, π_{2_i} is the slope for the ith student reflecting the effect of difficulty level on grades over mathematics courses completed, and e_{p_i} is within student error.

The π_{0_i}, π_{1_i}, and π_{2_i} varied significantly across students for the grade data and between-student models were constructed to predict variation in these parameters:

$$\pi_{0_i} = \beta_{00} + \sum_{p}^{P} \beta_{0_p} X_{p_i} + r_{0_i}. \tag{2}$$

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$$\pi_{1_i} = \beta_{10} + \sum_{p}^{P} \beta_{1_p} X_{p_i} + r_{1_i} \quad (3)$$

$$\pi_{2_i} = \beta_{20} + \sum_{p}^{P} \beta_{2_p} X_{p_i} + r_{2_i} \quad (4)$$

where β_{0_p} is a slope capturing the effect of the covariate X_{p_i} on intercepts, r_{0_i} is an error term, and β_{1_p} is a slope capturing the effect of X_{p_i} on growth trajectories (slopes). A generalized hierarchical proportional odds model for ordinal data was fitted to difficulty level in the same way as the grade data except that the within-student model did not contain the difficulty level covariate. Grand mean centering was used for between-student covariates.

The covariates in Equations (2) through (4) were curriculum cohort (commercially developed, Core-Plus), college major (STEM, humanities, other), ACT mathematics score, high school mathematics GPA, years of high school mathematics (3, 4, 5), sex, ethnicity (African American, Asian American, Hispanic, White), year enrolled (Fall 2003, Fall 2002), and difficulty level (2–4). Thus the main analyses consisted of fitting the within- and between-student models in Equations (1) through (4) separately to the grade- and difficulty-level dependent variables. Multilevel analyses were performed with the R 2.10.1 lme4 package (The R Foundation for Statistical Computing, 2009).

To check the role of high school characteristics, we fitted preliminary two level models to grade and difficulty level data that included, in the between-student model, the percentage of students eligible for a free or reduced-price lunch at a student's high school and the percentage of non-White students at a student's high school. None of these covariates was a significant predictor of college mathematics grades or course-taking and high school level variables were not subsequently considered. To control for compounding of Type I error rates in the multilevel analyses, we used an adjusted Type I error rate attributed to Sidak (1967). The adjusted error rate was computed as $\alpha' = 1 - (1 - \alpha)^{1/k}$, where α is the unadjusted Type I error rate, which was set equal to .10, k = number of statistical tests, and α' is the adjusted Type I error rate. Unadjusted error rates above the traditional value of .05, such as .10, are frequently recommended as part of a familywise error rate control strategy (Keselman, Cribbie, & Holland, 2002).

RESULTS

Descriptive Analyses

Among students in our sample 87% completed a commercially developed curriculum, and for high school outcomes these students tended to outperform students in the Core-Plus cohort. For example, scores on the ACT mathematics test for the commercially developed cohort were on average significantly higher ($p < .05$) compared with the Core-Plus cohort ($M = 26.04$, $SD = 5.1$ and $M = 24.70$, $SD = 4.9$, respectively). This pattern is consistent with results reported in Harwell et al. (2009) in which students completing a commercially developed curriculum in high school scored on average about one third of a standard deviation higher on a state-mandated mathematics test administered in eighth grade than students completing NSF-funded curricula.

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For college mathematics outcomes, differences between the cohorts tended to be small or statistically equal to zero. For example, the percentages of students completing different numbers of college mathematics courses showed that they were on the whole fairly equally distributed across the commercially developed and Core-Plus cohorts. For two courses completed, the percentages for the commercially developed and Core-Plus cohorts were 41.8% and 44.1%, respectively; for three courses, the percentages were 29.5% and 25.2%, respectively; for four courses, these percentages were 20% and 21%, respectively. Among the commercially developed and Core-Plus cohorts the percentages of students completing precalculus mathematics were 59.1% and 68.6%, respectively, and for Calculus I, these percentages were 43.1% and 57.5%, respectively. The cohorts on average completed similar numbers of college mathematics courses regardless of years of high school mathematics completed ($p > .05$ for all cohort comparisons).

The sample was predominately male (63.8%) and White (71.1%), with far fewer African American (6.7%) and Hispanic (3.3%) students. There was relatively little difference in college mathematics GPA between female ($M = 2.59$, $SD = 0.92$) and male ($M = 2.51$, $SD = 0.94$) students but much larger differences between African American ($M = 2.11$, $SD = 0.98$), Asian ($M = 2.35$, $SD = 0.89$), Hispanic ($M = 2.36$, $SD = 0.87$) and White ($M = 2.64$, $SD = .93$) students.

Bivariate correlations between curriculum cohort and other variables showed that the largest value was for ACT mathematics scores ($r = -.16$) and difficulty level of a student's first college mathematics course ($r = -.14$). Among covariates the largest correlations were between ACT mathematics scores and (a) difficulty of a student's first college mathematics course ($r = .70$), (b) whether a student was a STEM major ($r = .58$), (c) high school mathematics GPA ($r = .52$), (d) sex ($r = -.28$). All of these correlations were based on a sample exceeding 1,500 and were statistically significant at $p < .01$.

We also examined our data for evidence of the effect of missing values on covariates such as ACT mathematics score and high school mathematics GPA. Most covariates had small percentages of missing data (e.g., ACT mathematics scores, 3.6%) with the most missing data observed for the high school mathematics GPA variable (7.7%). Comparing students who did and did not provide information on our variables (e.g., ACT mathematics scores, college major) did not provide any evidence of differences.

Multilevel Analyses

Grades

The fixed effects results for grades are reported in Table 1. For the within-student models the linear grade trajectory for mathematics courses completed was −0.07, meaning that, on average, student mathematics grades decreased approximately 0.07 units on a 0–4 scale from one course to the next after statistically adjusting for the effect of difficulty level.

The most important finding in Table 1 is the absence of a relation between curriculum cohort and mathematics grade trajectory or between curriculum cohort and mathematics course difficulty trajectory represented by difficulty level slopes. That is, the linear pattern of college mathematics grades and the difficulty level of these courses were the same for the commercially developed and Core-Plus cohorts. Curriculum cohort was also unrelated to a student's first college mathematics grade as seen in the intercept model. Significant effects for the intercept model showed that higher

TABLE 1
Multilevel Results for Grade Data for the Commercially Developed Curricula and Core-Plus Cohorts

Within-student model		
Effect	B	SE B
Intercept	2.78*	.05
Math difficulty	−0.32*	.05
Linear	−0.07*	.03
Between-student model		
Effect	B	SE B
Intercepts (π_{0i})		
Years high school math	0.14	.07
American College Testing math	0.04*	.01
High school math grade point average	0.60*	.10
Gender	0.08	.10
African American	0.42	.36
Asian	−0.05	.12
Hispanic	−0.02	.37
STEM	−0.06	.19
Humanities	0.00	.19
Year a student enrolled in college	0.03	.09
Commercially developed curricula versus Core-Plus	−0.03	.12
Linear growth slopes (π_{1i})		
Years high school math	−0.03	.03
American College Testing math	0.01	.01
High school math grade point average	−0.02	.05
Gender	0.03	.05
African American	−0.16	.21
Asian	0.05	.06
Hispanic	0.15	.18
STEM	0.19	.10
Humanities	0.14	.10
Year a student enrolled in college	−0.03	.04
Commercially developed curricula versus Core-Plus	0.06	.06
Difficulty slopes (π_{2i})		
Years high school math	−0.02	.07
American College Testing math	−0.01	.01
High School math grade point average	0.08	.11
Gender	0.11	.10
African American	−0.11	.41
Asian	−0.18	.13
Hispanic	−0.46	.40
STEM	−0.29	.21
Humanities	−0.24	.20
Year a student enrolled in college	−0.07	.09
Commercially developed curricula versus Core-Plus	−0.01	.13

Note. * = significant at $\alpha' = 1 - (1 - \alpha)^{1/m} = .00142$ where $\alpha = .10$ is the unadjusted Type I error rate, m = number of statistical tests = 36, and α' is the adjusted error rate. Years of high school mathematics completed: 3, 4, 5; gender: 1 = male, 0 = female; African American: 1 = yes, 0 = no; Asian: 1 = yes, 0 = no; Hispanic: 1 = yes, 0 = no so White students served as the reference group; STEM: 1 = yes, 0 = no, humanities: 1 = yes, 0 = no so other served as the reference group; year a student enrolled in college: 2002, 2003; commercially developed: 1 = yes, 0 = no.

*$p < .009$ ($\alpha' = .0085$).

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ACT mathematics scores (.04 on a 4-point scale) and high school mathematics GPAs (.60) were associated with higher grades earned in a student's first college mathematics course.

Table 1 also shows that student-level covariates such as years of high school mathematics and whether a student was a STEM major were not significant predictors of students' linear growth slopes. In fact, none of the student-level covariates were related to the pattern of college mathematics course grades students earned.

Difficulty data

A generalized hierarchical proportional odds model for ordinal data was fitted to difficulty level (2 = precalculus mathematics; 3 = Calculus I; 4 = Calculus II/beyond Calculus II). The models fitted to the difficulty data were the same as those fitted to grade data, except that difficulty level was not a covariate in the within-student model.

The results for difficulty level are displayed in Table 2. For the within-student model, the linear slope difficulty trajectory (expressed in logits) for mathematics courses completed was 1.87. Exponentiating this value (Raudenbush & Bryk, 2002) such that $\exp(1.87) = 6.5$ means that students are, as would be expected, more likely (6.5 times) to enroll in a more difficult course relative to a less difficult course as they progress.

The results in Table 2 indicate that there was no relation between the high school mathematics curricula a student completed and the pattern of difficulty levels of the college mathematics courses they completed as captured with a linear slope. That is, the linear patterns of difficulty levels of college mathematics courses were the same for the commercially developed and Core-Plus cohorts. Also, curriculum cohort and whether a student was a STEM major were unrelated to the difficulty level of their first college mathematics course as seen in the intercept model. Among the significant effects for the intercept model the largest were for high school mathematics GPA and ACT mathematics scores. For example, Table 2 shows that the odds of enrolling in a more difficult course relative to the odds of enrolling in a less difficult course was $\exp(0.35) = 1.42$ for every one unit increase in ACT mathematics scores.

For the linear growth slope model the significant sex effect of $\exp(.49) = 1.63$ means that male students were 1.63 times more likely to enroll in a more difficult course relative to a less difficult course than were female students. An explanation of this pattern is difficult as the literature on sex and college mathematics performance is mixed (e.g., see Bridgman & Lewis, 1996; Hyde, Fennema, & Lamon, 1990; Hyde, Lindberg, Linn, Ellis, & Williams, 2008).

In quasi-experimental designs like the one we used, it is important to probe the findings for possible alternative explanations. Descriptive results presented earlier showed that the commercially developed cohort had on average higher ACT mathematics scores compared with the Core-Plus cohort. Although the effect of ACT mathematics scores was controlled for statistically through its inclusion in the multilevel analyses, we further explored its effect on the relation between curriculum cohort and college mathematics outcomes for additional insight. In particular, we polytimized ACT mathematics scores into three strata using cutoffs associated with being minimally college ready in mathematics ($\leq$22), ready for mathematics at the precalculus mathematics level ($\geq$23 and $\leq$27), or possessing skills consistent with Calculus I or a more difficult course ($\geq$28; ACT, 2007). The construction of strata is consistent with the "elimination" (pp. 213–214) strategy for data analysis suggested by Pedhazur and Schmelkin (1991) and helps to minimize the effect of preexisting differences in mathematics achievement. We then refitted our multilevel models (except for the ACT mathematics score variable) separately to the longitudinal grade and difficulty data in each

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TABLE 2
Multilevel Results for Difficulty Levels for Commercially Developed Curricula and Core-Plus Cohorts

Within-student model		
Effect	B	SE B
Intercept 2	−0.23*	.08
Intercept 3	−2.28*	.11
Linear	1.87*	.07
Between-student model		
Effect	B	SE B
Intercepts (π_{0i})		
Years high school math	0.29*	.12
American College Testing math	0.35*	.03
High school math grade point average	0.60*	.22
Gender	−0.38	.17
African American	0.31	.50
Asian	0.46	.24
Hispanic	0.74	.48
STEM	0.62	.40
Humanities	−0.55	.43
Year a student enrolled in college	0.09	.14
Commercially developed versus Core-Plus	−0.28	.19
Linear growth slopes (π_{1i})		
Years high school math	0.10	.09
American College Testing math	−0.09*	.02
High school math grade point average	0.18	.14
Gender	0.49*	.15
African American	0.25	.36
Asian	−0.35	.18
Hispanic	−0.64	.44
STEM	0.64	.28
Humanities	0.21	.26
Year a student enrolled in college	−0.20	.12
Commercially developed versus Core-Plus	0.12	.19

Note. * = significant at $\alpha' = 1 - (1 - \alpha)^{1/m} = .00421$ where $\alpha = .10$ is the unadjusted Type I error rate, m = number of statistical tests = 25, and α' is the adjusted error rate. Years high school math: 3, 4, 5; gender: 1 = male, 0 = female; African American: 1 = yes, 0 = no; Asian: 1 = yes, 0 = no; Hispanic: 1 = yes, 0 = no so White students served as the reference group; STEM: 1 = yes, 0 = no, humanities: 1 = yes, 0 = no so other served as the reference group; year a student enrolled in college: 2002, 2003; commercially developed: 1 = yes, 0 = no.
*$p < .009$ ($\alpha' = .0085$).

stratum. The overall pattern of findings was unchanged, providing additional evidence that our findings are not sensitive to the mathematics achievement of students before their entering college.

DISCUSSION

Our main conclusion is that students completing a commercially developed or Core-Plus high school mathematics curriculum were equally prepared for intense college mathematics coursework, a finding that was not affected by whether students completed a STEM major in college.

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Students who completed the commercially developed or Core-Plus curricula, on average, completed a first college mathematics course of similar difficulty and earned similar grades in this course. These students continued to complete mathematics courses of similar difficulty and earn similar grades across multiple college semesters.

That two of the student-level covariates (ACT mathematics scores, high school mathematics GPA) were significant predictors of students' first college mathematics grade but not their pattern of college mathematics grades suggests that their effect is relatively short-lived. A slightly different pattern emerged for the difficulty level data. One student-level covariate (ACT mathematics scores) was a significant predictor of the difficulty level of the first college mathematics course as well as the pattern of difficulty levels. However, the strongest effect appeared for male students, who, on average, took more difficult courses than did female students.

The absence of a relation between curriculum cohort and course difficulty levels or grades earned helps to refute criticism that the Core-Plus curriculum does not adequately prepare college-bound students (including those with a STEM major) for intense college mathematics coursework. We believe that the smaller degree of congruence between the Core-Plus curriculum, college mathematics placement tests, and traditional college mathematics instructional practices is important and suggests that students completing this curriculum might have performed better had closer alignment existed.

Our results also suggest that high schools should continue to make curriculum adoption decisions on the basis of factors believed to enhance mathematics learning and achievement such as available teaching expertise, state-mandated testing requirements, and resources, and not on differential preparation for college mathematics. Moreover, the findings suggest that college policies and practices related to advising students on enrollment in intense mathematics coursework should, other things being equal, not depend on whether a student completed a commercially developed or Core-Plus mathematics curriculum in high school.

Despite the pattern of nonsignificant findings for curriculum cohort our view is that the lack of evidence favoring one curriculum does not suggest that it would be wise to stay the course and to use only commercially developed curricula. That most students have completed a commercially developed curriculum over the past four decades is likely related to the disappointing mathematics achievement of U.S. students during this time (NCES, 1998; National Commission on Excellence in Education, 1983; Stevenson & Stigler, 1992).

We think that the Core-Plus curricula have two features that provide an attractive alternative to CD curricula. First, and most important, is that these reflect various cognitively oriented theories of the development of mathematics knowledge and the nature of student learning as we understand it today. We believe that this can promote greater mathematics learning for a larger and more diverse population of high school students. Second, these curricula are based on a variety of methodological and pedagogical approaches that are designed to initiate and sustain student interest including the seamless use of technology, expanded evaluation techniques, sophisticated use of real-world problem solving, and the use of cooperative approaches. However, it is important to emphasize that a compelling body of work specifically documenting the effect of these features is not currently available, although these variables are well established in the research literature embraced by the mathematics education community (English, 2002; Grouws, 1992; Lester, 2007).

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Limitations

It is important to recognize the limitations inherent in research of this nature. An important limitation is that the retrospective cohort design limits the causal inferences that can be drawn. Another limitation is that we did not have the data to examine the possibility of a relation between high school mathematics curricula and high school track (e.g., precollege track). The presence of such a relation might help to explain the relation between high school mathematics curricula and college mathematics course-taking and performance.

A third limitation is our inability to assess the role of the content and format of college mathematics placements tests in initial mathematics course enrollment or to explore the effect of academic counseling recommendations on the basis of placement test scores on course-taking patterns. We were also unable to formally assess the alignment of high school mathematics curricula and the content and structure of many college mathematics courses, although anecdotal evidence suggests that they are generally more consistent with commercially developed curricula. For example, a review of several Calculus I texts used in college courses (Himonas & Howard, 2003; Lial, Greenwell, & Ritchey, 2007; Stewart, 2007) provides evidence that these texts are consistent with the orientation and presentation of commercially developed mathematics curricula in high school. If true, this provides an additional advantage for commercially developed students in college mathematics. However, there is apparently no work studying the alignment of high school mathematics curricula and the content and structure of college mathematics texts and courses.

Teacher quality is also frequently cited as an important factor to consider in studying mathematics achievement, but the literature is mixed on key contributors to this effect and their magnitude (Wayne & Youngs, 2003). The teacher quality effect for our sample was minimized somewhat by the mathematics department's policy of using common student assignments, examinations, and a single course-specific distribution of scores for assigning grades. It is unfortunate that we had no direct data capturing the effect of teacher quality on student achievement. These limitations suggest directions for future research that examines factors related to curricula choices at the high school and college levels.

AUTHOR NOTES

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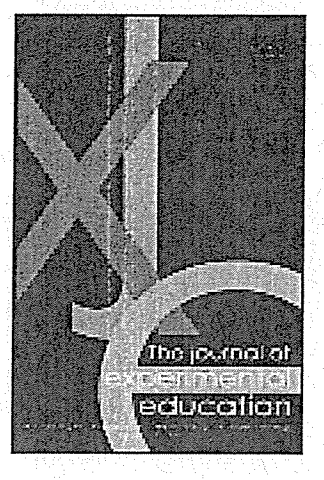

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Preparation of Students Completing a Core-Plus or Commercially Developed High School Mathematics Curriculum for Intense College Mathematics Coursework

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A MULTI-INSTITUTIONAL STUDY OF THE RELATIONSHIP BETWEEN HIGH SCHOOL MATHEMATICS ACHIEVEMENT AND PERFORMANCE IN INTRODUCTORY COLLEGE STATISTICS

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ABSTRACT

In this study we examined the effects of prior mathematics achievement and completion of a commercially developed, National Science Foundation-funded, or University of Chicago School Mathematics Project high school mathematics curriculum on achievement in students' first college statistics course. Specifically, we examined the relationship between students' high school mathematics achievement and high school mathematics curriculum on the difficulty level of students' first college statistics course, and on the grade earned in that course. In general, students with greater prior mathematics achievement took more difficult statistics courses and earned higher grades in those courses. The high school mathematics curriculum a student completed was unrelated to statistics grades and course-taking.

Keywords: *Statistics education research; Post-secondary education; Course-taking*

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1. INTRODUCTION

Statistics has received increasing recognition as an important part of both secondary and post-secondary education (we define post-secondary as institutions offering a Bachelor's degree). At the post-secondary level, an increasing number of students are taking courses in statistics, many of which satisfy the now common quantitative literacy graduation requirement for undergraduates. In fall 2005, an estimated 260,000 U.S. undergraduate students enrolled in a statistics course, an increase of over 40,000 students from 1995 (Lutzer, Rodi, Kirkman, & Maxwell, 2007). This number is likely an underestimate as it is limited to enrollment in courses offered by mathematics and statistics departments, and thus, does not count students who take statistics courses in other departments. These courses are often taught in psychology, biology, or economics departments and may differ from those typically offered by mathematics and statistics departments in the emphasis placed on understanding statistical theory and the importance of possessing a good working knowledge of calculus.

In this study, we explored the relationship between students' prior mathematics achievement in high school and achievement in their first college statistics course. Mulhern and Wylie (2005) identified six aspects of mathematical thinking relevant to success in statistics: calculation, algebraic reasoning, graphical interpretation, proportionality and ratio, probability and sampling, and estimation. Intuitively, student achievement in high school mathematics would impact these aspects of mathematical thinking and students' subsequent achievement in college statistics courses. Yet there has been little empirical study of the impact of students' prior mathematics achievement in high school on their achievement in college statistics courses. Moreover, the fact that the mathematics preparation of many college bound students is inadequate, as evidenced by numerous statistics; for example, 25% of all U.S. college freshman enroll in a developmental mathematics course (i.e., a non-credit bearing course that should have been completed in high school) (National Center for Education Statistics, 2002). This raises important questions about the adequacy of students' preparation for college statistics coursework.

To date, much of the research examining achievement in college statistics coursework has focused on factors such as student attitudes and anxiety toward statistics (see Baloglu & Zelhart, 2003, for a review), sex (e.g., Schram, 1996), and statistical reasoning (Garfield & Ben-Zvi, 2009; Zieffler, Garfield, delMas, & Reading, 2008). The role of classroom characteristics on statistics achievement has also been studied, and has included the influence of working in small-groups (e.g., Delucchi, 2007; Perkins & Saris, 2001) and of on-line learning environments (e.g., Everson, 2005). The absence of research examining the role of students' prior mathematics achievement on college statistics performance is, in part, attributable to arguments that mathematical reasoning and statistical reasoning are distinct cognitive processes, resulting from differences between mathematics and statistics as disciplines (Garfield & Gal, 1999). Gal and Garfield (1997) distinguish the discipline of statistics from that of mathematics in four ways: the role of context in interpreting data, the indeterminacy of data, the reduced need for accurate computation and execution of mathematical procedures in light of technological advances, and the need to make judgments and inferences and evaluate the quality of those judgments and inferences to solve statistical problems.

The Gal and Garfield (1997) framework provides an important jumping-off point for beginning to understand the nature of the relationship between mathematical and statistical reasoning. However, the absence of research examining the role of students' prior mathematics achievement on college statistics performance is surprising given the wide-spread recognition of the importance of mathematical preparation in other

quantitatively-oriented fields like economics and finance (Johnson & Kuennen, 2006). In short, our view is that the practice of statistics is more than a simple application of mathematics and requires reasoning skills not required in mathematics, but it seems unlikely that the ability to reason statistically (and be successful in college statistics courses) is unrelated to a student's ability to reason mathematically (reflected in prior mathematics achievement). However, as noted above, little work has been done to examine this relationship.

Studies that have examined mathematics preparation have largely been limited to examining student achievement in business statistics courses. For example, Johnson and Kuennen (2006) examined the impact of students' mathematical skills (basic and more advanced) and course-taking in college mathematics on the grades earned in an introductory course in business statistics. The most interesting finding was that students' scores on a basic skills mathematics test were a significant predictor of statistics grades, whereas scores on the mathematics portion of the ACT college entrance exam (www.act.org) and whether or not a student had taken calculus were not significant predictors. This suggests that a strong background in more advanced mathematical topics, like calculus, is not necessary to be successful in applied statistics but that the ability to perform more fundamental mathematical skills is important. In contrast, Green, Stone, Zegeye and Charles (2009) found that students who had taken a rigorous sequence of mathematical courses in college (including calculus) were twice as likely to receive an A in a required course in business statistics as students who had taken the minimum prerequisite mathematics courses (which did not include calculus). Collectively, this evidence suggests that some mathematical proficiency is needed to do well in college statistics but the role of more advanced mathematical skills is less clear.

An important, and to the best of our knowledge unstudied, aspect of student preparation for college work in statistics is the mathematical preparation students receive in high school. With the exception of students' scores on the mathematics portion of the ACT or SAT, variables reflecting students' high school mathematics preparation, such as the curriculum a student completed, have not been studied. This is surprising given the widely accepted view that high school mathematics should prepare college bound students for college mathematics (Mathematical Sciences Education Board, 2004; National Research Council, 2004), and the growing body of research examining this relationship (e.g., Harwell et al., 2009; Post et al., 2010; Schoen & Hirsch, 2003). This study contributes to this literature by examining the role of students' high school mathematics achievement, and the impact of completing a particular high school mathematics curriculum, on college statistics performance. The results have important implications for high schools preparing students for post-secondary study and for post-secondary institutions advising students on appropriate statistics coursework.

In some high schools, statistics is offered as a separate elective course; for example, the College Board's Advanced Placement (AP) statistics course, which is a non-calculus based introduction to college level statistics. In 2010, nearly 130,000 U.S. high school students took the AP statistics exam, with 34% of these students earning a score of four or higher (College Board, 2010a), qualifying them to receive college credits at many post-secondary institutions. More typically, statistics material is integrated into the high school mathematics curriculum. The National Council for Teachers of Mathematics (NCTM) outlined learning goals for statistics and probability (at all grade levels) in their document The Principles and Standards for School Mathematics (NCTM, 1989, 2000). It should also be noted that the mathematics portion of the SAT college entrance exam includes items assessing knowledge of data analysis, statistics, and probability (College Board, 2010b).

We identified three general categories of high school mathematics curricula: National Science Foundation (NSF)-funded curricula, the University of Chicago School Mathematics Project (UCSMP) curriculum, and a broad category of commercially developed (CD) curricula. The NSF-funded curricula were developed based on the Principles and Standards for School Mathematics (NCTM, 1989, 2000), and include topics in data analysis and probability as well as algebra, geometry, and topics in discrete mathematics at all grades levels. The data analysis and probability standard states that students should be able to

1) Formulate questions that can be addressed with data and collect, organize, and display relevant data to answer them;
2) Select and use appropriate statistical methods to analyze data;
3) Develop and evaluate inferences and predictions that are based on data, and;
4) Understand and apply basic concepts of probability. (NCTM, 2000, p. 4)

As an example, statistical topics covered in one NSF-funded curriculum, Contemporary Mathematics in Context (Core-Plus) (Coxford et al., 1998), include descriptive statistics, simulation, correlation and regression, sampling distributions, and probability (see Appendix A for a full list of statistical topics covered in the Core-Plus curriculum). Other NSF-funded curricula included in the present study are the Interactive Mathematics Program (IMP) (Fendel, Resek, Alper, & Fraser, 1998) and Mathematics Modeling Our World (MMOW) (Garfunkel, Godbold, & Pollack, 1998), both of which include similar statistical topics at all grade levels.

The UCSMP mathematics curriculum emphasizes problem solving, everyday applications, and the use of technology in the mathematics classroom at all grade levels (Usiskin, 1986). First developed in the early 1980s, UCSMP is currently in its third edition and has received both public (NSF) and private (Amoco) funding. In addition to courses in Algebra, Geometry, Advanced Algebra, and Precalculus & Discrete Mathematics, the UCSMP secondary curriculum includes a course in Functions, Statistics, & Trigonometry (FST). This course is typically taken after a student has completed the Advanced Algebra course and before they enroll in the Precalculus & Discrete Mathematics course. Statistical topics covered in the FST course include data collection, descriptive statistics, graphical representations of data, probability, simulation, and sampling distributions. Of the thirteen chapters in the current FST textbook, three are dedicated to these topics.

Although most CD curricula include statistical topics, the depth and breath of this coverage is limited. Generally, topics in data analysis and probability are included as a single chapter in Algebra I and II textbooks; for example, *Merrill Algebra 2 with Trigonometry* (Foster, Gordon, Gell, Rath, & Winters, 1995), often as the last chapter. In some cases statistical topics are included as a single unit in a chapter containing other units; for example *Algebra: Structure and Method Book One* (Brown et al., 1997). The most frequently addressed topics include measures of central tendency, measures of dispersion, and normal distributions as well as probability.

We hypothesize that because the NSF-funded curricula explicitly include statistical topics at all grade levels, students who completed one of these NSF-funded curricula in high school will be better prepared for college statistics than students who completed the UCSMP curriculum or a CD curriculum. To better understand the role of high school mathematics preparation on students' college statistics performance, we posed two research questions:

1) Is there a relationship between students' high school mathematics achievement and the (a) difficulty level of their first college statistics course, and (b) grade earned in their first college statistics course, controlling for background variables (sex, ethnicity, college major)?

2) Is there a relationship between the type of high school mathematics curriculum completed (NSF-funded, UCSMP, CD) and the (a) difficulty level of students' first college statistics course, and (b) grade earned in students' first college statistics course, controlling for background variables (sex, ethnicity, college major)?

Student high school mathematics achievement variables included ACT mathematics test score, years of high school mathematics completed, and high school mathematics GPA. We focus on the first college statistics course because the majority (90%) of students in our sample took a single statistics course at the college level, whereas 7% of students took two statistics courses, and the remaining 3% took three or more courses (up to eight courses).

2. METHOD

2.1 RESEARCH DESIGN

A retrospective, cohort, cluster design was used in which the difficulty level and grade of students' first college statistics course were examined. Based on their high school mathematics curriculum, students were categorized into one of three curriculum cohorts (NSF-funded, UCSMP, CD). To accurately categorize students, we contacted each of the approximately 300 high schools represented in the student sample to obtain descriptions of their mathematics programs, the mathematics courses offered, and the textbooks used. This information was then coordinated with each student's high school transcript. Although this effort required significant resources, it was necessary to ensure that categorizing a student as having completed a particular curriculum was accurate. All other student data, including their high school mathematics transcripts, college statistics course-taking and performance, and demographic information, was obtained from each student's post-secondary institution.

2.2 POPULATION AND SAMPLE

The target population consisted of college students in the United States who completed at least three years of high school mathematics in a CD, NSF-funded, or UCSMP curriculum and took at least one statistics course at the post-secondary level. The population was restricted to students who completed at least three years of high school mathematics because post-secondary students will generally be expected to have completed at least three years (Boyer, 1983) of high school mathematics, as reflected in the mathematics portions of the ACT and SAT exams (ACT, 2009; College Board, 2010c).

The sampled population consisted of $n = 5{,}308$ students from 23 purposively sampled post-secondary institutions (clusters). However, three institutions were excluded because their within-institution sample sizes were too small to provide adequate data for parameter estimation, reducing the number of students to 5,296. Sampled students enrolled in college during the fall 2002 and fall 2003 terms and had graduated from a Minnesota high school. These terms were selected because they represented the first groups of students who had the opportunity to complete four full years of the published (i.e., not pilot materials) NSF-funded curricula.

The remaining 20 four-year institutions were located in the upper midwest of the United States. Seven of the 20 post-secondary institutions were publicly-funded universities and are part of one of four states' college systems, whereas the remaining 13 institutions were privately-funded liberal arts colleges. All institutions are not-for-profit.

Four institutions are located in a large metropolitan area with the remainder located in suburban and rural settings. Eight institutions were classified as "most selective" according to the Carnegie Foundation's Undergraduate Profile Classification (Carnegie Foundation, 2009b), indicating that test scores for first year students place these institutions in the top-fifth of post-secondary institutions offering Bachelor's degrees. Eleven institutions were classified as "selective," which means that test scores for first year students place these institutions in the middle two-fifths of post-secondary institutions. The remaining institution was classified as "inclusive," which indicates that the institution either did not report test score data or does not make admissions decisions based on test score data.

Due to missing data, the sample was further reduced to n = 3,599 students. The variable with the most missing data was college major with 13% of students (n = 707) not providing values, most of whom were in "Other" majors, which included technologies, technicians, trades, military, and culinary arts. A missing values analysis of all variables revealed no significant differences in means between the data when all available values were included and the data set containing the n = 3,599 students who provided complete data. In addition, refitting our statistical models to subsets of students after filtering on missing values (e.g., including "Other" college majors) produced similar patterns of findings (regarding statistical significance and the direction of effects) to those based on all available data. Together, these analyses suggest that missing data did not seriously distort our findings.

Our sample is not random in the traditional sense (Cochran, 1977). However, we believe that following the purposive sampling arguments of Shadish, Cook, and Campbell (2002), and the similarity of the 20 sampled institutions to those of U.S. institutions represented in the undergraduate profile classification of the Carnegie Foundation (on institutional selectivity and nature of the student body), our results are generalizable to students attending post-secondary institutions beyond our sample.

2.3 VARIABLES

Archival data reflecting variables at the high school and college level were collected from each of the post-secondary institutions. Student-level predictor variables were the mathematics curriculum a student completed (NSF-funded 1 = yes, 0 = no; UCSMP 1 = yes, 0 = no so CD served as the reference group), ethnicity (African American 1 = yes, 0 = no; Hispanic 1 = yes, 0 = no; Asian 1 = yes, 0 = no; so Caucasian students served as the reference group), sex (1 = male, 0 = female), and college major (1 = STEM, 0 = non-STEM). Students' majors were classified as STEM (Science, Technology, Engineering or Mathematics) or non-STEM using the Carnegie Foundation's Undergraduate Instructional Program Classification (Carnegie Foundation, 2009a). Other predictors were students' ACT mathematics score, years of high school mathematics completed (3, 4, 5), and their GPA in high school mathematics courses. Years of high school mathematics completed was coded such that a value of three reflects completion of an Algebra II course, a value of four reflects completion of a Pre-calculus course, and a value of five reflects completion of a Calculus I course in high school. Students' high school mathematics GPAs were captured using a twelve-point scale ranging from A (scale value of 4.0), A- (scale value of 3.67), down to F (scale value of 0.0).

One dependent variable reflected the difficulty level of the first statistics course a student completed in college, allowing for the possibility that students with differing prior mathematics achievement, who completed different high school mathematics curricula, would enroll in courses of varying difficulty. This variable was constructed after a careful review of information about college statistics courses that included a course description,

prerequisites, a syllabus, and the department in which the course was taught (e.g., Statistics, Nursing). Combining this information produced the following difficulty scale for college statistics courses:

> Level 1: Least difficult (prerequisite is usually 1-2 yrs of high school algebra, content limited to simple methods and procedures like *t*-tests and ANOVA, little or no focus on statistical theory);
>
> Level 2: Moderately difficult (at least one college prerequisite, moderately-difficult content such as continuous/discrete probability distributions and mathematical expectation, moderate attention paid to statistical theory, procedures covered typically include multiple regression);
>
> Level 3: Most difficult (typically two college prerequisites, advanced content like moment generating functions and likelihood ratio test principle, considerable emphasis on statistical theory, linear models, and multivariate and Bayesian methods).

The students in our sample took 6,090 statistics courses across eight semesters of post-secondary coursework, 88.2% of which were taken at Level 1, 7.6% at Level 2, and 4.6% at Level 3. A review of the difficulty level of students' first course showed that 93.4% of students began at Level 1, 5.7% began at Level 2, and the remaining 0.9% began at Level 3. This led us to aggregate Level 2 and Level 3, producing the following dichotomous scale representing the difficulty level of students' first course in post-secondary statistics:

> Level 1: Applied introduction to statistics, prerequisite is usually 1-2 yrs of high school algebra and content is limited to simple methods (e.g., *t*-tests, ANOVA, correlation/regression, sampling distribution of the mean) with little or no focus on statistical theory, course is typically taught in a service department (e.g., psychology, public health, business);
>
> Level 2: Theoretical introduction to statistics, prerequisite is usually one semester of calculus and content includes advanced topics (e.g., moment generating functions, likelihood ratio test principle, multivariate and Bayesian methods) with greater emphasis on statistical theory, course is typically taught in a mathematics or statistics department but may be taught in a service department.

The difficulty variable was coded such that Level 2 = 1 and Level 1 = 0.

A second dependent variable reflected student achievement levels captured through the grade students earned in their first college statistics course. Grades were again captured using a twelve-point scale ranging from A (scale value of 4.0), A- (scale value of 3.67), down to F (scale value of 0.0), which was treated as showing an interval scale of measurement.

2.4 DATA ANALYSIS

We used descriptive analyses to describe patterns in the data and inferential analyses to address the research questions. To model the difficulty data, a hierarchical generalized linear model (Raudenbush & Bryk, 2002) was fitted, where students were treated as nested within colleges. Preliminary analyses revealed that only the intercepts varied across institutions, and, because of the small number of institutions ($j = 20$), no college-level covariates were included in the model. Slopes from the student level covariates were treated as fixed across colleges and the fitted model for the difficulty level outcome had the following form:

$$\log(y_{ij}) = \gamma_{00} + \sum_{p}^{P} \gamma_{0p} X_{pij} + u_{0j} \quad (1)$$

where, $\log(Y_{ij})$ is a vector of outcomes reflecting the log-odds of the difficulty level of the first course for the i^{th} (i = 1, 2, … , n_j) student in the j^{th} (j = 1, 2, … , J) post-secondary institution, γ_{00} is the precision-weighted average log-odds of beginning with difficulty Level 2, γ_{0p} is a precision-weighted average slope capturing the effect of the p^{th} covariate X_p, and u_{oj} is the random effect for intercepts.

To model the grade data, a hierarchical linear model (Raudenbush & Bryk, 2002) was fitted, where students were treated as nested within colleges. Again, because of the small number of institutions no college-level covariates were modeled. Likewise, preliminary analyses revealed that only the intercepts varied across colleges as such slopes for student-level covariates were treated as fixed across institutions. The fitted model for the grades outcome had the following form:

$$y_{ij} = \gamma_{00} + \sum_{p=1}^{P} \gamma_{0p} X_{pij} + u_{0j} + r_{ij} \quad (2)$$

where, Y_{ij} is a vector of outcomes reflecting the grade in the first course for the i^{th} (i = 1, 2, … , n_j) student in the j^{th} (j = 1, 2, … , J) post-secondary institution, γ_{00} is the precision-weighted average grade, γ_{0p} is a precision-weighted average slope capturing the effect of the p^{th} covariate X_p, u_{oj} is the random effect for intercepts, and r_{ij} is the student level residual term. The student-level covariates included in the models fit to the grade and difficulty outcomes were identical except that the difficulty variable was used as a covariate in the grade model. The grade data effect sizes were calculated following

$$\delta_p = \frac{\delta_{0p}}{\sqrt{\tau_{00} + \sigma^2}} \quad (3)$$

where, δ_p is the standardized effect size for the p^{th} covariate, γ_{0p} is the precision-weighted average slope capturing the effect of the p^{th} covariate, τ_{00} is the variance component associated with the unconditional random intercepts, and σ^2 is the variance of the unconditional student-level error term (Raudenbush, Spybrook, Congdon, Liu, & Martinez, 2009). An a priori power analysis using equation (3) and the Optimal Design Software (Raudenbush et al.) revealed a statistical power of approximately .75 to detect a fixed-effect for high school mathematics curriculum of moderate size (.40) (Spybrook, Raudenbush, Congdon, & Martinez, 2009).

To control for compounding of Type I error rates we used an adjusted Type I error rate attributed to Sidak (1967). The adjusted error rate was computed as $\alpha' = 1 - (1 - \alpha)^{1/k}$, where α is the unadjusted Type I error rate, k = number of statistical tests, and α' is the adjusted Type I error rate. For the two-level models that follow the adjusted error rate was $\alpha' = 1 - (1 - \alpha) = .008$, where $\alpha = .10$, and $k = 11$ for the grade outcome and 10 for the difficulty level outcome. All of the multilevel analyses were performed using the lme4 package in R 2.10.1 (The R Project for Statistical Computing).

3. RESULTS

2.5 DESCRIPTIVE ANALYSES

We began by performing descriptive analyses to explore patterns in the data. As noted previously, 90% of the students in our sample took one statistics course at the college level, whereas 7% of students took two courses and the remaining 3% took three or more courses. Regarding the difficulty of the first statistics course taken, 93% of students began with a course equal to difficulty Level 1 (applied introduction) and 7% began with a course equal to difficulty Level 2 (theoretical introduction).

Of the students in our sample, 67% completed a CD curriculum in high school, 19% completed the UCSMP curriculum, and 14% completed a NSF-funded curriculum. Descriptive statistics for variables in our statistical models can be found in Table 1. Table 2 presents descriptive statistics for enrollment, 25th and 75th percentiles of ACT mathematics scores for first year students, percentage of African American students, and percentage of STEM majors across institutions. Within institution samples sizes (n_j) ranged from 14 to 1,213 (M = 265, SD = 284). Table 3 contains bivariate correlations between variables, most of which are relatively small.

Table 1. Descriptive statistics for student variables

	Mean[a]	Std Dev	Minimum	Maximum
Difficulty level of first course	0.07	0.249	0.00	1.00
Grade of first course	3.05	0.834	0.67	4.00
ACT mathematics score	24.23	4.476	11.00	36.00
High school mathematics GPA	3.25	0.643	0.66	4.00
Years of high school mathematics	4.07	0.740	3.00	5.00
UCSMP	0.19	0.400	.00	1.00
NSF-funded	0.14	0.349	.00	1.00
African American	0.03	0.156	.00	1.00
Asian	0.07	0.255	.00	1.00
Hispanic	0.02	0.129	.00	1.00
Male	0.41	0.493	.00	1.00
STEM major	0.30	0.459	.00	1.00

Note. Difficulty level of the first course 1 = Level 2, 0 = Level 1; Curriculum variables (UCSMP, NSF-funded) were coded so that CD students served as the reference group; Ethnicity variables were coded so that Caucasian students served as the reference group; Male 1 = yes, 0 = no; STEM major 1 = yes, 0 = no.

[a]Mean values that are less than 1 can be interpreted as percentages, such as 19% of the sample completed a UCSMP high school mathematics curriculum.

Table 2. Descriptive statistics for institution variables (j = 20)

	Mean	Std Dev	Minimum	Maximum
Enrollment	6384.15	8170.60	440.00	37383.00
ACT math 25th percentile	21.85	2.54	19.00	28.00
ACT math 75th percentile	26.70	2.39	23.00	32.00
Percentage of African American students	2.15	2.46	.00	7.69
Percentage of STEM majors	7.22	4.90	.00	19.55

Table 3. Correlations between student variables

	1	2	3	4	5	6	7	8	9	10	11	12
Difficulty level	1.0											
Grade	-.00	1.0										
ACT math	.14**	.34**	1.0									
HS math GPA	.10**	.41**	.49**	1.0								
Years HS math	-.05**	.21**	.44**	.28**	1.0							
UCSMP	.03	.01	.07**	.01	.02	1.0						
NSF-funded	-.05**	-.04**	-.15**	-.03	-.04**	-.20**	1.0					
African Amer	-.03*	-.08**	-.17**	-.08**	-.07**	-.02	.05**	1.0				
Asian	-.00	-.07**	-.11**	-.01	.02	-.02	.05**	-.04**	1.0			
Hispanic	-.02	-.02	-.05**	-.05**	-.04**	-.00	.02	-.02	-.04**	1.0		
Male	-.02	-.09**	.15**	-.09**	.09**	.02	-.00	.02	-.01	-.01	1.0	
STEM major	.28**	.11**	.29**	.24**	.13**	.05**	-.07**	-.02	.02	.01	.07**	1.0

Note. HS = high school; Difficulty level 1 = Level 2, 0 = Level 1; Curriculum variables (UCSMP, NSF-funded) were coded so that CD students served as the reference group; Ethnicity variables were coded so that Caucasian students served as the reference group; Male 1 = yes, 0 = no; STEM major 1 = yes, 0 = no.
*$p < .05$. **$p < .01$.

2.6 INFERENTIAL ANALYSES

Difficulty level of the first college statistics course First, an unconditional model was fitted to the difficulty outcome. The average (across institutions) log-odds of a college student beginning with a course of difficulty Level 2 (theoretical introduction) as opposed to Level 1 (applied introduction) was -4.33. This means that for a "typical" institution (i.e. with a random effect of 0) the expected log-odds of taking a first statistics course of difficulty Level 2 is -4.33, which corresponds to an odds ratio of 0.01.

The results for the conditional model (see Table 4) revealed that the type of high school mathematics curriculum a student completed and variables reflecting prior mathematics achievement (with one exception) were unrelated to the log-odds of students beginning their statistics coursework at difficulty Level 2, as were the ethnicity and sex variables. ACT mathematics score and whether a student was a STEM major were significant predictors of the log-odds that a student would begin their statistics coursework with a course of difficulty Level 2. Being a STEM major was associated with an expected increase in the log-odds of a student beginning their statistics coursework at difficulty Level 2 of 1.11, with the other predictors held constant. Converting to an odds ratio produces 3.03. Thus, the odds that a student begins their coursework at difficulty Level 2 is three times higher for STEM majors than non-STEM majors. The effect of ACT mathematics score on the odds a student begins with a statistics course at Level 2 is considerably smaller; for example, at no point on the ACT mathematics scale does the predicted probability of a student beginning with a course of difficulty Level 2 exceed 0.50.

Grade earned in the first college statistics course In order to determine whether a multilevel model was necessary to analyze the grade outcome, an unconditional model was fitted first. This initial analysis showed that there was enough variation in average grades between the post-secondary institutions to continue modeling ($\hat{\rho}_{ICC} = .04$). Next we constructed a predictive model to account for the variation in grades. The results of this

analysis are presented in Table 5. Effect sizes for all statistically significant findings are presented.

Table 4. Multilevel results of the difficulty level of the first college statistics course

Between-Student Model				
Effect	*b*	SE(*b*)	*z*-value	*p*-value
Intercept	-8.908*	1.067	-8.35	<0.001
ACT math score	0.176*	0.025	7.03	<0.001
Years of HS math	0.019	0.134	0.14	0.888
HS math GPA	-0.122	0.173	-0.70	0.482
UCSMP	-0.048	0.205	-0.24	0.814
NSF-funded	-0.111	0.313	-0.35	0.723
African American	-0.034	0.823	-0.04	0.967
Asian	0.337	0.311	1.08	0.279
Hispanic	-0.057	0.698	-0.08	0.935
Male	-0.072	1.176	-0.41	0.680
STEM	1.109*	0.197	5.64	<0.001

Note. Difficulty level of the first course 1 = Level 2, 0 = Level 1; Curriculum variables (UCSMP, NSF-funded) were coded so that CD students served as the reference group; Ethnicity variables were coded so that Caucasian students served as the reference group; Male 1 = yes, 0 = no; STEM major 1 = yes, 0 = no.
*$p < .008$.

Table 5. Multilevel results for the grade earned in the first college statistics course

Between-Student Model				
Effect	*b*	SE(*b*)	*t*-value	Effect Size
Intercept	0.978*	0.102	9.59	----
Difficulty of first course	-0.165*	0.025	-6.68	-0.197
ACT math score	0.038*	0.004	10.28	0.045
Years of HS math	0.053	0.019	2.74	
HS math GPA	0.406*	0.023	17.72	0.486
UCSMP	-0.012	0.031	-0.40	
NSF-funded	-0.008	0.036	-0.21	
African American	-0.092	0.081	-1.14	
Asian	-0.197*	0.048	-4.08	-0.236
Hispanic	-0.028	0.092	-0.30	
Male	-0.127*	0.026	-4.89	-0.152
STEM	-0.001	0.031	-0.04	

Note. Difficulty level of the first course 1 = Level 2, 0 = Level 1; Curriculum variables (UCSMP, NSF-funded) were coded so that CD students served as the reference group; Ethnicity variables were coded so that Caucasian students served as the reference group; Male 1 = yes, 0 = no; STEM major 1 = yes, 0 = no.
*$p < .008$.

These results indicate that the type of high school mathematics curriculum a student completed was unrelated to the grade earned in the first course. Students' prior mathematics achievement was the strongest predictor of the grade earned in the first college statistics course, with both ACT mathematics score (b = 0.05) and high school mathematics GPA (b = 0.41) significant predictors. Despite ACT mathematics score being a statistically significant predictor, its practical significance seems to be relatively small. Students with ACT mathematics scores greater than 21 (the U.S. average) are predicted to receive a B (3.0) or higher in their first college statistics course. Further, at no point on the ACT mathematics scale are students predicted to fail their first college statistics course by earning less than a C-. Another significant predictor was the difficulty level of the first course (b = -0.17), with students tending to earn lower grades in more difficult courses. Asian students (b = -0.20) also earned lower grades in the first course than Caucasian students, and male students (b = -0.13) earned lower grades than female students.

4. DISCUSSION

In the present study we sought to examine the relationship between students' high school mathematics preparation (achievement and curriculum) and performance in their first college statistics course. In response to our research questions, our findings can be summarized in two ways. First, students with stronger prior mathematics achievement generally took more difficult statistics courses and earned higher grades in those courses than students with weaker prior mathematics achievement. This finding suggests that a stronger mathematical background benefits students in college statistics. Whereas ACT mathematics score was a significant predictor of the difficulty outcome, the size of the effect was small, with students who earned a perfect score on the ACT mathematics exam (i.e., 36) only having a predicted probability of taking a theoretical introductory course of 0.25. None of the other prior mathematics achievement variables were significant predictors of the difficulty outcome.

In fact, the best predictor of a student beginning their statistics coursework at difficulty Level 2 was college major. As previously noted, the odds of beginning with a theoretical introductory course were three times higher for STEM majors than their non-STEM major peers. This result suggests that a student's major (STEM vs. non-STEM) plays a key role in the type of introductory statistics course with which they begin. Research on the impact of college major on mathematics course-taking shows that over 90% of students take the minimum mathematical requirement(s) of their major (Harwell, et al., 2009). This finding may explain the current finding in that STEM majors take more difficult college statistics courses, because their major requires it, compared to non-STEM majors.

Our results also show that high school mathematics GPA and ACT mathematics scores were significant predictors of students' grades in their first college statistics course. Students' high school mathematics GPA appeared to be the strongest predictor with a medium size effect of d = .49, supporting the inference that prior mathematics achievement (reflected in success in high school mathematics courses) is related to college statistics achievement. Although scores on the ACT mathematics test had a statistically significant relationship with students' grades, the practical significance of that relationship was small. This result is consistent with work by Johnson and Kuennen (2006) showing that scores on the ACT mathematics test did not predict grades in an introductory business statistics course and suggests that the mathematical skills measured by the ACT test only modestly overlap with those needed in college statistics. The ACT mathematics test is designed to measure mathematical reasoning and contains 60 items

divided into three subtests covering topics in pre-algebra, elementary and intermediate algebra, coordinate and plane geometry, and trigonometry. As such only half of the items on the ACT mathematics test measure aspects of mathematical reasoning that would seem to be related to achievement in a statistics course.

Similarly, the number of years of high school mathematics a student completed was unrelated to both the difficulty level of the course taken and the grade subsequently earned, which is consistent with previous research (Johnson & Kuennen, 2006) and suggests that exposure to more advanced mathematics courses (e.g., calculus) is not necessary to do well in statistics. For the difficulty variable this finding is particularly interesting because it suggests that students who take calculus in high school do not necessarily enroll in calculus-based statistics courses in college. In addition, the non-significant findings regarding ethnicity (with one exception for the grade data) suggest that students of different ethnic backgrounds are equally prepared for college statistics. The finding that female students earned higher grades than their male peers in college statistics is consistent with previous research regarding sex effects in statistics achievement (Schram, 1996). This finding is contrary to the sex effect traditionally observed in mathematics, where males have generally been found to outperform their female peers. This may be related to the composition of statistics courses, with less relative emphasis on formal abstraction and a greater reliance on contextually based applications.

Second, the type of high school mathematics curriculum a student completed was unrelated to the difficulty level of a student's first college statistics course as well as the grade earned in that course. Thus, greater exposure to statistical content in high school does not lead to students taking more difficult statistics courses in college or earning better grades in those courses. The non-significant findings for curriculum may in part be due to factors we were not able to observe such as the fidelity of implementation of the curricula or variations in teaching quality and assessment practices within- and across-institutions. Likewise, we do not have data assessing the extent to which teachers (high school, college) actually covered the statistical content present in the curricula. However, we have anecdotal information suggesting that many high school mathematics teachers do not feel comfortable teaching statistics and frequently skip this material. Another possibility is that the integrated nature of the NSF-funded curricula may make it difficult for students to learn distinct statistical concepts, leaving them as prepared for college work in statistics as their peers who likely received little exposure to statistical concepts in high school. Of course, it is also possible that the statistical material embedded in the NSF-funded curricula do not promote more understanding than the statistical material appearing in the CD and UCSMP curricula.

In sum, these findings do not resolve the debate between those who believe that mathematical and statistical reasoning are distinct types of reasoning and those who do not, because the grades students' earn in a statistics course are a measure of many things, only some of which can be attributed to a student's ability to reason statistically. But these results do provide preliminary evidence that there is a relationship between students' mathematics achievement and statistics achievement, although the exact nature of this relationship remains unclear.

That being said, the present findings regarding the effect of curriculum are consistent with previous results regarding the impact of high school mathematics curricula on performance in students' first college mathematics course (e.g., Harwell et al., 2009; Post et al., 2010). In general, studies to date examining the impact of high school mathematics curricula and the difficulty of a student's first college mathematics class (defined as a course whose difficulty level equals or exceeds College Algebra/Pre-Calculus) or the grade earned in that class have not found any relationship. Other work (Harwell, LeBeau,

Post, Dupuis, & Medhanie, 2011) has found that students completing an NSF-funded curriculum are more likely to begin college with a developmental mathematics course compared to students completing a CD or UCSMP curriculum. The current results extend these previous findings to include performance in college statistics coursework.

Future research should further examine the amount and nature of mathematical preparation necessary to be successful in statistics, particularly the importance of higher-level mathematical skills. For example, the literature remains unclear about the advantages gained in statistics from exposure to calculus. In the present study, students who had taken calculus in high school did not outperform their peers in university statistics work (even in calculus-based statistics courses), however other studies (e.g., Green et al., 2009) found that students who had a more rigorous sequence of mathematical courses (including calculus) did outperform their peers who had experienced a less rigorous sequence of courses that did not include calculus. It is possible that the mathematical skills that students gain in a calculus course are not directly applicable to an introductory statistics course (particularly in an applied course), but that taking calculus is related to a student having a positive attitude about and less anxiety toward mathematics, which translates into that student having less anxiety toward statistics than students with less advanced coursework in mathematics. Similarly, future research would benefit from an examination of the aspects of mathematical proficiency that contribute to a better understanding of specific statistical concepts. For example, it seems plausible that a good working knowledge of calculus would not be related to a student's understanding of measures of central tendency, but that it may be related to a student's understanding of sampling distributions. Studies that measure specific aspects of both mathematical and statistical thinking would significantly contribute to our understanding of the relationship between mathematical and statistical reasoning.

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The Role of the ACCUPLACER Mathematics Placement Test on a Student's First College Mathematics Course

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Abstract

The first college mathematics course a student enrolls in is often affected by performance on a college mathematics placement test. Yet validity evidence of mathematics placement tests remains limited, even for nationally standardized placement tests, and when it is available usually consists of examining a student's subsequent performance in mathematics courses. This study expands on existing literature by considering whether a nationally standardized college mathematics placement test (ACCUPLACER) contributes to the prediction of enrollment and success in developmental and nondevelopmental mathematics courses above and beyond prediction associated with the ACT mathematics test. Results for a sample of more than 1,300 students from 20 postsecondary institutions suggest that ACCUPLACER does not contribute to either the prediction of enrollment or subsequent success in such courses, and that comparable information is provided by using the ACT mathematics score alone. Implications of these findings are discussed.

Keywords

ACCUPLACER, postsecondary education, developmental mathematics, placement

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Research in mathematics education suggests that many U.S. students struggle in mathematics. Not only is there evidence of unsatisfactory performance in mathematics in high school (U.S. Department of Education, National Center for Education Statistics, Institute of Education Sciences [NCES], 2010), with only 26% of 12th graders at or above proficiency, but there is also evidence of unsatisfactory preparation for college level mathematics (NCES, 2006, 2009).

The most dramatic evidence of inadequate preparation is perhaps the percentage of students who complete at least one developmental (remedial) mathematics course in college (which do not usually result in credits toward graduation), defined as a course intended to lead to a student attaining the minimum expected mathematics competency of entering college freshmen (Bahr, 2007). For example, Attewell, Lavin, Domina, and Levey (2006) reported that 28% of traditional college students (i.e., 18-22 years old, full-time enrollment, etc.) took a developmental course in mathematics in college. Other reports cite developmental mathematics course-taking rates of 28% (NCES, 2004) and 17% (NCES, 2008). These percentages are still considerable when taking a college's selectivity into account (typically assessed using SAT/ACT scores for a freshman class), for example, 11.5% and 27% for more selective (higher ACT/SAT scores) versus less selective institutions, respectively (Bettinger & Long, 2009).

Whatever the precise estimate, the number of students enrolling in college in the United States taking developmental coursework is staggering. In Fall 2008, approximately 12 million students were enrolled in 4-year institutions (NCES, 2010) and using the Attewell et al. (2006) value of 28% implies that more than 3,000,000 students completed at least one developmental mathematics course. Of course, the impact on students is substantial. Requiring students to take a developmental mathematics course as a result of their performance on a placement test has the effect of requiring those students to pay for noncredit courses, and extend the number of courses required toward acquiring a degree.

Initial placement in a mathematics course is usually related to performance on a college mathematics placement test. The goal of a placement test is to determine whether or not a student can be successful in a particular course (McFate & Olmsted, 1999). Naturally, the accurate placement of students into an appropriate college mathematics course is key to the success of students in a course. Students who enroll in mathematics courses that they are unprepared for are not likely to do well, and students who enroll in courses for which they are overprepared may underperform because of a lack of motivation (Norman et al., in press). In short, there is a need to ensure that students are accurately placed in their first college mathematics course.

Despite the important role placement tests play in initial college mathematics course-taking and performance, validation evidence for these tests is typically limited to examining a student's subsequent performance in college mathematics courses. This study examines the extent to which a nationally standardized placement test (ACCUPLACER) predicts student enrollment in their first college mathematics course and performance in that course above and beyond the ACT mathematics test. The

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results have important implications for colleges attempting to appropriately match a student's mathematics preparation and their first college mathematics course.

The College Mathematics Placement Process

The process by which students are matched to courses is referred to as the placement process (Frisbie, 1982). Colleges and universities often make use of several pieces of information when moving students through the placement process. Institutions vary in the amount of collateral information used during the placement process (Rueda & Sokolowski, 2004), but typically use high school performance information (e.g., high school percentile rank, grades) and nationally standardized achievement tests (such as the ACT or SAT). A majority of institutions (61%) also use placement tests (NCES, 2004). Those that use placement tests may either use a locally constructed placement test or one developed by an outside agency. Institutions may also differ in their response to placement test performance. That is, some institutions may require the student to enroll in the course suggested by the placement process whereas others present the student with a recommendation, allowing them to take a more or less difficult course than the one recommended.

Concerns over the mathematics placement of students in colleges and universities as well as disagreement on the purpose of college mathematics placement tests have existed for more than 30 years (Mathematical Association of America [MAA], 2010). Still, one widely agreed on goal of a college mathematics placement test is to measure the extent to which students possess the necessary skills and knowledge needed for specific mathematics courses (MAA, 2010). In keeping with the purpose of placement tests, validation of these tests often consists of examining student performance in subsequent college mathematics courses in light of their placement test score.

Validating Placement Tests

Assuming that postsecondary mathematics course grades are valid measures of mathematical achievement, there are four possible outcomes from the combination of a placement test and subsequent performance in a mathematics course (Sawyer, 1996). Suppose that a placement test for determining whether a student has the necessary skills and knowledge to begin college mathematics at the college algebra level is available.

Ideally, a successful outcome is one in which a placement test would place students that score above a particular score (i.e., score cutoff) into college algebra who are subsequently successful in college algebra. Also, ideally this placement test would place students that score below the cutoff into a lower level mathematics course (e.g., developmental mathematics) who would have been unsuccessful in college algebra. These two performance by outcome combinations represent a "correct" placement decision. Additionally, the placement test would ideally neither have students score above the cutoff that would be unsuccessful in college algebra nor have students score

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below the cutoff that would have been successful in college algebra. An important feature of the performance by outcome combinations is that students that score below the cutoff do not typically have the opportunity to either be successful or unsuccessful in the course into which they attempted to place. In any event, evidence of correct placement reflects the validity of a placement test.

Comprehensive validation studies of placement tests should include multiple types of evidence. Sawyer (1996) describes one measure of placement test validity as the conditional probability of success for a given test score on the placement test (i.e., a valid test is one in which the conditional probability of success in a course increases around the cutoff score on the placement test). Methods suggested by Frisbie (1982) for evaluating a placement program incorporate, among other things, the extent to which students enroll in the recommended course. Frisbie also stated that the validity of placement tests is often explored through student and instructor opinions. That is, the validity of a placement test can be assessed, in part, by students believing they were placed correctly, as well as whether instructors thought students were adequately prepared and motivated (evidence of face validity).

Several factors can influence the relationship between the placement test and subsequent performance in a course including student motivation, the number of hours students work, and the number of hours per week students study. Factors at the instructor level include grading practices across instructors of the same course at a single institution and the quality of instruction. These factors can have the effect of decreasing the predictive validity associated with a placement test when the outcome of interest is success in a particular course.

ACCUPLACER

One of the most widely used placement tests in the United States is ACCUPLACER with more than 1,300 institutional users of the online testing system (College Board, 2009). The ACCUPLACER placement tests consist of six tests (Sentence Skills, Reading Comprehension, Arithmetic test, Elementary Algebra, College-Level Math [CLM] test, and Written Essay; College Board, 2003). The ACCUPLACER is an untimed computerized adaptive test administered online and is typically taken by students entering college. A test taker usually needs less than 75 minutes to take the entire test battery (College Board, 2003), in part because test takers are not required to take all tests within the ACCUPLACER test battery. The mathematics tests consist of multiple choice items: The Arithmetic test consists of 17 questions that measure ones' ability to perform basic arithmetic operations; the Elementary Algebra test consists of 12 questions that measure ones' ability to perform operations on algebraic expressions, solving linear equations, solving verbal problems using algebra, and so on; and the CLM test consists of 20 questions assessing intermediate algebra through precalculus. A calculator is provided through a pop-up window for mathematics items in which the use of a calculator would not supplant the knowledge necessary to answer the question (College Board, 2007). Scores for ACCUPLACER tests range from 0 to 120 and are

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computed directly from the Item Response Theory (IRT) models used to calibrate items and calculate ability estimates. ACCUPLACER is typically taken during the first year of college. A thorough explanation of the tests and their content as well as abilities being assessed can be found in the technical manual (College Board, 2003).

The 2003 ACCUPLACER online technical manual presents information on the validity of the three mathematics portions of the test. The correlations between scores on the Arithmetic test and student grades earned in college courses in general mathematics, arithmetic, elementary algebra, and intermediate algebra ranged between .31 and .38. The correlations between the Elementary Algebra test and student grades earned in college classes in elementary algebra, intermediate algebra, college algebra, precalculus, and calculus ranged between .19 and .38. The pattern changed somewhat for the CLM test, with reported correlations for grades in elementary algebra, intermediate algebra, college algebra, precalculus, and calculus between .32 and .49.

ACT Mathematics Test

The ACT mathematics test consists of 60 items. Unlike the ACCUPLACER, the ACT mathematics test is timed, and students are given 60 minutes to complete the test. Additionally, the ACT mathematics test is not a computerized adaptive test. The test contains multiple choice items assessing six content areas: prealgebra/elementary algebra (24 items), intermediate algebra/coordinate geometry (18 items), and plane geometry/trigonometry (18 items). The ACT mathematics test covers content areas considered to be prerequisites for success in entry-level college mathematics courses (ACT, 2007) and is typically taken by students during 11th or 12th grade. Calculator use is permitted during the ACT mathematics test.

The ACT technical manual (ACT, 2007) states that the median accuracy rate for the mathematics test (when used as a placement test) across institutions ranged from 69% to 75% (from intermediate algebra through calculus). This statement is supported by evidence that the ACT mathematics test can be useful in placing students. Ang and Noble (1993) examined the incremental validity of the ACT test above high school information (i.e., high school subject GPA, high school GPA, and number of high school courses taken in the subject) and found that, in general, the inclusion of ACT mathematics score in models predicting success in the first college mathematics course provided additional predictive power beyond a model including high school performance information alone. Their results were consistent with a previous study conducted by Noble (1991) on this topic.

The ACCUPLACER CLM test and the ACT mathematics test are both designed to assess proficiency in intermediate algebra through precalculus whereas the ACT mathematics test also covers geometry and trigonometry. Therefore, ACCUPLACER CLM test scores and ACT mathematics test scores should correlate as they overlap in both content and goals (i.e., assess proficiency in prerequisite mathematics skills necessary for success in entry-level college mathematics courses).

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However, little is known about the predictive validity of ACCUPLACER scores after taking into account collateral information (e.g., ACT scores). James (2006) used a sample of participants from a single postsecondary institution to examine the predictive validity of the ACCUPLACER online testing system on the grades of students in three courses (basic math, introductory algebra, and precalculus). The results indicated that there was a statistically significant relationship between the scores on two ACCUPLACER mathematics tests examined (Arithmetic and Elementary Algebra) and student mathematics GPAs computed for college mathematics courses. When scores from both tests were included as predictors in the regression equations for the basic math, introductory algebra, and precalculus courses, the model correlation coefficients (i.e., the square root of the R^2 values) were .52, .55, and .60, respectively.

The College Board released a meta-analysis summarizing evidence of the predictive validity of the ACCUPLACER scores for course placement (College Board, 2009). This meta-analysis used data from 47 placement validity studies that took place between 2001 and 2006 across 17 institutions. The outcomes of interest in the meta-analysis were correlations between ACCUPLACER placement tests and course success. In estimating the population correlation coefficients, the authors applied corrections for sampling error, range restriction, measurement reliability, unreliability of the predictor and unreliability of the criterion. The average correlations between the Arithmetic, Elementary Algebra, and CLM tests and success in college mathematics courses (C or better) were .37, .47, and .55 based on 13, 34, and 9 studies, respectively. The authors concluded that this provided evidence of the validity of ACCUPLACER for the purpose of placing students into the appropriate college mathematics course.

In sum, although there is evidence of the predictive validity of the mathematics portions of the ACCUPLACER placement test, there are still important gaps in our understanding. First, the grades students earn in college mathematics courses used in existing validity studies are subject to measurement error, and perhaps in significant amounts. Correlating the ACCUPLACER CLM test scores with ACT mathematics scores (known to possess comparatively less measurement error) would enhance our understanding of existing validity evidence for ACCUPLACER. That is, nonnegligible correlations between CLM test scores and ACT mathematics scores would enhance arguments of the validity of using the CLM test scores to place students rather than (or in addition to) ACT mathematics scores; low or near-zero correlations would suggest a quite different conclusion and would raise the question of what the CLM test is measuring.

Second, the added predictive value offered by the CLM test above and beyond that provided by a standardized achievement test such as the ACT mathematics test has not been studied. This is important because many postsecondary institutions currently (and, apparently, unofficially) use ACT or SAT mathematics scores to help place students (Merisotis & Phipps, 2000). Moreover, if the CLM test contributes little to college mathematics placement after considering the ACT mathematics test, institutions may want to consider using only the ACT mathematics test to place students into college-level mathematics courses.

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Third, it appears that most college mathematics placement tests are locally developed, often by a department of mathematics (Norman, 2008). Many institutions lack the resources to undertake a thorough test development process similar to that connected to the ACCUPLACER and ACT tests, and as a result may have psychometrically unsound placement instruments. In such instances, the use of the ACT mathematics test may prove to be an appropriate substitute for locally constructed placement tests.

Research Questions

Research Question1: Is the ACCUPLACER College-Level Math test a significant predictor of whether a student (a) initially took a developmental mathematics course and (b) passed that course (earned a grade of C or better) after controlling for the student's mathematics proficiency via ACT mathematics test scores?

Research Question 2: Among students who began their college mathematics course-taking with a nondevelopmental course, is the College-Level Math test a significant predictor of (a) students' initial mathematics course and (b) whether they passed that course (C or better) after controlling for the student's mathematics proficiency via ACT mathematics test scores?

We distinguished between developmental and nondevelopmental mathematics courses in studying the placement process for two reasons. First, the test developers of ACT and ACCUPLACER explicitly distinguish between developmental and nondevelopmental mathematics courses in describing the placement process. Second, the institutions included in our sample are part of a consortium whose governing board has established as policy the use of placement tools that aid in the proper placement of students into either developmental or college-level courses. This led us to separate developmental and nondevelopmental mathematics courses in studying the role of the ACCUPLACER test in the placement process.

The first research question assesses the predictive ability of the CLM test to identify students who completed a developmental mathematics course and the likelihood they passed that course, after taking into account students' ACT mathematics test scores. According to the College Board (2003), the CLM test assesses "proficiency in intermediate algebra through precalculus," which includes developmental coursework. The second research question examines the ability of the CLM test to identify students who did not begin college with a developmental mathematics course, and the likelihood they will pass that course, after taking into account students' ACT mathematics scores. It is important to emphasize that our inferences are comparative; that is, we do not attempt to assess the accuracy of the ACCUPLACER CLM test per se but rather compare the predictive abilities of the CLM and ACT mathematics tests. Similar evidence of prediction would suggest that either could be used as part of a placement decision or recommendation; evidence of different prediction would suggest that one test offers better predictive power.

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Method

Population and Sample

The sampled population consisted of college freshman that graduated from a Midwestern high school and subsequently attended a public college or university. The data used in this study came from the Minnesota Mathematics Achievement Project (MNMAP), a National Science Foundation (NSF)-funded program of research at the University of Minnesota examining various aspects of the relationship between high school mathematics curricula and college mathematics performance. MNMAP collected data on students that graduated from high school and immediately enrolled in a college or university in Fall 2002 or Fall 2003. A total of 28 colleges or universities provided ACCUPLACER placement test scores for a total of 6,471 students from approximately 300 high schools.

The variables of interest in this study were ACCUPLACER CLM test scores, ACT mathematics test scores, the difficulty level of a student's first college mathematics course, and the corresponding grade in that course. The sample was restricted to students who took at least one college mathematics course and had both an ACT mathematics score and a CLM test score, which lowered the sample size to 1,305 students from 20 postsecondary institutions (four 4-year institutions and sixteen 2-year institutions). Ninety-five percent of the sample was Caucasian, and 3% were Asian American. Approximately 57% of the sampled students were female.

Preliminary analyses revealed that the sample of 1,305 students was slightly less mathematically proficient than those excluded from our analyses (N = 5,166). Specifically, students included in the analyses tended to take fewer mathematics courses in high school, 3.43 (SD = 0.90) versus 3.68 (0.71), Cohen's d = .30, and had lower high school mathematics GPAs, 2.75 (0.77) versus 2.95 (0.65), d = .27. One explanation for this result based on anecdotal evidence is that many institutions consider a "large" ACT mathematics score as evidence of college-level mathematics preparedness in which case students can bypass the CLM test. In our sample students provided both an ACT mathematics score and a CLM score.

Variables

The dependent variables in this study are the difficulty level of students' first college mathematics course and the grade a student received in his or her first college mathematics course. Difficulty was originally measured on a 5-point Likert-type scale with the following levels:

Level 1: This level includes courses that should have been completed at the high school level (i.e., developmental);
Level 2: This level includes courses that would be considered precalculus mathematics and includes college algebra, finite mathematics, and coursework at a similar level.

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Level 3: This would be the typical entry level for mathematically well-prepared high school students, who would start their college mathematics coursework with beginning calculus.

Level 4: Calculus II. This would be the typical entry level for very well prepared high school students who have successfully completed a high school AP Calculus course, normally with a score of 4 or 5.

Level 5: This level would be considered an extension of one full year of calculus and would include course work required for a major (or minor) in mathematics. Course titles include differential equations, linear algebra, multivariable calculus, advanced calculus, and so on. Students at this level would normally be science, technology, engineering, or mathematics (STEM) majors.

These levels overlap with those used in Teitlebaum's (2003) analysis of the National Educational Longitudinal Study data in which high school mathematics courses were categorized. College mathematics courses taken by students in our sample were classified into one of these five categories through a process in which three MNMAP staff members (each with advanced degrees in mathematics education) examined the course titles and descriptions of each mathematics course and categorized them into one of the five difficulty levels. Minor differences were negotiated. This process ultimately resulted in complete agreement about the classification of each of the college mathematics courses represented in the sample.

The 5-point Likert-type scale reflecting the difficulty level of the mathematics courses was truncated into a variable with two levels (Level 1 vs. Levels 2-5) to reflect whether a student began with a developmental or nondevelopmental mathematics course. This was used as an outcome for Research Question 1. For Research Question 2, only those students that began with a course of difficulty Level 2 or higher were retained, and difficulty Level 2 was coded as 0, and Levels 3 and higher were coded as 1. This procedure reflected whether a student began with a course such as college algebra or finite mathematics rather than calculus I or higher.

Grade in students' first college mathematics course was measured on a scale from 4.0 to 0.0, and was dichotomized into 1 = *grade of C or higher* and 0 = *grade lower than a C*. The dichotomous grade variable served as another outcome of interest. The dichotomization of grade based on a cutoff of a C is consistent with previous placement validation studies (Ang & Noble, 1993; Latterell & Regal, 2003; Sawyer, 1996) and has the added advantage of decreasing the possible effect of instructor grading practices, a potential source of measurement error.

Other variables used in this study were students' ACT mathematics score and CLM test score on the ACCUPLACER placement test. Although we did not have the exact date on which either the ACT mathematics test or the CLM test was taken, as noted above the ACT test is typically taken during 11th or 12th grade of high school whereas the ACCUPLACER is typically taken during the first year of college. As such, the lag in time between having taken the ACT and enrollment in the first college mathematics course is longer than the lag between the CLM test and enrollment in the first college

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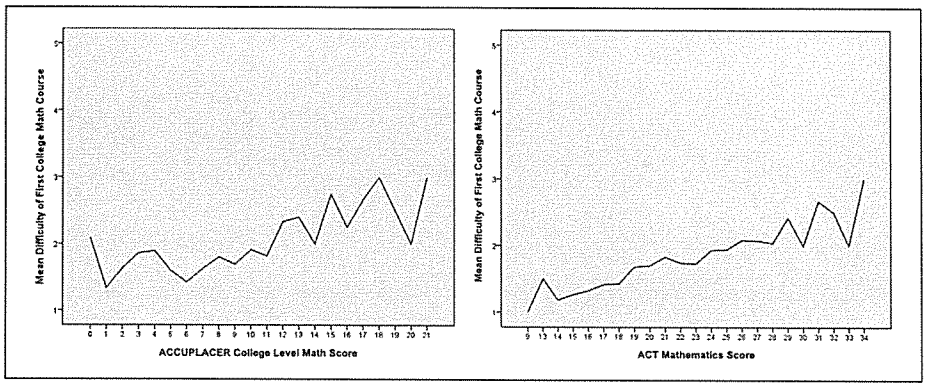

Figure 1. Average difficulty of first college mathematics course by CLM (College-Level Math) test score and ACT mathematics test score

mathematics course for all students in our sample. Whether this lag affects student performance and, if so, how much is unknown. To control for possible differences in the difficulty of, and the grade earned in, a student's first college mathematics course between different types of institutions, a variable reflecting the type of institution attended (1 = *4-year institution*, 0 = *2-year institution*) was included in all the inferential analyses.

Results

Descriptive Analyses

Figure 1 depicts the mean difficulty level (in its original 5-point scale) of students' first college mathematics course by CLM test and ACT mathematics test scores. For the purpose of illustration (for both Figure 1 and Figure 2), the CLM test score was rescaled so that original test scores (which ranged from 0 to 110) were grouped into intervals of 5 points (i.e., 106-110). This was done because many points on the original scale had very few students. However, for the analyses that follow, the original scale of the placement score was used.

Figure 1 illustrates that the difficulty of a student's first college mathematics course tends to increase, on average, as the corresponding CLM test and ACT mathematics test scores increase. This suggests that students are following the recommendation of the placement test, which in the case of the CLM test would be that a student should enroll in a college-level course (difficulty Level of 2 or higher). The ACT mathematics test shows a similar pattern. It is important to point out that prior to 2006, the institutions in our sample varied in the way they enforced placement. Some institutions required students to follow the course recommendation, whereas others did not.

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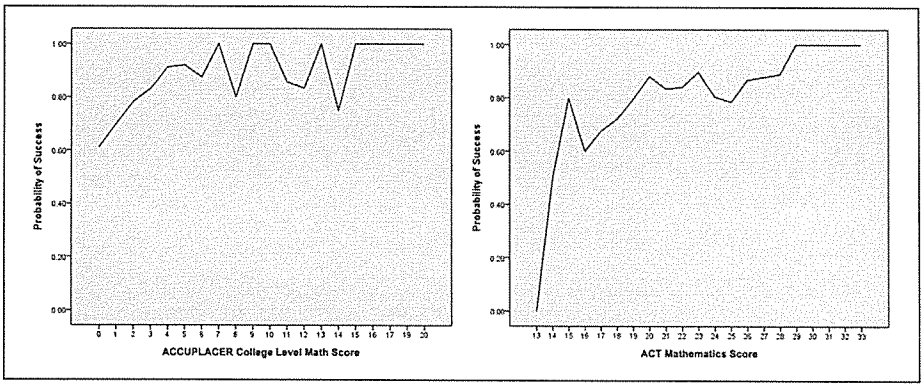

Figure 2. Proportion of successes (C or greater) for students that began in college algebra/ precalculus by CLM (College-Level Math) test score and ACT mathematics test score. The proportions of success for the ACT mathematics test scores of 13 and 14 are based on sample sizes of 1 and 2, respectively

However, since 2006, students at these institutions have been required to take the mathematics course into which they are placed.

Figure 2 depicts the proportion of students that scored C or higher for students that began college in a course of difficulty Level 2 (this is the lowest nondevelopmental/ college-level mathematics course in the coding scheme, i.e., precalculus, college algebra, etc.) by both the CLM and ACT mathematics test scores.

Figure 2 illustrates that, for the CLM test, as students' scores increased, the proportion of students receiving a grade of C or higher fluctuated somewhat but showed an upward trend as would be expected. The minimum probability of success as a function of the CLM test score was approximately .60, meaning that, regardless of the score on the CLM test, at least 60% of students were successful in a college algebra/precalculus course. The graph depicting the relationship between ACT mathematics test score and the proportion of students obtaining a grade of C or higher in a Level 2 college mathematics course also shows an upward (increasing) pattern. However, for ACT mathematics test scores, the proportions of success for scores of 13 and 14 are based on sample sizes of 1 and 2, respectively, which helps to explain the erratic behavior of the graph at the lower end of the continuum.

As a preliminary analysis, the mean test scores (ACT mathematics, and CLM) were compared for students who began with either a developmental (N = 401) or nondevelopmental course (N = 904). The mean ACT mathematics scores for these two groups were 19.45 (SD = 3.07) and 22.39 (3.37), respectively, while the mean CLM test scores for these two groups were 16.76 (9.74) and 20.39 (14.43), respectively. The corresponding Cohen's d values of .90 for the ACT mathematics test and .28 for the CLM test suggest that much larger differences were observed between the developmental and nondevelopmental students using the ACT mathematics score,

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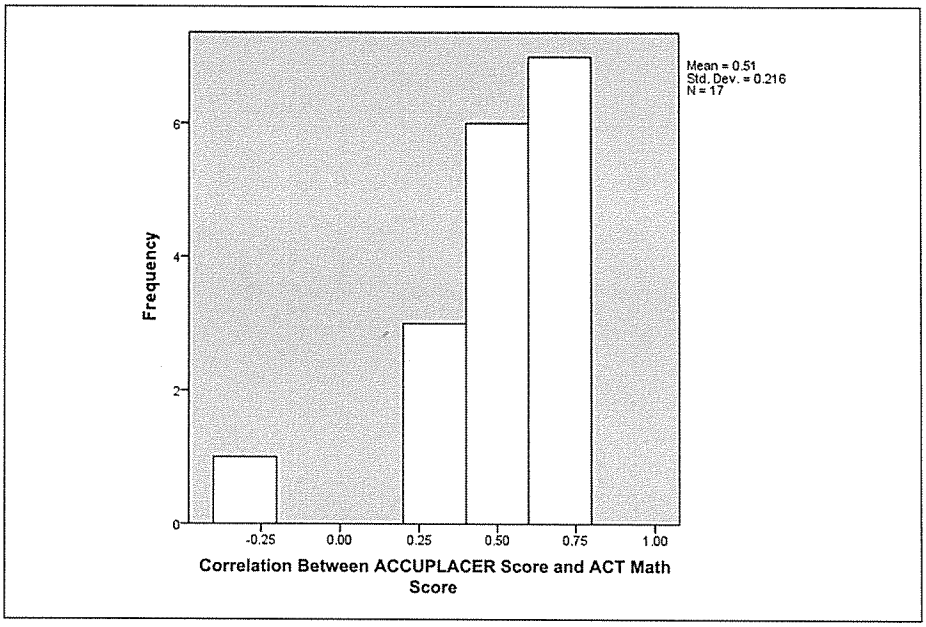

Figure 3. Histogram of correlations between ACCUPLACER CLM (College-Level Math) test score and ACT mathematics test score

compared with the CLM test score. To further examine the relationship between ACT mathematics test scores and CLM test scores, correlations of these two variables were generated by institution for those institutions that contributed at least 10 students (17 institutions). These correlations ranged from −.23 to .69 and are presented in a histogram in Figure 3. The weighted (by sample size) average correlation of .43 suggests some overlap in the information provided by these two test scores.

Inferential Analyses

Marginal logistic regression models using the generalized estimation equation (GEE; Liang & Zeger, 1986) were fit to our two outcome variables for Research Question 1: (a) student took a developmental mathematics course (0 = *college-level mathematics course*, 1 = *developmental mathematics course*) as their first college mathematics course, (b) student obtained a C or higher (0 = *less than C*, 1 = *C or higher*) in their first developmental mathematics course. The marginal model framework was used because inferences about the population were of interest. These models were fit using the geeglm function in geepack (Højsgaard, Halekoh, & Yan, 2005; Yan, 2002; Yan & Fine, 2004), a package used in the open source statistical software program R (The R Foundation for Statistical Computing, n.d.).

The ACT mathematics test score and then the CLM test score were entered into the model predicting whether a student began with a developmental mathematics course

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Table 1. Marginal Logistic Regression Results Predicting the First Developmental Course

Parameters	Fixed effect	Odds ratio	*SE*
Intercept	5.01***	149.90	0.70
Four-year institution	−1.28	0.28	0.87
ACT Math	−0.20***	0.81	0.03
CLM score	−0.04*	0.96	0.02

Note. Four-year institution 1 = *yes*, 0 = *no*; CLM score = ACCUPLACER college-level math test score. Odds ratio is obtained by exponentiating the fixed effect (log odds), for example, exp(−1.28) = 0.28.
*$p < .05$. ***$p < .001$.

Table 2. Marginal Logistic Regression Results Predicting Success in a Developmental Course

Parameters	Fixed effect	Odds ratio	*SE*
Intercept	−1.76*	0.17	0.76
Four-year institution	−0.28	0.76	0.87
ACT math	0.18***	1.20	0.05
CLM score	0.00	1.00	0.02

Note. Four-year institution 1 = *yes*, 0 = *no*; CLM score = ACCUPLACER college-level math test score. Odds ratio is obtained by exponentiating the fixed effect (log odds), for example, exp(−0.28) = 0.76.
*$p < .05$. ***$p < .001$.

(along with a predictor indicating whether a student's institution was a 2- or 4-year school). The results for the analysis of developmental course taking are presented in Table 1.

Table 1 shows that ACT mathematics test score was a significant predictor of whether a student began with a developmental mathematics course. This fixed effect estimate of −0.20 is in the anticipated direction, with students having higher ACT mathematics test scores less likely to start with a developmental mathematics course. More precisely, for each one unit increase in ACT mathematics test score students were exp(−0.20) = 0.82 times as likely to start with a developmental mathematics course. The CLM test of the ACCUPLACER was a significant predictor of this outcome after taking into account the ACT mathematics test score ($\beta = -0.04$, $p = .049$). The term reflecting the type of institution (4-year vs. 2-year) was not statistically significant for this or any of the inferential analyses and will not be discussed further.

The second part of Research Question 1 (Is the CLM test a significant predictor of passing a developmental course after controlling for ACT mathematics test score?) was answered by first restricting the sample to students that began with a developmental mathematics course and then using the dichotomous outcome of course success (0 = *less than a C*, 1 = *C or higher*). Results for this model are presented in Table 2.

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Table 3. Marginal Logistic Regression Results Predicting the First Nondevelopmental Course

Parameters	Fixed effect	Odds ratio	*SE*
Intercept	−9.02***	0.00	1.30
Four-year institution	−0.16	0.85	0.66
ACT math	0.25***	1.28	0.05
CLM score	0.02	1.02	0.01

Note. Four-year institution 1 = *yes*, 0 = *no*; CLM score = ACCUPLACER college-level math test score. Odds ratio is obtained by exponentiating the fixed effect (log odds), for example, exp(−0.16) = 0.85.
***$p < .001$.

Table 2 shows that ACT mathematics score was a significant predictor ($\beta = 0.18$, $p < .001$) in the anticipated direction, with students with higher ACT mathematics test scores predicted to be more successful in the developmental mathematics course. The CLM test score was not a statistically significant predictor above and beyond the ACT mathematics test score ($\beta = 0.00$, $p = .84$).

Next, Research Question 2 was answered by restricting the sample to students that began with a nondevelopmental course. Because very few students began with calculus II or higher (9 total), the difficulty variable was recoded such that those students who began with difficulty Level 2 (e.g., college algebra, finite mathematics, etc.) had difficulty recoded to 0, whereas students who began with difficulty Level 3 or higher (calculus I or higher) had difficulty recoded to 1. This served as the first outcome for Research Question 2. The results for this analysis are presented in Table 3.

For students that began with a nondevelopmental mathematics course, ACT mathematics test was a significant predictor of whether they began with calculus I or higher relative to courses such as college algebra or finite mathematics ($\beta = 0.25$, $p < .001$). The CLM test score was not a significant predictor of this outcome after taking into account the ACT mathematics test score ($\beta = 0.02$, $p = .06$).

The final outcome of interest was whether students were successful in their first nondevelopmental college mathematics course (earned a grade of C or higher). This model incorporated the dichotomous outcome variable of 0 = *college algebra/finite mathematics and so on,* 1 = *calculus I or higher* from Table 3 as a predictor in the model to account for students beginning with a more difficult course when modeling success. The results from this analysis are presented in Table 4.

The slope for difficulty level was nonsignificant ($\beta = -1.09$, $p = .11$), suggesting that the difficulty of a student's first nondevelopmental mathematics course does not predict whether they will be successful in that course (receive a grade of C or higher). This may be because of the relatively low criterion of C or higher for success. Although ACT mathematics test was a significant predictor of students' success in a nondevelopmental mathematics course ($\beta = 0.10$, $p < .001$), the CLM test was not ($\beta = 0.01$, $p = .24$).

To assess the sensitivity of these results to the use of the grade cutoff of C or higher, a cutoff of B or higher was also used (this distinction is only relevant for the models

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Table 4. Marginal Logistic Regression Results Predicting Success in a Nondevelopmental Course

Parameters	Fixed effect	Odds ratio	*SE*
Intercept	−0.44	0.64	0.54
Four-year institution	−0.58	0.56	0.48
Difficulty	−1.09	0.34	0.67
ACT math	0.09***	1.09	0.02
CLM score	0.01	1.01	0.01

Note. Four-year institution 1 = *yes*, 0 = *no*; Difficulty 0 = *college algebra/finite mathematics*, 1 = *calculus I or higher*; CLM score = ACCUPLACER college-level math test score. Odds ratio is obtained by exponentiating the fixed effect (log odds), for example, exp(−0.58) = 0.56.
*$p < .05$. ***$p < .001$.

predicting success). The results for the models using the B cutoff led to the same conclusions as those models using the C cutoff. Additionally, two correlation structures were assessed for the models fit above (exchangeable and independence). The results under both correlation structures were similar and lead to the same conclusions, suggesting that estimates from the models were stable.

To assess the sensitivity of our results to the inclusion of the CLM score after taking into account the ACT mathematics score, we reversed the order of entry of these predictors, putting the CLM score in the model first followed by the ACT mathematics score. For both success outcomes (success in a developmental or nondevelopmental course), the CLM score was not a significant predictor prior to the inclusion of the ACT mathematics score and remained nonsignificant after the inclusion of the ACT mathematics score. Although the CLM score was a significant predictor of beginning in a nondevelopmental course prior to the inclusion of the ACT mathematics score, CLM was no longer statistically significant after including the ACT mathematics score in the model. Thus, the order of entry of ACT mathematics scores and CLM scores did not alter the conclusions of this study.

A final sensitivity analysis involved the inclusion of a curriculum variable in the marginal logistic regression models as well as two interaction terms (between curriculum and ACT mathematics test score, between curriculum and CLM test score). The curriculum variable consisted of two dummy coded predictors (UCSMP [University of Chicago School Mathematics Program] 1 = *yes*, 0 = *no*; NSF 1 = *yes*, 0 = *no*) indicating whether a student had completed the UCSMP curriculum in high school or an NSF-funded mathematics curriculum in high school. The reference group was commercially developed (CD) curricula. CD curricula represent the most common high school mathematics curricula and are characterized by the separation of mathematics into algebra, geometry, algebra II, and for college intending students, precalculus. In contrast to CD curricula, NSF-funded curricula integrate these and other mathematics topics in each year. The UCSMP curriculum is often characterized as a hybrid of both CD and NSF-funded curricula. For a more thorough explanation of these curricula, see Harwell et al. (2009).

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The main effect for curriculum and the interaction terms were included to determine whether the relationship between the CLM test and the outcomes of interest differed by the curriculum a student completed. Results from these sensitivity analyses were similar to the results presented without these additional terms (Tables 1-4), suggesting that the effect of the CLM test did not differ by curriculum, and, thus, ruling out a potential confound. This provides evidence of the credibility of our findings.

Discussion

This study examined the performance of the CLM test score for placing students into mathematics courses when ACT mathematics test scores are available. The results suggest that the CLM test did not contribute to the prediction of whether a student took a developmental course as their first course above and beyond the ACT score. When the outcome of interest was the difficulty level of a nondevelopmental course (e.g., college algebra/finite mathematics vs. calculus I or higher), the ACCUPLACER CLM test did contribute slightly to the prediction of where a student began, with a one unit increase in the CLM test score associated with a decreased odds of beginning with a developmental course, $\exp(-0.04) = 0.96$. However, regardless of where a student began his or her mathematics course taking, the CLM test did not contribute to the prediction of success in those courses above and beyond the ACT mathematics test.

Our results have at least four overlapping implications. First, the correlations between CLM test and ACT mathematics test scores provide some evidence of the validity of the former, but the rather wide range in these correlations across institutions suggests that a single summary statement of validity of CLM test scores for placement decisions may be inappropriate. Of course, the small samples in some institutions may play a role in this variation.

Second, our results speak to the practice of using the ACT mathematics test as a de facto placement test. The evidence suggests that little (if any) predictive power is lost in relying on ACT mathematics test scores to place students in their first postsecondary mathematics course. Third, the similar predictive power of the CLM and ACT mathematics tests and the strong psychometric properties and relatively widespread availability of the ACT mathematics test suggests that this instrument is preferred to many locally developed college mathematics placement tests that have not undergone a rigorous construction and evaluation process.

Fourth, our ability to generalize our results across postsecondary institutions varying in size and educational profile enhances the applicability of our findings. This is important because the study of the performance of placement systems is often restricted to a single institution (Armstrong, 2000; Donovan & Wheland, 2008; Latterell & Regal, 2003; Marshal & Allen, 2000; Rueda & Sokolowski, 2004).

In short, our results suggest that the ACCUPLACER CLM test provides little additional prediction of the effectiveness of the placement process above and beyond the ACT mathematics test. In many cases, accurate placement of students into either a developmental or college-level mathematics course may be accomplished simply by

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using the ACT mathematics test score. This does not mean that locally constructed college mathematics placement tests should never be used; to the contrary, the additional information gleaned from the CLM test may be important when placement tests need to be tailored specifically toward the mathematics courses offered at a particular institution (MAA, 2010). However, such locally constructed tests would ordinarily not have strong validity or reliability evidence supporting their use.

Limitations

This study examined the effect of a specific college-level placement test in predicting the mathematics courses students began with as well as their success in those courses. There are other sources of placement test validity that were not considered in this study, such as student and instructor questionnaires, which gauge the extent to which students felt prepared for the course they were directed into. For students that did not enroll in a postsecondary institution immediately after graduation from high school, collateral information such as the ACT or high school mathematics performance may not be available. Such students were not part of this study.

Another limitation stems from the use of ACCUPLACER information by the institutions.

The data used in this study come from students who enrolled in a postsecondary institution in either Fall 2002 or Fall 2003. At that time, guidelines for the use of the ACCUPLACER outlining score thresholds for making determinations about placement were not in place but now are. As such, there was variability in the way in which the placement test was used across institutions, and we did not have data capturing this information.

Authors' Note

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

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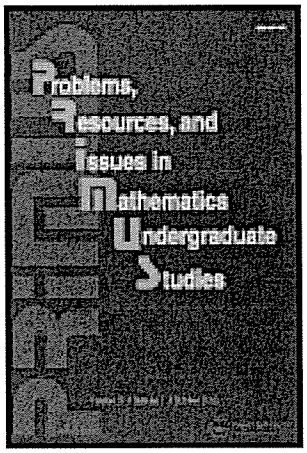

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High School Mathematics Curricula, University Mathematics Placement Recommendations, and Student University Mathematics Performance

Ke Wu Norman, Amanuel G. Medhanie, Michael R. Harwell, Edwin Anderson, and Thomas R. Post

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Abstract: Recent "math wars" have drawn attention to how well various high school mathematics curricula prepare students for college-level mathematics. The purpose of this study was to investigate the relationship between the high school mathematics curricula and students' post-secondary mathematics placement recommendation, specifically how students responded to the mathematics placement recommendations and the students' performance in the first college mathematics class.

The results showed no relationship between students' participation in a particular high school mathematics curriculum and mathematics placement recommendation, or between student high school mathematics curriculum and students' responses to a university mathematics placement recommendation. However, students who took a more/less difficult class than what was recommended achieved significantly lower/higher grades than those who followed the recommendation. The findings have implications for high school mathematics curricula selection, post-secondary student placement, and future research in this area.

Keywords: High school mathematics curricula, mathematics placement, post-secondary mathematics achievement.

1. INTRODUCTION

For most high school graduates in the United States, the bridge between high school and post-secondary mathematics coursework is the mathematics

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placement test. However, research on the relationships between high school mathematics curricula and

1. college mathematics recommendations,
2. student response to the recommendation, and
3. the impact of students' responses on performance in their initial university mathematics class,

has just started to appear in the literature.

Mathematics placement tests are used at many post-secondary institutions to assess students' readiness for mathematics, especially their preparation for calculus. Results of mathematics placement tests are used to guide decisions about the mathematics class students enroll in initially. Students who begin their post-secondary mathematics study with a course appropriate for their background increase the likelihood of succeeding in their initial class as well as subsequent mathematics courses [7]. On the other hand, misplacement of a student into an initial mathematics course can have negative effects on attitude, self-image, and college mathematics performance. College mathematics placement tests may also play a role in the enrollment of about one-quarter of all freshmen in at least one post-secondary remedial mathematics course that they should have completed in high school.

The common practice in universities and colleges is to gather information on students' academic history to help make decisions on admission and mathematics course placement recommendations. This information includes students' high school mathematics courses and grades as well as their mathematics scores from various aptitude exams such as the Scholastic Assessment Test (SAT) and/or the ACT. These factors usually serve as reliable predictors of university mathematics performance, and are used to place students into different levels of initial mathematics courses.

Many post-secondary institutions also use mathematics placement exams developed by the department of mathematics to provide course recommendations for incoming local or other freshmen (e.g., University of Wisconsin system, Iowa State University, University of Minnesota, Washington state post-secondary institution system, and the University of North Dakota). These universities are representative of large-sized research I universities in the United States. Available evidence suggests that mathematics placement exams used by colleges and universities generally focus on traditional topics such as algebra, trigonometry, and calculus. For instance, the placement exam used by the University of Wisconsin system contains three broad categories:

1. mathematics basics (e.g., arithmetic, intuitive geometry),
2. algebra (e.g., linear, quadratic, exponential, and logarithm functions, and geometry), and
3. trigonometry [57].

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Juxtaposed against the current use of college mathematics placement tests are the "math wars," which have pitted advocates of various high school mathematics curricula against one another. Fueling the math wars are concerns that students are not adequately prepared by their high school mathematics curricula for college mathematics coursework [4, 32, 45].

The two extreme sides of the "math wars" are: the defenders of traditional commercially developed (CD) mathematics curricula and advocates of National Science Foundation-funded (NSFF) curricula [50]. Since the 1990s many new NSFF curricula have been introduced into American K–12 school systems. These curricula were designed to meet the National Council of Teachers of Mathematics (NCTM) 1989 *Standards* [37]. In 2000, the NCTM published a revised document, *Principles and Standards for School Mathematics* [38], that expanded upon and updated the 1989 NCTM *Standards* document [37].

Compared with CD curricula, the NSFF curricula cover broader topics in mathematics including statistics, probability, and discrete mathematics [25, 51]. In general, CD mathematics curricula stress traditional algorithms and procedures, while NSFF curricula focus on developing student conceptual reasoning abilities and skills in modeling-based problems situated in real life contexts with a commensurate de-emphasis on procedural skills [25, 51].

Advocates of CD curricula have expressed their concerns that using NSFF curricula leads to poor college mathematics preparation [23, 59]. On the other hand, proponents of NSFF curricula often point to the disappointing mathematics achievement of all students, most of whom have completed a CD curriculum [22, 49]. They argue that changes in the development and implementation of curricula are needed. A number of studies of the impact of high school mathematics curricula on students' postsecondary mathematics achievement have appeared since 2000 [17, 18, 23, 41, 48, 54]. However, it is important to address the role that post-secondary mathematics placement tests play in the impact of high school mathematics curricula on college mathematics performance.

The purpose of this study is to investigate whether there is a relationship between the high school mathematics curriculum a student completes, and

1. their college mathematics placement recommendation,
2. students' responses to the college mathematics placement recommendation, and
3. students' performance in their first mathematics class as a function of whether they followed the placement recommendation.

2. LITERATURE REVIEW

The literature on post-secondary mathematics placement tests is fairly recent. Researchers have generally found low predictive validity represented by

modest to small correlations between placement test scores and the grade students earn in their initial mathematics course [9, 15, 27, 44].

Researchers have also considered other factors in examining placement recommendations. These include students' SAT or ACT math scores, high school mathematics grades, percentile rank in high school graduating class, gender, ethnicity, and age. A common practice is to generate models that produce an overall (predicted) score that is the basis of the placement recommendation [13, 27, 30, 40, 46, 47, 52, 59].

Other research has focused on various features of the placement recommendation process, such as the format used to administer the placement test. Some post-secondary institutions use paper-and-pencil tests, which delay the recommendation, while others use computerized adaptive tests that give placement suggestions immediately after the test is completed [1, 6, 53]. While there is substantial evidence in the psychometric literature that test format matters in a variety of ways [3, 29], there is no such evidence one way or the other in the mathematics literature.

Studies of the relationship between students' high school mathematics curriculum experience, and college mathematics study experience started to appear in the literature in the 1990s [10, 33, 54]. There are conflicting findings from different studies about how well NSFF students perform on college placement tests. Huntley, Rasmussen, Villarubi, Sangtong, and Fey [28] reported that students who completed three years of CORE-Plus (one of the NSFF curricula) did not do as well as their CD counterparts on symbolic manipulation items on the placement exam. In their study, students did not have access to technology (calculators) when they took these test items.

However, Schoen and Hirsch [48] found no significant differences on the algebra sections of a college placement test among students with four years of CORE-Plus, and individuals who completed four year of a CD curriculum. For the calculus readiness placement exam, Schoen and Hirsch did find significant differences favoring the CORE-Plus students. Students were allowed to use graphing calculators on both placement exams in Schoen and Hirsh's study. It appears that the use of calculators may play a role on CORE-Plus students' performance on the placement exam.

Importantly, there appears to be no available research on the relationship between high school mathematics curriculum and post-secondary mathematics placement recommendations, students' responses to the recommendation, or on the impact on performance, in their initial mathematics class, of students' responses. The results of this study should add to our understanding of the role of college mathematics placement tests as a moderating variable of the relationship between the high school mathematics curricula students complete, and the college mathematics courses students enroll in initially and their performance in those courses.

In this study, we posed the following three research questions:

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1. Is there a relationship between the high school mathematics curriculum students completed, and the university calculus placement test recommendation, with various student background factors held constant, and, if so, what is its nature and magnitude?
2. Is there a relationship between the high school mathematics curriculum students completed, and student enrollment patterns after taking the university calculus placement recommendation, and, if so, what is its nature and magnitude? (i.e., are students from a particular curriculum more likely to take a more difficult than recommended course?)
3. Is there a relationship between students' responses to the university calculus placement recommendations, and performance in their initial mathematics courses, with various students' background factors held constant, and, if so, what is its nature and magnitude?

Examination of (1) should shed light on whether the high school mathematics preparation of college-bound students is related to the level of mathematics classes in which they are advised to enroll initially. The results of Harwell et al. [18] suggest that students from NSFF curricula tended to start with less difficult mathematics classes than their CD and UCSMP counterparts. Answering our first question may help to explain why this happened. Our results will also provide evidence relevant to a fundamental criticism of NSFF curricula: that these curricula do an inadequate job of preparing students for success in post-secondary mathematics [31]. The reason that we chose placement course recommendations as the key variable, instead of placement test scores, is that the critique is really on the course recommendations, not the test score itself. In general, placement recommendations are made based on the position of resulting score relative to certain cut-off scores. It is possible that students with difference placement test scores are placed into the same mathematics courses based on the break scores.

The study of (2) will compare student responses to the placement recommendations as a function of high school mathematics curricula (i.e., do students follow the recommendation). We included this question because of anecdotal evidence that students who completed a NSFF curriculum in high school were less likely to go along with the placement recommendation than were students completing a CD curriculum in high school. If corroborated, this would suggest that there may be features of these high school mathematics curricula that are related to motivation and expectations students have of themselves in mathematics.

The study of (3) will provide evidence of the appropriateness of the placement recommendation with respect to students' performance in their initial mathematics classes. A finding that students who ignored a placement recommendation, and instead enrolled in a higher level course than recommended, had lower grades than those in the same course that followed the recommendation, thus providing support for the initial placement recommendation. On

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the other hand, a finding that students who chose to follow a recommendation to begin with calculus, and who subsequently performed poorly in this class, would provide evidence that the initial recommendations were somehow flawed.

3. METHODOLOGY

This study employed a retrospective quasi-experimental design for cross-sectional data.

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3.1. Population and Sampling

The target population consisted of university students in a Midwestern state. Students graduated from one of approximately 325 public high schools and completed at least three levels of high school mathematics in a traditional commercially developed (CD), National Science Foundation-funded (NSFF), or University of Chicago School Mathematics Project (UCSMP) curricula, up to a maximum of five levels, with the fifth level including coursework beyond first-year calculus. A high school level was defined to be the content equivalent of a year-long course taught one hour per day, and has been commonly referred to as a Carnegie unit.

The sample in this study consisted of 1,391 students from 100 high schools who enrolled at a single public research I university during the Fall 2002 or Fall 2003 terms, and took the calculus readiness placement exam and at least one mathematics class in their first year of university study. Students who withdrew from the courses were not included in this study. Fall 2002 was chosen as the first term to study post-secondary performance because it allowed significant numbers of students, who had completed at least three levels of a NSFF high school mathematics curriculum, to be sampled.

3.2. Data

Archival data were collected from both high school and university sources. Data on student high school transcripts were obtained from the university in an electronic form. Variables extracted from this information included: GPA in high school mathematics classes, levels of high school mathematics completed (3, 4, or 5), gender, ethnicity (Native American, African American, Asian, Hispanic, or White), and type of high school mathematics curriculum (CD, UCSMP, NSFF).

The university maintained electronic records of the high school courses taken by each applicant and the associated grades, but we quickly detected discrepancies in how particular mathematics courses were coded in university records and the correct descriptors. For example, some high schools

reported "Integrated 1, 2, 3, and 4" as their high school mathematics courses, information which the university then recorded. We use the descriptor "integrated" to denote detailed study of algebra, geometry, probability, and statistics each year. This is sometimes used interchangeably with "*Standards*-based," the term sometimes used to denote NSFF curricula. In some cases, the courses recorded as "integrated" were actually a traditional publisher's text series that had a subtitle with the word "integrated" in it. In other cases, districts using NSFF (CORE-Plus) curricula recorded students' course titles as Algebra I, Geometry, and Algebra II because of parental pressure.

These experiences led us to contact each of the 100 high schools to independently obtain descriptions of their mathematics programs—including the mathematics courses offered and the textbooks used—to allow accurate categorizations of students enrolling in the university as having completed a CD, UCSMP, or NSFF mathematics curriculum in high school. Contacts at the high schools were usually a mathematics curriculum coordinator, senior or master mathematics teacher, principal, or assistant principal. After categorizing high school mathematics curricula, four NSFF curricula were represented in our sample: Interactive Mathematics Program (IMP), Mathematics: Modeling Our World (MMOW), Contemporary Mathematics in Context (CORE-Plus), and Mathematics Connections. These curricula were not differentiated in the data analysis.

University variables collected included: when a student enrolled at the university (Fall 2002, Fall 2003), course or placement recommendation based on the calculus placement test score, ACT mathematics score, and name and grade of mathematics courses in the first year of university study. An important variable was the academic unit a student was admitted to, which we categorized as the following:

1. Science-, Technology-, Engineering-, and Mathematics-related colleges or departments (STEM),
2. Humanities (e.g., sociology, literature, political science), and
3. General College.

General College admits students who were academically not ready to be admitted by any other academic units at the university. These three academic units tend to have different mathematics course-taking requirements that need to be taken into account. For example, STEM students are generally required to take more, and more difficult, mathematics classes than students in Humanities. This implies differences among students in the kind of starting mathematics class recommended.

Another university variable was a student's mathematics grade, which was based on a 4-point grade scale—ranging from A (scale value of 4.0) to F (scale value of 0)—that was treated as showing a course interval scale of measurement. Under university rules, students have the grading option of

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receiving a Satisfactory/Unsatisfactory or, at the discretion of the instructor, an Incomplete, but these options appeared in only a few student records in our sample and were not examined.

The university calculus readiness exam was developed by the Department of Mathematics. The criterion-related validity coefficient of the exam (correlation between placement test scores and ACT mathematics scores) is 0.546, and the predictive validity coefficient of the exam (correlation between placement test scores and mathematics course grades earned) is 0.603[39], which are among the typical range of validity statistics of placement tests [9, 15, 27, 44, 59]. The test is offered online, and students are not allowed to use calculators.

Among the 30 multi-choice items on the university calculus placement exam, 23 (76.7%) test knowledge in Algebra/Functions while the remaining items test content in Geometry/ Trigonometry [39]. The category of Algebra/Functions includes knowledge in linear, exponential, power, polynomial, logarithmic, rational, and periodic functions. The category of Geometry/Trigonometry includes patterns with regard to shape, size, location; representational patterns with drawings, coordinates, or vectors; organization of geometric facts, and relationships through deductive reasoning.

There are two other placement exams offered by the university: an algebra readiness exam and the General College placement exam. In this study, we limited our focus to recommendations provided by the university placement test in calculus for two reasons. First, calculus and its immediate antecedents are viewed by many in the mathematics community as critical to post-secondary mathematics learning, understanding, and success, and by some as a key difference in the preparation of students from CD and NSFF high school mathematics curricula. Second, narrowing the focus to students who took the calculus readiness placement test imposes a homogeneity on the sample that should enhance inferences by removing groups of students with wildly different mathematics backgrounds (e.g., students who need significant remedial mathematics work, and those who are ready to begin their post-secondary mathematics studies with advanced classes such as Calculus II or III).

After taking the calculus readiness placement test, enrollment in one of the following is recommended to students:

1. Precalculus I (review of high school algebra),
2. Precalculus II (refresher course in preparation for calculus which includes trigonometric functions, DeMoivre's Theorem, and solutions of linear and nonlinear systems of equations),
3. Short Calculus (differential and integral calculus in one variable, and differential calculus in two variables), or
4. Calculus I.

On the basis of the descriptions of these courses we decided to group the recommendations into two levels: Precalculus I and II formed one level, and

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Short Calculus and Calculus I the other level. We treated Precalculus I and II recommendations as similar, and the Short Calculus and Calculus I recommendations as similar, but treated the differences in mathematics preparation between Precalculus (I, II) and Calculus (Short, Calculus I) as sufficient to detect the effects of interest.

There were three possible student responses to the calculus placement recommendation at the university:

1. followed the recommendation,
2. did not follow the recommendation and took a more difficult mathematics course, and
3. did not follow the recommendation and took a less difficult mathematics course.

University precalculus and calculus classes are offered in various academic units but are always taught by instructors in the Department of Mathematics. Variation across sections of precalculus and calculus classes was curtailed somewhat by the Department of Mathematics' policy of standardizing course syllabi, using common student assignments and examinations, and using a single course-specific distribution of scores for assigning grades.

3.3. Data Analyses

We began with various descriptive analyses to detect important patterns in the data. Then we addressed the first research question (Is there a relationship between high school mathematics curricula and university placement test recommendations?) using a binary logistic regression model in which high school mathematics curriculum was the key independent variable, and the calculus placement recommendation served as the dependent variable. To answer the second research question (Is there any relationship between high school mathematics curricula and student response to the university placement recommendation?) we used a multinomial logistic regression model in which high school mathematics curriculum was the independent variable of interest and student response to the recommendation the dependent variable.

The third research question (Is there any relationship between students' responses to the placement recommendations and their grades earned in the first university mathematics course?) was answered by using linear regression analyses. Here the key independent variable was student response to the calculus placement recommendation (followed; did not follow and took a more difficult mathematics course; did not follow and took a less difficult mathematics course) and grade in a student's first mathematics class, the dependent variable. All statistical hypotheses were tested using $\alpha = 0.05$.

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3.4. Limitations of the Study

Data in this study were collected from a single large Midwestern public university. Variation in how placement information is used by academic counselors is of particular concern since it is poorly understood. For example, it is unclear whether academic counselors in Humanities and STEM departments and programs make the same placement recommendations based on the same information, or whether the nature of the counseling is similar (i.e., supportive and interactive versus prescribed and abrupt). There is some anecdotal evidence that different units at this university (Humanities, STEM, and General College) handle placement decisions differently. One of the reasons we think this may matter is that the students are, generally, just out of high school and unfamiliar with post-secondary education and, in many cases, unsure about their mathematical preparation. This may make them more likely to accept the "official" placement recommendation even if they disagree with it.

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4. RESULTS

4.1. Descriptive Analyses

A total of 1,391 of the students in our sample took the calculus readiness placement test and received a course recommendation. Based on the course names recommended by the placement test compared with the actual courses students took, among those who took the calculus readiness placement test, 63.3% (N = 880) followed the placement recommendation, 11% did not follow the recommendation and took a less difficult mathematics course, while 25.7% did not follow the placement recommendation and took a more difficult mathematics class.

Next, we examined the relationships between when a student enrolled (Fall 2002, Fall 2003) and the high school variables ACT mathematics, high school mathematics GPA, ethnicity, and gender, which were either not significant or significant, but quite weak in magnitude, with less than 1% of the variance explained. Thus, students who enrolled in Fall 2002 (50.1%) or Fall 2003 (49.9%) appeared to have similar backgrounds, and were combined and treated as a single group for the remaining analyses.

4.2. Binary Logistic Regression Analysis

A binary logistic regression with calculus placement (precalculus, calculus) as the dependent variable was used to help answer the first research question: Is there a relationship between high school mathematics curricula (CD, UCSMP, NSF) and the mathematics course placement recommendation generated by

the university calculus placement exam. The variables included in the logistic regression equation were: gender, high school curriculum (UCSMP, CD, and NSF, with the latter as the reference group), academic units (STEM, Humanities, and General College, with the latter as the reference group), ACT mathematics scores, high school mathematics GPA, high school mathematics levels, and ethnic groups (African American, Asian, Hispanic, and White, with the latter as the reference group).

The significant results in Table 1 indicate that students with stronger prior mathematics background are more likely to be placed into university calculus. For example, with other variables being held constant, students who completed more high school mathematics levels were 1.64 times more likely to be recommended to start in calculus relative to precalculus, compared to students who completed fewer high school mathematics levels. Similarly, students with higher ACT mathematics scores and high school mathematics GPAs were

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Table 1. Binary Logistic Regression Predicting Calculus Placement Recommendation

Effect	B	SE B	Exp (B)
Intercept	−17.1	1.21	
Gender	−0.28	0.15	
UCSMP	0.47	0.27	
CD	0.26	0.23	
STEM	0.61	0.42	
Humanities	0.04	0.43	
ACT Math*	0.43	0.03	1.54
HS Math GPA*	0.92	0.18	2.51
HS Math levels*	0.50	0.11	1.64
African American	0.84	0.59	
Asian	0.43	0.22	
Hispanic	0.37	0.55	

Gender (0 = male, 1 = female), CD = Commercially Developed, UCSMP = University of Chicago School Mathematics Project, STEM = Academic units in science, technology, engineering or mathematics, HS Math levels = high school math levels completed, HS Math GPA = high school mathematics GPA, African American (0 = no, 1 = yes), Asian (0 = no, 1 = yes), Hispanic (0 = no, 1 = yes). B = Beta is the unstandardized coefficient, SE B = standard error of Beta, Exp(B) = exponentiated Beta.

*Chi-square tests were used for the tests of significance and the results were significant at $P < 0.05$. ACT Math, HS MATH GPA, and HS Math levels were significantly related to the likelihood of being recommended to a calculus course as opposed to a precalculus course. For instance, with other variables held constant, one unit higher on the ACT Math score was associated with a student being approximately one and one-half times (1.54) more likely to be placed into a calculus course as opposed to a precalculus course.

more likely (by factors of 1.54 and 2.51, respectively) to be recommended for calculus as opposed to precalculus, compared to students with lower ACT mathematics scores or high school mathematics GPAs. There was no significant difference between male and female students on the likelihood of being placed into university calculus as opposed to precalculus.

The key result in Table 1 is the absence of a significant relationship between high school mathematics curricula and the calculus placement recommendation (precalculus, calculus), after controlling all the other variables. This means that a student's high school mathematics curriculum was not related to his or her mathematics placement recommendation at the university. In other words, none of the high school mathematics curricula (CD, UCSMP, and NSFF) sufficiently or insufficiently prepared students for placement in university calculus courses.

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4.3. Multinomial Logistic Regression Analysis

To investigate whether there was a relationship between students' high school mathematics curricula and their responses to the placement recommendation (followed, did not follow and took a more difficult mathematics course, did not follow and took a less difficult mathematics course), with other factors held constant, we used a multinomial logistic regression model. Covariates in this model were the same as those in the binary logistic model with one exception. Hispanic students were omitted because only one Hispanic student took a mathematics course of lower than recommended difficulty. The dependent variable was student response to the recommendation.

The results presented in Table 2 indicate that female, non-STEM students, and students who took fewer high school mathematics classes are likely to take an initial mathematics classes of difficulty lower than that recommended by the calculus placement test. For instance, females were one and one-half times more likely ($B = 0.42$, $\text{Exp}(B) = 1.53$, $p = 0.04$) to take a lower difficulty than recommended course than male students.

The results for the model predicting the likelihood of a student taking a mathematics class of higher than recommended difficulty level, relative to following the recommendation, indicated that being in a Humanities program made the student a third as likely ($B = -1.25$, $p = 0.00$) to take a mathematics class of higher than recommended difficulty, compared to those not in Humanities. Students with a stronger prior mathematics background (higher average ACT mathematics scores, high school mathematics GPA, and high school mathematics levels) were more likely to take a mathematics class more difficult than that recommended by the calculus placement exam. There were no significant differences between male and female students, and among different ethnic and curricula groups.

Students' responses to the placement recommendation (follow, not follow and took a more difficult course, not follow and took a less difficult course)

Table 2. Multinomial Logistic Regression Predicting Response to Calculus Placement Test Recommendation

Effect	B	SE B	Exp (B)
Took lower			
Intercept	1.72	1.08	
Gender*	0.42	0.21	1.53
UCSMP	−0.44	0.36	
CD	−0.14	0.30	
STEM*	−1.37	0.42	0.25
Humanities	−0.60	0.42	
ACT Math	−0.01	0.03	
HS Math GPA	0.02	0.23	
HS Math Levels*	−0.55	0.15	0.58
African American	0.63	0.54	
Asian	−0.10	0.29	
Took higher			
Intercept*	−3.69	0.91	
Gender	−0.26	0.16	
UCSMP	−0.01	0.27	
CD	0.24	0.24	
STEM	−0.21	0.41	
Humanities*	−1.25	0.43	0.29
ACT Math*	0.06	0.02	1.06
HS Math GPA*	0.70	0.19	2.02
HS Math levels*	−0.25	0.11	0.78
African American	−0.34	0.60	
Asian	0.16	0.20	

Analysis does not include Hispanic students due to small n for analysis of log odds of taking a lower difficulty course. Gender (0 = male, 1 = female), CD = Commercially Developed, UCSMP = University of Chicago School Mathematics Project, STEM = Academic units in science, technology, engineering or mathematics (0 = non-STEM major, 1 = STEM major), HS Math levels = high school math levels completed, HS Math GPA = high school mathematics GPA, African American (0 = no, 1 = yes), Asian (0 = no, 1 = yes). B = Beta is the unstandardized coefficient, SE.B = standard error of Beta, Exp (B) = exponentiated Beta.

*Chi-square tests were used for the tests of significance and the results were significant at $P < 0.05$. Gender, STEM major, and HS Math levels were significantly related to the likelihood of a student who did not follow the course recommendation based on the placement test and took a course less difficult than recommended. For instance, with other variables held constant, female students were approximately one and one-half times (1.53) more likely to take less difficult courses than male students; and STEMmajored students were one-fourth (0.25)more likely to take less difficult courses than non-STEM students. Humanities, ACT Math, HS GPA, and HS Math levels were significantly related to the likelihood of a student who did not follow the course recommendation based on the placement test and took a course more difficult than recommended. For instance, Humanities-majored students were approximately one-third (0.29) more likely to take more difficult courses than non-Humanities students; and one unit higher on HS GPA was associated with twice (2.02) the likelihood of taking more difficult courses.

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may be related to students' motivation/expectation toward mathematics and their selfconfidence in mathematics. Table 2 shows that students who completed different high school mathematics curricula (CD, UCSMP, NSFF) did not respond to the placement recommendations differently, meaning that none of the curricula had more specific features that related to students' motivations and expectations of themselves in mathematics. However, male students and students with stronger prior mathematics knowledge were more likely not to follow the placement recommendations.

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4.4. Linear Regression Analysis

The third research question, "is there a relationship between students' responses to the university calculus placement recommendation (followed, did not follow and took a more difficult mathematics course, did not follow and took a less difficult mathematics course) and students' performance (grade) in their initial mathematics course, with various background factors held constant," was tested using linear regression. Covariates in this model include those in the logistic regression models, and the difficulty level of the initial university mathematics class a student took, as well as the student's response to placement recommendation, which was represented with dummy-coded predictors that used followed-the-placement recommendation as the reference group. The dependent variable was grade earned in a student's first university mathematics course. The results are presented in Table 3.

The key findings in this table are the significant slopes for students who took less difficult or more difficult mathematics classes than recommended. For the former, the slope of 0.28 indicates that, with other predictors held constant, students who took less difficult mathematics classes than recommended earned, on average, a grade that was 0.28 units (0–4 point scale) above that earned by students who followed the recommendation or took a more difficult mathematics class than recommended.

5. DISCUSSION

The result of no gender difference on mathematics placement (calculus, precalculus) is similar to the findings of some studies of gender differences on mathematics achievement [5, 35], but contradicts other studies which found that, in general, males outperformed females in mathematics [2, 19, 42]. This study showed that female students were more likely to take mathematics courses that were less difficult than those recommended by the placement test. This may be due to societal influences that lead to female students' negative attitudes toward mathematics and a lower self-confidence in subject [14, 16, 43, 56].

Table 3. Linear Regression Analysis for Grade in First University Mathematics Class

Effect	B	SE B
Intercept	−1.40	0.334
Gender	0.01	0.06
Difficulty	0.05	0.07
UCSMP	−0.09	0.10
CD	0.07	0.09
STEM	−0.07	0.14
Humanities	0.01	0.14
ACT Math*	0.06	0.01
HS Math GPA*	0.56	0.07
HS Math levels*	0.13	0.04
African American	0.51	0.19
Asian	0.08	0.08
Hispanic	0.12	0.21
Took Lower*	0.28	0.10
Took Higher*	−0.17	0.07

Model $R^2 = 0.15$, Gender (0 = male, 1 = female), CD = Commercially Developed, UCSMP = University of Chicago School Mathematics Project, STEM = Academic units in science, technology, engineering, or mathematics; HS Math levels = high school math levels completed, HS Math GPA = high school mathematics GPA, African American (0 = no, 1 = yes), Asian (0 = no, 1 = yes), Hispanic (0 = no, 1 = yes), Took Lower (1= took less difficult mathematics class than recommended, 0 = did not), Took Higher (1 = took more difficult mathematics class than recommended, 0 = did not). B = Beta is the unstandardized coefficient, SE B = standard error of Beta.

*Chi-square tests were used for the tests of significance and the results were significant at $P < 0.05$. For instance, with other variables held constant, one unit increase on the ACT Math score was associated, on average, with 56 units (on scale of 0–4 points) of higher grade in the first university mathematics class.

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Although many research studies have shown that African-American and Hispanic students are more likely to have lower standardized test scores than White students [11, 12, 34, 36, 55], it is important to note that the achievement gap has been closing over the past 25 years [8, 20, 21]. In this study, the results showed no significant differences on the mathematics placement recommendations among different ethnic (African-American, Asian, Hispanic, White) groups.

This study contributes to the literature in the area of connecting students' high school mathematics experiences and their initial college mathematics course placement, their responses to the course recommendation, and their

performance in the course. It was motivated by an ongoing and important discussion within the mathematics community on the ability of various high school mathematics curricula to prepare students to succeed in college mathematics coursework.

The main finding of the study is that students' high school mathematics curricula and their calculus readiness placement test recommendations (precalculus, calculus) were unrelated. This implies that, insofar as the university calculus readiness placement test is concerned, the high school mathematics programs completed by college-bound students are not related to the initial level of mathematics classes recommended to them. Our data suggest that students are not disadvantaged by any of the three high school mathematics curricula (CD, UCSMP, NSFF) studied with respect to university mathematics placement recommendation. Advocates of particular high school mathematics curricula are likely to be disappointed because these results do not provide clear evidence of an advantage for any one curriculum over another [4, 23, 25, 32, 45, 50, 51, 59, 60].

Along the same lines, these findings provide evidence that any advantage that CD students may have on the university placement test, or the disadvantage that NSFF students may have on the placement test because of content, appear to be minimized. That is, neither NSFF, CD, nor UCSMP students have any advantage with respect to the university placement recommendations. However, it remains an open question as to what course recommendation NSFF students would receive if the placement test, which emphasizes standard algorithms and repetitive procedures consistent with CD curricula, reflected a more balanced mixture of items illustrating an expanded domain of topics and more clearly paralleling NSFF curricula.

In other words, considering the disadvantage that NSFF students have due to less content alignment between NSFF curricula and the placement test, the lack of a difference in placement recommendations, compared to CD and UCSMP students, is important. Similarly, it is unclear how NSFF students would perform in university mathematics precalculus and calculus courses structured to be more consistent with an NSFF high school curriculum, as opposed to their current alignment with commercially developed high school curricula and instruction. The learning environment, pedagogy, and technology used in college mathematics courses may be very different from a typical high school classroom using a NSFF mathematics curriculum. For example, LaBerge, Sons, and Zollman [33] interviewed 30 mathematics faculty members from 11 universities and colleges in the Midwest, and found that 28 interviewees used lecture most of the time or always; 23 stated that they never or seldom asked students to write about mathematics in their "own language," and 19 faculty members mentioned that they rarely have students work in groups. This contrasts greatly with the teaching style recommended by the NSFF high school mathematics curricula [24, 58].

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Future work could include investigation of the following:

1. students' longitudinal mathematics achievement and course-taking patterns at college as a function of their high school mathematics curricula,
2. the way that various academic counselors at universities use placement information and the nature of the counseling process, and
3. identifying psychometric properties of placement tests.

These are all areas of possible study that would add significantly to our knowledge of college mathematics placement tests and their various ramifications.

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Ke Wu Norman is an assistant professor in the Department of Mathematical Sciences at the University of Montana. Her research focuses on teacher education and professional development, and student achievement in science, technology, engineering, and mathematics, at post-secondary institutions, as a function of their mathematics experiences in high school. She enjoys teaching content and methods courses to preservice teachers and professional development courses for in-service teachers.

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The Impact of Prior Mathematics Achievement on the Relationship Between High School Mathematics Curricula and Postsecondary Mathematics Performance, Course-Taking, and Persistence

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This retrospective study examined the impact of prior mathematics achievement on the relationship between high school mathematics curricula and student postsecondary mathematics performance. The sample ($N = 4,144$ from 266 high schools) was partitioned into 3 strata by ACT mathematics scores. Students completing 3 or more years of a commercially developed, University of Chicago School Mathematics Project, or National Science Foundation?funded curriculum comprised the sample. Of interest were comparisons of the difficulty level and grade in their initial and subsequent college mathematics courses, and the number of mathematics courses completed over 8 semesters of college work. In general, high school curriculum was not differentially related to the pattern of mathematics grades students that earned over time or to the difficulty levels of the students' mathematics course-taking patterns. There also was no relationship between high school curricula and the number of college mathematics courses completed.

Key words: College mathematics; Integrated curriculum; Large scale studies; Longitudinal studies; Quasi-experimental design

Ongoing discussions regarding the appropriate nature of high school mathematics curricula have not been adequately informed by longitudinal research that examines the effects of various curricula on students' performance in college mathematics, despite calls for such research (National Research Council [NRC], 2004). The fact

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that approximately 25% of all college students enroll in at least one college mathematics course that should have been completed in high school (Parsad & Lewis, 2003) illustrates that many entering students are not prepared for college-level mathematics. As a result, colleges devote considerable time, effort, and expense providing what is essentially remedial high school mathematics to many of their students. Nor is the lack of preparation confined to introductory mathematics courses. More than 2 decades ago, the Mathematical Association of America (MAA) surmised that less than one half of the students in the typical college calculus sequence were likely to complete that sequence (Kasten, 1988). Further, in some instances, only about one half of the students enrolled in a college algebra course in college are likely to complete this course successfully despite the fact that college algebra is essentially mathematics that a well-prepared student should have completed in high school (Hofacker, 2006).

The roots of these difficulties in college mathematics courses appear in high school. For example, the Nation's Report Card (Planty et al., 2008) indicated that only 23% of 12th-grade students performed at or above the proficient level. Each state has its own story, but in Minnesota, generally noted for the quality of its educational programs and the achievement levels of its students, only 18% of entrants to 2-year state colleges were deemed ready to enroll at the college algebra level, and only 38% of students entering a 4-year state university were judged ready to start at that level (Baer, 2001).

Although concerns about students' college mathematics performance have been raised for some time (National Commission on Excellence in Education, 1983), the release in the late 1990s and early 2000s of NSF-funded (NSFF) mathematics curricula at the elementary, middle grades, and high school levels generated concerns and criticism that these new curricula could potentially cause problems with mathematics performance nationwide (Klein, 2000; Wu, 1997). This seems unlikely for at least two reasons. First, as noted previously, these concerns were raised before substantial data were available concerning the effectiveness of curricula funded with NSF support (Schoenfeld, 2004). Second, our estimate is that nationally only about 5% of high school students have been or are enrolled in one of the NSF-funded secondary curricula, too few to account for the current problems in college mathematics.

Still, it is important to examine the relationship between high school curricula—including those supported by NSF—and students' college mathematics performance to inform the ongoing discourse about the nature of appropriate mathematics for college-intending high school students. This longitudinal study is designed to explore the impact of prior mathematics achievement on the relationship between high school mathematics curricula and college mathematics performance.

The NRC (2004) recently published criteria for evaluating the effectiveness of curricular material. The authors of that document suggested that:

> Clearly knowing how effective a particular curriculum is, and for whom and under what conditions it is effective, represents a valuable and irreplaceable source of information to the decision makers, whether they are classroom teachers, parents, district

curriculum specialists, school boards, state adoption boards, curriculum writers and evaluators, or national policy makers. (NRC, 2004, p. 1)

To this list we add college-level mathematicians and mathematics educators. The MSEB called for more comparative studies involving traditional and NSFF curricula to deepen our understanding of the impact of these curricula. Our study contributes to this effort.

The phrase *Commercially Developed curricula* (CD) is used in this study to denote those programs that embody contemporary (as of 2002 and 2003), yet traditional, textbook approaches to mathematics curricula. The University of Chicago School Mathematics Project curriculum (UCSMP) (UCSMP, 1996) is considered by the authors to be a hybrid program containing aspects of both the CD and NSFF curricula. UCSMP has received funding from private sources (AMOCO) as well as NSF support. UCSMP has evolved through several editions over the last several decades. The phrase *NSFF curricula* denotes curricula that were funded by the National Science Foundation, and in this study, NSFF curricula refers to three secondary programs: Contemporary Mathematics in Context (Core-Plus) (Coxford et al., 1997), Interactive Mathematics Project (IMP) (Fendel, Resek, Alper, & Fraser, 1997), and Mathematics Modeling Our World (MMOW) (Garfunkel, Godbold, & Pollak, 1999). These latter curricula are demonstrably different, and their approaches represent a departure from CD and UCSMP approaches.

THEORETICAL UNDERPINNINGS

Principles and Standards for School Mathematics (NCTM, 2000), a publication of the National Council of Teachers of Mathematics, was a renewed attempt to bring together the best practices from the descriptive and research literature in mathematics education accumulated over the past several decades. Classroom teachers, mathematics educators, administrators, researchers, and mathematicians were involved in the development of the document. In many ways, *Principles and Standards* was a theory-based document, although not all practices were directly linked to research literature. Connections to the research literature, however, were perceived (see Harwell et al., 2007).

A previous version, *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989), was used as the mandated template for the development of each of the proposals for the new curriculum projects to be funded by NSF in the early 1990s. The six Principles, five Content Standards, and the five Process Standards of *Principles and Standards* closely parallel the curriculum standards of the earlier *Curriculum and Evaluation Standards* document. Each attempted to reflect the theoretical underpinnings represented in the descriptive and research literature. To the extent that both are theory-based documents, the three NSF-funded curricula considered here are as well, because the original NSF mandate was that each be developed to adhere to these same sets of overarching principles.

The Relationship Between High School Mathematics Curricula and High School and College Mathematics Performance

Studies of the relationship between high school mathematics curricula and high school and college mathematics performance can be divided into two major areas, those concerned with school achievement patterns at the middle school and high school levels, and those concerned with college mathematics performance.

Many studies focused on the middle school and high school levels have been concerned with specific NSFF curricula. For example, in examining algebra topics found in the Core-Plus curricula, Huntley, Rasmussen, Villarubi, Sangtong, and Fey (2000) found that Core-Plus students lagged behind students using traditional curricula in algebraic fluency with algorithms and procedures, but surpassed those students in solving nonroutine problems and making sense of more complex mathematical situations.

Post et al. (2008) found that achievement levels of students enrolled in classes using middle-grades NSFF curricula (Connected Mathematics Project [Lappan, Fey, Fitzgerald, Friel, & Difanis Phillips, 1998] and Six Through Eight Mathematics Project—currently MathThematics program) on the Stanford 9 (three subtests: problem solving, procedures, and open ended, with results compared to national norms) and on the New Standards Test (three subtests: skills, concepts, and problem solving, with results also compared to national norms) were higher on the open-ended and problem-solving subtests than on the procedures subtests, a finding consistent with other attempts to evaluate achievement outcomes of students using NSFF curricula. In a similar vein, Harwell et al. (2007), using a purposeful sample chosen to represent a variety of demographics and school contexts, found that students enrolled in the IMP, MMOW or Core-Plus curricula, when examined on topics as contained in the Stanford 9 and the New Standards test, scored at or above the national mean on the various subtests, with suburban districts achieving at unusually high levels. These studies and other published research suggest that comparisons of student achievement levels for middle school and high school students enrolled in classes using NSFF curricula were similar: students using NSFF curricula lag somewhat in their ability to carry out routine procedures and standard algorithms but outperform students using traditional curricula on nonroutine tasks. In general, on the basis of relatively small, nonrandom samples, students using these NSFF curricula are performing well within the usual parameters of middle school and high school mathematics including standardized tests and localized investigations by a variety of researchers (Billstein & Williamson, 2003; Garfunkel, 2007; Ridgway, Zawojewski, Hoover, & Lambdin, 2003; Schoen & Hirsch, 2003; Webb, 2003).

Such findings however do not provide evidence about the performance of students using NSFF curricula outside of the high school context. Few studies have addressed the longitudinal connection between high school curricula and college mathematics performance, persistence, and course-taking patterns. In a cross-sectional study, Schoen, Hirsch, & Ziebarth (1998) examined data from students

from two similar high schools who later enrolled in the first calculus course at the University of Michigan. These authors found that students who completed an NSFF curriculum (Core-Plus) had higher overall grade point averages in that first course and in subsequent mathematics courses when compared with students who completed a CD program.

Harwell et al. (2009) compared students' first college mathematics class selection and achievement level for students who completed at least 3 years of a CD, UCSMP, or NSFF curriculum. A high school year was defined to be the content equivalent of a yearlong course taught 1 hour per day and has been commonly referred to as a Carnegie unit (Boyer, 1983). The major findings showed that (a) differences in mathematics achievement levels favoring students from CD curricula over those from NSFF curricula emerged in eighth grade and persisted into college; (b) students from NSFF curricula tended to enroll initially in less difficult college mathematics classes than those who completed a UCSMP or CD curriculum, but there was no difference in average grades earned in the first mathematics class when background factors (high school mathematics GPA, number of years of high school mathematics, high school percentile rank, ACT scores, etc.; see covariates listed in Tables 5–9) were taken into account; and (c) grades of students whose first college mathematics class was Calculus I were significantly higher for students from NSFF curricula than those who completed a CD curriculum. This mirrors the results of Schoen and Hirsch (2003), who also found that students from a particular NSFF curriculum had higher grades in college calculus courses.

Hill and Parker (2006) used a sample of more than 3,000 students in studying performance in mathematics course work at a large midwestern university, and reported that students who completed the Core-Plus mathematics curriculum in high school began their university mathematics course work with less difficult courses and subsequently completed even less difficult mathematics courses than students who did not complete Core-Plus in high school. Because the researchers did not have access to students' high school transcripts, the precise identification of the specific set of high school mathematics course materials completed by students in these schools was impossible to determine. Recognizing this situation, Hill and Parker state that "The lack of information from the Core-Plus schools made it impossible to ascertain precisely which students had completed the Core-Plus program and which had also taken non Core Plus mathematics courses such as AP calculus" (p. 918).

Previous research studies in this domain have provided a "patchwork" view usually focusing on a single dependent variable. Although useful, these efforts were not designed to provide a holistic perspective necessary to understand the bigger picture. The current study is a longitudinal effort covering all 4 years of college work and focuses on the impact of prior mathematics achievement on students' initial and subsequent mathematics course taking, persistence, and achievement. By partitioning students into comparable thirds on the basis of ACT scores, an important source of variation was reduced allowing a comparison of students with similar prior mathematical proficiency on each of the dependent variables of

interest. Earlier studies have not dealt with the longitudinal or the prior mathematical proficiency issues to the extent embodied here. This research therefore contributes new information and extends our understandings of other previous efforts to document the impact of NSFF curricula on various aspects of college-level mathematics performance.

COMPLICATING FACTORS IN STUDYING THE IMPACT OF PRIOR MATHEMATICS ACHIEVEMENT ON THE RELATIONSHIP BETWEEN HIGH SCHOOL MATHEMATICS CURRICULA AND COLLEGE MATHEMATICS

Steering More Mathematically Proficient Students to CD or UCSMP Curricula

Studying the impact of prior mathematics achievement on the relationship between high school mathematics curricula and college mathematics performance is complicated by several factors. One factor is preexisting differences in mathematics achievement between students. Harwell et al. (2009) reported differences exceeding one third of a standard deviation on the Minnesota statewide Basic Skills Mathematics Test administered in eighth grade between students who eventually completed an NSFF or CD curricula. These differences favoring CD students can be traced at least partly to high schools offering two mathematics curricula, approximately 14% of the schools and 25% of the sampled students in Harwell et al. (2009). Students who demonstrated greater mathematics achievement in eighth grade eventually self-selected into, or were disproportionately steered toward, CD or UCSMP curricula in high school. The exact nature of the steering process is unclear, but teachers, counselors, principals, and parents likely play key roles.

This steering process is important because it represents a selection bias that favors students who completed a CD or UCSMP curriculum as they move into college mathematics course work, and may initially disadvantage students who completed an NSFF curriculum. Because of the possibility that preexisting differences in mathematics achievement among students completing different curricula could persist into college mathematics, it is important to take these differences into account (see the data analysis that follows).

Mathematics Placement Test Considerations

Another factor complicating study of the impact of prior mathematics achievement on the relationship between high school mathematics curricula and college mathematics performance is college mathematics placement tests. Although there is variation across institutions, placement information is typically used by a student's academic advisor or an academic counselor in conjunction with other information, such as a high school transcript and ACT or SAT scores, to direct the student toward an appropriate mathematics course. Understanding this process is important in assessing the relationship between the high school mathematics curriculum that a student completed and the student's first college mathematics course.

Norman, Medhanie, Harwell, Post, and Andersen (in press) found evidence of a selection bias involving academic counselors who dismissed or misunderstood the mathematics achievement of students whose high school transcripts indicated that they completed an NSFF curriculum. Using qualitative methods to explore the attitudes of academic counselors toward, and understanding of, NSFF curricula, Norman and colleagues found that counselors generally were unfamiliar with NSFF curricula labels. For example, a student whose transcript indicated that their highest completed course was "Integrated Mathematics IV" (Core-Plus) was generally recommended to start with a precalculus/college algebra mathematics course, even though Core-Plus IV is a precalculus course and the student should have been advised to enroll in Calculus I. Norman and colleagues also found that academic counselors viewing a student's transcript reflecting a CD curriculum in which the highest mathematics course completed was listed as precalculus were more likely to recommend that a student initially enroll in Calculus I.

A related factor is the match between the mathematics content and assessment of that content linked to various high school mathematics curricula and college mathematics placement tests. Other things being equal, students completing a high school mathematics curriculum whose content and assessment format (e.g., multiple-choice items) match that of a college mathematics placement test would be expected to perform better on that test compared to students for whom this was not the case.

It is our view that the content and assessment format of college mathematics placement tests is more consistent with that of a CD curriculum than with the format of an NSFF curriculum. The University of Minnesota Twin Cities campus uses a Calculus Readiness Test and an Algebra Readiness Test to help place students. Our examination of these tests showed that approximately seven eighths of the items covered traditional algebraic manipulation and functions. Only a handful of the items were "word" problems reflecting real-world situations, and there were no items on probability, statistics, or discrete mathematics. On the whole, these tests, like the College Board's widely used Accuplacer test, are focused on traditional algebraic algorithmic skills as emphasized in traditional high school mathematics curricula. Unfortunately, we had no placement data or data relating to the academic counseling process that allowed us to examine the impact of these tests or processes on course-taking patterns. This is an important limitation of the study.

Fidelity of Implementation

A third important factor is fidelity of implementation (FOI). It has been noted that new curricula should be implemented as recommended by the curriculum developers to permit a credible assessment of their efficacy (Mowbray, Holter, Teague, & Bybee, 2003). However the literature on fidelity of implementation of educational curricula has not yet reached consensus on precisely what FOI is or how to assess it in a valid and cost-effective way (see, e.g., Ertle, Ginsburg, & Lewis, 2006; NRC, 2004; O'Donnell, 2008; Remillard, 2005; Reys, Reys, Tarr, & Chávez, 2006).

Evidence of FOI in this study comes largely from teachers who participated in professional development activities directly related to implementing the various NSFF curricula in the late 1990s. Initiated in 1997 with a Local Systemic Change grant from the National Science Foundation, the Minneapolis and St. Paul Merging to Achieve Standards Project $(MASP)^2$ was a teacher development program that involved 21 Minneapolis and St. Paul metropolitan area K–12 school districts. The program created a critical mass of over 1,100 teachers and an infrastructure for the implementation of NSFF curricula.

In 2001, there were over 74,000 middle grades and secondary students who were using these curricula daily as a direct result of interaction with this project. The teachers of these students each received at least 120 hours of professional development targeted directly to the NSFF middle-grades and high school curriculum that was being adopted by their district. In 2000–2001 there were 637 Core-Plus, 278 IMP, and 107 MMOW classrooms enrolling over 28,000 students representing the majority of the classrooms in the state that were using NSFF curricula.

Because of the retrospective nature of this study, we were not able to visit individual classrooms to ascertain the fidelity with which these curricula were implemented. However, through the Local Systemic Change project a large majority of the teachers whose students are represented in this study received relatively extensive targeted professional development that included a minimum of 20 hours of in-class support by a $(MASP)^2$ project mentor teacher. It therefore seems reasonable to posit a moderate degree of FOI of these three NSFF curricula during the period for which student data were collected. Here, moderate should be interpreted to mean that teachers understood the philosophy underlying the curricula and were inclined to make a realistic effort to implement the curriculum as the authors intended. Of course, one can expect significant variability. Such variability of implementation would not be restricted to NSFF curricula but would also be expected in UCSMP and CD classrooms.

Variation in FOI can relate to both content knowledge and pedagogical style. Classroom practices intended by the NSFF authors are well articulated in the NSFF curricula and, in general, conform to the *Curriculum and Evaluation Standards* (1989) and *Principles and Standards* (2000) documents. In addressing the issue of FOI at the middle grades in a 2-year study, Tarr et al. (2008) used a teacher-targeted instrument and identified five classroom events that the authors associated with "*Standards*-based instruction." Originally suggested by Romberg and Shafer (2003), these included: student opportunity for mathematical conjectures, focus on conceptual understanding, use of student verbal explanations, utilizing multiple mathematical perspectives, and teachers valuing student comments. These pedagogical "moves" were found to be more prevalent in classrooms using an NSFF curriculum. Tarr et al. reported that approximately 30% of teachers using NSFF curricula had low scores on the instrument used, suggesting variability among the sample of teachers. Because of the extensive amount of targeted professional development provided to most of the teachers in $(MASP)^2$, we speculate that the use of the Tarr et al. instrument with these teachers would produce less variability

than that found in the Tarr et al. study. Still, the difficulties in identifying, with precision, the degree of FOI make this an important limitation of this study.

RESEARCH QUESTION

It is apparent that NSFF program characteristics depart substantially from more traditional approaches found in CD curricula, as well as the UCSMP curriculum, but to a substantially lesser degree. Studying the relationship between these curricula and college mathematics performance, while taking into account prior mathematics achievement, will significantly add to our understanding of the ability of these curricula to prepare students for university mathematics.

The main research question was this:

> What is the nature and magnitude of differences, if any, in university mathematics achievement, course-taking patterns, and persistence in these classes among students with similar prior mathematical proficiency who completed at least 3 years of high school mathematics in commercially developed, the University of Chicago School Mathematics Project, or National Science Foundation–funded curricula?

METHOD

Research Design

A retrospective cohort design was employed in which students were categorized into one of five curriculum cohorts: CD, UCSMP, or NSFF, which included the IMP, MMOW, and Core-Plus curricula. Curriculum cohort was the independent variable of greatest interest. The importance of taking prior mathematics achievement into account and of minimizing the impact of selection bias linked to students completing a particular high school mathematics curriculum in schools that offered two curricula led us to (a) use ACT mathematics scores, known to provide accurate evidence of a student's achievement (ACT Inc., 2007) to estimate students' prior mathematics achievement; and (b) partition ACT mathematics scores into three strata (low: ≤ 22; middle: 23–26; and top: ≥ 27) to study the impact of high school curricula on college mathematics performance for students with similar prior mathematics achievement. These cutoffs generated an approximately equal number of students within strata. Analyses were performed separately for achievement, course-taking pattern, and persistence data collected for 8 semesters for each ACT stratum.

The construction of strata is consistent with the "elimination" strategy for data analysis suggested by Pedhazur and Pedhazur Schmelkin (1991), and it offers many methodological and substantive advantages. One such advantage is that it permits the identification of differences among students with comparable levels of prior achievement. When viewed in this manner, many whole-group differences are substantially modified, enabling a more accurate determination of the impact of various predictor variables on the dependent variable. This strategy helps to minimize the impact of preexisting differences in mathematics achievement in studying

the relationship between curriculum and college mathematics performance because each analysis involves students with similar prior mathematics achievement.

We created strata of students using their ACT mathematics score with the intent of analyzing the college mathematics performance and course-taking of students within each stratum separately. Doing so should strengthen inferences about the impact of high school mathematics curriculum because of the similarity of students' mathematics proficiency within each stratum. However, a concern in creating strata is that high school mathematics curriculum is confounded with ACT mathematics scores in the sense that the curriculum a student completed could be partly responsible for their ACT mathematics scores, and, by extension, their categorization into an ACT stratum (e.g., a student categorized into the lowest ACT stratum may be there, in part, because of his or her high school mathematics curriculum). In this case, inferences about the impact of curriculum on college mathematics and course taking would likely be biased because of the confounding of curriculum and ACT mathematics scores.

To respond to this possibility, we provide three pieces of evidence that curriculum and ACT mathematics scores are not confounded in a way that biases inferences about the impact of curriculum on college mathematics. First, there is evidence that mathematics proficiency is a fairly stable trait over time (e.g., Ding & Davison, 2005), suggesting that the mathematics achievement gap between students in NSFF curricula and those in CD or UCSMP curricula is unlikely to change dramatically over time and, thus, that curriculum effects and ACT mathematics scores are not likely to be seriously confounded. Our data support this, with a correlation between students' 8th-grade Basic Skills Mathematics Test (BSMT) scores and their ACT mathematics test scores from 11th grade of .65 for a sample of more than 2,400.

The BSMT scores provide information about mathematics achievement prior to students entering high school and, as such, cannot be confounded with curriculum. However, we chose ACT mathematics scores as our measure of prior mathematics achievement for two reasons. First, the BSMT is essentially a criterion referenced test with limited variation in scores (e.g., mean BSMT in our sample was approximately 94% correct out of 100%). The ACT, on the other hand, represents a state-of-the-art, norm-referenced psychometric tool for assessing mathematics achievement. Second, BSMT scores were unavailable for 59% of our sample, and students for whom these scores were available appeared to differ systematically from students for whom these scores were unavailable (see Harwell et al., 2009). Specifically, students in our sample with BSMT scores tended to have higher ACT mathematics scores and completed more levels of high school mathematics than those for whom BSMT scores were unavailable. This led us to use ACT mathematics scores as our primary measure of prior mathematics achievement in assessing the impact of high school mathematics curriculum on college mathematics, although we did use BSMT scores in a few exploratory analyses (see the following description of those analyses).

Second, the creation of strata should minimize the relationship between ACT mathematics scores and other indicators of mathematics achievement within strata.

Our data support this, with both descriptive and inferential analyses providing evidence of zero or near-zero relationships, with correlations of curriculum and ACT mathematics scores within strata of .143, .016, .042, respectively. Two of these correlations are statistically significant at the .05 level because of the large sample sizes ($N > 1{,}100$), but squaring the correlations suggests that the effects are negligible. A similar pattern of findings appears within strata for BSMT scores, high school mathematics GPA, and number of levels of high school mathematics completed.

Third, the greatest impact of a confounding of curriculum and ACT mathematics scores on inferences would occur if students were incorrectly classified in a stratum due to the impact of curriculum on ACT mathematics scores. Other things being equal, the likelihood of misclassification due to the impact of curriculum on ACT scores would increase with evidence that other student indicators of mathematics achievement are inconsistent with the stratum into which students are classified (e.g., if average BSMT scores and high school mathematics GPAs were similar for students in the middle and lower ACT strata).

To address the likelihood of misclassification, we conducted descriptive and inferential analyses that examined the differences between ACT strata on various indicators of mathematics achievement. These analyses treated the ACT strata as an independent variable and BSMT scores, high school mathematics GPA, and number of levels of high school mathematics completed as dependent variables.

The results of these analyses indicated that, on average, the ACT strata significantly and substantially differed from one another on all the dependent variables. Computing Cohen's *d* for each pair of ACT strata on the BSMT and high school mathematics GPA variables produced values ranging from 0.43 *SD* to 1.57 *SD* in favor of the higher stratum. The Kruskal-Wallis rank test was used to analyze the number of high school mathematics levels completed and was statistically significant, indicating that 17% of the variance in ranked levels completed was attributable to ACT strata. Higher ACT strata showed higher average ranks, a pattern consistent with the BSMT and high school mathematics GPA results.

Collectively, these results suggest that students categorized into a particular stratum based on their ACT mathematics score show a pattern of mathematics achievement consistent with that stratum, meaning that it is unlikely that curriculum affected ACT scores in a way that produced misclassification. Thus, separate analysis of the college-level data of each stratum allows us to obtain accurate estimates of the relationship between high school mathematics curriculum and college mathematics performance and course taking.

Populations and Sampling Mechanism

The target population consisted of university students in the upper midwest in the United States who had completed at least 3 years (Carnegie units) of high school mathematics in a CD, UCSMP, or NSFF curriculum and had completed at least one university mathematics course. Our sampled population, on the other hand,

consisted of over 4,000 students who enrolled at one large public university during the Fall 2002 or Fall 2003 terms, had graduated from one of 266 high schools in the state, and had completed at least one university mathematics course.

The median number of students from a single high school enrolled at the University was 6, with a maximum of 112, and several schools with 1 student (10.5% of the sampled high schools), 2 students (12.8%), or 3 students (9%). Without exception, high schools with 3 or fewer students enrolled at the University were located in rural areas of the state. We kept these students and high schools in our sample to enhance our generalizations.

The admission policies at this University include preference for students who are in the top quartile of their high school graduating class, and who have completed 3 years of high school mathematics. Of the 4,144 students in our sample, 125 (3%) did not take any credit-bearing collegiate mathematics class with a mathematics course designator. This level of course work is defined here as college algebra, finite mathematics, or precalculus. Thus, the majority of the 4,144 students have enrolled in mathematics course work at some point in their college career. Those 125 who did not enroll in any credit-bearing mathematics course included 87 (3.0%) from the CD curriculum,15 (2.3%) from UCSMP, and 23 (3.8%) from the three NSFF programs.

Because of the selective nature of the student body, findings must be tempered by the fact that students included in this study do not represent a random sample of the population of college-intending high school students. This is a limitation of the current study, because it does not support generalizing our findings to a larger population of students who enrolled in colleges and universities varying in mission, size, and selectivity. All student data collected were archival in nature and included information from each student's high school transcript and other university sources.

High School Variables

The major high school variable was the mathematics curriculum a student completed. In order to be included in this study, a student must have completed at least 3 years of CD, UCSMP, or one of the NSFF curricula. Once data collection began, we quickly detected discrepancies in the electronic files (courses taken and grades) that the University maintains on each of its students. Some high schools labeled students' courses as "integrated 1, 2, 3, and 4," information that was then recorded by the University. We use the descriptor *integrated* to refer to the study of topics in algebra, geometry, probability, and statistics each year. As noted earlier, integrated is sometimes used interchangeably with "*Standards*-based," a term normally used to denote NSFF curricula. In a significant number of cases, the courses recorded as integrated actually referred to a CD text series that had the word *integrated* in its subtitle.

To ensure an accurate matching of students to the appropriate high school mathematics curricula, we contacted each high school in the sample to obtain

descriptions of their mathematics programs and a listing of their courses and the texts used during Spring 2002 and Spring 2003, the two terms immediately prior to their students' enrollment at the University. By this means, and the subsequent inspection of each student's transcript for course title, it was possible to accurately match each student with that student's high school mathematics curriculum.

Although labor-intensive, this procedure validated the curriculum assignment process and ensured the integrity of the major independent variable in this study, the high school mathematics curriculum a student completed. Of the 266 sampled high schools, most (65.8%) offered a CD curriculum exclusively, 13.5% offered UCSMP, and 6% offered a single NSFF curriculum. A total of 37 high schools (14%) offered two curricula, with Core-Plus and CD being the most frequent combination (18 high schools, or 6.8%).

Other student high school variables included grade point average (GPA) in mathematics classes; percentile rank; number of years of mathematics completed (3, 4, or 5; AP Calculus or AP Statistics was coded as a 5th year); graduating class size, which served as a proxy for high school enrollment; and SAT or ACT mathematics score. SAT mathematics scores were converted to ACT mathematics scores following Dorans (1999).

University Variables

One of the University variables indicated the academic college to which a student was admitted (e.g., College of Liberal Arts, Institute of Technology, General College, etc.). This information was collected because most admitted students, by University policy, need to have graduated in the top 25% of their high school class to be eligible for admission to the academic college of their choice. Applicants not satisfying this requirement are sometimes admitted, usually to General College, in which they have the opportunity to strengthen their skills before transferring to another college within the University. Students admitted to General College typically take different courses in their first 2 semesters of study from those taken by students admitted to other colleges in the University, and this will impact findings since these students will generally enroll in lower level mathematics classes. The academic college to which a student was admitted was coded using a trichotomous variable representing major (STEM, Humanities, Other). Students in the "Other" major category were often those students admitted into General College.

For University mathematics classes, the initial sample consisted of 5,057 students who enrolled in the University in Fall 2002 or Fall 2003. Fall 2002 was chosen as the first term to study postsecondary performance because it was the 1st year that allowed for significant numbers of students to be sampled who had completed at least 3 years of the first edition (not draft versions) of an NSFF high school mathematics curriculum. Students who had not completed at least 3 years of high school mathematics ($N = 128$) or provided no information about the number of years of high school mathematics ($N = 300$) were eliminated from the sample. Students for whom no information about high school mathematics curriculum could be obtained

from their high school files ($N = 309$ students from 98 high schools) were also omitted, as were those who failed to identify their ethnicity ($N = 75$). Students who identified themselves as American Indian ($N = 36$) or who participated in the NSFF Math Connections high school mathematics curriculum ($N = 9$) were excluded from the analyses because of the small samples. Finally, $N = 56$ students were omitted who enrolled in a mathematics class but then withdrew. The resulting sample size was 4,144.

Because of the possibility that missing data might bias inferences (Little & Rubin, 2002), we compared available data for students omitted from the sample against those of students retained in the sample on variables such as ACT mathematics scores, high school mathematics GPA, and grade earned in the student's first university mathematics course. We found no significant differences in means or variances between students omitted from or retained in our sample, suggesting that our findings are not seriously biased by missing data.

The outcome variables were based on university records and reflected student achievement, course-taking patterns, and persistence. Achievement levels were captured by grades earned in mathematics classes over time. The University 4-point grade scale ranges from A (scale value of 4.0) to F (scale value of 0.0) and was treated as an interval measurement scale. Under University rules, students have the option of receiving a Satisfactory/Unsatisfactory or, at the discretion of the instructor, an Incomplete, but these options appeared in only a few student records in our sample, and those records were not examined.

Course-taking patterns were captured using a variable reflecting the difficulty levels of university mathematics classes and was treated as an ordinal measurement scale. This variable was constructed for this study and consisted of a 5-point Likert scale, which was measured at each semester for each student and had the following values:

Level 1: This level would include courses that should have been and typically are completed in high school (i.e., (a) Introductory Algebra, including rational and real numbers, inequalities. exponents, graphs, linear functions; (b) Geometry, including proof, properties of plane and solid figures; and (c) Intermediate Algebra, including quadratic, exponential, and logarithmic functions, rational expressions, complex numbers, and probability).

Level 2: This level includes courses that students would have completed if they had satisfactorily completed 4 years of college preparatory mathematics in high school. College credit may be offered for these courses when taken at the college level. Courses at this level could be categorized in general as trigonometry, college algebra, discrete mathematics, and/or precalculus mathematics.

Level 3: Calculus I. This would be the typical entry level for well-prepared high school students who would begin postsecondary mathematics course work with beginning calculus.

Level 4: Calculus II. This would be the typical entry level for very well-prepared high school students who have successfully completed an AP Calculus course, normally with a score of 4 or 5.

Level 5: This level would be considered an extension of 1 full year of calculus and would include course work required for a major (or minor) in mathematics. Course titles include differential equations, linear algebra, multivariable calculus, advanced calculus, etc. Students at this level would normally be science, technology, engineering, or mathematics (STEM) majors.

In almost all cases, students enrolled for only one mathematics class per semester, and for those few cases in which a student enrolled for two mathematics classes in the same semester, the data for the more advanced class were used.

Persistence was assessed by examining the total number of mathematics classes a student completed during 8 semesters of study. To be included in the sample, students had to have completed at least one University mathematics course over 8 semesters.

Data Analyses

For the inferential analyses, two-level hierarchical linear models (HLM) (Raudenbush & Bryk, 2002) and Poisson regression models were used. Two sets of two-level HLMs were used for analyzing the grade and difficulty data. The first level consisted of either the grade or difficulty data collected over 8 semesters that were treated as repeated measures nested within students. This allowed a student's University grade performance and difficulty levels over time to be studied.

The second level of the HLM was between students and allowed explanatory models to be constructed to explain variation in student grade and difficulty level trajectories over time. For example, if student linear slopes (trajectories) capturing the pattern of their grades in mathematics classes over time varied across students, explanatory models were constructed to model this variation. To control for the compounding of Type I error rates, an adjusted Type I error rate attributed to Šidák (1967) was used. The adjusted error rate for the HLMs was computed as $\alpha' = 1 - (1\ \alpha)^{1/k}$, where α is the unadjusted Type I error rate, k = number of statistical tests, and α' is the adjusted Type I error rate. We chose an overall $\alpha = .15$ for each set of HLM analyses, which resembles a familywise error rate strategy often associated with planned comparisons in ANOVA (actual error rates appear in the HLM tables).

The role of high school curriculum on persistence in college mathematics courses was studied using cross-sectional data in which a Poisson regression model was fitted with number of college mathematics courses taken as the dependent variable. All HLM analyses were performed using the mixed-model procedure in SAS version 9.1.

RESULTS

Descriptive Analyses

Among the sample of 4,144 students who completed at least one University mathematics course, 2,124 (51.3%) enrolled at the University in Fall 2002 and 2,020 in Fall 2003. A total of 19.6% of the sample completed 3 years of high school mathematics, 45.5% completed 4 years, and 34.9% completed a 5th year, usually AP Calculus or AP Statistics.

The 2002 and 2003 groups, when contrasted in their entirety, differed slightly in average high school percentile rank, years of high school mathematics completed, ACT mathematics score, and size of the graduating high school class, with the 2003 group tending to come from high schools with larger graduating classes. However, less than 1% of the variance was attributable to when a student initially enrolled at the University, suggesting that the impact of students' year of initial enrollment was minimal. Nevertheless, year of enrollment was included as a covariate in subsequent analyses. For the descriptive analyses described subsequently, the Fall 2002 and Fall 2003 data were combined.

Most of the sampled students completed a CD curriculum (69.4%), with UCSMP next most frequent (16%); most were White (74.4%) or Asian American (17.1%). There were (proportionally) more African American students completing an NSFF curriculum compared to White students (24.3% and 12.9%, respectively). There also was evidence that students from each curriculum cohort were disproportionately represented in the ACT mathematics strata. For example, the rounded percentages of the CD, UCSMP, IMP, MMOW, and Core-Plus cohorts that were in the lowest ACT stratum were 32, 23, 88, 45, and 43, respectively; for the middle stratum 28, 30, 11, 33, and 33, respectively; and for the high ACT stratum 40, 47, 1, 21, and 23, respectively. This is consistent with our previous finding (Harwell et al., 2007) that, based on students' 8th-grade mathematics scores on a basic skills test, a larger proportion of lower achieving students were directed into NSFF curricula. These previously identified differences appear to have persisted throughout high school.

Across the five high school curricula, there appears to be parity in the mathematics achievement within each stratum (see Table 1). For example, the average high school mathematics GPA for the CD, UCSMP, IMP, MMOW, and Core-Plus cohorts in the lowest stratum were 3.0, 3.1, 2.9, 3.1, and 3.0, respectively. A similar pattern emerged in the middle and high strata.

The highest attained difficulty level of college mathematics course work was compared within each of the prior mathematics achievement strata across the five types of curriculum (see Table 2 on pp. 292–293). Entries are listed by the number of college mathematics courses taken.

The data suggest comparable college mathematics achievement levels for students, regardless of the high school curricula completed. This is true for all three strata. It should be noted that the small sample sizes of IMP and MMOW students preclude making meaningful inferences about these two high school curricula, and

Table 1
Descriptive Statistics on Achievement by Curricula and ACT Strata

		Curriculum																	
		CD			UCSMP			IMP			MMOW			Core-Plus			NSFF		
		M	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*
Lowest 1/3 on ACT math (<=22)	ACT math score	19.0	2.4	903	19.5	2.2	149	17.1	2.2	64	18.6	2.3	34	18.7	2.4	196	18.3	2.4	294
	HS math GPA	3.0	.7	836	3.1	.7	138	2.9	.8	59	3.1	.7	29	3.0	.8	181	3.0	.8	269
	No. of college math courses completed	2.0	1.3	903	2.0	1.4	149	1.8	1.4	64	2.3	1.8	34	2.0	1.2	196	2.0	1.3	294
	No. of years of HS math	3.8	.7	903	3.9	.4	149	3.7	.7	64	3.9	.6	34	3.7	.7	196	3.7	.7	294
Middle 1/3 on ACT math (23–26)	ACT math score	24.6	1.1	793	24.6	1.1	200	24.1	1.0	8	24.6	1.1	25	24.6	1.1	151	24.6	1.1	184
	HS math GPA	3.4	.6	721	3.4	.5	189	3.5	.5	8	3.5	.5	21	3.5	.5	141	3.5	.5	170
	No. of college math courses completed	2.0	1.3	793	2.3	1.5	200	1.8	1.8	8	2.8	2.0	25	2.2	1.5	151	2.3	1.6	184
	No. of years of HS math	4.2	.7	793	4.1	.5	200	3.9	.8	8	4.2	.6	25	4.1	.7	151	4.1	.7	184
Top 1/3 on ACT math (>=27)	ACT math score	29.6	2.3	1148	29.6	2.2	312	27.0	.	1	29.3	1.8	16	29.3	2.1	105	29.3	2.1	122
	HS math GPA	3.7	.4	1071	3.6	.4	287	3.0	.	1	3.6	.4	14	3.7	.4	94	3.7	.4	109
	No. of college math courses completed	2.5	1.5	1148	2.4	1.5	312	3.0	.	1	2.6	1.6	16	2.6	1.3	105	2.6	1.4	122
	No. of years of HS math	4.5	.6	1148	4.4	.6	312	4.0	.	1	4.4	.5	16	4.6	.6	105	4.6	.6	122

Note. NSFF = IMP, MMOW, & Core-Plus, *M* = Mean, *SD* = Standard Deviation, *N* = Sample Size

henceforth the Core-Plus, IMP, and MMOW cohorts are combined and referred to as the NSFF cohort.

Table 2 indicates that among students in the top stratum that took 2 college mathematics courses, approximately 70% of all NSFF students, 68% of CD students, and 72% of UCSMP students successfully completed course work at the Calculus II level or beyond. Similar results appear for students who took 3, 4, or more mathematics courses and for students in the lower and the middle strata.

As would be expected, the strongest differences occur between strata, with students in the middle and top strata generally more successful in attaining the level of Calculus II or beyond Calculus II in their college mathematics course taking. A surprising feature of Table 2 is the large percentage of students in the top stratum who completed only 2 college mathematics courses (noted previously) but who completed those courses at Level 4 or Level 5. These students are likely to have had an accelerated and/or a very substantial high school mathematics experience and who began their college career with Calculus I, Calculus II, or beyond. Tables 3 and 4 depict, by curriculum, the difficulty level of students' initial and last university mathematics class, respectively.

As noted previously, Harwell et al. (2009) found significant differences favoring the CD curricula in the difficulty level of the mathematics course in which students initially enrolled. Tables 3 and 4 (pp. 294–295) suggest that, after this initial course experience, differences between curricula in the difficulty level of subsequent (highest) mathematics courses essentially disappear when viewed by ACT strata. For example, in Table 4 the percentages of students who completed a CD, UCSMP, or NSFF curriculum who completed courses at the Calculus II level and beyond are approximately 12%, 13%, and 9%, respectively, for students in the lowest stratum; 27%, 33%, and 33% for the middle stratum ; and 63%, 63%, and 64% for the highest stratum.

When we further examined by semester and curriculum cohort the average difficulty levels of the mathematics courses in which students enrolled, small but significant differences favoring students in the CD and UCSMP curricula were found for the low ACT stratum in the difficulty level of their initial college mathematics course. These curriculum cohort differences did not appear for the middle and high strata. However, the middle and high ACT strata began and ended at higher levels of course difficulty, with the middle stratum generally beginning with Level 2 course work (college algebra) and the high stratum beginning with Level 3, the first course in calculus. As the semesters progressed, the difficulty levels predictably increased for all strata through semester 6, when no discernable pattern became evident for the very small numbers of mathematics students remaining in semesters 7 and 8. For all three curriculum cohorts and for each stratum the percent of students enrolled in a mathematics course in semester 1 was between 35% and 40%. For semester 2, about 25% of the students in each curricular cohort enrolled in a mathematics class regardless of ACT stratum. For semester 3, the percentage decreased another 10% to about 15%, again regardless of curricular cohort or stratum. By

Table 2
Highest Difficulty Level Attained and Number of Mathematics Courses Taken by ACT Strata

		ACT Mathematics Strata											
		Low (<=22)				Middle (23–26)				High (>=27)			
No. of college math courses completed	Curriculum	Highest difficulty level of math course taken				Highest difficulty level of math course taken				Highest difficulty level of math course taken			
		1–3		4 or 5		1–3		4 or 5		1–3		4 or 5	
		N	Row %	*N*	Row %	*N*	Row %	*N*	Row %	*N*	Row %	*N*	Row %
1	CD	372	98.7	5	1.3	339	96.3	13	3.7	299	80.6	72	19.4
	UCSMP	65	97.0	2	3.0	66	97.1	2	2.9	82	77.4	24	22.6
	IMP	30	100.0	0	.0	4	100.0	0	.0	0	.0	0	.0
	MMOW	14	100.0	0	.0	8	88.9	1	11.1	3	75.0	1	25.0
	Core-Plus	74	100.0	0	.0	61	100.0	0	.0	25	100.0	0	.0
	NSFF	118	100.0	0	.0	73	98.7	1	1.4	28	96.6	1	3.5
2	CD	223	94.1	14	5.9	156	79.6	40	20.4	86	32.0	183	68.0
	UCSMP	33	94.3	2	5.7	40	67.8	19	32.2	23	28.4	58	71.6
	IMP	16	100.0	0	.0	1	50.0	1	50.0	0	.0	0	.0
	MMOW	8	100.0	0	.0	3	60.0	2	40.0	2	66.7	1	33.3
	Core-Plus	59	98.3	1	1.7	26	76.5	8	23.5	6	25.0	18	75.0
	NSFF	83	98.8	1	1.2	30	73.2	11	26.8	8	29.6	19	70.4

Table 2 (continued)
Highest Difficulty Level Attained and Number of Mathematics Courses Taken by ACT Strata

		N	Row %	*N*	Row %	*N*	Row %	*N*	Row %	*N*	Row %	*N*	Row %
3	CD	101	79.5	26	20.5	50	46.7	57	53.3	19	11.5	147	88.6
	UCSMP	15	75.0	5	25.0	9	40.9	13	59.1	5	11.4	39	88.6
	IMP	10	100.0	0	.0	0	.0	0	.0	1	100.0	0	.0
	MMOW	4	80.0	1	20.0	2	50.0	2	50.0	0	.0	4	100.0
	Core-Plus	25	75.8	8	24.2	13	56.5	10	43.5	3	12.5	21	87.5
	NSFF	39	81.3	9	18.8	15	55.6	12	44.4	4	13.8	25	86.2
4 or more	CD	59	48.4	63	51.6	14	12.4	99	87.6	6	1.9	313	98.1
	UCSMP	9	42.9	12	57.1	15	31.9	32	68.1	1	1.3	75	98.7
	IMP	2	50.0	2	50.0	0	.0	1	100.0	0	.0	0	.0
	MMOW	2	33.3	4	66.7	0	.0	7	100.0	1	25.0	3	75.0
	Core-Plus	11	52.4	10	47.6	1	3.6	27	96.4	1	3.5	28	96.6
	NSFF	15	48.4	16	51.6	1	2.8	35	97.2	2	6.1	31	93.9

Note. NSFF = IMP, MMOW, & Core-Plus, N = Sample Size. 1–3 = Course work through Calculus 1, 4–5 = Calculus 2 and beyond

Table 3
Difficulty Level of Initial College Mathematics Class by Curricula and ACT Strata

		CD		UCSMP		NSFF	
	Level	*N*	Col %	*N*	Col %	*N*	Col %
Low 1/3 on ACT mathematics (<=22)	1	319	37.0	41	28.5	127	45.2
	2	502	58.2	98	68.1	149	53.0
	3	34	3.9	3	2.1	5	1.8
	4	5	.6	1	.7	0	.0
	5	3	.4	1	.7	0	.0
Middle 1/3 on ACT mathematics (23–26)	1	46	6.0	13	6.6	13	7.3
	2	518	67.5	123	62.8	127	71.4
	3	174	22.7	54	27.6	33	18.5
	4	28	3.7	5	2.6	5	2.8
	5	2	.3	1	.5	0	.0
High 1/3 on ACT mathematics (>=27)	1	10	.9	0	.0	0	.0
	2	308	27.4	81	26.4	39	32.8
	3	500	44.4	141	45.9	55	46.2
	4	190	16.9	49	16.0	18	15.1
	5	118	10.5	36	11.7	7	5.9

Note. NSFF = IMP, MMOW, & Core-Plus, N = Sample Size, Col % = Column percentage, Level 1 = classes typically taken in high school, Level 2 = College algebra/precalculus, Level 3 = Calculus 1, Level 4 = Calculus 2, Level 5 = beyond Calculus 2.

semester 8, less than 2% of the students in the CD, UCSMP or NSFF curricula, or in any of the three strata, were enrolled in a mathematics course.

HLM Analyses

Analyses using HLM with longitudinal grade and difficulty level data for each ACT mathematics stratum were used to help answer the major research question:

> What is the nature and magnitude of differences, if any, in university mathematics achievement, course-taking patterns, and persistence in these classes among students with similar prior mathematical proficiency who completed at least 3 years of high school mathematics in commercially developed, the University of Chicago School Mathematics Project, or National Science Foundation?funded curricula?

Grades earned and the course difficulty levels attained by students in mathematics classes over semesters were treated as repeated measures that were nested within students (Level 1). Two key features of this approach are that constant measurement of occasions is not required, allowing students with 8 semesters worth of data to be analyzed together, and that students do not have to provide complete data across

Table 4
Difficulty Level of Last College Mathematics Class by Curricula and ACT Strata

		CD		UCSMP		NSFF	
	Level	*N*	Col %	*N*	Col %	*N*	Col %
Low 1/3 on ACT mathematics (<=22)	1	153	17.7	16	11.1	75	26.7
	2	494	57.2	97	67.4	159	56.6
	3	110	12.8	12	8.3	21	7.5
	4	54	6.3	15	10.4	13	4.6
	5	52	6.0	4	2.8	13	4.6
Middle 1/3 on ACT mathematics (23–26)	1	17	2.2	3	1.5	5	2.8
	2	395	51.4	84	42.9	85	47.8
	3	148	19.3	44	22.5	29	16.3
	4	93	12.1	31	15.8	22	12.4
	5	115	15.0	34	17.4	37	20.8
High 1/3 on ACT mathematics (>=27)	1	1	.1	0	.0	0	.0
	2	213	18.9	54	17.6	24	20.2
	3	200	17.8	60	19.5	19	16.0
	4	198	17.6	67	21.8	25	21.0
	5	514	45.7	126	41.0	51	42.9

Note. NSFF = IMP, MMOW, & Core-Plus, *N* = Sample Size, Col % = Column percentage, Level 1 = classes typically taken in high school, Level 2 = College algebra/precalculus, Level 3 = Calculus 1, Level 4 = Calculus 2, Level 5 = beyond Calculus 2

occasions, assuming that the missing data are missing at random (Little & Rubin, 2002). The assumption of missing at random is made here, meaning that data on the dependent variable is assumed to be missing as a function of observed variables included in the HLM analyses that follow (specifically, the variables major, years of high school mathematics, and high school mathematics GPA). The HLM analyses were performed separately for each ACT stratum in accordance with Pedhazur and Pedhazur Schmelkin's (1991) "elimination" strategy discussed previously.

Analyses of grade data. For the grade variable, the first model fitted was

$$Y_{t_i} = \pi_{0_i} + \pi_{1_i} a_{t_i} + \pi_{2_i} b_{t_i} + e_{t_i},$$

where Y_{t_i} the mathematics grade of student i in semester t; a_{t_i} indicates a particular semester (i.e., the predictor variable—Semester); b_{t_i} is the time-varying predictor, reflecting the difficulty of a particular mathematics class in a given semester that served as a control variable; (π_{0_i}) is the intercept for student i at $a_t = 0$, which reflects their first mathematics grade earned; π_{1_i} is the slope of student i reflecting the linear

trajectory of grades over semesters adjusted for difficulty; π_{2_i} is the slope of student i reflecting the impact of difficulty on grades over semesters; and e_{t_i} is the within-student error

$$[e_{t_i} \sim N(0, \sigma_t^2)].$$

Models that allowed unequal variances over time were fitted.

Level 2 was between-students and the corresponding models use intercepts and slopes from level 1 as outcomes. The Level 2 models have the general form:

$$\begin{aligned} \pi_{0_i} &= \beta_{00} + \sum_q \beta_{0q} X_{qi} + r_{0_i} \qquad (1) \\ \pi_{1_i} &= \beta_{10} + \sum_q \beta_{1q} X_{qi} + r_{1_i} \\ \pi_{2_i} &= \beta_{20} + \sum_q \beta_{2q} X_{qi} + r_{2_i} \end{aligned}$$

X_{qi} is a student-level predictor used to explain variation in intercepts (π_{0_i}) and slopes (π_{1_i}, π_{2_i}); r_{0_i}, r_{1_i}, and r_{2_i} represent error terms of these equations for student i and are assumed to follow a multivariate-normal distribution with variances τ_{00}, τ_{11}, and τ_{22}, respectively.

Initially the Level 2 models had no predictors (so-called unconditional model) to assess whether intercepts and slopes varied significantly between students, meaning the Level 2 models were

$$\begin{aligned} \pi_{0_i} &= \beta_{00} + r_{0_i} \qquad (2) \\ \pi_{1_i} &= \beta_{10} + r_{1_i} \\ \pi_{2_i} &= \beta_{20} + r_{2_i} \end{aligned}$$

Fitting fully unconditional models revealed that the intercepts and slopes differed significantly for the low and high ACT strata. For the middle stratum, neither the intercepts nor the slopes differed significantly across students. As a result, no predictors were included in the grade model for the middle ACT stratum. The results for the conditional grade models by the high and low strata are presented below. Preliminary work exploring variation among the sampled high schools indicated that high school characteristics, such as the percentage of students receiving a free or reduced priced lunch or the percentage of students of color, were not predictors of grades earned in college mathematics courses or the difficulty of those courses. As a result, high-school-level variables were not used in the models we report. However, results for student-level variables obtained from high school records such as mathematics GPA and ethnicity are reported.

Results from the longitudinal analysis of grades in mathematics courses taken across semesters indicate that curriculum cohorts do not differ in terms of either their initial grade in a mathematics course or in their linear trajectory of grades across semesters. There were other predictors that were significant for the lowest and highest ACT strata. For example, for the low and high ACT mathematics strata,

years of high school mathematics was positively related to the grade a student received in that student's initial mathematics course ($B = .31$ and .28, respectively).[1] That is, after controlling for other predictors, students with more high school mathematics courses tended to have higher grades in their initial college mathematics course. Another finding related to the longitudinal grade models for the low and high ACT mathematics strata was that high school mathematics GPA was a significant predictor of the impact of mathematics difficulty on grade ($B = .24$ and $B = .16$, respectively), meaning that students with higher high school mathematics GPAs tended to get better grades as they took more difficult courses.

College major was also related to grade outcomes in the longitudinal models. For example, for the low ACT mathematics stratum, after controlling for other predictors, being in a STEM major was associated with a lower grade in the initial college mathematics course ($B = -1.01$) and an increase in grades across semesters ($B = .17$) relative to nonSTEM majors. Major was a significant predictor of the intercepts and linear slope for the low ACT mathematics stratum but not the high ACT mathematics stratum. As noted previously, it is important to ensure that the findings are not confounded by preexisting differences in mathematics achievement among the curriculum cohorts. Performing the HLM analyses for each ACT stratum separately ensures that prior mathematics achievement is taken into account. It is also important to ensure that these differences do not confound findings related to the college mathematics course in which a student enrolls initially and their mathematics achievement thereafter. To do this, we repeated each of the HLM analyses for grades by conditioning on difficulty level of initial college mathematics course. That is, we examined longitudinal mathematics achievement for students who began their college mathematics course taking with a course of similar difficulty, for example, college algebra (difficulty Level 2). This strategy ensures that examinations of the relationship between curriculum cohort and achievement are not confounded by students beginning their college mathematics course taking at different difficulty levels as a result of preexisting achievement differences. Performing separate HLM analyses of grades for the three ACT strata for each of five difficulty levels produced a total of 9 analyses. There were insufficient data to perform 6 of these analyses (e.g., there were no students in the high stratum whose first college mathematics class was of difficulty Level 1 or 2).

The separate HLM analyses for each of the three strata, conditioned on initial difficulty level, were similar to those reported in Table 5 with one exception: High stratum students who completed an NSFF curriculum earned, on average, higher grades over semesters than students who completed a CD curriculum (B = .50). These findings indicate that the high school mathematics curriculum that a student completed was generally not a predictor of initial college mathematics course grade or the pattern of grades earned over semesters. The exception to this general conclusion is that NSFF students in the high ACT stratum earned, on average, higher grades across semesters than CD students.

[1] B is a regression coefficient (slope).

Table 5

HLM Results for Mathematics Grade Data for Students in the Low ACT Stratum

Within-Student Model

Effect	*B*	*SE B*
Intercept	1.98*	0.48
Mathematics Difficulty	−0.58	0.27
Linear	−0.20	0.12

Between-Student Model

Effect	*B*	*SE B*
Intercepts (π_{0_i})		
Years of high school mathematics	0.31*	0.11
High school mathematics GPA	0.01	0.12
Gender	−0.03	0.15
African American	−0.19	0.20
Asian	0.02	0.17
Hispanic	0.10	0.34
STEM	−1.01*	0.31
Humanities	−0.45	0.20
Year enrolled	−0.19	0.15
UCSMP	0.05	0.24
NSFF	−0.25	0.18
Linear-Order Slopes (π_{1_i})		
Years of high school mathematics	0.05	0.03
High school mathematics GPA	−0.04	0.03
Gender	−0.09	0.04
African American	−0.01	0.05
Asian	−0.03	0.04
Hispanic	0.03	0.08
STEM	0.17*	0.06
Humanities	0.18*	0.05
Year enrolled	0.00	0.03
UCSMP	−0.01	0.06
NSFF	−0.02	0.04
Math-Difficulty Slopes (π_{2_i})		
Years of high school mathematics	−0.14	0.06
High school mathematics GPA	0.24*	0.07
Gender	0.04	0.08
African American	0.02	0.11
Asian	−0.05	0.09
Hispanic	−0.10	0.19
STEM	0.29	0.15
Humanities	0.11	0.11
Year enrolled	0.06	0.08
UCSMP	0.00	0.13
NSFF	0.04	0.10

Note. CD is the reference group. Therefore, UCSMP and NSFF curricula are compared to CD.

*$p < .004$ ($\alpha' = .0046$)

Analyses of difficulty level data. Tables 7–9 (pp. 300–303) present the HLM results for mathematics difficulty level data for the ACT strata. The predictors were those used in the grade HLM analysis with the exception of the difficulty level variable, which served as the outcome.

For the upper two ACT mathematics strata, high school curriculum was unrelated to the likelihood of taking more difficult initial mathematics courses or taking more difficult mathematics courses across semesters. However, for the lowest stratum (ACT mathematics ≤ 22), a student who completed an NSFF curriculum was less likely, compared to CD students, to take a more difficult initial college mathematics course ($B = -.48$, see Table 7) and was approximately $\exp(-.48) = e^{-0.48} = .62$ times as likely to take a more difficult initial college mathematics course rather than a less difficult course. For all three strata, the pattern of taking increasingly difficult mathematics courses across semesters was the same across curriculum cohorts.

High school mathematics GPA was a significant predictor in every fitted model, with higher high school mathematics GPAs associated with taking more difficult initial college mathematics courses for each stratum ($B = .55$, .85, and .72, respectively). Males were more likely to take more difficult initial college mathematics courses for each stratum ($B = .44$, .40, and .69, respectively), and students with a STEM or humanities major were more likely to take more difficult initial college mathematics courses relative to students in other majors.

Analyses of persistence data. The last question of interest was whether students from the different curriculum cohorts differed in their persistence in college mathematics courses. Persistence was measured by examining the number of college mathematics courses students took over their first 8 semesters of postsecondary study. The dependent variable was the number of college mathematics courses taken. A single-level Poisson regression was used to model this dependent variable for each ACT stratum.

Persistence is heavily influenced by a student's college major, with STEM majors in general taking a larger number of mathematics courses. Thus, it is important to include college major as a predictor in the regression analyses. We used the same predictors in the Poisson regression analyses as were used in the longitudinal analyses (Years of High School Mathematics, High School Mathematics GPA, Gender, Ethnicity, College Major, Year Enrolled, and Curriculum).

The results indicated that high school mathematics curriculum was not a significant predictor of the number of college mathematics courses taken, but high school mathematics GPA was a significant predictor of the number of college mathematics courses taken. This is consistent with previous findings, suggesting that more mathematically proficient students tended to take more mathematics courses during their postsecondary experience. Other analyses (not reported here) found that, after controlling for other variables, males tended to take more mathematics courses during their postsecondary experience.

Table 6

HLM Results for Mathematics Grade Data for Students in the High ACT Stratum

Within-Student Model

Effect	*B*	*SE B*
Intercept	1.56	0.69
Mathematics Difficulty	−0.79*	0.23
Linear	0.12	0.15

Between-Student Model

Effect	*B*	*SE B*
Intercepts (π_{0_i})		
Years of high school mathematics	0.28*	0.10
High school mathematics GPA	0.33	0.16
Gender	0.05	0.16
African American	1.32	0.84
Asian	0.07	0.22
Hispanic	0.90	0.58
STEM	−0.82	0.33
Humanities	−0.79	0.32
Year enrolled	0.07	0.13
UCSMP	0.12	0.17
NSFF	0.10	0.25
Linear-Order Slopes (π_{1_i})		
Years of high school mathematics	0.01	0.02
High school mathematics GPA	−0.06	0.03
Gender	−0.05	0.03
African American	0.32	0.26
Asian	0.07	0.04
Hispanic	−0.03	0.13
STEM	0.09	0.08
Humanities	0.07	0.08
Year enrolled	−0.06	0.03
UCSMP	0.07	0.03
NSFF	0.06	0.05
Math-Difficulty Slopes (π_{2_i})		
Years of high school mathematics	−0.07	0.03
High school mathematics GPA	0.16*	0.05
Gender	−0.02	0.05
African American	−0.32	0.28
Asian	−0.04	0.06
Hispanic	−0.20	0.17
STEM	0.41*	0.13
Humanities	0.39*	0.13
Year enrolled	0.00	0.04
UCSMP	−0.09	0.05
NSFF	−0.05	0.08

Note. CD is the reference group. Therefore, UCSMP and NSFF curricula are compared to CD.

*$p < .004$ ($\alpha' = .0046$)

Table 7
HLM Results for Mathematics Difficulty Data for Students in the Low ACT Stratum

Within-Student Model

Effect	*B*	*SE B*
Intercept1	−12.39*	0.59
Intercept2	−11.29*	0.56
Intercept3	−9.88*	0.54
Intercept4	−5.31*	0.47
Linear		0.25

Between-Student Model

Effect	*B*	*SE B*
Intercepts (π_{0_i})		
Years of high school mathematics	0.74*	0.10
High school mathematics GPA	0.55*	0.11
Gender	0.44*	0.13
African American	0.06	0.19
Asian	0.26	0.15
Hispanic	0.11	0.31
STEM	3.62*	0.25
Humanities	2.93*	0.21
Year enrolled	−0.22	0.13
UCSMP	−0.02	0.22
NSFF	−0.48*	0.15
Linear-Order Slopes (π_{1_i})		
Years of high school mathematics	−0.02	0.05
High school mathematics GPA	0.18*	0.06
Gender	0.06	0.07
African American	0.16	0.11
Asian	−0.06	0.08
Hispanic	−0.28	0.19
STEM	−0.56	0.14
Humanities	−0.50	0.09
Year enrolled	−0.16	0.07
UCSMP	−0.06	0.10
NSFF	−0.07	0.09

Note. CD is the reference group. Therefore, UCSMP and NSFF curricula are compared to CD.
*$p < .006$ ($\alpha' = .0060$)

Table 8
HLM Results for Mathematics Difficulty Data for Students in the Middle ACT Stratum

Within-Student Model

Effect	*B*	*SE B*
Intercept1	−13.34*	0.84
Intercept2	−11.78*	0.82
Intercept3	−10.16*	0.80
Intercept4	−4.95*	0.67
Linear	0.58	0.35

Between-Student Model

Effect	*B*	*SE B*
Intercepts (π_{0_i})		
Years of high school mathematics	0.72*	0.11
High school mathematics GPA	0.85*	0.17
Gender	0.40*	0.14
African American	0.59	0.50
Asian	0.30	0.18
Hispanic	0.52	0.38
STEM	3.68*	0.45
Humanities	2.54*	0.41
Year enrolled	0.20	0.14
UCSMP	0.35	0.17
NSFF	−0.21	0.21
Linear-Order Slopes (π_{1_i})		
Years of high school mathematics	0.01	0.06
High school mathematics GPA	0.17	0.08
Gender	0.17	0.08
African American	0.07	0.17
Asian	0.11	0.10
Hispanic	0.36	0.24
STEM	−0.14	0.15
Humanities	−0.57*	0.13
Year enrolled	−0.14	0.08
UCSMP	−0.16	0.09
NSFF	0.06	0.12

Note. CD is the reference group. Therefore, UCSMP and NSFF curricula are compared to CD.
*$p < .006$ ($\alpha' = .0060$)

Table 9
HLM Results for Mathematics Difficulty Data for Students in the High ACT Stratum

Within-Student Model

Effect	*B*	*SE B*
Intercept1	−8.45*	0.86
Intercept2	−6.92*	0.85
Intercept3	−4.96*	0.84
Intercept4	−0.36	0.89
Linear	−0.35	0.45

Between-Student Model

Effect	*B*	*SE B*
Intercepts (π_{0_i})		
Years of high school mathematics	0.12	0.09
High school mathematics GPA	0.72*	0.15
Gender	0.69*	0.13
African American	−0.77	0.52
Asian	0.61*	0.17
Hispanic	1.30*	0.45
STEM	2.92*	0.53
Humanities	1.38	0.53
Year enrolled	0.26	0.11
UCSMP	0.04	0.13
NSFF	−0.38	0.20
Linear-Order Slopes (π_{1_i})		
Years of high school mathematics	0.22*	0.06
High school mathematics GPA	0.12	0.09
Gender	0.02	0.10
African American	1.31*	0.30
Asian	−0.23	0.11
Hispanic	−0.90*	0.33
STEM	0.19	0.19
Humanities	−0.15	0.18
Year enrolled	−0.20	0.09
UCSMP	−0.15	0.10
NSFF	−0.10	0.20

Note. CD is the reference group. Therefore, UCSMP and NSFF curricula are compared to CD.
*$p < .006$ ($\alpha' = .0060$)

DISCUSSION

This study provided evidence of the longer-term influence of high school mathematics curricula on achievement, course-taking patterns, and persistence in mathematics classes in a large midwestern university for students of similar prior mathematics achievement. Students completed a commercially developed (CD), University of Chicago School Mathematics Project (UCSMP), or one of three NSF-funded (NSFF) curricula (Core-Plus, IMP, MMOW) in high school. The relatively large numbers of subjects in this study (between 1,950 and 4,100 students depending on the particular analysis from 266 high schools) provided a rich dataset that allowed various relationships to be explored. The availability of 8 semesters of University transcript data allowed us to study the long-term collegiate-level effects of CD, UCSMP, and NSFF high school mathematics curricula on university mathematics course work.

The most important finding was that high school mathematics curriculum was not differentially related to student achievement (grades), course-taking patterns (difficulty level of completed courses), or persistence (number of mathematics courses completed) in college mathematics classes across strata. For the low ACT stratum, a difference in the difficulty level of initial course enrollment favoring CD and UCSMP students was detected. On the whole, though, the CD, UCSMP, and NSFF curricula appear to do a comparable job of preparing students with similar prior mathematics achievement for college mathematics.

These findings add to our understanding of the ability of various high school mathematics curricula to adequately prepare students for college mathematics. These results also help to respond to the largely anecdotal criticisms of the NSFF curricula that have often revolved around the supposition that they would leave middle and high ability students poorly prepared for college mathematics. These findings suggest that when schools (students, parents, teachers, and administrators) make curriculum adoption decisions, it is no longer appropriate to make those decisions on the expectation that students will not do well in college mathematics if they participate in one of the NSFF high school *Standards*-based curricula. Our findings suggest that this is not the case.

The specific issue addressed in this article is whether these approaches adequately prepare students with similar prior mathematics achievement for college mathematics. The transition to college mathematics is significant and often problematic for many high school students, but we believe more so for students who have used NSFF curricula, given the greater discontinuities that exist between mathematical content, pedagogical perspectives, and the nature of course expectations as they enter their first college mathematics course.

With the exceptions of the lower difficulty level of the initial course for the lowest ACT stratum, and the superior performance of high stratum NSFF students when compared to high stratum CD students for grades earned over semesters, given that they began their college mathematics at the level of Calculus I, there were no curriculum-related differences for course difficulties, grades, and number of college mathematics courses completed. Why might this have occurred?

Swafford (2003), commenting on her overall impressions of the NSFF curricula, concludes that

> Taken as a group, these studies offer overwhelming evidence that the reform curricula can have a positive impact on high school mathematics achievement. It is not that students in these curriculum [sic] learn traditional content better, but that they develop other skills and understandings while not falling behind on traditional content. (Swafford, 2003, p. 468)

It seems evident that students using NSFF curricula have learned about many (but not all) of the traditional topics often included on procedurally oriented placement tests that have been considered fundamental to success in college mathematics. The contexts within which these topics were explored in NSFF classrooms were different, both in the manner in which the content was introduced and in the classroom organizational methods used. We also note that the amount of classroom time spent by NSFF students on developing and sustaining a variety of algorithmic procedures was, by design, considerably less.

Another possible factor accounting for so few differences was increased tenacity or the willingness to "stay with" a task as observed initially by many classroom teachers in our previous Local Systemic Change project. Such tenacity is a prerequisite for success in college mathematics for all but the most precocious of students, and increased levels of it would serve all students well in college-level mathematics courses.

Limitations

Four general limitations of our study should be kept in mind in interpreting our findings. First, there is evidence that less mathematically proficient students were disproportionately steered into NSFF curricula, producing a gap among the curriculum cohorts that persisted through high school. Although we presented evidence of preexisting differences among students in the various high school curricula, we had no data capturing the nature or intensity of the steering process that lead to these preexisting differences.

Second, the use of ACT mathematics scores as a measure of prior mathematics achievement raises concerns that these scores and curriculum were confounded. Our analyses using the ACT strata provided evidence that curriculum did not impact ACT mathematics scores in ways that would bias inferences, but we cannot be certain that such confounding has been completely removed.

Third, we had no data capturing the process leading to a student's enrollment in his or her initial college mathematics course, such as might be associated with the academic counseling that a student received or how college mathematics placement tests were used.

Fourth, our use of a sample of students who enrolled at a large, research-oriented public university with relatively high entrance standards implies that our findings cannot automatically be generalized to students in colleges or universities varying in mission, size, or selectivity.

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The Preparation of Students From National Science Foundation–Funded and Commercially Developed High School Mathematics Curricula for Their First University Mathematics Course

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The selection of K–12 mathematics curricula has become a polarizing issue for schools, teachers, parents, and other educators and has raised important questions about the long-term influence of these curricula. This study examined the impact of participation in either a National Science Foundation–funded or commercially developed mathematics curriculum on the difficulty level of the first university mathematics course a student enrolled in and the grade earned in that course. The results provide evidence that National Science Foundation–funded curricula do not prepare students to initially enroll in more difficult university mathematics courses as well as commercially developed curricula, but once enrolled students earn similar grades. These findings have important implications for high school mathematics curriculum selection and for future research in this area.

KEYWORDS: mathematics education, postsecondary education, achievement

There is widespread agreement among teachers, parents, school administrators, mathematics educators, mathematicians, and others of the importance of providing K–12 students with rigorous instruction that promotes mathematical understanding, but there is less agreement on how to achieve this goal. A manifestation of the lack of agreement is the "math wars" that have pitted defenders of commercially developed (CD) mathematics curricula against those advocating curricula supported by the National Science Foundation (NSF; Schoenfeld, 2004). In general, CD mathematics curricula stress traditional algorithms and procedures, while National Science Foundation–funded (NSFF) curricula focus to a greater degree on problem solving and other higher order thinking skills such as synthesis, generalization, and evaluation. Specifically, NSFF curricula refer to those curricula that were funded from a solicitation of proposals (RFP NSF 91-100) through the NSF in the early 1990s (Senk & Thompson, 2003) that were designed to be aligned with the National Council of Teachers of Mathematics' (NCTM; 1989) *Curriculum and Evaluation Standards for School Mathematics.*

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Disagreement over which mathematics curriculum is most appropriate can be linked indirectly to the introduction of the NCTM (1989) standards and directly to curricula that have been designed to conform to those standards. It is important to emphasize that the 1989 standards were the result of the efforts of hundreds of professionals and received near unanimous acceptance and acclaim from all sectors of academia and society. In 2000 the NCTM published a revised document, *Principles and Standards for School Mathematics*, that expanded upon and updated the 1989 NCTM document and again involved literally hundreds of professionals in its development. Endorsees of the 2000 document voiced their support for the "quality mathematics curricula and assessment criteria" and included 15 major mathematical societies such as the Mathematical Association of America and the American Mathematical Society.

In a very real sense these two NCTM documents, although widely supported, challenged the hegemony of CD mathematics curricula in U.S. schools. In response to this challenge, advocates of CD curricula have taken their case to a variety of constituencies, especially parents. The advocates have argued that teaching mathematics using NSFF curricula leads to poor mathematics performance with potentially dire consequences such as poor student performance on the mathematics portions of standardized tests taken by college-bound students (Wu, 1997) or that these curricula do not adequately prepare students for calculus and other mathematically oriented courses as taught at the university level.

On the other hand, proponents of NSFF curricula often point to published reports of disappointing mathematics achievement of students, most of whom completed a CD curriculum (Hiebert, 2000; Schoen, Fey, Hirsch, & Coxford, 1999), and argue that large-scale changes in the way we think about, develop, implement, and assess school mathematics programs are needed. For example, results from the Trends in International Mathematics and Science Study (1996) indicated that U.S. 8th grade and 12th grade students performed below international averages. The *National Assessment of Educational Progress Report* (Braswell et al., 2001) indicated that 17% of high school seniors in the United States were proficient in mathematics in terms of reaching a minimum level of competency. The report of the Senior Assessment Panel of the International Assessment of U.S. Mathematical Sciences (1998) characterized U.S. K–12 mathematics instruction as weak and pointed out that this has several adverse effects, including the fact that the majority of mathematics PhD students at U.S. universities hail from other countries.

Regional evidence of problems are reflected in a 1999 report by the Minnesota State College and University System that found only 38% of students entering a postsecondary institution in the State College System, almost all of whom would have completed a CD curriculum, were considered ready to begin postsecondary mathematics at the level of college algebra. The corresponding figure for 2-year colleges was 18%. Moreover, fewer than 7% of eighth-grade students will take 4 full years of college-intending mathematics

coursework in high school. Similar findings have been reported in many states, including Wisconsin, Iowa, Illinois, North and South Dakota, and Washington.

These findings challenge the assumption that CD curricula, which have dominated K–12 mathematics instruction in Minnesota and elsewhere in the United States for many years, have adequately prepared students for university-level mathematics. However, little is known about the preparation of students who completed an NSFF mathematics curriculum in high school for university mathematics.

Theoretical Models Underlying NSFF Curricula

The 1989 and 2000 NCTM *Standards* documents incorporated components of various cognitively oriented theories about the development of mathematics knowledge and the nature of student learning. There is also reason to believe that the fundamental principles that underpin the development of the NSFF curricula should promote mathematics learning for a diverse population of student learners.

Until recently, principles of behavioral (Skinner, 1938) and neobehavioral (Gagne, 1963, 1985) psychology have dominated curriculum development and evaluation in school mathematics. Widespread use of behavioral objectives, task analysis, stimulus response theory, and operant conditioning are artifacts of this psychological perspective. More recently, cognitive psychological perspectives (Bruner, 1960; Dienes, 1960; Dienes & Golding, 1971; Piaget, 1960; Piaget & Inhelder, 1958) have received considerable attention as psychologists and educators realized that learning is primarily internal to the individual and its measurement involves more than documenting a series of observable behaviors that may be related only to the mastery of isolated skills and procedures.

Research on student conceptual development during the 1970s and beyond (e.g., Bruner, 1964; Bruner, Olver, & Greenfield, 1966; Dienes, 1960; Lesh, 1979; Shulman & Sparks, 1992) has been strongly influenced by these cognitive perspectives. The NCTM *Standards* documents (1989, 2000) were also influenced by these perspectives. This in turn had a decided impact on the direction of the development of NSFF curricula because the NSF required that each proposal document specify how the planned project would reflect the principles espoused in the 1989 *Standards* document.

Development of NSFF Mathematics Curricula

Fearing for the future levels of technological literacy in this country and for the future of our ability to lead in this area, the NSF funded 13 full mathematics programs (3 elementary, 5 middle grades, and 5 at the high school level) at a cost of approximately US$100 million, beginning in the early 1990s. Within each program, 30 to 40 persons (mathematics educators, mathematicians, teachers, etc.) worked to complete the writing and pilot testing

of each of these curricula, which generally took 5 or 6 years. At the high school level, all programs were committed to considering algebra, geometry, probability, statistics, and topics in discrete mathematics every school year. This was a radical departure from the traditional Algebra 1, Geometry, Algebra 2, Precalculus (college algebra) sequence.

The NSFF curricula were for the most part problem centered, focusing on problems embedded in realistic settings that took several class periods to explore and resolve. The mathematical content and its development were in part motivated by the problems considered. Small group and cooperative student activities, along with a de-emphasis on the teacher lecture format, were strongly encouraged and considered fundamental to the success of the programs. Extensive teacher support materials and professional development opportunities, also funded by the NSF, were provided to school districts adopting these programs.

Although no curriculum can ever be totally supported by research, these materials were the most "researched" school mathematics curriculum materials of any in history. They represent the "best practices" of the time and in general reflect accumulated research results as to the nature of student mathematical conceptualization and learning. Thus, these new curricula reflect, for the most part, contemporary cognitive approaches to mathematics teaching, learning, and assessment. These curricula are also consistent with a broader school reform movement emphasizing the role of students as active participants in a learning process that involves problem solving, small group work, and connections to the world outside the classroom (Newmann, Marks, & Gamoran, 1996; Schoenfeld, 2004). Last, these curricula correspond to the positions of the NCTM, the main professional organization in the field.

Evidence of the Performance of Middle School and High School Students in NSFF Curricula

Research on the achievement of students learning from NSFF curricula in Grades K–12 provides evidence of the levels of student learning (Senk & Thompson, 2003). In general, students who completed an NSFF curriculum do as well as or better than students who completed a CD curriculum on nonstandardized achievement measures that typically consist of classroom problem sets (Cichon & Ellis, 2003; Huntley, Rasmussen, Villarubi, Sangtong, & Fey, 2000; Lott et al., 2003; Schoen & Hirsch, 2003; Webb, 2003).

Research has also indicated that NSFF curricula appear to satisfactorily prepare students to succeed on standardized tests, including the Stanford Achievement Test (Abeille & Hurley, 2001; Harwell et al., 2007; Post et al., 2008), the Iowa Test of Educational Development Ability to Do Quantitative Thinking (Abeille & Hurley, 2001), the Iowa Test of Basic Skills (Schoen & Hirsch, 2003), the Scholastic Assessment Test Mathematics subtest (Webb, 2003), the Comprehensive Test of Basic Skills (Webb, 2003), and the Preliminary Scholastic Aptitude Test (Cichon & Ellis, 2003; Lott et al., 2003).

The performance of students completing an NSFF curriculum on standardized tests is in many ways surprising because such tests are not aligned with the content or focus of NSFF curricula, suggesting that these students are at a disadvantage. Despite strong encouragement in the literature to link mathematics content and assessment in new ways (Le et al., 2006; National Research Council, 1998; NCTM, 2000; Pellegrino, 2006), the evidence is that tests sensitive to NSFF curricula such as the New Standards Reference Examination in Mathematics (New Standards, 1997) are used far less often than their standardized counterparts.

Research of NSFF high school mathematics curricula has also shown that students completing these curricula generally fare as well as or better than students who completed a CD curriculum on a variety of nonachievement outcomes related to educational success. These outcomes have included the tendency of students completing an NSFF curriculum to take more mathematics classes in high school than students studying from CD mathematics curricula (Cichon & Ellis, 2003; Webb, 2003), report higher ratings in their confidence and the usefulness of mathematics, and display more positive attitudes toward mathematics (Schoen & Pritchett, 1998).

Another factor that has been shown to be related to mathematics achievement in middle school and high school is socioeconomic status (SES; Coleman et al., 1966; Reys, Reys, Lapan, Holliday, & Wasman, 2003; Riordan & Noyce, 2001; Sirin, 2005), a relationship that appears to be unaffected by participation in an NSFF curriculum (Harwell et al., 2007; Post et al., 2008). Teacher professional development (Schoen, Cebulla, Finn, & Fi, 2003) and fidelity of implementation of an NSFF mathematics curriculum (Schoen et al., 2003) have also been shown to be related to mathematics achievement. The impact of teacher quality on student achievement is often cited as an important factor to consider, but the literature is mixed on key contributors to this impact or the magnitude of the effect (Wayne & Youngs, 2003).

The data across different geographical areas and various curricula show that middle school and high school students studying from NSFF curricula perform similarly on standardized measures of mathematics achievement and on the mathematics portions of nonstandardized tests when factors that can affect mathematics performance are taken into account. Some of these factors are wholly or partly under the control of schools (e.g., fidelity of implementation of NSFF curricula and teacher professional development), while others, like prior mathematics achievement and SES, are not. This in turn suggests that these factors need to be taken into account in examining the preparation of students completing NSFF and CD high school mathematics curricula for university-level mathematics.

Evidence of the Effectiveness of NSFF Mathematics Curricula to Prepare Students for University-Level Mathematics

Despite evidence of success, NSFF curricula have come under intense criticism from Web-based sources[1] and other venues. Much of the criticism

claims that students completing an NSFF curriculum are poorly prepared to succeed in calculus and other university-level mathematics courses. Since 70% of the students who were eighth graders in 1988 attended some type of postsecondary institution by 1994 (National Center for Education Statistics [NCES], 1998), the preparation of students completing NSFF and CD high school mathematics curricula for university-level mathematics is important (Kuenzi, Matthews, & Mangan, 2006; Riley, 1998). Students who enroll at a university underprepared often must first take courses they should have completed in high school, and about one quarter of all freshmen enroll in at least one postsecondary remedial mathematics course (Parsad & Lewis, 2003). Documenting the preparation of students for university-level mathematics as a function of their high school mathematics curriculum is essential to remedying the problem of underprepared students.

Somewhat surprisingly, disappointing results on national and international assessments and the low percentage of high school graduates ready to begin their college mathematics careers with college algebra or calculus have not prompted much research on how well NSFF or CD high school mathematics curricula prepare students for university-level mathematics. In fact, we are aware of only two peer-reviewed published research studies addressing the relationship between high school mathematics curricula and student success in university-level mathematics.

Schoen and Hirsch (2003) examined the grade point averages (GPAs) of students from two similar high schools who later enrolled in Calculus I and above at the University of Michigan. Graduates of the NSFF Contemporary Mathematics in Context (Core-Plus) curriculum had a higher overall GPA in Calculus I and other more advanced mathematics courses when compared with students in a CD curriculum. Hill and Parker (2006) used a sample of more than 3,000 to study student performance in university mathematics coursework and reported findings that were critical of NSFF curricula. In particular, these authors reported that students who completed the NSFF Core-Plus mathematics curriculum in high school began their university mathematics coursework with less difficult courses and subsequently completed even less difficult mathematics courses than a group of students who did not complete Core-Plus in high school. One important difficulty with this study was that the specific mathematics curriculum completed by these students was, as acknowledged by the authors, in question and could have included students who completed other NSFF curricula, University of Chicago School Mathematics Project (UCSMP) curricula, or even a CD curriculum.

O'Neil (2002), in an unpublished thesis, studied the relationship between high school mathematics curriculum (CD and NSFF) and grade in the first college mathematics course for a small sample of students ($N = 48$) who had completed 4 years of the NSFF math curriculum Math Connections. Completion of a CD or NSFF curriculum (Math Connections) did not have a statistically significant relationship with the grade a student earned in the first college mathematics course.

In response to the sparseness of this literature, the Mathematical Sciences Education Board (2004) has called for formal studies of this relationship. The current study examined the preparation of students for university-level mathematics as a function of the high school mathematics curriculum they completed. The research question addressed was as follows:

> Do university students who completed at least three levels of high school mathematics in an NSFF curriculum differ from those who had similar exposure to a commercially developed curriculum in the difficulty level of their first university mathematics course or in the grade they earned in that course, when taking into account background factors such as prior mathematics achievement, ethnicity, and gender?

A high school level was defined to be the content equivalent of a year-long course taught 1 hour per day and has been commonly referred to as a Carnegie unit (Boyer, 1983). Addressing this question should lead to a better understanding of the similarity or differences in the preparation of students who completed an NSFF or a CD curriculum in high school for university-level mathematics.

Method

The methodology in this study was informed in part by *On Evaluating Curricular Effectiveness: Judging the Quality of K–12 Mathematics Evaluations* (Mathematical Sciences Education Board, 2004). This document provides criteria for effective evaluation of curriculum material, and several of its recommendations were incorporated into this research.

Research Design

This study employed a quasi-experimental design for cross-sectional data (Pedhazur & Schmelkin, 1991) in which the high school mathematics curriculum (group) a student completed was the independent variable of most interest. Difficulty of students' first university-level mathematics course and the grade earned in that course served as dependent variables.

This quasi-experimental design allowed effects corresponding to the research question to be examined but did not automatically permit strong causal inferences because of the absence of random assignment (Pedhazur & Schmelkin, 1991). As a result, our goal was to statistically control for important differences between the two curriculum groups that could bias our inferences.

Sampling

Two populations were of interest. The target population consisted of post-secondary students in a midwestern state in the United States who had graduated from 1 of the more than 325 high schools in the state. Our sampled population, on the other hand, consisted of students enrolled at a single large

public university in a midwestern state in the United States during the fall of 2002 or fall of 2003 terms who had graduated from 1 of 85 high schools for which at least 10 students enrolled at the university, which helped to ensure reasonable high school–based statistical estimates. The fall of 2002 was chosen as the first term to study postsecondary performance because it allowed for significant numbers of students to be sampled who had completed at least three levels of an NSFF high school mathematics curriculum. The fall of 2002 term was also the point at which fairly extensive student admission information was available in an electronic form, greatly simplifying data collection. Student data for two consecutive semesters were available: fall of 2002 and spring of 2003 for one cohort and fall of 2003 and spring of 2004 for a second cohort.

Data

Archival data from three sources were collected.

State. Data for the state-mandated Basic Skills Test (BST) for mathematics (Minnesota Department of Education, 1998) administered in eighth grade served as a measure of prior mathematics achievement. This test covers mathematics content up through and including pre-algebra, and its development followed traditional procedures for establishing psychometric criteria of validity and reliability and for ensuring that different forms of the test offered from one year to the next possessed items of the same difficulty. However, its use resembles a criterion-referenced test since students must exceed a predetermined cutoff to be eligible for high school graduation (students falling below the cutoff can retake the test). The percentage of students eligible for a free or reduced-price lunch at a high school when the sampled students were high school seniors was also obtained from the Minnesota Department of Education and was used as a high school–level measure of SES.

University. Information on a student's application to the university was also tapped, including a student's transcript data reflecting his or her high school coursework. Among the variables extracted from these records were ACT Mathematics score, GPA in high school mathematics courses, overall high school GPA, high school percentile rank, number of levels of high school mathematics completed (three, four, or five), sex, SES (eligibility for a free or reduced-price lunch), and ethnicity (Native American, African American, Asian, Hispanic, or White). Other variables collected from the university included when a student enrolled (fall of 2002 or fall of 2003) and mathematics grade and course-taking information.

Altogether, 5,410 university student records were available initially. Because BST scores resided in an external data file, it was necessary to match records electronically to link each student's BST score with his or her university record, a process that was successful for 70% (3,788) of the students but unsuccessful for 1,622 students. Among students for whom records could

not be matched, there was evidence that some ethnic subgroups were disproportionately affected. Among African American students in the original sample, 52% did not have BST scores, followed by Asian (40.4%), Hispanic (35.1%), and White (24.6%) students. Further examination of the ethnic subgroups showed that students for whom BST scores were not available on average showed slightly lower mathematics achievement. For example, average ACT Mathematics scores for African American and White students for whom BST scores were not available were 17.7 (SD = 4.61) and 25.1 (SD = 4.16), respectively, but among students with BST scores ACT Mathematics means for these two ethnic subgroups were 18.8 (SD = 4.35) and 25.6 (SD = 4.58), respectively.

Among students for whom BST Mathematics scores were available (N = 3,788), a total of 32% (1,225) did not take a mathematics course in their first year of university study and therefore provided no difficulty or grade data. Among the 1,225, 81.7% had completed a CD curriculum in high school, quite similar to the percentage of students completing a CD curriculum in the original sample (82.1%). While the tendency of students who completed NSFF and CD curricula to enroll in a mathematics course during their first year of university study was similar (67.1% and 65.6%, respectively), the group of 1,225 consistently showed slightly lower average prior mathematics achievement compared to those who enrolled for a mathematics course.

Missing data further reduced the pool of available students and was most pronounced for the ethnicity and ACT Mathematics variables. There were too few Native American students in our sample with sufficient data to ensure credible estimation of parameters connected to this ethnic group, and these students were omitted. Similarly, two students who classified themselves as international students were also omitted.

We further restricted the sample to students from high schools offering a single curriculum, all of which offered a CD curriculum, and to students who completed an NSFF curriculum in a high school offering both NSFF and CD curricula. The result was that the 513 students who completed a CD curriculum and graduated from a high school offering both NSFF and CD curricula were omitted from the analyses. This was done to allow mathematics curriculum (NSFF and CD) to be a high school–level variable; that is, all students from a particular high school in the sample completed an NSFF or a CD mathematics curriculum.

In sum, many students from the original sample of 5,410 were omitted because (a) BST Mathematics scores were not available, (b) students completed a CD mathematics curriculum in a high school offering both CD and NSFF mathematics curricula, (c) there were too few Native American students, or (d) students did not enroll in a mathematics course during their first year of university study. In addition, missing data on variables such as ACT Mathematics and ethnicity reduced the pool of available students in some analyses; these students were slightly more likely to be non-White and to have slightly lower scores on variables reflecting academic success such as high school percentile rank and high school GPA. With these omissions,

the remaining sample consisted of 371 students who completed an NSFF curriculum and 1,296 students who completed a CD curriculum.

The net effect of the reduction in sample size was to limit inferences to a population of students (a) whose prior mathematics achievement was less variable and slightly higher than that of the original sample and (b) who enrolled in at least one mathematics course in their first year of university study. Still, the resulting 1,667 students from 85 high schools used in the analyses ensured good precision of estimation and adequate statistical power to detect effects of interest.

High schools. Another important source of data were the high schools, which provided information about the kind of mathematics curriculum a student completed in high school (i.e., had successfully completed at least three levels). Although the university maintains electronic records of the high school courses taken by each applicant and the associated grades, we quickly detected discrepancies in how particular mathematics courses were coded in university records and the correct descriptors in the data collection. For example, some schools reported "Integrated 1, 2, 3, and 4" as their high school mathematics courses, information that the university then recorded. We use the descriptor *integrated* to denote detailed study of algebra, geometry, probability, and statistics each year. This is sometimes used interchangeably with *Standards-based*, the term sometimes used to denote NSFF curricula. In some cases the courses recorded as integrated were actually a traditional publisher's text series that had a subtitle with the word integrated in it.

These initial experiences confirmed our pre-study expectation that accurate categorizations of students as having completed an NSFF or a CD mathematics curriculum in high school required us to contact each high school to independently obtain descriptions of their mathematics programs, including the mathematics courses offered and the textbooks used. Although labor intensive, this was necessary to ensure credible inferences about the NSFF and CD groups. Among the sampled high schools, 32 offered both an NSFF and a CD mathematics curriculum and 53 offered a CD curriculum exclusively, the latter including students exposed to the UCSMP curriculum (none of the high schools offered an NSFF curriculum exclusively). NSFF curricula in our sample of high schools were the Interactive Mathematics Program, Mathematics: Modeling Our World, and Core-Plus, which, given our research questions, were not differentiated in the data analyses.

The maximum number of students from a single high school in our sample was 77, with a median of 16 students, and these students were assumed to be representative of a population that completed at least one university mathematics course. Evidence that mathematics achievement patterns frequently differ among urban, suburban, and rural high schools (Finn, Gerber, & Wang, 2002; NCES, 2004) led us to obtain information on high school location. An examination of the sample showed that large urban and suburban high schools were well represented and smaller, typically rural high schools less so.

Variables

Independent variables. The high school mathematics curriculum a student completed (NSFF or CD) was the independent variable of major interest. Other independent variables in the data analyses that served as control variables were sex; ethnicity; BST Mathematics score; ACT Mathematics score; high school percentile rank; high school mathematics GPA; overall high school GPA; high school location; size of graduating high school class, which served as a proxy for high school size; and the proportion of students eligible for a free or reduced-priced lunch at each high school, which was a proxy for SES. When a student enrolled at the university and when he or she enrolled in the first university mathematics course also served as independent variables. The latter was used to take into account the possibility that students who waited to enroll in a mathematics course in the spring might perform differently because they had the benefit of a semester's experience.

Dependent variables. The dependent variables were the difficulty level of a student's first mathematics course and the grade earned in that course. If completion of a particular high school curriculum better prepares students mathematically, those students would be expected to initially enroll in more difficult (advanced) mathematics courses and to have better grades. Examining mathematics course descriptions at the university led to the construction of a 4-point Likert scale to capture the level of course difficulty:

Level 1: This level includes courses that should have been completed in high school. They often do not carry college credit. Students do enter the university without the required or recommended high school background, and these courses are designed to provide a foundation for students to be successful in future mathematics courses.

Level 2: This level includes courses for students who satisfactorily completed four levels of college-intending mathematics in high school. Courses at this level could be categorized in general as college algebra and precalculus mathematics.

Level 3: This is the typical entry level for well-prepared high school students who begin postsecondary mathematics coursework with beginning calculus (several types of calculus are available at the university, such as short calculus, Calculus I, or calculus with a biological emphasis).

Level 4: This level is considered an extension of beginning calculus. Titles include Calculus II, Differential Equations, Linear Algebra, Multivariable Calculus, and Sequences and Series.

University Mathematics Placement Tests and Mathematics Teaching

In studying the difficulty of the first university mathematics class a student enrolls in and his or her grade in that class, it is important to consider the role of mathematics placement tests. Other things being equal, we might expect students who completed a particular mathematics curriculum in high school to perform well on a placement test whose content and structure tend

to match the high school mathematics curriculum, but there appears to be little evidence one way or the other. O'Neill (2002) reported a significant difference favoring students who completed a CD curriculum over those who completed an NSFF curriculum on the College Board's Accuplacer Test, whose content is generally consistent with that of CD curricula.

The university uses three mathematics placement tests; one is administered by a campus unit designed for students who are not yet ready to succeed academically in college-level mathematics coursework and the other two by academic units with significant instructional service functions. Two of the exams test for general readiness for postsecondary mathematics, and the other focuses on readiness for calculus. A perusal of these exams showed that they concentrate on algebraic manipulation, with only a few items on trigonometry and probability.

Once a student has taken one or more of these placement tests, he or she meets with an academic counselor, who may also be the student's advisor. The exact structure of the advising process varies across academic units at the university but in general involves the counselor using placement test information along with other relevant information, such as ACT Mathematics score, high school grades, and Advanced Placement Calculus scores if available, to advise a student on which mathematics class to enroll in initially. Any such recommendation is nonbinding, although previous analyses indicated that approximately 88% of freshmen in our sample followed the placement recommendation. The variability in the counseling process and the way that mathematics placement information is used can mean different initial course placement recommendations for identical student profiles. For example, a student could be advised to begin with calculus but opt to enroll in a precalculus course or vice versa, and both cases could impact the difficulty and grade dependent variables. Significant difficulties with obtaining credible placement data led us to omit this information from our analyses, and this is an important limitation of the study.

Variation across university mathematics instructors was curtailed somewhat by the university mathematics department's policy of standardizing course syllabi, using common student assignments and examinations, and using a single course-specific distribution of scores for assigning grades. Still, we had no data capturing the impact of teacher quality on student achievement, and this too is an important limitation of the study.

Data Analyses

Initially, descriptive analyses were performed to explore patterns in the data, followed by two-level hierarchical linear modeling (HLM; Raudenbush & Bryk, 2002) that treated students as nested within high schools. To control for compounding of Type I error rates, we used an adjusted Type I error rate attributed to Sidak (1967) of the form $\alpha' = 1 - (1 - \alpha)^{1/k}$, where α is the unadjusted Type I error rate, k is the number of statistical tests, and α' is the adjusted Type I error rate. We chose to compute adjusted Type I error rates for

each table of statistical test results that we report based on an overall $\alpha = .10$, which resembles a family-wise error rate strategy often associated with planned comparisons in ANOVA. For example, Table 1 reports 28 statistical test results for four student-level background variables. Based on $\alpha = .10$, each of the 28 statistical hypotheses are tested at $\alpha' = .0037$.

Results

Descriptive Analyses

Several descriptive analyses were performed to explore patterns in the data, and a representative subset of these results is reported. First, data for students who entered the university in the fall of 2002 ($N = 1{,}075$, 49.3%) and the fall of 2003 ($N = 1{,}105$) were examined. The percentage of students who completed an NSFF curriculum entering in the fall of 2002 and the fall of 2003 increased slightly, from 21.3% to 23%, respectively, while the corresponding CD percentages were 78.7% and 77%, respectively.

Descriptive statistics reflecting students' academic achievement and background variables like SES and ethnicity are reported in Table 1. A number of these relationships were found to be statistically significant, and the variance in a dependent variable explained by an independent variable was used as an index of effect size, with larger effects associated with stronger relationships. Following Cohen's (1988) guidelines, the effects in Table 1 ranged from quite weak (.02) to large (.16).

Correlations among several student variables are reported in Table 2 and show that the relationships between first university mathematics grade and other variables range from weak to moderate. Among the strongest correlations are those involving the number of levels of high school mathematics, overall high school GPA, ACT Mathematics scores, and the correlation between BST and ACT scores.

Table 3 reports high school mathematics achievement, difficulty of the first university mathematics course by curriculum group, and the number of levels of high school mathematics completed. As expected, increasing numbers of levels of high school mathematics were associated with higher average mathematics grades for both curriculum groups.

HLM Analyses

HLM was used to examine the effect of high school mathematics curriculum on the difficulty level of the first university mathematics course a student enrolled in and the grade earned in that course. The two dependent variables were analyzed separately. Students were treated as clustered within high schools to permit within–high school dependency among students to be estimated. If present, modeling this dependency with HLM should ensure more credible statistical test results than would ordinarily be possible with traditional regression modeling. The difficulty of a student's first university mathematics course was analyzed assuming an ordinal scale of measurement.

Table 1

Descriptive Results for Student Socioeconomic Status (SES), Sex, Ethnicity, and High School Location

Variable	ACT Math	BST Math	High School GPA	High School Math GPA	High School Percentile Rank	First University Math Grade	Difficulty of First University Math Course
SES	$\hat{\eta}^2 = .02^*$	$\hat{\eta}^2 = .05^*$	$\hat{\eta}^2 = .02^*$	$\hat{\eta}^2 = .02^*$	$\hat{\eta}^2 = .01^*$		$\hat{\eta}^2 \leq .01^*$
Low (6.3%)							
M	22.5	83.5	3.23	3.10	70.8	2.50	1.97
SD	5.6	14.3	0.60	0.76	20.2	1.21	0.99
n	100	104	104	97	99	104	104
High (93.7%)							
M	25.4	91.9	3.51	3.45	78.9	2.74	2.44
SD	4.7	8.1	0.47	0.58	16.9	1.02	0.87
n	1,519	1,558	1,551	1,424	1,523	1,558	1,558
Sex	$\hat{\eta}^2 = .09^*$	$\hat{\eta}^2 = .05^*$			$\hat{\eta}^2 \leq .01^*$		$\hat{\eta}^2 = .06^*$
Female (45.1%)							
M	23.8	89.2	3.53	3.4	79.9	2.70	2.18
SD	4.5	10.2	0.48	0.58	16.8	1.05	0.77
n	735	751	747	682	733	751	751
Male (54.9%)							
M	26.5	93.1	3.47	3.43	77.4	2.75	2.60
SD	4.6	7.1	0.49	0.58	17.5	1.04	0.93
n	887	913	910	841	900	913	913
Ethnicity	$\hat{\eta}^2 = .12^*$	$\hat{\eta}^2 = .16^*$	$\hat{\eta}^2 = .10^*$	$\hat{\eta}^2 = .08^*$	$\hat{\eta}^2 = .06^*$	$\hat{\eta}^2 = .016^*$	$\hat{\eta}^2 = .06^*$
African American (4.7%)							
M	19.2	77	2.87	2.73	62	2.15	1.53
SD	4.1	16.8	0.56	0.72	19	1.19	0.66
n	77	79	79	75	76	79	79
Asian American (15.1%)							
M	23.8	88.8	3.49	3.42	77.2	2.77	2.34
SD	4.9	10.9	0.47	0.61	18.2	1.04	0.96
n	244	256	256	235	253	256	256

(continued)

Table 1 (continued)

Variable	ACT Math	BST Math	High School GPA	High School Math GPA	High School Percentile Rank	First University Math Grade	Difficulty of First University Math Course
Hispanic (3.1%)							
M	21.9	87.6	3.21	3.12	70.6	2.65	2.04
SD	4.9	9.7	0.64	0.73	21.7	0.99	0.95
n	49	52	50	50	51	52	52
White (76.8%)							
M	26.1	92.9	3.54	3.48	80.1	2.75	2.49
SD	4.4	6.3	0.44	0.54	16.1	1.02	0.84
n	1,254	1,280	1,275	1,166	1,256	1,280	1,280
High school location	$\hat{\eta}^2 = .026$*	$\hat{\eta}^2 = .057$*	$\hat{\eta}^2 = .027$*	$\hat{\eta}^2 = .032$*	$\hat{\eta}^2 = .014$*	$\hat{\eta}^2 \leq .01$*	$\hat{\eta}^2 = .016$*
Rural (16.8%)							
M	25.7	92.6	3.57	3.51	81.3	2.58	2.42
SD	4.5	7	0.43	0.55	15.4	1.0	0.83
n	268	273	273	248	270	273	273
Suburban (65.8%)							
M	25.5	92.1	3.52	3.45	78.4	2.79	2.47
SD	4.5	7.5	0.46	0.56	17.3	1.01	0.88
n	1,039	1,066	1,060	981	1,054	1,066	1,066
Urban (17.4%)							
M	23.5	86.6	3.30	3.21	74.3	2.58	2.58
SD	5.8	13.1	0.59	0.71	18.4	1.13	1.13
n	272	282	281	255	260	282	282

Note. BST = Basic Skills Test; GPA = grade point average; *M* = mean; *SD* = standard deviation; *n* = sample size; $\hat{\eta}^2$ = percentage of variance in a dependent variable (e.g., ACT Mathematics) explained by an independent variable (e.g., SES).

*Indicates statistically significant at $\alpha' = 1 - (1 - \alpha)^{1/k} = .0037$, where $\alpha = .10$ is the unadjusted Type I error rate, k is the number of statistical tests = 28, and α' is the adjusted error rate. To investigate the relationship between the background variables and the difficulty of a first university mathematics course, a Kruskal-Wallis test for ordered contingency tables was performed; for all others an F test was used. Difficulty of first university mathematics course was coded on a 4-point scale (1 = *courses that would typically have been completed in high school*; 4 = *beyond Calculus I*). The value in parentheses for each subgroup reports the relative size of that subgroup; for example, 6.3% of the sample was categorized as eligible for a free/reduced-price lunch.

Table 2
Correlations Among Student Variables

Variable	1	2	3	4	5	6	7	8	9
1. ACT Math score	—								
2. High school math GPA	.506*	—							
3. BST Math (% correct)	.643*	.469*	—						
4. High school math levels	.564*	.417*	.394*	—					
5. High school percentile rank	.439*	.764*	.393*	.366*	—				
6. High school overall GPA	.493*	.878*	.451*	.423*	.860*	—			
7. Sex	−.291*	−.002	−.204*	−.143*	.062	.063	—		
8. Difficulty of first university math course	.719*	.521*	.486*	.593*	.512*	.534*	−.237*	—	
9. Grade of first university math course	.270*	.276*	.173*	.212*	.204*	.270*	−.023	.131*	—

Note. Listwise N = 1,455. BST = Basic Skills Test; GPA = grade point average.
*Indicates statistically significant at $\alpha' = 1 - (1 - \alpha)1/k = .0029$, where $\alpha = .10$ is the unadjusted Type I error rate, k is the number of statistical tests = 36, and α' is the adjusted error rate.

All analyses were performed with the HLM6 software (Raudenbush, Bryk, & Congdon, 2005).

Difficulty data. A generalized hierarchical proportional odds model for ordinal data was fitted to difficulty (1 = *courses ordinarily completed in high school*; 2 = *courses up to but not including Calculus I*, such as college algebra and precalculus; 3 = *Calculus I*; 4 = *beyond Calculus I*). Analysis of ordinal data is typically done by reexpressing the m categories of the ordinal variable in the form of $m - 1$ cumulative logits, represented by the log of the cumulative odds of each level of the variable (Raudenbush & Bryk, 2002, pp. 317–322). For example, for the first category of the difficulty variable, the corresponding logit had the form $\log\{[\text{Prob}(R_{ij} = 1)]/[1 - \text{Prob}(R_{ij} = 1)]\}$, where $\text{Prob}(R_{ij} = 1)$ represents the probability that the ith student from the jth high school enrolled in a university mathematics course of Level 1 difficulty and $1 - \text{Prob}(R_{ij} = 1)$ is the probability that a student enrolled in a more difficult mathematics course. For the second category of mathematics courses up to but not including Calculus I, the corresponding logit had the form $\log\{[\text{Prob}(R_{ij} \leq 2/[1 - \text{Prob}(R_{ij} \leq 2)]\}$, and so on.

We began by fitting an unconditional model to the logits that produced a statistically significant between-school variance of .61, indicating that students from different high schools tended to enroll in initial mathematics courses of varying difficulty.

Next, a model was fitted with the student-level control variables BST (eighth-grade BST in mathematics); number of levels (full-year equivalents) of high school mathematics completed (three, four, or five); GPA in high school mathematics classes; overall high school GPA; high school percentile rank; ACT Mathematics score; sex; ethnicity as captured by dichotomous

Table 3

High School and University Mathematics Achievement by Number of Levels of Mathematics Completed in High School and Curriculum Group

High School Math Levels Completed	Curriculum Group	ACT Math	BST Math	High School GPA	High School Math GPA	Grade of First University Math Course	Difficulty of First University Math Course
3	CD						
	M	21.1*	84.7*	3.11	2.86	2.34	1.68*
	SD	4.1	12.3	0.6	0.75	1.12	0.6
	n	190	194	194	175	194	194
3	NSFF						
	M	18.7	78.2	2.91	2.81	2.33	1.35
	SD	3.3	15.4	0.6	0.75	1.13	0.51
	n	65	68	68	62	68	68
4	CD						
	M	24*	90.6*	3.44*	3.31	2.56	2.06*
	SD	3.9	8.3	0.44	0.56	1.05	0.57
	n	626	641	639	587	641	641
4	NSFF						
	M	21.3	87.7	3.31	3.27	2.53	1.76
	SD	3.7	10.2	0.56	0.68	1.07	0.57
	n	150	153	153	142	153	153
5	CD						
	M	28.1*	94.2*	3.66	3.6	2.91	2.99*
	SD	3.9	6.3	0.35	0.45	0.99	0.79
	n	943	974	965	903	974	974
5	NSFF						
	M	26.3	92.8	3.61	3.54	2.92	2.73
	SD	4.3	6.8	0.36	0.46	0.94	0.85
	n	146	150	149	134	150	150

Note. *M* = mean; *SD* = standard deviation; *n* = sample size; CD = commercially developed high school mathematics curriculum; NSFF = National Science Foundation–funded high school mathematics curriculum. Difficulty of first university mathematics course was coded on a 4-point scale (1 = *courses that would typically have been completed in high school*; 4 = *beyond Calculus I*). To investigate the relationship between curriculum group and difficulty of first university mathematics course, a Kruskal-Wallis test for ordered contingency tables was performed; for all others an *F* test was used.

*Indicates statistically significant at $\alpha' = 1 - (1 - \alpha)^{1/k} = .0058$, where $\alpha = .10$ is the unadjusted Type I error rate, k is the number of statistical tests = 18, and α' is the adjusted error rate. The largest $\hat{\eta}^2$ values were for ACT (.10) and BST (.07) for three high school math levels completed and ACT for four levels (.09); the remaining $\hat{\eta}^2$ values were .03.

predictors for African American (yes, no), Asian, and Hispanic students; year a student enrolled at the university (fall of 2002, fall of 2003); and whether the student took a mathematics course in the first semester.

The school-level model included the curriculum offered at a high school (CD or NSFF) as a predictor in the random intercepts model in which average school logits served as the outcome. Other predictors added to the random

Table 4
Hierarchical Linear Modeling Fixed Effects for University Mathematics Course Difficulty Data

Fixed Effect	Coefficient
Threshold1	4.72*
Threshold2	7.08*
Level 2: high schools	
Intercept	–3.598*
Size	0.001
Rural	–0.219
Suburban	0.183
Socioeconomic status	1.977
High school curriculum	0.717*
Between-school variance	0.611*
Level 1: students	
Year enrolled	–0.079
ACT Math score	–0.078*
High school percentile rank	–0.012
High school math GPA	–0.026
Female	0.970*
BST Math	–0.065*
High school overall GPA	–2.165*
TookFall	0.218
High school math level	–1.560*
African American	–0.284
Asian	0.115
Hispanic	–0.086

Note. High school size = size of high school graduating class; High School Location 1: 0 = *nonrural*, 1 = *rural*; High School Location 2: 0 = *nonsuburban*, 1 = *suburban*; high school socioeconomic status = percentage of high school students eligible for a free/reduced-price lunch; high school curriculum: 0 = *CD*, 1 = *NSFF*; year enrolled: 0 = *fall of 2002*, 1 = *fall of 2003*; sex: 0 = *males*, 1 = *females*; BST Math = Basic Skills Test in Mathematics; TookFall = took first mathematics course in the fall term: 0 = *no*, 1 = *yes*; high school math level = number of high school mathematics levels completed (three, four, or five); African American: 0 = *non-African American*, 1 = *African American*; Asian: 0 = *non-Asian*, 1 = *Asian*; Hispanic: 0 = *non-Hispanic*, 1 = *Hispanic*.
*Indicates statistically significant at $\alpha' = 1 - (1 - \alpha)^{1/k} = .0050$, where $\alpha = .10$ is the unadjusted Type I error rate, k is the number of statistical tests = 21, and α' is the adjusted error rate.

intercepts model to serve as control variables were size of graduating class (*Mdn* = 380), location (urban, suburban, or rural), and percentage of students eligible for a free/reduced lunch (SES; *Mdn* = 18%). All predictors were grand mean centered.

High school–level random effects were assumed to be normally distributed, and modest evidence of nonnormality did not appear to be large enough to distort the results. The HLM results for difficulty are reported in Table 4.

With respect to our first research question, the key result in Table 4 is the significant slope for curriculum group (.717), meaning that curriculum is

Table 5
Hierarchical Linear Modeling Fixed Effects for University Mathematics Course Grade Data

Fixed Effect	Coefficient
Level 2: high schools	
Intercept	2.661*
Size	0.000
Rural	0.088
Suburban	-0.119
Socioeconomic status	-0.445
High school curriculum	0.068
Between-school variance	0.047*
Level 1: students	
Year enrolled	0.006
ACT Math score	0.010
High school percentile rank	0.001
High school math GPA	-0.004
Female	-0.071
BST Math	0.012*
High school overall GPA	0.807*
TookFall	-0.138
High school math level	0.182*
African American	-0.162
Asian	-0.010
Hispanic	0.075
Difficulty Level 4 vs. Level 1	1.348*
Difficulty Level 3 vs. Level 1	-0.018
Difficulty Level 2 vs. Level 1	0.544

Note. High school size = size of high school graduating class; High School Location 1: 0 = *nonrural*, 1 = *rural*; High School Location 2: 0 = *nonsuburban*, 1 = *suburban*; high school socioeconomic status = percentage of high school students eligible for a free/reduced-price lunch; high school curriculum: 0 = *CD*, 1 = *NSFF*; year enrolled: 0 = *fall of 2002*, 1 = *fall of 2003*; sex: 0 = *males*, 1 = *females*; BST Math = Basic Skills Test in Mathematics; TookFall = took first mathematics course in the fall term: 0 = *no*, 1 = *yes*; high school math level = number of high school mathematics levels completed (three, four, or five); African American: 0 = *non-African American*, 1 = *African American*; Asian: 0 = *non-Asian*, 1 = *Asian*; Hispanic: 0 = *non-Hispanic*, 1 = *Hispanic*; Difficulty Level 4 vs. Level 1: 0 = *Level 1*, 1 = *Level 4*; Difficulty Level 3 vs. Level 1: 0 = *Level 1*, 1 = *Level 3*; Difficulty Level 2 vs. Level 1: 0 = *Level 1*, 1 = *Level 2*.
*Indicates statistically significant at $\alpha' = 1 - (1 - \alpha)^{1/k} = .0048$, where $\alpha = .10$ is the unadjusted Type I error rate, k is the number of statistical tests = 22, and α' is the adjusted error rate.

related to the difficulty level of a student's first university mathematics course. Exponentiating the curriculum slope exp(.717) = 2.04 means that with other predictors held constant, a student who completed an NSFF curriculum in high school and who is "average" on all predictors is twice as likely to enroll in less difficult courses as not, compared to an average student who completed a CD curriculum in high school. None of the other high school–level predictors of intercepts were significant.

Following Raudenbush and Bryk (2002, pp. 317–325), the model-implied probabilities of a student enrolling in a mathematics course of a given difficulty can be estimated. Specifically, for an average student who completed a CD curriculum in high school, the model-implied probabilities of enrolling in a university mathematics course of Difficulty 1 (least difficult), 2, 3, or 4 (most difficult) are .03, .72, .24, and .01, respectively; for an average student who completed an NSFF curriculum in high school, these probabilities are .05, .81, .13, and .01, respectively. These probabilities suggest that average students completing a CD or an NSFF curriculum are about equally likely to enroll in the least difficult mathematics courses (.03 vs. .05, respectively). Students who completed an NSFF curriculum are somewhat more likely to enroll in a mathematics course of Level 2 difficulty (e.g., precalculus) than those completing a CD curriculum (.81 vs. .72, respectively), but both curriculum groups have large model-implied probabilities that an average student will enroll in a course that should have been completed in high school. The major difference in the model-implied probabilities between the two curriculum groups is for enrolling in a mathematics course of Level 3 difficulty, where students who completed a CD curriculum are almost twice as likely to enroll in a course of the difficulty of Calculus I compared to students who completed an NSFF curriculum (.24 vs. .13, respectively).

Other statistically significant effects reported in Table 4 included the number of levels of high school mathematics (–1.56), with students completing fewer high school mathematics levels more likely to enroll in less difficult courses, and sex (.97), with male students who were average on all predictors 2.63 times more likely to enroll in more difficult courses as not, compared to female students. The significant slope for ACT Mathematics exp(–.078) = .92 reflects the change in the log odds of a student enrolling in a less difficult mathematics course relative to a more difficult mathematics course for a one-unit increase in ACT Mathematics. Put another way, for higher levels of mathematics achievement, as defined by the ACT, students are increasingly likely to enroll in more difficult university mathematics courses relative to enrolling in less difficult courses. A similar interpretation can be given to the significant and large effect for overall high school GPA.

Grade data. An unconditional model was first fitted to the grade data to estimate variation between and within high schools on this variable and the dependency of grades for university students who graduated from the same high school. The average grades that students earned in their first university mathematics course varied significantly across the sample of high schools, producing an estimated between-school variance of .047. The associated intraclass correlation of .054 means that the dependency among students' first university grade in a mathematics course who had attended the same high school was relatively small. However, if such dependency is not modeled, estimates can be biased (Raudenbush & Bryk, 2002); therefore, we continued to use HLM. There was evidence of moderate nonnormality, but refitting the model after omitting cases that appeared to be responsible did not

substantially impact the findings, and the results reported below are based on all available cases.

Next, the models including student- and high school–level predictors used for mathematics course difficulty were fitted to the grade data. The only difference was that dummy-coded predictors capturing mathematics course difficulty were added as student-level predictors to control for this variable. The fixed-effects HLM results are reported in Table 5.

With respect to our first research question, the key result in Table 5 is that the slope for curriculum group in the intercept model is not significant (.068, p = .418), meaning that with the other predictors held constant, the mathematics curriculum a student who was average on other predictors completed while in high school was not related to the grade earned in the first university mathematics course. The significant slope for the dummy-coded predictor for Level 4 versus Level 1 difficulty (1.348) means that students enrolling in more difficult mathematics courses tended to obtain higher grades on average than those enrolling in the least difficult courses, and the significant slope for number of high school mathematics levels completed (.182) means that students who completed more levels tended to get higher grades.

In sum, the HLM results provided evidence that students completing a CD mathematics curriculum are more likely to begin their university mathematics work in a more difficult mathematics course than students who completed an NSFF curriculum once various background factors were taken into account. However, there was no relationship between the high school mathematics curriculum a student completed and the grade earned in that course.

Discussion

There is abundant evidence that many students are not adequately prepared for university mathematics in the United States, with approximately one quarter of all U.S. students completing at least one mathematics course that should have been completed in high school. Concerns about the preparation of students for university-level mathematics have focused attention on high school mathematics.

For decades, high school mathematics has been dominated by CD curricula, which stress traditional algorithms and procedures in which students tend to be passive learners. Beginning in the 1990s, standards-based curricula that were informed by cognitive principles of learning were developed with the support of the NSF. These curricula emphasized problem solving and small group work and treated students as active participants in the learning process. Despite the investment in NSFF curricula, little is known about their ability to prepare students for university mathematics. This study adds two primary findings to this literature.

First, there was no relationship between a student's high school mathematics curriculum and the grade that student earned in the first university

mathematics course, suggesting that the two curricula prepare students equally in terms of proficiency attained in a university mathematics course. Our grade results are consistent with findings reported by O'Neill (2002) but differ from those reported by Schoen and Hirsch (2003), who found that students completing an NSFF curriculum had higher grades.

Second, CD curricula appeared to do a better job of preparing students to initially enroll in a more difficult mathematics course than NSFF curricula, for example, Calculus I. However, the model-implied probabilities that students will enroll in a mathematics course of Level 2 difficulty, reflecting courses that should have been completed in high school, are large regardless of the high school mathematics curriculum a student completed.

The model-implied probabilities also differ substantially from the NCES (Parsad & Lewis, 2003) finding that approximately 25% of all U.S. students take at least one college mathematics course that should have been completed in high school. However, in comparing these results it is important to recognize that the NCES finding is based on the definition of adequate preparation for college mathematics used by each of the more than 1,100 sampled institutions. We defined adequate student preparation for college mathematics as initially enrolling in Calculus I or higher, but it is likely that some and perhaps many of the institutions sampled by the NCES employed a different definition. Different definitions of adequate mathematics preparation will impact the percentage of students identified as needing remedial college mathematics coursework and the interpretation of related findings. For example, had we chosen to treat students who were ready to begin their college mathematics work with college algebra or precalculus as adequately prepared, the interpretation of our course-taking findings would be quite different.

In short, our results suggest that if the focus of high school mathematics for college-bound students is to prepare them to begin their college work with Calculus I, the percentage of students in need of remedial mathematics in college may be higher (and perhaps much higher) than suggested by the NCES finding.

It is also important to emphasize that the tendency of students who completed an NSFF curriculum to enroll in a less difficult university mathematics course emerged in our findings after statistically controlling for several variables, including high school mathematics achievement. Thus, explanations for this finding do not appear to include students' mathematical proficiency but may include factors at the high school level, the university level, or both.

High School Factors

An important high school factor to consider in framing our grade and difficulty results is that students in the fall of 2002 or fall of 2003 cohorts who completed NSFF curricula were likely taught by high school teachers with limited experience with these materials, in many cases only 1 to 2 years.

Effective instruction in any curriculum requires that the teacher know what content students have already been exposed to and ideally where the elaboration of various topics will appear in the future. The relatively limited experience of teachers with these curricula may have impacted our findings, but we had no data with which to explore this possibility.

A second high school factor that may be relevant to understanding our findings is the fidelity with which teachers using NSFF curricula implemented these materials in terms of both content and method. We did not have access to students' high school teachers or administrators; thus, we were unable to ascertain with certainty the fidelity with which teachers of these students implemented these curricula. However, we are optimistic about the fidelity with which NSFF curricula were implemented because of previous experiences with the NSF Local System Change (LSC) Project at the University of Minnesota (1997–2001). This project provided 130 or more hours of targeted professional development to over 1,100 middle-grade and secondary mathematics teachers. As part of the 130 hours, teachers received 20 hours of mentor on-site assistance based on mentor impressions. We were satisfied that most of the teachers implemented new NSFF curricula with fidelity. These experiences provide indirect evidence that high school mathematics teachers who taught students who completed NSFF curricula, an unknown number of whom also participated in the LSC Project, implemented these materials with fidelity. Still, limited resources precluded our ability to make credible assessments of this important variable.

University-Level Factors

One university-level factor that may help to explain the tendency of graduates of NSFF curricula to enroll in less difficult mathematics courses initially relates to the content of the mathematics placement tests. The first mathematics course a student enrolls in is guided by the results of a mathematics placement test and input or recommendations from the student's academic advisor, but for students in our sample these are not binding. Thus, students may have elected to enroll in more or less difficult mathematics courses, and available data suggest that about 12% of our sample did so.

Still, students for whom the content of these tests is consistent with their high school mathematics curriculum would be expected to perform better than students for whom the content of these tests does not match their high school mathematics curriculum particularly well. Our experience with university mathematics placement tests suggests that their content generally favors students completing a CD curriculum because of its emphasis on traditional algorithms and procedures and represents a significant obstacle for students who completed an NSFF curriculum in which traditional algorithms and procedures were not emphasized. A mismatch between the content of the university mathematics placement tests and that of NSFF curricula may help to explain the tendency of these students to start with less difficult university mathematics courses.

Another factor that may be related to the tendency of students from NSFF curricula to start with less difficult mathematics courses is the familiarity (or unfamiliarity) of university academic counselors with these curricula. For example, a counselor examining the high school transcript of a student who completed a CD curriculum and who saw the label "Precalculus" would likely be more familiar with the content of this course, compared to seeing the label "Core-Plus 3" on the transcript of a student who completed an NSFF curriculum. It seems plausible that counselors unfamiliar with the high school mathematics courses a student completed might tend to recommend a less difficult university mathematics course.

A second university factor to consider in framing our findings is the orientation of mathematics instruction in examining the grade a student earned in the first university mathematics course. Students for whom instructional content and method in a university mathematics course are similar to that which they experienced in high school may be predisposed to perform better. At the university from which students were sampled, it is fair to say that in general mathematics instruction is much closer in content and practice to that in CD curricula than NSFF curricula, a common phenomenon (Steen, 1987). This similarity may help students who experienced a CD curriculum in high school, but we had no data to shed light on this important factor.

In sum, one or more high school– and university-level factors may have influenced our results, and our inability to control for these factors in our statistical models is a clear limitation, in particular the role of university mathematics placement tests and academic counselors. Moreover, student-oriented factors such as motivation and self-confidence may have impacted our findings. Our findings about the efficacy with which the high school mathematics curricula prepare students for their first university mathematics course need to be framed against these limitations.

Implications for Future Curricular Decisions and Future Research

These results should advance the ongoing debate regarding the adequacy of NSFF curricula in helping students prepare for subsequent university-level mathematics. Curricular decisions by school districts toward CD mathematics curricula and away from NSFF curricula that appear to be occurring in many states, including Minnesota, need to be guided by credible research evidence. Our findings add to this evidence but certainly do not provide a sufficient basis for such decisions.

As evidence of the ability of these curricula to prepare students for university mathematics increases, teachers, administrators, counselors, researchers, and the public must engage in discussions concerning CD and NSFF high school mathematics curricula and which option is more appropriate for their schools. Such discussions will almost inevitably identify several questions in need of further study, such as (a) What are the patterns in student performance, course taking, and persistence in later university science, technology, engineering, and mathematics courses, and where do students end their

university mathematics course taking? (b) Do these results generalize to 2-year colleges and less selective 4-year institutions? (c) Do these results generalize to students from small high schools where they are less likely to have competing types of high school mathematics curricula? and (d) What is the nature and role of university mathematics placement tests and academic counselors on students' university mathematics course taking?

Notes

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[1]For example, see Mathematically Correct at http://www.mathematicallycorrect.com and New York City HOLD at http://www.nychold.com.

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Standards-based Mathematics Curricula and Middle-Grades Students' Performance on Standardized Achievement Tests

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Approximately 1400 middle-grades students who had used either the Connected Mathematics Project (CMP) or the MATHThematics (STEM or MT) program for at least 3 years were assessed on two widely used tests, the Stanford Achievement Test, Ninth Edition (Stanford 9) and the New Standards Reference Exam in Mathematics (NSRE). Hierarchical Linear Modeling (HLM) was used to analyze subtest results following methods described by Raudenbush and Bryk (2002). When *Standards*-based students' achievement patterns are analyzed, traditional topics were learned. Students' achievement levels on the Open Ended and Problem Solving subtests were greater than those on the Procedures subtest. This finding is consistent with results documented in many of the studies reported in Senk and Thompson (2003), and other sources.

Key words: Achievement; Assessment; Curriculum; Middle grades, 5–8; Multivariate techniques; Reform in mathematics education

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This study examined achievement patterns of middle school students enrolled in *Standards*-based curricula, in particular those curricula that were funded from a solicitation of proposals through the National Science Foundation (NSF) in the early 1990s (NSF RFP 91-100). The focus was on traditional topics in mathematics as measured by two nationally normed achievement tests.

This study builds on and extends our existing understanding of student achievement in *Standards*-based programs in three significant ways. First, we examine the impact of published editions of these curricula on student understanding as contrasted to earlier studies using field-test versions. Second, we focus on *Standards*-based curricula used as part of district-wide curricula adoptions. By examining adopted versions of these curricula, we study the achievement of students whose teachers are required to, and have not necessarily volunteered to, teach *Standards*-based curricula. This provides a more accurate overall picture of expected student achievement as many earlier field-test teachers were volunteers and therefore may not be typical of all teachers who implement a *Standards*-based curriculum. Third, we employ hierarchical linear modeling (HLM) to account for the wide variability between classrooms and the inter-dependency of students within the same classroom. Thus, HLM allows us to consider student and classroom results simultaneously.

The present study adds yet another brushstroke to the emerging picture of mathematics achievement in classrooms using curricula directly and fundamentally influenced by the *Curriculum and Evaluation Standards for School Mathematics* (NCTM, 1989) and similar documents (National Research Council, 1989). An earlier study (Harwell et al., 2007) discussed similar issues for secondary students. Schoenfeld (2002) reviewed this emerging body of work and concluded that there is growing support for the success of such programs in terms of problem solving or other in-depth measures. This characterization is largely consistent with research on *Connected Mathematics Project* (CMP) (Reys, Reys, Lapan, Holliday, & Wasman, 2003; Ridgway, Zawojewski, Hoover, & Lambdin, 2003; Riordan & Noyce, 2001) and *MATHThematics* (MT) (Billstein, 1998; Reys et al., 2003). Kilpatrick (2003), when referring to the 13 chapters in Senk and Thompson (2003), concluded that "[the] studies reported in this volume offer the best evidence we have that *Standards*-based reform works" (p. 487).

The research on student achievement in *Standards*-based curricula with regard to facility with both arithmetic and symbolic manipulation procedures is mixed over the short and long terms. For instance, Ridgway et al. (2003) found that sixth-grade CMP students started 1 year behind non-CMP students on the Iowa Test of Basic Skills (ITBS), and at the end of Grade 6 were 1.5 years behind the other group. The CMP Students who started .52 standard deviation (*SD*) behind non-CMP students were .61 SD behind after 1 year. At the end of eighth grade, however (3 years later), CMP students were .32 *SD* ahead. In related investigations, the authors concluded that there is no immediate short-term advantage to CMP, but that the longer view is promising, with CMP students making large gains on a broad range of curriculum topics and processes when compared to non-CMP students.

There is research suggesting that the benefits from a *Standards*-based curriculum extend beyond increases in mathematics achievement on Open Ended and Problem Solving subtests. Billstein and Williamson (2003) found that students who used MT improved in their attitudes toward mathematics and had higher scores on the language achievement subtest of the ITBS than a comparable group of students studying from other mathematics curricula.

Research suggests that curriculum is only one of the factors that influences student achievement: "Whereas improved curriculum materials can provide rich activities that support students' mathematical investigations, in and of themselves such materials may not be sufficient enablers of instruction that affords pursuit of conceptual issues" (Gearhart et al., 1999, p. 309; cf. Ball & Cohen, 1996). Briars and Resnick (2000) looked at fidelity of implementation in *Everyday Mathematics,* a K–6 program used in the Pittsburgh schools. They found that schools with high fidelity of implementation scored two to five times higher on skills, problem solving, and concepts on the New Standards Reference Examination (NSRE). McCaffrey et al. (2001) also found that *Standards*-based teaching was positively related with student achievement but only made a significant impact when a *Standards*-based curriculum was also in place. Weiss, Banilower, Overstreet, and Soar (2002) found that classrooms using a *Standards*-based curriculum were rated higher on a scale measuring inquiry-oriented teaching practices when compared to classrooms with traditional mathematics curricula. These findings suggest that, although a *Standards*-based curriculum alone can positively influence teacher pedagogy, the results are especially promising if combined with high fidelity of implementation and effective instruction of these new materials.

Another variable in this complex interaction between curriculum and achievement is the students themselves. The ways that students react to and interface with the curriculum in the classroom can affect implementation of the curriculum (Cooney, 1985; Henningsen & Stein, 1997). In addition, it has long been known that characteristics that students bring with them to the classroom help to shape their achievement. For instance, with respect to the School Mathematics Study Group (SMSG), Begle (1973) stated,

> Even a casual inspection of the results of this study of predictors reveals two clear generalizations. The first of these is that the best predictor of mathematics achievement is previous mathematics achievement. . . . The second generalization is this: The best predictors of computational skill at the end of the school year are generally computational skills at the beginning of the school year. On the other hand, the best predictors of performance at the high cognitive levels of understanding, application, and analysis seldom include computational skills. (pp. 213–214)

Student SES has also been shown to play a role in how students interact with *Standards*-based curricula (Lubienski, 2000). Past research on student achievement in *Standards*-based classrooms has used SES either as a variable in matching groups for comparison purposes (Reys et al., 2003; Riordan & Noyce, 2001) or as a predictor of student achievement in regression analyses (Schoen et al., 2003). School environment also affects successful implementation of any curriculum (cf.,

Cohen & Ball, 2001; Eisenhart et al., 1993). Schoen et al. (2003) found that professional development that is related to the curriculum is positively correlated with student achievement.

METHODOLOGY

Selection of the Districts

In the mid to late 1990s, the NSF, in an attempt to provide much needed professional development for school districts that adopted one or more of the new NSF-funded *Standards*-based curricula, created the Local Systemic Change through Teacher Enhancement Initiatives (LSCs). The 47 funded LSCs (NSF 95–145) were "designed to engage entire school districts in the reform of science, mathematics and technology education, . . . to provide 47,000 teachers with professional development and . . . reach 1.6 million students in 240 school districts nationally" (NSF, 1997, p. 5).

The "Minneapolis and St. Paul Merging to Achieve Standards Project" $(MASP)^2$, one of these 47 projects, provided professional development to over 1100 middle-grades and secondary teachers in 21 districts between 1997 and 2000. These teachers then provided *Standards*-based mathematics instruction to over 74,000 students in the 2000–2001 school year, and slightly larger numbers of students each year thereafter.

Of these 21 school districts, 5 were invited to participate in the study of student achievement reported here. The districts were selected by $(MASP)^2$ project managers to provide a range of district types while remaining within the budget that was available for the testing phase of the NSF grant. The districts contained different combinations of middle school and senior high NSF-funded curricula and represented all types of districts in the project—urban, suburban, and boundary districts. Purposive sampling of districts served two purposes. First, it provided information to key groups of constituents connected to the project. Second, following the arguments of Cochran (1983), it allowed generalizations to be made to a target population of similar school districts.

The 5 districts included in our sample used either CMP (Lappan, Fey, Phillips, & Anderson, 1998) or MT (Billstein & Williamson, 1998) at the middle school level. These curricula differ from each other in various ways; the length of the units, the reality of the contexts, and the emphasized content are a few examples. Because both curricula responded to the same NSF Request for Proposals (1989), they also share many similarities: recurring integration of topics within grade levels, extended explorations, and a decreased emphasis on paper-and-pencil computation. Although we recognize the danger of combining similar curricula (Davis, 1990), our research questions referred to students in broadly defined *Standards*-based mathematics classrooms, not those studying from a specific *Standards*-based curriculum. Therefore, after initially finding no significant differences in our descriptive results, we pooled student results from these two curricula.

The location, curriculum, rationale for choice, and assessment for each district are shown in Table 1. There were sharp differences among some of the districts in geographic location, student enrollment, and student characteristics. The purely urban school district had substantially greater enrollment than the others and showed the greatest diversity in student ethnicity, eligibility for free or reduced lunch, percentage of English language learners, and special education status. In contrast, the remaining four districts had only modest variation on student demographic variables, and their students were predominately White native-English speakers who were not eligible for free or reduced lunch. We note that the criteria for a student to be classified as eligible for free or reduced lunch, as a nonnative speaker, or as a special education student are state-mandated and so are the same across the five school districts.

Table 1
Location, Curriculum, Rationale for Choice, and Assessment for Each District

District	Geographic location	Middle-grades curriculum assessed in this study	Rationale for choice of district	Assessment used
A	Urban-suburban boundary	MT	High fidelity of implementation with much parental support	Stanford 9
B	Urban	CMP	Large urban population—variable implementation	Stanford 9
C	Suburban	CMP	Wholesale adoption sabotaged by a few faculty dissenters and parents	Stanford 9
D	Suburban	MT	Wholesale adoption with district authorized supplements	Stanford 9
E	Suburban	CMP	Enthusiastic adoption	Stanford 9 & NSRE

Data Collection

Cross-sectional data for two groups of eighth-grade middle school students, one tested in the Spring of 2001, and the other in the Spring of 2002, were collected. These students had been studying from either CMP or MT for a total of 3 years. There were no theoretical reasons to consider the cohorts of eighth-grade students tested as separate. Similarly, the results of analyses such as ANOVA and HLM in which cohort served as a predictor indicated that there were no empirical reasons to treat the groups as separate. As a result, they were combined into a single group for analysis purposes.

The sample for this study consisted of approximately 1600 *Standards*-based middle school students, most of whom (85%) took the Stanford Achievement Test, Ninth Edition (Stanford 9), which consisted of Open Ended, Problem Solving, and

Procedures subtests. Approximately 25% of the students took both the Stanford 9 and the New Standards Reference Exam in Mathematics (NSRE). However, the statistical analyses were based on fewer than 1600 students, primarily because of missing data on the Stanford 9 subtests. Overall, the percentages of missing data on the Open Ended, Problem Solving, and Procedures subtests were 8.9%, 10.3%, and 11.2%, respectively. Also, 9% of the students who failed to provide data for the Open Ended subtest also failed to provide data for Problem Solving, 10.9% failed to provide data for the Open Ended and Problem Solving subtests, 3.5% failed to provide data for the Problem Solving and Procedures subtests, and 5% failed to provide data for any of the Stanford 9 subtests.

Proceeding with statistical analyses in the presence of missing data would typically invoke the listwise deletion option popular in data analysis software. Listwise deletion requires all subjects with missing data be eliminated from the analysis. Listwise deletion also requires that the missing data be missing completely at random. Under this condition, the missing data (if obtained) would convey the same information as the available data, or, more formally, the distributions of the missing and available data would be identical. If the assumption that the missing data are missing in a completely random fashion is not satisfied, the statistical results will be biased by an amount depending on the extent to which this assumption is not true. If a biasing effect is present, it will be exaggerated for groups with disproportionately greater amounts of missing data through its impact on statistics computed for those groups. Accordingly, we examined the data for evidence that particular demographic groups had substantially different amounts of missing data.

In general, the amount of missing data on the Stanford 9 subtests appeared to be similar across the student demographic variables. The median percentages of missing data across the Stanford 9 subtests for nonnative and native-English speakers were 10% and 9%, respectively; for students who were or were not eligible for free or reduced lunch, these percentages were 12% and 8%, respectively; for Black, Asian, Hispanic, and White students, the median percentages of missing data across the Stanford 9 subtests were 16%, 5%, 10%, and 8%, respectively.

The largest difference in the amount of missing data was for Black and Asian students (16% and 5%, respectively). However, given the numbers of Black and Asian students in our sample (248 and 124, respectively), and the average Stanford 9 scores for these groups, it is unlikely, at least in terms of mean differences, that the missing data introduced serious bias. For example, the Open Ended mean for Black students was 42.6 ($N = 204$), whereas for Asian students this mean was 50.9 ($N = 118$). The 44 missing Black students would need to score on average 89 for the entire sample to produce an Open Ended mean equal to that of the Asian students (i.e., 50.9), and an average of 61 to produce an Open Ended mean halfway between 42.6 and 50.9 (i.e., 46.7). Similarly, if the missing Asian students each scored 1 on the Open Ended subtest, the mean for the entire sample of Asian students would still be much larger than 42.6. Similar results appear for the Problem Solving and Procedures subtest. Thus, the largest observed difference in missing Stanford 9 data among demographic groups is unlikely to change the basic findings.

On the whole, these results provide evidence that missing Stanford 9 data were not disproportionately located in a particular Stanford 9 subtest or student demographic group, which suggests that the reasons data were missing cannot be explained by which subtest a student may have failed to take or by a student's demographic profile. This in turn makes biased results due to missing data less likely. However, we cannot be certain that the results reported in this article would be similar to those obtained if the missing data had been available, and it would be prudent to interpret our findings in light of this potential bias.

Design

A nonexperimental design with clustering was used. Students were considered to be clustered (nested) within classrooms, which in turn were clustered within school districts. Information was obtained for each level of clustering in the sample, but the focus was on students and classrooms. No control group of students who had experienced a traditional curriculum existed since we were testing students in districts that had adopted, in wholesale fashion, a reform curriculum in each of their middle schools. The lack of experimental manipulation means that study results support inferences about relationships among variables and their magnitude but do not generally support strong causal inferences.

We determined that an appropriate comparison would be the standards established by the publisher of the standardized test instrument used in the assessment—the Stanford 9 or the NSRE. The Normal Curve Equivalent (NCE) mean of 50 was selected as our benchmark because it reflected average performance on the Stanford 9 based on national norms. By definition, NCEs are "normalized standard scores with a mean of 50 and a standard deviation of 21.06. The standard deviation of 21.06 was chosen so that NCEs of 1 and 99 are equivalent to percentiles of 1 and 99" (Harcourt Assessment, Inc., 2005, p. 4).

Instruments

This research project began with questions from districts, teachers, and parents concerning achievement of students enrolled in *Standards*-based classes in schools for which $(MASP)^2$ provided professional development. We had a responsibility to document achievement patterns of students related to national norms on traditionally oriented standardized tests. The school districts wanted testing instruments with national norms, and after reviewing several instruments from national publishers and consulting with our districts, we narrowed the list to the Stanford 9 and the NSRE. Both sets of tests were published and scored by Harcourt Brace.

The mathematics portion of the Stanford 9 has three subtests. The Problem Solving subtest contains 30 multiple-choice problems that require students to solve problems set within real-world and mathematical contexts. The Procedures subtest has 20 multiple-choice questions that require students to perform one of the four basic arithmetic operations with whole numbers, integers, and fractions. The Open Ended subtest uses realistic problems to evaluate students' concepts and skills

(Harcourt Brace & Company, 1997). Each of the Stanford 9 subtests tests the content areas of number, measurement, geometry, algebra, functions, statistics, and probability as deemed appropriate for each grade level. The two multiple-choice subtests combined, and the Open Ended subtest, each takes two 50-minute school periods to administer. Calculators are allowed on all subtests except for Procedures. The Stanford 9 reports both scale scores and NCEs, and it is important to emphasize that NCEs are monotonically related to, but are not identical to, percentiles. Most of the data analysis results are reported in NCEs because of their familiarity and interpretability.

The mathematics portion of the NSRE consists of three parts. The first consists of 20 multiple-choice questions that are a subset of Stanford 9 multiple-choice test items. In addition, this first section contains short tasks. The Stanford 9 questions enable this portion of the test to be compared to national norms. The student has 20 minutes to complete the multiple-choice questions and 35 minutes for the short tasks. Students then spend 55 minutes on the second section, which is made up of long and medium-length tasks. The third section requires 55 minutes to complete and covers both short and long tasks. Short tasks are constructed-response items; medium and long tasks are extended-response items that require detailed answers.

The NSRE is a criterion-referenced test that sets levels of constructed-response performance in three areas: Skills, Concepts, and Problem Solving. The performance levels (achieved with honors, achieved the standard, nearly achieved, below the standard, and little achievement) are derived from national Standards developed by a conglomerate of assessment-related organizations (Wiley & Resnick, 1998). The content and process areas assessed include number and operations, geometry and measurement, algebra and function, mathematics skills, problem solving and reasoning, and mathematical communication.

Other studies have noted (Begle, 1973; Reys et al., 2003; Riordan & Noyce, 2001) that prior achievement in mathematics is an important predictor of student achievement. As is often the case, different districts administered different mathematics tests to students. At the middle school level, the ITBS, Northwest Achievement Level Test (NALT), Minnesota Comprehensive Assessment (MCA), Metropolitan Achievement Test (MAT7), and Terra Nova were used. In three of the districts, a subsample of students had scores on two of these mathematics tests. We only had access to total scores for these tests.

Because we wished to have a single (common) prior mathematics achievement score for each student, we began by examining the characteristics of these tests. We felt there was considerable overlap in the objectives, content, and format of the various tests, providing strong evidence for treating these tests as reflecting a single construct of mathematics proficiency.

Next we examined available correlation evidence of student performance on these tests. A subset of students had scores on two prior mathematics tests, producing a Pearson correlation of .79 between the MCA and ITBS ($N = 172$), .50 for the MAT7 and MCA ($N = 108$), and .69 for the MCA and NALT ($N = 302$). These correla-

tions provided empirical support for the conclusion that these tests assess a common construct of mathematics proficiency.

We also fitted multiple regression models to the Stanford 9 student data within each district using the available prior mathematics measures as a predictor, along with other student-level predictors like gender, attendance, SES, and native versus nonnative English speaker. The results of these analyses produced similar percentages of explained variance attributable to prior mathematics achievement when the effects of the other predictors were held constant.

The logical and empirical analyses above led us to treat the different prior achievement measures as commensurable. We then created a combined, across-district prior achievement measure by treating the NCEs associated with these varied measures as equivalent. For example, students with an NCE of 70 on any of these tests were assumed to possess approximately the same mathematics proficiency. A plausible criticism of this assumption is that it suggests more precision than is justified. That is, students with the same NCE score from different mathematics tests probably have similar prior knowledge but perhaps less than that implied by having the same NCE score.

To examine the effect of using the NCE metric of 1–99 for the combined measure versus another representation of this metric, some of the statistical analyses reported below were also performed using a polytomized form of the NCEs. NCEs were replaced by a value indicating student membership in a particular decile of NCE performance. For example, the NCE performance of students in the sixth decile exceeded approximately 60% of the remaining students but was lower than approximately 40%. The similarity of results in using the combined prior mathematics scores in their original NCE metric of 1–99, versus replacing these scores with a value reflecting a student's decile membership, suggests that our findings are not overly sensitive to the metric of the combined prior mathematics achievement variable (i.e., our findings are similar regardless of whether the original [combined] prior mathematics scores or their deciles are used).

Student and Classroom Samples

Students in 43 *Standards*-based classrooms were tested. The teachers in these classrooms had participated in three types of professional development provided through the $(MASP)^2$ LSC. (NSF mandated 130 hours of targeted professional development experience for all LSC teachers.) First, teachers participated in 2 weeks (80 hours) of summer training related to a particular *Standards*-based curriculum. This usually entailed working through activities of the curriculum while teaching strategies were modeled by experienced $(MASP)^2$ staff members. In-depth consideration of some of the mathematics underlying these activities was also provided by $(MASP)^2$ staff. Second, during the school year teachers participated in sessions (30 hours) focused on more general topics such as facilitating cooperative learning in mathematics classrooms, current research on the brain and its implications for mathematics classroom instruction, and meetings with teachers and administrators to discuss administrative issues and with their counselors relating to the scheduling

of students. Last, $(MASP)^2$ employed district personnel experienced in the curriculum to serve during the school year as mentors to teachers newly implementing *Standards*-based curricula. The 20-hour mentoring component consisted primarily of classroom observations followed by one-on-one debriefings and, in some cases, demonstration lessons. Teachers requested professional development beyond year 1 that was designed for the next level of the curriculum to be used. Higher Education Eisenhower funds provided for additional professional development in years 2 and 3. Middle-grades teachers in this study had completed an average of 162 professional development hours over a 3-year period.

The number of students that could be tested within a district was limited by the cost of test administration and grading and was allocated in proportion to the number of mathematics teachers in the various schools. District administrators, in collaboration with project personnel, selected a number of classrooms whose totals contained the requested number of students. Administrators were provided direction to purposively select classes containing students who were representative of the entire spectrum of the student body in that school. Administrators were also asked to select classes that were perceived to have a high fidelity of curriculum implementation; this judgment of fidelity was confirmed by the assigned mentor teacher. Assessing the fidelity of implementation can be difficult for a variety of reasons. Reys and colleagues (2003) found that teaching strategies consistent with the reform text author's suggestions were used from 10% to 90% of the time in the middle-grades *Standards*-based classrooms they observed.

As part of our effort to determine implementation levels, the assigned $(MASP)^2$ mentors for teachers provided their informal assessment of their teacher's implementation level. This assessment was done for all teachers. Another effort was to send selected mentors into a sample of classrooms using an instrument, "Core Evaluation Classroom Observation Protocol" (Lawrenz, Huffman, & Appeldoorn, 2002), developed under another NSF grant and modeled after "Inside the Classroom: Observation and Analytic Protocol" (Weiss, Pasley, Smith, Banilower, & Heck, 2003). The authors of the protocol trained the observers in its use. These observers then practiced using videos of various classrooms. Eleven of the 43 classrooms tested in our sample were observed using this protocol. On a Likert scale of 1 to 5, with 5 representing exemplary implementation, all observed classrooms fell in the 3 to 5 range. Finally, we relied on knowledge of individual teacher's classrooms possessed by district evaluation personnel and district curriculum directors. While any set of efforts cannot guarantee full fidelity, these efforts and associated evidence of their success indicate that the curricula were implemented in our test classrooms at a satisfactory level.

RESULTS

Descriptive Summaries of District Performance

As shown in Table 2, students across all five districts performed above the national norm on the Problem Solving subtest. Only the large urban district had a

mean below 50 on the Open Ended subtest, but for the Procedures subtest four of the five districts scored below the national mean. Recall that this subtest of the Stanford 9 covers the four basic operations with whole numbers, integers, and fractions within purely symbolic settings and sometimes with one-step word problems solved without calculators.

Table 2
Stanford 9 Open Ended, Problem Solving, and Procedures Sample Sizes, Means and Standard Deviations by District

	Subtest		
	Open Ended		
District	*N*	Mean	*SD*
A	579	58.7	18.3
B	385	47.2	24.8
C	113	57.3	17.7
D	161	63.4	16.1
E	128	76.5	16.8
	Problem Solving		
A	584	62.9	19.3
B	399	52.6	23.1
C	120	60.3	15.8
D	162	63.3	20.0
E	123	84.9	14.8
	Procedures		
A	565	37.1	16.2
B	386	36.7	20.3
C	120	49.9	18.0
D	158	40.1	15.4
E	123	59.2	17.8

The results for District E, the district that also administered the Mathematics portion of the NSRE, a subsample of the Stanford 9, are in Table 3. The results show that 86% (57 + 29) of the students tested in District E achieved or exceeded the mathematical skills standard, whereas 33% (11 + 22) of the students at the national level performed at this level. The above-average performance of District E students who used CMP extends also to the Mathematical Concepts and Mathematical Problem Solving subtests. Within Mathematical Concepts, 71% of students achieved or exceeded the standard as compared to 20% nationally. On Mathematical Problem Solving, 44% achieved or exceeded the standard, whereas only 11% did so nationally.

Although these results come from an advantaged suburban district, it should be kept in mind that the norming group for the NSRE instrument comes from the Northeastern part of the United States, an area that typically has higher scores than other geographical areas (National Center for Education Statistics, 2003).

Table 3
Percentage and Number of Students Meeting New Standards Achievement Levels—District E

	Mathematical skills		Mathematical concepts		Mathematical problem solving	
Achievement levels	Percentage of students (*N*)	National norm %	Percentage of students (*N*)	National norm %	Percentage of students (*N*)	National norm %
Achieved the standard with honors	57% (137)	11%	29% (69)	6%	2% (5)	0%
Achieved the standard	29% (70)	22%	42% (101)	14%	42% (101)	11%
Nearly achieved the standard	9% (21)	24%	19% (45)	17%	17% (41)	14%
Below the standard	4% (10)	24%	6% (15)	26%	30% (73)	27%
Little evidence of achievement	1% (2)	19%	4% (10)	37%	8% (20)	48%

There were sharp differences among some of the districts in student characteristics. Figure 1 shows that District B had just under 70% non-White students and the remaining districts approximately 20% or less. Similarly, District B had approximately 21% of its middle school students classified as nonnative English speakers, whereas the remaining districts had values ranging from 0% to 6%. The percentage of special education students varied from 2% to 12%, with the highest value attached to a suburban district. SES showed comparatively more variability across all districts. One district had more than 60% of its middle school students eligible for free or reduced lunch (low SES on Figure 1), two districts had 18% to 20% eligible, another about 15%, and in one district 3% of the middle school students were eligible.

Average performances for the Stanford 9 mathematics subtests and prior mathematics achievement are displayed by district in Figure 2 and show considerable variability. The outcome showing the greatest variability was Problem Solving, with 27 NCE points separating the highest and lowest performing school districts. The Open Ended subtest produced almost as much variability (25), followed by Procedures (15) and prior mathematics achievement (18). Collectively, this variation suggests that there are large differences in average mathematics proficiency and prior mathematics knowledge across the districts. However, as shown in the analysis to follow, these differences shrink when various demographic variables are statistically partialled out.

There was also evidence of substantial variability in average mathematics performance at the classroom level. Figure 3 shows the classroom means for the Stanford

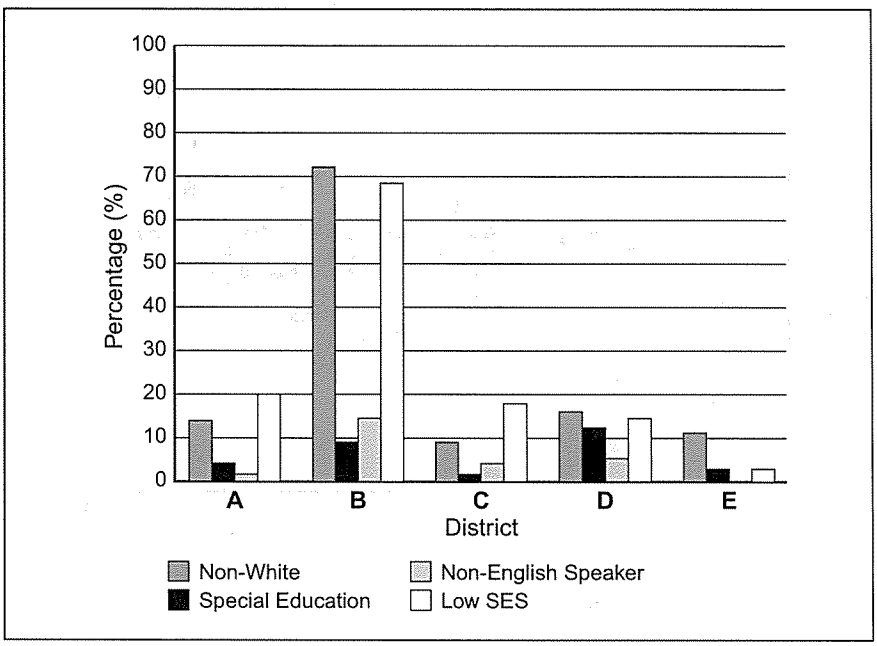

Figure 1. District demographic data.

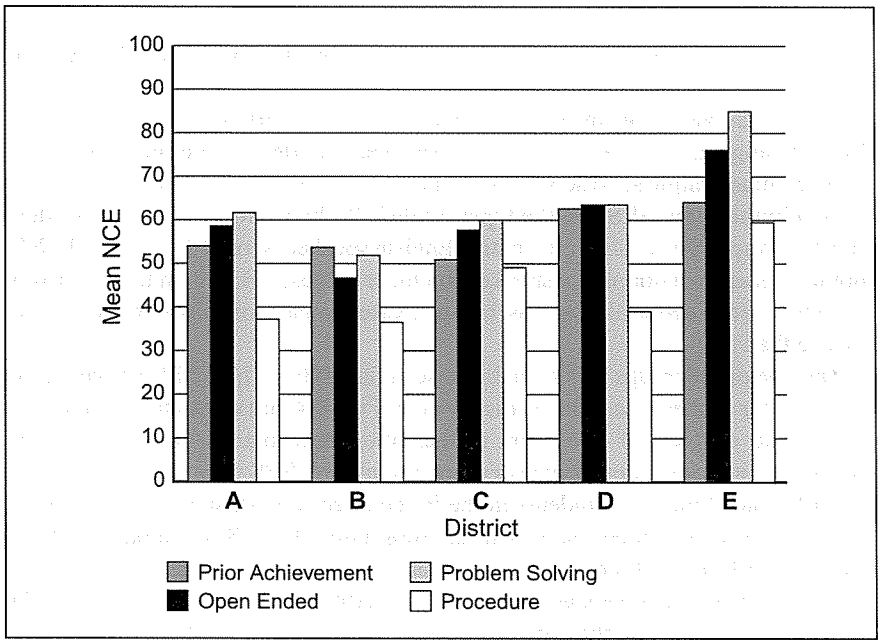

Figure 2. District achievement data

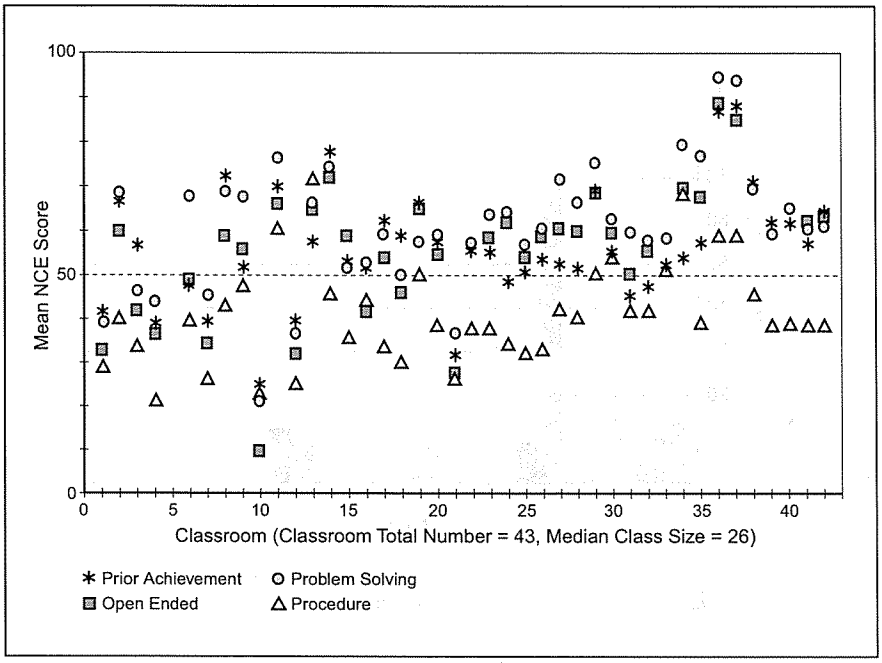

Figure 3. Mean NCE Open Ended, Problem Solving, Procedure, and Prior Achievement by individual classroom.

9 subtests. Similar classroom performance on these subtests would produce approximately a straight line.

There was variation in average mathematics performance across SES and English language (native vs. nonnative speaker). Students not eligible for free or reduced lunch (high SES) scored on average 17, 17, and 7 NCE points higher than those eligible (low SES) on the Open Ended, Problem Solving, and Procedures subtests, respectively. Similarly, native English speakers scored 26, 23, and 11 NCE points higher than nonnative speakers on these subtests. Based on the descriptive statistics, English-speaker status had a greater impact on mathematics performance than SES.

The role of subgroups generated by these variables (e.g., high SES/native English speaker) for each ethnic group is displayed in Figure 4 and shows that their combination had differential effects on mathematics performance. In general, native speakers outperform nonnative speakers in the same SES group. With the exception of Asian American students in the Procedures subtest, subtest scores largely mimic prior achievement scores in all subgroups. Low SES, nonnative White students performed at the lowest level (note small *N*).

Additional descriptive statistics, including effect sizes, appear in Table 4. The effect sizes help to quantify differences apparent in Figure 4. Following Hedges and Olkin (1985, p. 78), effect sizes were computed as the difference in two

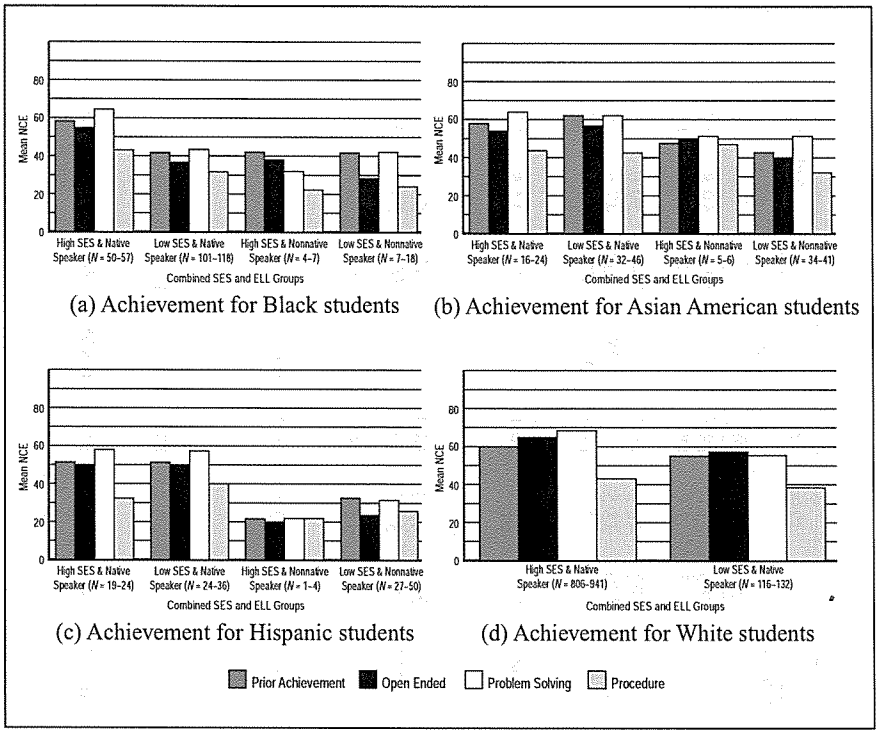

Figure 4. Achievement by ethnicity.

means divided by the estimated pooled standard deviation of the difference. For example, low SES students scored on average .88 standard deviations lower on the Open Ended subtest than high SES students.

Several patterns are apparent among the effect sizes. One is the lower average performance for low SES, non-White, urban, nonnative English speakers, and special education students. These effect sizes ranged from –.12 to –1.24 standard deviation units. The only demographic variables not showing a statistically significant effect size were gender (not reported) and Asians versus Whites. With one exception, each of the remaining 23 effect sizes was statistically significant. Another pattern apparent in Table 4 is that Problem Solving subtest NCE means were higher than those for the Open Ended subtest across all subgroups; the Procedures subtest produced the lowest NCEs for all subgroups.

In sum, there is ample evidence of variability among the school districts in student demographic characteristics and in average mathematics performance. The suburban districts, which included the district on the urban-suburban boundary, tended to have far smaller percentages of non-White, low SES, and nonnative English speakers. The distribution of special education students showed little relationship with the suburban or urban location of districts.

Table 4
Descriptive Data and Effect Sizes for Various Subgroups

Group		Open Ended				Problem Solving				Procedures				Prior Achievement			
		N	Mean	*SD*	Effect size	*N*	Mean	*SD*	Effect size	*N*	Mean	*SD*	Effect size	*N*	Mean	*SD*	Effect size
SES	**High**	982	62.9	18.5		985	67.1	20.0		966	43.0	18.8		1044	59.8	17.9	
	Low	444	45.3	23.0	–0.88*	463	50.1	20.9	–0.84*	445	35.1	18.5	–0.42*	342	48.5	19.2	–0.62*
Ethnicity	**White**	1004	63.3	18.2		1011	66.4	19.9		992	42.5	18.3		1068	59.6	17.9	
	Black	204	42.6	24.1	–1.07*	208	49.5	22.0	–0.84*	200	35.5	20.0	–0.38*	169	47.6	20.4	–0.66*
	Asian	118	50.9	18.5	–0.65*	120	59.4	17.8	–0.36*	115	40.7	18.8	–0.12	87	53.9	17.1	–0.31*
	American																
	Hispanic	111	39.7	23.0	–1.24*	120	46.3	23.4	–1.04*	115	32.3	19.1	–0.57*	71	44.5	17.2	–0.83*
Location	**Suburban**	981	61.6	18.7		989	65.4	20.0		966	42.0	18.3		1033	58.3	17.7	
	Urban	456	48.8	24.6	–0.28*	470	54.3	23.1	–0.23*	456	37.7	20.3	–0.62*	362	53.5	21.3	–0.58*
Language status	**Native English**	1315	59.8	20.4		1326	64	20.7		1294	41.6	18.7		1315	58.1	18.6	
	Non-native English	122	33.4	20.1	–0.91*	133	40.3	19.2	–0.90*	128	29.1	18.3	–0.68*	79	39.9	13.8	–1.03*
Special Education	**Non-special Ed.**	1356	58.5	21.1		1374	63	21.3		1340	41.2	19.0		1310	58.1	18.2	
	Special Ed.	81	41.3	23.4	–0.73*	85	43	19.2	–0.95*	82	29.6	14.7	–0.83*	84	39.8	20.4	–1.00*

*$p < 0.05$

HLM Analyses of Student and Classroom Data

The Stanford 9 mathematics subtests data were analyzed with HLM following the methods described in Raudenbush and Bryk (2002). Treating students as clustered within classrooms permitted within-classroom dependency among student mathematics test scores to be modeled, and allowed student- and classroom-level questions to be answered simultaneously. This in turn helped to ensure more credible statistical test results than would ordinarily be possible with traditional regression modeling.

Student-level regression models containing prior mathematics achievement, attendance, SES, and gender were fitted to each middle school classroom's data. Because of missing data, the total number of students was reduced to approximately 1050–1200, depending on the outcome. For each outcome (Open Ended, Problem Solving, Procedures), three models were fitted. First an unconditional model of the form

$$\begin{aligned} Y_{ij} &= \beta_{0j} + r_{ij} \qquad (1) \\ \beta_{0j} &= \gamma_{00} + u_{0j}, \end{aligned}$$

was fitted, in which Y_{ij} is the mathematics score of the ith student in the jth classroom, β_{0j} is the average mathematics score (intercept) for the jth classroom, γ_{00} is the average mathematics performance across classrooms, r_{ij} is a student-level residual, and u_{0j} represents the unique effect of the jth classroom. The unconditional model results tell us whether average outcomes differ across classrooms. Next, we fitted a student-level model of the form

$$\begin{aligned} Y_{ij} = \beta_{0j} &+ \beta_{1j}(\text{attendance}_{1ij} - \overline{X}_{1j}) \\ &+ \beta_{2J}(\text{SES}_{2ij} - \overline{X}_{2j}) \\ &+ \beta_{3j}(\text{gender}_{3ij} - \overline{X}_{3j}) \qquad (2) \\ &+ \beta_{4j}(\text{prior}_{4ij} - \overline{X}_{4j}) + r_{ij}, \end{aligned}$$

in which β_{1j} is the student level slope capturing the effect of attendance on mathematics with other predictors held constant, $\overline{X}_{1j}$ is the mean attendance in the jth classroom, and prior_{4ij} is the prior mathematics knowledge predictor. We also tested whether slopes for the predictors varied across classrooms.

Classroom-level predictive models for intercepts (average mathematics achievement) and, where appropriate, slopes, were then developed. That is, for instances when the effect of a student-level predictor like SES on a Stanford 9 subtest varied across classrooms, we constructed a predictive model to try to explain variation in these slopes with classroom-level predictors. Key classroom predictors included Class SES (percentage of students eligible for free or reduced lunch in a classroom) and average prior mathematics knowledge in a classroom. Other classroom predictors that we examined were the effect of different concentrations of various ethnic groups, nonnative English speakers, special education students, and female students in a classroom. Average classroom attendance and predictors capturing school district membership were also used. Preliminary analyses showed that average class-

room attendance and the percentage of female students in a classroom could be removed because they did not contribute to explaining variation in classroom mathematics means (intercepts) or slopes. These analyses also indicated that differences across the five districts could be captured by a single predictor indicating whether or not the classroom was in the urban district.

The classroom model for intercepts fitted in most analyses was

$$\begin{aligned}\beta_{0j} = \gamma_{00} &+ \gamma_{01}\,(\text{Class SES}_{1j} - \overline{W}_1) + \gamma_{02}\,(\text{Class Black}_{2j} - \overline{W}_2) \\ &+ \gamma_{03}\,(\text{Class Asian}_{3j} - \overline{W}_3) + \gamma_{04}\,(\text{Class Hispanic}_{4j} - \overline{W}_4) \\ &+ \gamma_{05}\,(\text{Class nonnative English speakers}_{5j} - \overline{W}_5) \\ &+ \gamma_{06}\,(\text{Class special education}_{6j} - \overline{W}_6) \\ &+ \gamma_{07}\,(\text{district}_{7j} - \overline{W}_7) + \gamma_{08}\,(\text{professional development}_{8j} - \overline{W}_8) \\ &+ \gamma_{09}\,(\text{prior}_{9j} - \overline{W}_9) + u_{0j},\end{aligned} \tag{3}$$

in which where γ_{01} is the classroom-level slope capturing the effect of class SES (percentage of low SES students) on average mathematics performance, $\overline{W}_1$ is class SES averaged across classrooms, Class Black is the percentage of Black students in a classroom, and so on. The percentage of White students in a classroom was not used as a predictor because doing so would have introduced a dependency among the ethnicity predictors.

In a few cases, student-level slopes varied randomly across classrooms, and models similar to those for intercepts tried to account for this variation. The deviance test described in Raudenbush and Bryk (2002, pp. 59–61) was used to test for model fit, allowing us to discriminate among models with more or less explanatory power. Model-fitting was followed by extensive model-checking to help to ensure validity of inferences. Cases in which normality, homoscedasticity, or linearity appeared to be suspect were examined in detail, and various remedies (e.g., modeling unequal classroom variances) were employed. The analyses reported below are based on fitted models in which these assumptions appeared to be at least approximately satisfied.

An initial difficulty with several of the classroom-level predictor variables, such as the percentage of nonnative English speakers in a classroom, was their ragged and discontinuous nature. For example, about 40% of the classrooms had less than 3% nonnative English speakers, another 25% of the classrooms had values between 5% to 7%, and, at the other end of the distribution, 10 of the classrooms had values ranging between 14% to 96%. We explored various transformations of these variables with the goal of representing their variation in a more succinct form, and polytomized the distributions into quartiles (see Table 5). Thus, each of the classroom predictors above was transformed to a scale in which each was represented with four values corresponding to the Table 5 quartiles.

The HLM5 software (Raudenbush, Bryk, Cheong, & Congdon, 2001) does not permit missing data when fitting the student level model in equation (3) to each classroom's data, employing listwise deletion to ensure that only complete student datasets are analyzed. This raises the possibility of biased statistical findings due

Table 5
Definitions of Classroom Level Predictors

Variable	Quartile	*N*	Range (R)
Class SES	1	10	$R \leq 15.38\%$
	2	12	$15.38\% < R \leq 23.53\%$
	3	11	$23.53\% < R \leq 69.57\%$
	4	10	$R > 69.57\%$
Class English language status	1	15	$R = 0\%$
	2	7	$0\% < R \leq 5.36\%$
	3	11	$5.36\% < R \leq 14.29\%$
	4	10	$R > 14.29\%$
Class special education	1	15	$R = 0\%$
	2	7	$0\% < R \leq 4.17\%$
	3	11	$4.17\% < R \leq 10.26\%$
	4	10	$R > 10.26\%$
Class Black	1	19	$R \leq 5.27\%$
	2	9	$5.27\% < R \leq 10.71\%$
	3	11	$10.71\% < R \leq 33.3\%$
	4	11	$R > 33.3\%$
Class Asian	1	15	$R = 0\%$
	2	7	$0\% < R \leq 3.85\%$
	3	11	$3.85\% < R \leq 8.7\%$
	4	10	$R > 8.7\%$
Class Hispanic	1	17	$R = 0\%$
	2	3	$0 < R \leq 3.33\%$
	3	12	$3.33\% < R \leq 9.52\%$
	4	11	$R > 9.52\%$
Professional development hours	1	6	$R = 65$
	2	8	$65 < R \leq 130$
	3	15	$130 < R \leq 158$
	4	14	$R > 158$

to omitting cases showing missing data systematically differing from cases that provided data. Earlier results suggested that the percentages of missing data were generally similar across demographic groups and that average Stanford 9 scores were not seriously affected by the presence of missing data. We continued to explore possible effects of missing data by focusing on HLM5's use of listwise deletion. We also fitted the model in equation (3) to the student Stanford 9 data using the AMOS5 (SmallWaters Corporation, 2003) structural equation modeling software (AMOS5 does not perform hierarchical linear modeling). Rather than employing listwise deletion, AMOS5 uses whatever data students provide in estimating regression parameters for a classroom. Although HLM5 and AMOS5 use different methods to estimate parameters (Ordinary Least Squares and Full Information Maximum Likelihood, respectively), they should produce similar estimated parameters except when class sample sizes are quite small, which we did not think was a serious problem given the median classroom size of 26 in our data.

For the Open Ended subtest, we then compared the attendance, SES, gender, and prior slopes estimated using HLM5 and AMOS5 for each classroom. Similar slopes

suggest that HLM5's use of listwise deletion did not have a large effect on the analysis. We summarized the differences by subtracting the HLM5 estimated slope for each student predictor from that produced by AMOS5, and then computed the median of these differences. For example, the median difference between the HLM5 and AMOS5 estimated slopes for attendance was .16, meaning that, on average, there was a very small difference among the estimated attendance slopes (< 1 NCE point).

The median difference between the HLM5 and AMOS5 estimated slopes for the gender and prior predictors was –.004 and –.04, respectively, again suggesting little impact of HLM5's use of listwise deletion. For SES slopes, the median difference was noticeably larger at –3.6, meaning that, on average, the HLM5 estimated SES slopes under listwise deletion were smaller than those estimated by AMOS5 by 3.6 NCE points. Put another way, the SES slopes produced by HLM5 on average underestimated the effect of this predictor on Open Ended scores. It is hard to pinpoint the source of this dampening effect in HLM5 other than its use of listwise deletion, but the consequence is that the impact of SES on Open Ended scores is probably greater than suggested by the findings reported below.

We performed the same analyses with the Problem Solving and Procedures scores, and found little difference between the HLM5 and AMOS5 estimated slopes. For example, the median difference for SES slopes for Problem Solving was –.84, indicating that HLM5 on average underestimated SES slopes by less than 1 NCE point; for Procedures the median difference for SES slopes was –.07. On the whole, these results suggest that the only bias among the estimated slopes as a result of HLM5's use of listwise deletion is for SES for the Open Ended subtest. This in turn suggests that HLM5's use of listwise deletion did not have much effect when estimating the impact of attendance on subtest scores. We interpret the HLM results presented below accordingly.

The HLM cross-sectional results are summarized in Table 6 using a Type I error rate of $a = .05$ for each statistical test. Several general findings emerged across the Stanford 9 subtests. First, there was substantial between-classroom variation in the Open Ended, Problem Solving, and Procedures subtest scores with variation between classroom means of 34%, 38%, and 34%, respectively. Second, at the student level (level 1), prior mathematics knowledge was a statistically significant predictor in every model, although its effect expressed in NCE units tended to be modest (< 1). Student-level SES was a statistically significant predictor of Open Ended scores, and its effect is probably underestimated. Student SES was also a significant predictor of Problem Solving scores, producing a moderate effect. Gender was never a statistically significant student-level predictor, and student attendance was only occasionally a significant (and weak) predictor of mathematics performance.

Third, results for the Procedures subtest were somewhat different from those for the others in that there were fewer significant effects. Fourth, there was evidence of a difference in average classroom performance (level 2) between the large urban district and the remaining districts even when demographic variables (e.g., SES and

Table 6
Results of HLM

Dependent variable	HLM results
Open Ended	There was significant between-classroom variation in means (34%). 1. There were significant within-classroom student effects for SES, which produced an average slope of –3.38 and is likely an underestimate, prior mathematics knowledge (.72), and attendance (.27). Classroom Open Ended means and prior knowledge slopes varied significantly across classrooms. 2. There was a significant difference between urban and suburban districts (–7.8) favoring suburban districts. This effect accounted for approximately 30% of the variance in classroom Open Ended means. 3. Concentrations of low SES students, expressed in quartiles, was a significant predictor of classroom Open Ended means (–3.14), meaning that shifting from the first quartile (£ 15.38% eligible for a free or reduced price lunch) to the second quartile (15.38% < R £ 23.53%) produces an approximate decline of 3 points in classroom Open Ended means. Classroom prior mathematics knowledge (.72) was also a significant predictor of classroom means, along with special education (–1.57) and Asian students (1.28). 4. Significant variation in classroom Open Ended means remained unexplained.
Problem Solving	1. There was significant between-classroom variation in means (38%). 2. There were significant within-classroom effects for SES (–5.1) and prior mathematics knowledge (.63). Intercepts and slopes for all student level predictors varied across classrooms. 3. There was a strong district effect (–11.4) for classroom means, accounting for approximately 19% of the variance in classroom means. 4. Concentrations of Asian students in a class (2.58), expressed through quartiles, was also significant. Classroom prior mathematics knowledge (.81) was a significant predictor of classroom Problem Solving means. 5. The slope capturing the effect of concentrations of special education students on prior mathematics knowledge slopes was .09, meaning that increases in the percentages of these students (expressed through quartiles) in a classroom was associated with a slightly greater impact of prior mathematics knowledge on Problem Solving scores. Second, concentra tions of low SES students had a significant effect on attendance slopes (.79), meaning that increases in this variable tended to be associated with classrooms in which the effect of attendance on Problem Solving scores was weaker. Increasing concentrations of nonnative speakers in a classroom was also a significant predictor of attendance slopes (–.38), meaning that increases in nonnative speakers exacerbated the effect of attendance on Problem Solving scores. 6. Significant variation in classroom Problem Solving and the gender, SES, and prior mathematics knowledge slopes remained unexplained.
Procedures	1. There was significant between-classroom variation in means (34%). 2. The only significant within-classroom predictor was prior mathematics knowledge (.46). Procedures means showed significant variation within classrooms. 3. The largest classroom effect for classroom means was the concentration of special education students (–3.9), which accounted for approximately 44% of the variability in classroom Procedures means. Classroom prior mathematics knowledge (.53) was also a significant predictor. Classroom means continued to show significant variability. There were no significant between-district effects.

prior mathematics achievement) were held constant for the Open Ended and Problem Solving subtest analyses. Fifth, there was no evidence of contextual effects for SES or prior mathematics knowledge (Raudenbush & Bryk, 2002, pp. 149–141), meaning, for example, that the effect of SES on mathematics achievement was the same at the student level and classroom level.

Other results were specific to particular subtests. For the Open Ended subtest, student-level SES was a significant predictor with an average slope of –3.36, meaning that, with the other predictors held constant, students eligible for free or reduced lunch tended to score slightly more than 3 NCE points below those not eligible. There was a strong district effect in classroom means of –7.8, meaning that urban classrooms tended to score on average about 8 NCE points lower than suburban classrooms. Other classroom effects were for SES (–3.14), prior mathematics knowledge (.72), and the concentration of special education (–1.57) and Asian (1.28) students. The latter finding means that, with other predictors held constant, increasing the concentration of Asian students by one quartile (e.g., using Table 5, from 0% to 3.85%) in a classroom was associated with average increases in Open Ended means of approximately 1.28 NCE points.

For Problem Solving, SES again had a pronounced effect at the student level (–5.1), whereas at the classroom level there was an even larger difference between urban and suburban classrooms (–11.7). Another statistically significant classroom effect was for the concentration of Asian students in a classroom (2.58), meaning that a one quartile increase in this variable was associated with an increase in classroom Problem Solving means of 2.58 NCE points.

Three small cross-level interaction effects for Problem Solving also emerged. The slope capturing the effect of special education on the prior knowledge slopes was .09, meaning that increases in the concentration of special education students in a classroom tended to be associated with a (slightly) greater impact of prior mathematics knowledge on Problem Solving scores. Second, increasing concentrations of students eligible for free or reduced lunch in a class had a significant effect on attendance slopes (.79), meaning that increases in this variable tended to be associated with classrooms in which lower attendance was associated with lower Problem Solving scores. Third, increasing concentrations of nonnative English speakers in a classroom was also a significant predictor of attendance slopes (–.38), meaning that increasing the concentration of nonnative speakers in a classroom tended to exacerbate the effect of attendance on Problem Solving scores.

For Procedures, only prior mathematics knowledge was a significant predictor at the student level (.46), whereas at the classroom level the concentrations of special education students (–3.9) and prior mathematics knowledge (.53) were significant predictors of classroom means. There were no between-classroom effects for Procedures.

In sum, the HLM cross-sectional results for the Open Ended and Problem Solving subtests were similar. Student's prior mathematics knowledge was a consistent predictor of mathematics proficiency although its effect was modest, but student SES was a stronger predictor, and results presented above relating to the effect of

HLM5's use of listwise deletion suggests its impact on Open Ended scores may be even more substantial. Strong differences between the urban and suburban classrooms also emerged for the Open Ended and Problem Solving subtests (–7.8 and –11.7 points, respectively), along with other, smaller, classroom effects such as the concentration of Asian students and students eligible for free or reduced lunch. For the Procedures subtest, only prior mathematics knowledge and the concentration of special education students in a classroom were significant predictors.

SUMMARY

Descriptive Data

The descriptive results reported in Table 2 show that on the Stanford 9 subtests, which are designed to measure traditional content, students enrolled in *Standards*-based mathematics curricula performed above the NCE national mean of 50 on the Open Ended and Problem Solving subtests. Students were generally below the NCE mean of 50 on the Procedures subtest. These results suggest that students are learning traditional topics but are also lacking in paper-and-pencil procedural skills. This result parallels the findings of other studies of a similar nature (Schoen et al., 2003; Senk &Thompson, 2003).

Although students were, on the whole, performing at or above expectations on two of the three Stanford 9 subtests, the performance of various subgroups differed sharply. Students in the high-SES and native-English-speaker groups on average scored substantially higher than those in the low-SES and nonnative-English-speaker groups. Among the ethnic groups represented in the sample, White students uniformly produced the highest averages, with Black and Hispanic students scoring significantly lower. There were no differences among male and female students on any of the Stanford 9 subtests. There was substantial variation in average Stanford 9 performance among classrooms, however.

One set of district results is noteworthy. The results reported in Table 3 show that students enrolled in the *Standards*-based mathematics curricula in one of our districts far outperformed national norms. The NSRE, which is more closely aligned to the NCTM *Standards*, purports to measure conceptual development, problem solving, and traditional skills and admits the use of calculator technology. District E was the only one that elected to use the NSRE and was the highest achieving and most affluent of our districts.

HLM Across-District

The HLM results for the Stanford 9 Open Ended and Problem Solving subtests indicated that prior mathematics knowledge was a consistent predictor of mathematics performance at the student and classroom levels, although the effect was modest in size. SES was a relatively strong predictor of mathematics performance at both the student and classroom levels, with higher SES linked to higher performance. Whether a classroom was in an urban or suburban school also impacted

achievement, with strong differences favoring suburban classrooms emerging. Higher student and classroom prior mathematics scores were also associated with higher Procedures scores.

There is long-standing documentation of the effect of prior achievement on future performance (e.g., Begle, 1973). Our analysis again suggests that prior mathematics achievement is a significant predictor (though small, less than 1 NCE point on average) of student achievement on all three Stanford 9 subtests. This finding underscores the position that achievement gaps need to be seriously addressed early in students' academic careers.

Boaler (2003) suggested that gaps between low and high SES decrease over time when students are involved in an open-ended project-based curriculum. The present study found results favoring the high SES group on both the Open Ended and Problem Solving subtests of the Stanford 9. This effect was moderate, consisting of 3.38 NCE points on the Open Ended subtest and 5.1 NCE points on the Problem Solving subtest. Student achievement on the Procedures subtest, however, was not significantly associated with student level SES. Our explanation for this finding is that the more context-bound and language intensive Problem Solving and Open Ended assessments may be more difficult for students of low SES for reasons described by Lubienski (2000), whereas the Procedure subtest relates only to paper-and-pencil algorithmic calculations.

High percentages of nonnative speakers, low SES students, and high percentages of minority students are commonly associated with urban schools (Grant & Tate, 2001). When these independent variables were accounted for in the HLM model employed, large significant differences between urban and suburban classrooms on the Open Ended and Problem Solving subtests remained. There apparently are other student or classroom predictors associated with urban and suburban schools that were not accounted for in our model (e.g., class size, degree of parental involvement).

Much research has been conducted on the importance of school and classroom culture when considering factors that affect student achievement (Finnan, 2000; Pang, 2003). Examining the effect of classroom level predictors on student achievement enables us to describe this classroom culture quantitatively. For example, as the percentage of special education students increases in the classroom, this tends to increase the association between prior mathematics knowledge and Problem Solving. That is, as the percentage of special education students in the classroom increases, prior knowledge plays a bigger role in predicting student achievement on Stanford 9 subtest scores.

BROADER IMPLICATIONS

These results indicate that those middle-grades students in the five districts discussed, each of whom have been involved with either the MathThematics or the Connected Mathematics Project (CMP) *Standards*-based mathematics curricula for 3 consecutive years, demonstrated achievement patterns on the Open-Ended and

Problem-Solving subtests of the Stanford 9 that exceeded the means of the national samples upon which the Stanford 9 was normed. Results on the Stanford 9 Procedures subtest showed that four out of five of the districts scored below the mean of the national norming group. Given the decreased amount of attention to algorithmic development in these curricula, these results may reflect a time-on-task result.

The results are very promising on the NSRE, with roughly 2.5 to 4 times the percentage of students in our sample meeting or exceeding the standards relating to Mathematical Skills, Mathematical Concepts, and Mathematical Problem Solving when compared to the means of the national norming group. It must be noted that these results were from our highest achieving district, the only district in our sample that used take the NSRE.

Students in this study were not mathematically disadvantaged in the topics contained in the Open Ended and Problem Solving Stanford 9 subtests nor on the topical areas represented in the NSRE. Student performance on computational items has always been a problem area for students, teachers, parents, and educators. It seems that no one is ever fully satisfied with student performance in the procedural area. The evidence presented here does not suggest a resolution to this concern.

The Stanford 9 and the NSRE, although not the most "conservative" measures available, are primarily attuned to traditionally oriented content. There are topics that *Standards*-based students have studied that are either inadequately assessed or are absent altogether from the two assessments used here. There is published evidence from other studies suggesting that when a *Standards*-based curriculum is fully implemented with fidelity, students achieve at a rate that is significantly higher than in classrooms in which teachers simply sample, supplement, and significantly modify the *Standards*-based mathematics curriculum (Briars & Resnick, 2000; Senk & Thompson, 2003).

The HLM results document that when prior mathematics knowledge and several demographic variables were taken into account there continued to be significant achievement differences between the urban and suburban classrooms. There really are no surprises in our data relating to the impact of demographic variables on student achievement. That is, urban, low SES, nonnative speakers, and low levels of prior achievement are all associated with lower achievement levels on the standardized tests.

FUTURE DIRECTIONS

Consensus on Curriculum Content

There presently is no agreed upon core of essential mathematics for school mathematics programs. Indeed one of the primary reasons for the "math wars" is disagreement about the role of speed and accuracy with the manipulation of standard arithmetic and algebraic algorithms. The NCTM's *Standards* documents

(1989, 2000) have provided such a roadmap, but its implementation as embodied by the various NSF supported *Standards*-based curricula is a divisive issue. It is the relative balance between algorithmic mastery, applications, and understanding the more structural aspects of the subject (place value, rational number, proportionality, Cartesian space, etc.) that are the bones of contention.

Our experience has also suggested that there are operational vocabulary differences among the groups. For example, some colleagues in the mathematics department were inclined to call every type of mathematics activity "problem solving," whereas mathematics educators tend to differentiate problem solving from algorithmic activity. Semantic clarification would serve all parties well.

Education is a cooperative venture that requires support from a broad range of school and extra-school constituencies, including of course teachers, students, parents, school administrators, and counselors. Individuals who are external to the school, such as legislators, business representatives, curriculum developers, and others involved in teacher professional development including mathematics educators and mathematicians, also must be invited to the negotiating table. This was essentially what NCTM did over the past 20 or so years as preparation for the development of the two Standards documents. It is no small task to get everyone on the same page, given our traditional interest in maintaining local control over our schools and their curricula. It is not essential that there be unanimity of agreement, but it will be important that there be a substantial core of confluence. It is our belief that additional and more concerted efforts must be made to inform the uninformed and to negotiate the parameters of desired mathematical outcomes with the various parties, as mentioned above. This seeming impasse is ironic in that all parties ostensibly want the best possible mathematics education for all students.

These research results provide yet another platform for teachers, researchers, and the public to discuss what exactly is valued in a middle school mathematics curriculum. Can schools do it all? For all persons? What computational skills are basic? Who decides what mathematics society should value? What mathematical abilities does the average citizen need to function well in our society? Is there a separate set of computational skills that future mathematics-oriented students need for success? If so, what are they?

> Between 4 and 5 percent of an age group will major in Mathematics, Science, or Engineering, one of the traditional mathematics-intensive disciplines. This percentage has been fairly constant since the 1950's through both mathematics reform and "back-to-basics" movements. Majors grow or shrink by reapportioning students in this 5 percent group. (Mathematical Association of America, 2000, p. 3)

Can society afford to implement a curriculum whose primary purpose is to prepare the 4% to 5% who pursue mathematics-related careers?

Assessment Tools and Needs

Valid and reliable instruments that adequately measure the new content and processes inherent in *Standards*-based curricula are challenging to develop and may

be cumbersome to administer and time-consuming to score. The variety of instruments to be used in future No Child Left Behind assessments hold little promise in this regard, as they will of necessity focus on low-level factual knowledge and procedural skills. The next step in ongoing research efforts in this area should paint a portrait of the content and processes that students in *Standards*-based curricula learn that are above and beyond traditional mathematical topics considered at the grade levels of interest. In a parallel effort, it will be important to conduct studies that document the related situation or what traditional students are learning that *Standards*-based students are not. It will then be possible to ask and answer the question, "What kind of student mathematical outcomes do we value, and what are the dimensions of programs most likely to produce them?"

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Standards-based Mathematics Curricula and Secondary Students' Performance on Standardized Achievement Tests

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The current study examined the mathematical achievement of high school students enrolled for 3 years in one of three NSF funded *Standards*-based curricula (IMP, CMIC, MMOW). The focus was on traditional topics in mathematics as measured by subtests of a standardized achievement test and a criterion-referenced test of mathematics achievement. Students generally scored at or above the national mean on the achievement subtests. Hierarchical linear modeling results showed that prior mathematics knowledge was a significant but modest predictor of achievement, student SES had a moderate effect, and increasing concentrations of African American students in a classroom were associated with a stronger effect of attendance on achievement. No differences on the standardized achievement subtests emerged among the *Standards*-based curricula studied once background variables were taken into account. The two suburban districts providing data for the criterion-referenced test achieved well above the national norm.

Key words: Cross sectional studies, Integrated curriculum, Multivariate techniques, Program/project assessment, Secondary mathematics

The current study examined achievement patterns of high school students enrolled in *Standards*-based curricula, with a focus on traditional topics in mathematics as measured by two nationally normed achievement tests. *Standards*-based curricula are curricula that were funded from a solicitation of proposals through the National Science Foundation (NSF) in the early 1990s (RFP NSF 91-100). These curricula were designed to be aligned with the National Council of Teachers of Mathematics'

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(NCTM) *Curriculum and Evaluation Standards for School Mathematics* (1989). Five secondary level mathematics curricula were funded by NSF: Contemporary Mathematics in Context (CMIC or Core-Plus) (Coxford et al., 1998), Interactive Mathematics Program (IMP) (Fendel, Resnick, Alper, & Fraser, 1997), Math Connections: A Secondary Mathematics Core Curriculum (Connections) (Berlinghoff, Sloyer, & Hayden, 2000), Mathematics: Modeling Our World (MMOW or ARISE) (Garfunkel, Godbold, & Pollak, 1998), and SIMMS Integrated Mathematics: A Modeling Approach Using Technology (SIMMS) (Lott & Burke, 1996). Schools represented in this study used CMIC, IMP, or MMOW.

This study builds on and extends our existing understanding of student achievement in *Standards*-based programs in three significant ways. First, we examine the impact of published editions of these curricula on student understanding as contrasted to earlier studies using field-test versions. All students in our study had participated in three levels of one of three *Standards*-based curricula. A high school level was defined to be the content equivalent of a yearlong course taught 1 hour per day (commonly referred to as a Carnegie unit). The three curricula (IMP, CMIC, MMOW) share major similarities including the recurring integration of topics within grade levels and a decreased emphasis on procedures and algorithms. They also differ from each other in various ways; the length of the units, the reality of the contexts, and the emphasized content are a few examples.

Second, this research concerns the use of *Standards*-based curricula as part of district-wide curricula adoptions. By examining the achievement of students in district-wide adoptions, we gain insights into more realistic situations where teachers are required, and have not necessarily volunteered, to teach *Standards*-based curricula. This provides a more accurate overall picture of expected student achievement, since many earlier field-test teachers were volunteers and therefore cannot be considered typical of all teachers who will eventually be expected to implement a *Standards*-based curriculum. Third, in studying student achievement, we employ hierarchical linear modeling (HLM) to account for the variability between classrooms and the interdependency of students within the same classroom (Kilpatrick, 2003). HLM allows us to consider student and classroom results simultaneously and to examine the role of classroom-level variables such as a teacher's hours of professional development and the *Standards*-based curriculum that a student experienced with various background variables taken into account.

Kilpatrick (2003) recommended that the next generation of curriculum research should focus on comparing different *Standards*-based curricula against one another on dependent variables measuring common goals. Following this recommendation, this study sought to deepen our understanding of the achievement of secondary students in *Standards*-based programs vis-à-vis multiple standardized-test measures.

THEORY UNDERLYING STANDARDS-BASED CURRICULA

In 1978, the National Council of Supervisors of Mathematics (NCSM), under the leadership of Ross Taylor, produced a document that redefined the "basic skills of

school mathematics." This document departed from current thinking at the time in that it identified the basics in broad strokes rather than specific and "behavioral" objectives that focused on skills that were computational in nature. Its fundamental concepts were incorporated into the NCTM's Agenda for Action (1980). As a result of these efforts, problem solving, spatial visualization and awareness, mathematics as communication, technological literacy, and appropriate computational skills became the new basics for school mathematics.

In 1989, the NCTM published *Curriculum and Evaluation Standards for School Mathematics;* in 2000, NCTM published a revised document, *Principles and Standards for School Mathematics*. This second document expanded upon and updated the 1989 *Curriculum and Evaluation Standards* and received wide and enthusiastic acceptance for reflecting best practices and current thinking in the field. The 1989 and 2000 *Standards* documents received wide endorsement for the integrity of the process used in their design and implementation and in the vision of school mathematics embodied therein. Among the many professional organizations endorsing these documents were the American Mathematical Society, Conference Board of the Mathematical Sciences, Mathematical Association of America, Mathematical Sciences Education Board, Society for Industrial and Applied Mathematics, American Statistical Association, Association for Women in Mathematics, and the Association of State Supervisors of Mathematics.

The 1989 and 2000 NCTM *Standards* documents were both broadly based on various cognitively oriented theories about the development of mathematics knowledge and the nature of student learning. Dienes suggested that mathematics knowledge should include the structural relationships within mathematical concepts, complemented by attention to their applications in the real world. The learning of mathematics involves the "apprehension of such relationships together with their symbolization, and the acquisition of the ability to apply the resulting concepts to real situations" (Dienes, 1969, p. 19). Related theories certainly include Piaget's theory of cognitive development about which much has been written (Furth, 1970; Ginsburg & Opper, 1988; Meece, 1997; Piaget, 1970). Vygotsky's sociocultural theory (Kozulin, 1986; Tudge & Winterhoff, 1993) and Dewey's positions (Dewey, 1916) emphasize that children are active learners, that they construct their own knowledge and that the social environment is a key setting for cognitive development and learning (Bruner, 1964; Bruner, Olver, & Greenfield, 1966). Lesh (1979) and Dienes (1969) extended these understandings by suggesting the importance of representation in the development and presentation of school mathematics curricula. One can see these and many other 'theoretical' positions reflected in the pedagogical guidelines as discussed within the process and curriculum standards.

Although the *Standards* documents are not completely research-based in a formal sense, they do reflect implications of the research conducted during the period preceding their publication. In 1989, the extent of the existing research was not adequate to completely inform a document with such a wide scope and extensive set of recommendations. The 2000 *Standards* document attempted to integrate existing research literature to a greater extent. Still, *Standards*-based curricula

cannot and do not claim to be completely informed by research, as they were developed to be consistent with several widely acclaimed documents (NCTM, 1989) that are only partially influenced by the results of empirical research. There is agreement, however, that these documents reflect best teaching practices and contemporary thinking about approaches to mathematics content development and the promotion of student learning.

STUDENT ACHIEVEMENT

Schoen and Hirsch (2003) found that CMIC students enrolled in Course 1 (9th grade) outperformed a group of similar students[1] studying prealgebra, algebra, and accelerated geometry on problems set within real-world contexts. Students in Course 2 (10th grade) outperformed comparable students enrolled in prealgebra and algebra on contextualized problems. There were no significant differences between Course 2 students and a matched group studying from accelerated advanced algebra. Huntley, Rasmussen, Villarubi, Sangtong, and Fey (2000) found that grade 11 students enrolled in CMIC performed significantly better than students enrolled in traditional curricula on problem-solving measures situated within a context.

Other studies (Cichon & Ellis, 2003; Lott et al., 2003) found that students in the *Standards*-based curricula Connections and SIMMS, respectively, outperformed students not in a *Standards*-based curriculum (i.e., a traditional curriculum) on contextual, open-ended problems. Webb (2003) found that students in the IMP curriculum performed significantly better than comparable groups of students learning from traditional mathematics programs on performance assessments involving statistics, probability, and combinatorics.

Standardized Achievement Tests

In order to satisfy concerns of parents and other stakeholders, evaluations of *Standards*-based curricula have also been conducted using standardized achievement tests. Schoen and Hirsch (2003) used the Iowa Test of Educational Development (ITED) Ability to do Quantitative Thinking, or ITED-Q (Feldt, Forsyth, Ansley, & Alnot, 1993) to assess student performance. CMIC Course 1 students performed $d = .17$ standard deviations higher on this test than a comparison group enrolled in prealgebra, and .57 standard deviations higher than a group enrolled in algebra, which were statistically significant differences. (Following Hedges & Olkin, 1985, p. 78, d is an effect size representing the difference in the means of two groups divided by the pooled within-groups standard deviation.) There were no significant differences between CMIC Course 2 students and those in algebra, geometry, or Accelerated Advanced Algebra. Webb (2003) found no significant differences between traditional and IMP students on the standardized test, Comprehensive Test of Basic Skills.

Abeille and Hurley's (2001) final evaluation report on the MMOW curriculum

[1] Students were matched based upon prior achievement, school, and gender.

describes student achievement data on two different standardized tests, the Stanford Achievement Test (SAT-9) and the ITED-Q. The authors found statistically significant increases in the number of MMOW students who scored in the top quartile of the ITED-Q when compared to students who had used traditional mathematics curricula at the same school in previous years. In another district, students learning from MMOW produced a mean score of 62 for the SAT-9 in 1999, expressed in normal curve equivalents (NCEs), and a mean of 60 in 2000. Both of these means were significantly above the national average of 50 for this test.

Symbolic Manipulation

Despite the successes of these curricula as described above, intense criticism from Web-based sources (e.g., Mathematically Correct—www.mathematicallycorrect.org) and others (Wu, 1997) has been leveled against *Standards*-based mathematics curricula. One of the areas of criticism has been whether *Standards*-based curricula provide students with sufficient practice to develop facility with symbolic manipulation tasks. This may be a valid criticism, for as Schoen, Fey, Hirsch, and Coxford (1999) point out, CMIC focuses on conceptual understanding with a decreased emphasis on symbolic manipulation.

There is mixed evidence with regard to this assertion that may be related to the use of technology in solving these types of problems. For instance, Huntley et al. (2000) found that 11th grade CMIC students did less well on symbolic manipulation tasks without the assistance of graphing calculators when compared to students enrolled in traditional mathematics programs, scoring on average .54 standard deviations lower. However, Schoen and Hirsch (2003) found no differences between students who had 4 years of CMIC and traditionally prepared students on a college mathematics placement exam emphasizing symbolic manipulation when graphing calculator technology (without symbolic manipulation capabilities) was available.

College Aptitude Tests

Lott et al. (2003) found no significant differences between SIMMS students and students in a traditional mathematics curriculum on the mathematics portion of the Preliminary Scholastic Aptitude Test (PSAT-M). Connections students scored just as well as traditional students on the PSAT-M, with prior achievement used as a covariate in the analysis (Cichon & Ellis, 2003). On the mathematics portion of the Scholastic Aptitude Test (SAT-M), there were no differences between Connections students and traditional students.

One of the strongest findings with regard to college aptitude tests is from Schoen and Hirsch (2003). They randomly assigned students to either CMIC or a traditional mathematics program in grade 9. Students' prior knowledge of mathematics at grade 9 in both groups was equivalent as measured by the Iowa Test of Basic Skills (ITBS). In grade 11, there were no statistically significant differences on the SAT-M between CMIC and traditional students. In another experimental study involving random assignment of students to classrooms, no significant differences were

detected between CMIC and traditional students on the American College Testing examination (ACT Math) in grade 11.

Webb (2003) found no significant differences on the SAT-M between 11th-grade students who had 3 years of IMP or a traditional mathematics program. One of the concerns of parents about *Standards*-based curricula is that higher achieving students will be disadvantaged by the program. Webb addressed this concern by examining the grade 11 SAT-M scores of IMP and traditional mathematics students who scored at or above the 76th percentile on the CTBS in grade 7 and found no significant differences between these two groups.

Overall, the data suggest that students studying from different *Standards*-based curricula do not perform differentially on the mathematics portions of the PSAT, SAT, or ACT college tests when compared to students learning from traditional curricula. This finding is relatively robust as it involves students from different geographical regions and in some cases involved random assignment of students to different curricula.

Ancillary Benefits of Standards-*based Curricula*

The focus of research regarding *Standards*-based mathematics programs has been on student achievement as summarized in the aforementioned studies. However, other benefits of studying in these programs have been documented. For instance, Webb (2003) found that students studying from IMP took more high school mathematics courses than students studying from traditional mathematics programs. Cichon and Ellis (2003) followed a group of 30 students in an inner-city school where 3 years of mathematics were required of all students. Although it was required that students study 3 years of mathematics, they found that 20 out of 30 students who had studied from Connections for 3 years elected to take mathematics as seniors. Moreover, 19 out of those 20 students chose to pursue postsecondary education. Typically, only 74% of students from this school continue on after high school.

The relationship between mathematics attitudes and achievement has been well researched (e.g., McLeod, 1994; Schoenfeld, 1989). Cichon and Ellis (2003) found that Connections students had significantly higher ratings in confidence and usefulness of mathematics on the Fennema-Sherman Mathematics Attitude Scale. Schoen and Pritchett (1998) found that students learning from CMIC Course 1 and 2 had more positive attitudes toward mathematics than a comparison group using a traditional mathematics program. For example, significant differences in favor of CMIC students were found on confidence in solving problems, believing that mathematical ideas make sense, and believing that mathematical ideas are inherently interesting.

Teacher Factors

There is much research indicating that teachers do not implement *Standards*-based curricula with fidelity (e.g., Lambdin & Preston, 1995; Truitt, 1998; Wilson & Lloyd, 1995). In order to better understand the relationships between implementation and student achievement, Schoen, Cebulla, Finn, and Fi (2003) used regres-

sion techniques to relate student achievement on the ITED-Q to particular teacher characteristics measured through teacher surveys and classroom observations. Nearly 1,500 students from 40 different teachers' classrooms participated in the Schoen et al study. The percentages of class time devoted to teacher presentation and whole-group discussion were both significantly negatively correlated with student achievement. Replacing activities or homework with skill supplementation and replacing CMIC (Coxford et al., 1998) tests with assessments that were less open-ended and more focused on symbolic manipulation were also significantly negatively correlated with student performance. On the other hand, summer workshop training, using individual interviews, the number of extending tasks assigned, the observer's rating, and collaboration between teachers were all positively significantly correlated with student achievement. These findings suggest that the commonly used teacher-dominated approaches to instruction, whose focus is on narrow and replicable routines, may not be the most efficient way to promote high levels of student achievement.

METHODOLOGY

Sampling

In the mid to late 1990s, the NSF, in an attempt to provide much needed professional development for school districts that adopted one or more of the new NSF funded *Standards*-based curricula, created the Local Systemic Change through Teacher Enhancement Initiatives (LSCs). The 47 funded LSCs (NSF 95-145) were "designed to engage entire school districts in the reform of science, mathematics and technology education, . . . to provide 47,000 teachers with professional development and . . . reach 1.6 million students in 240 school districts nationally" (Introduction, http://www.nsf.gov/pubs/1997/nsf97145/intro.htm).

The LSC discussed here, "Minneapolis and St Paul Merging to Achieve Standards Project" $(MASP)^2$, was one of these 47 Projects. $(MASP)^2$ provided professional development targeted to new NSF funded *Standards*-based curricula to over 1,100 middle grades and secondary teachers in 21 districts between 1997 and 2002. These teachers then provided *Standards*-based mathematics instruction to over 74,000 students in the 2000–2001 school year, and slightly larger numbers of students in the years since. Data from $(MASP)^2$ were used in this study. Although districts naturally varied by location (urban, suburban), especially striking differences emerged in enrollment and student characteristics, with mostly urban districts showing quite large enrollments and diverse student populations, and suburban districts tending to have smaller enrollments and relatively homogeneous student populations.

Following the distinction drawn by Cochran (1983), our target populations consisted of urban and suburban U.S. high school classrooms and students in a midwestern state who had completed at least three levels of high school mathematics in one of three *Standards*-based curricula. Our sampled populations, on the other hand, consisted of students and classrooms in seven purposively sampled urban and

suburban districts from those in (MASP)[2] who were subsequently invited to participate in the study of achievement reported here. Because of missing data in one district, six districts ultimately provided the high school data used in this study. These six districts were chosen for reasons of variation in location (urban and suburban), demographic variables, and school context. Mentors and project staff observed that there were high degrees of implementation and satisfaction with the curricula in some districts, whereas in others, the secondary mathematics faculty actively tried to undermine efforts to use *Standards*-based curricula.

The location, curriculum, rationale, assessment instrument, and enrollment for each of the sampled districts are shown in Table 1. One school district had substantially greater enrollment than the others and was the only one located in a purely urban setting. Another district was located on an urban-suburban boundary, whereas the remaining districts were located in suburban settings. The largest district also showed the greatest diversity in student ethnicity and the highest percentages of students eligible for a free or reduced-price lunch, nonnative English speakers, and special education students. In contrast, students in the remaining districts were predominately White native English speakers, most of whom were not eligible for a free or reduced-price lunch, and not involved in Special Education. The result of our purposive sampling is a dichotomy in enrollment and student characteristics in

Table 1
Location, Curriculum, Rationale, Assessment, and Enrollment for Sampled Districts

District	Geographic location	High school curriculum assessed in this study	Rationale for choice of district	Assessment used	Enrollment
A	Urban suburban boundary	MMOW	High fidelity of implementation with much parental support	SAT-9	11, 675
B	Urban	IMP	Large urban population—variable implementation	SAT-9	48,850
C	Suburban	CMIC	Wholesale adoption sabotaged by a few faculty dissenters and parents	SAT-9	3,466
D	Suburban	CMIC	Wholesale adoption with district authorized supplements	SAT-9	6,768
E	Suburban	CMIC	Enthusiastic adoption	SAT-9, NSREM	8,625
F	Suburban	MMOW	Partial adoption	NSREM	7,528

Note. SAT-9 = Open Ended and Multiple Choice subtests, NSREM = New Standards Reference Examination in Mathematics.

our samples of students and classrooms that we believe is largely representative of that existing in the target populations.

Data Collection

Cross-sectional data for two cohorts of 11th-grade students were gathered, one in Spring 2001 and the other in Spring 2002. There were no theoretical reasons to consider the cohorts of 11th-grade students tested as different. The results of descriptive analyses, as well as regression analyses including HLM in which cohort served as a predictor, indicated that there were no empirical reasons to treat the cohorts as separate. As a result, they were combined into a single group for analysis purposes. The initial samples for this study consisted of 972 high school students in 27 classrooms using a *Standards*-based curriculum; however, missing data reduced these sample sizes, especially for the HLM, which was based on 25 classrooms and 560 students. Table 1 shows that District F, a very affluent district, provided only criterion-referenced test data and hence was not included in the HLM analyses, which were based on standardized achievement test data.

A nonexperimental design was used in which students were considered to be clustered (nested) within classrooms, which in turn were clustered within school districts. Information was obtained for each level of clustering, but the focus was on students and classrooms. The lack of experimental manipulation means that study results support inferences about relationships among variables and their magnitude but do not generally support strong causal inferences (Rosenbaum, 2002, pp. 342–344).

This research project began with questions from districts, teachers, and parents concerning achievement of students enrolled in *Standards*-based classes in schools for which the $(MASP)^2$ LSC provided professional development. Our partner school districts wanted to assess students with instruments having national norms, despite the fact that most of these instruments were not designed to assess the effects of *Standards*-based curricula. After reviewing several instruments from national publishers and consulting with our partner districts, we narrowed the list to two: the Stanford Achievement Test, Ninth Edition (SAT-9) developed by Harcourt Brace Educational Measurement and the New Standards Reference Examination in Mathematics (NSREM) developed by the Learning Research and Development Center of the University of Pittsburgh and the National Center on Education and the Economy. Both sets of tests were supplied and scored by Harcourt Brace, and were administered under the supervision of project staff. The NSREM was designed to reflect *Standards*-based curricula, whereas the SAT-9 is a traditional standardized norm-referenced test that was not specifically designed to be sensitive to *Standards*-based curricula.

Specifically, the NSREM is a criterion-referenced test that assesses performance in three areas: Mathematical Skills, Mathematical Concepts, and Problem Solving. The performance levels are derived from national standards developed by a conglomerate of assessment-related organizations (Wiley & Resnick, 1998). The content and process areas assessed include number and operations, geometry and

measurement, algebra and function, mathematics skills, problem solving and reasoning, and mathematical communication.

Operationally, the first section of the NSREM in mathematics is composed of 20 multiple choice questions, which are a subset of SAT-9 multiple-choice test items, enabling this portion of the test to be compared to national norms. In addition, the first section contains short tasks. The student has 20 minutes to complete the multiple-choice questions and 35 minutes for the short tasks. Students then spend 55 minutes on the second section, which comprises long and medium length tasks. The third section requires an additional 55 minutes to complete and covers both short and long tasks. Short tasks are constructed-response items, whereas the medium and long tasks require detailed answers and are considered extended-response items.

The NSREM test reports student scores in a number of different ways, but we chose to examine student performance levels categorized as *achieved with honors, achieved the standard, nearly achieved, below standard,* and *little achievement.* Sample items for the NSREM are located in Appendix A.

The mathematics portion of the SAT-9 has two subtests. The Open Ended subtest consists of nine items with a total raw score value of 27 points. These items require students to construct extended responses and explain their solutions. "Although each item in the mathematics subtests can be solved by reasoning and an understanding of general mathematics, topics from algebra, geometry, trigonometry, statistics, discrete mathematics, and pre-calculus are included" (Harcourt Brace, 1996, p. 27). The Multiple Choice subtest consists of 30 items, each contributing 1 raw score point, which require students to solve problems set within real world and symbolic mathematical contexts. Calculators are allowed on both subtests, which each take two 50-minute time periods to administer. Sample items for the SAT-9 subtests are in Appendix B.

The SAT-9 reports both scaled scores and normal curve equivalents (NCEs). By definition, NCEs are normalized standard scores with a mean of 50 and a standard deviation of 21.06. The standard deviation of 21.06 was chosen so that NCEs of 1 and 99 are equivalent to percentiles of 1 and 99. (Stanford Technical Data Report, 1997). Although NCEs and percentiles are similar they are not identical. The data analysis results are generally reported in NCEs because of their familiarity and interpretability.

Table 2 presents information aligning NCE scores and SAT-9 percentiles with their corresponding scaled and raw scores. In the middle of the distribution of raw scores, a change of a single point (e.g., from 13 to 14 on Open Ended) results in a change of 12 percentile points and a change of about 6 NCE and scaled score points. Similarly, in the tails of the distribution of raw scores, a change of 1 point corresponds to a change of 3 to 5 percentile points and as many as 8 scaled score points. Thus, small changes in raw scores on the SAT-9 produce correspondingly larger changes in percentiles and scaled scores.

Teacher Professional Development

Teachers participated in three types of professional development. In one type, teachers participated in 2 weeks (approximately 80 hours) of summer training

Table 2
Selected SAT-9 Percentiles, NCEs, Scaled Scores, and Raw Scores (Spring 11th Grade Norms)

Open Ended				Multiple Choice			
%tile	NCE	SS	Raw	%tile	NCE	SS	Raw
94	82.7	714	20	98	93.3	781	26
90	77.0	704-705	19	90	77.0	750-751	22
85	71.9	696-697	18	85	71.8	740-741	21
60	55.3	672	14	61	55.9	710	15
48	48.9	666	13	56	53.2	705	14
39	44.1	661	12	51	50.5	700	13
13	26.3	636	7	17	29.9	671	8
10	23.0	631	6	12	25.3	665	7
7	18.9	625	5	8	20.4	657	6

Note. %tile = percentile, NCE = normal curve equivalent, SS = scaled scores.

related to a particular *Standards*-based curriculum. This usually entailed working through activities of the curriculum while teaching strategies were modeled by experienced $(MASP)^2$ staff members. In addition, mathematical content discussions were held each day, which were integrated into the work with student activities. In the second type, teachers participated in a series of sessions (approximately 30 hours) during the school year that focused on more general topics such as facilitating cooperative learning in mathematics classrooms, current research on the brain and its implications for mathematics classroom instruction, meetings with teachers and administrators to discuss administrative issues, and meetings with counselors relating to the scheduling of students. In the third type, $(MASP)^2$ employed district personnel experienced in the curriculum to serve during the school year as mentors to teachers newly implementing *Standards*-based curricula. The 20-hour mentoring component consisted primarily of classroom observations followed up with one-on-one debriefings and, in some cases, demonstration lessons. High school teachers in this study had completed an average of 144 professional development hours over a 3-year period.

An important consideration in a study examining the effectiveness of *Standards*-based curricula is the fidelity with which these curricula were implemented. Briars and Resnick (2000) determined that schools who implemented the Everyday Mathematics curriculum, as intended by the authors, scored higher on skills, problem solving, and concepts on the New Standards Reference Examination. *Standards*-based teaching was related positively to student achievement, especially when accompanied by a *Standards*-based curriculum (McCaffrey et al 2001). Weiss, Banilower, Overstreet, and Soar (2002) found that inquiry-oriented teaching practices were more prevalent when *Standards*-based classrooms were compared to traditional classrooms.

In a retrospective study such as the one reported here, assessing the fidelity of implementation can be difficult for a variety of reasons but is important because

of variation among teachers. Reys & Reys (2004, personal communication) found that teaching strategies consistent with the *Standards*-based text authors' suggestions were used from 10% to 90% of the time in the *Standards*-based classrooms they observed.

One factor promoting fidelity of implementation is that teachers whose classrooms were used in the present study participated in the $(MASP)^2$ professional development activities linked to *Standards*-based instruction. Other evidence of fidelity was provided by $(MASP)^2$ mentors assigned to provide an informal assessment of their teacher's implementation level in all classrooms between 1997 and 1999 and by information about individual teachers' classrooms as possessed by district evaluation personnel and district curriculum directors. Although no set of efforts can guarantee full fidelity, these considerations have helped to document that *Standards*-based curricula were generally implemented faithfully in our $(MASP)^2$ project classrooms.

Data Analyses

We used two methods to study students' achievement patterns. First, descriptive analyses were used to examine district- and classroom-level characteristics and achievement. As noted above, districts were concerned about how their students' scores compared to nationally normed groups, and we compared each district's mathematics performance to norms established by the test publisher. Second, we used HLM to model variation in mathematics achievement using both student- and classroom-level predictors. Student-level predictors included SES (not eligible for a free or reduced-price lunch, eligible), gender, and attendance, whereas classroom-level predictors included type of *Standards*-based curriculum (CMIC, IMP, MMOW), ethnic composition, and SES (percentage of students eligible for a free/reduced price lunch). Students were treated as clustered within classrooms (too few districts were tested to include this level of clustering). In a few cases, a single teacher taught two classes of students. To avoid dependency among classrooms with the same teacher we collapsed these two groups and henceforth considered them to be one classroom.

As other studies have noted (Begle, 1973; Reys, Reys, Lapan, Holliday, & Wasman, 2003; Riordan & Noyce, 2001), prior achievement in mathematics is an important predictor of student achievement. Accordingly, we needed a measure of prior achievement for use in several statistical analyses. All districts are mandated by the state to administer the mathematics portion of the Minnesota Basic Skills Test (MBST) in 8th grade, and the mathematics scores of this test, which were available in an NCE metric, served as our measure of Prior Mathematics Knowledge for both the student- and classroom-level regression models.

Background student data (e.g., ethnicity, attendance, student eligibility for a free or reduced-price lunch, English language learner status) were gathered from each district. As noted above, because of missing data most of the statistical analyses were based on fewer classrooms and students (25 and 560, respectively) than those orig-

inally sampled (27 and 972, respectively), with a substantial percentage of the missing values occurring for Prior Mathematics Knowledge and the SAT-9 subtests. The higher percentage of missing data for these variables is at least in part due to the 3-year gap between when these measures were administered (8th grade, 11th grade). Among the most plausible explanations for the greater percentage of missing scores for the Prior Mathematics Knowledge measure are that some students transferred into a district after completing 8th grade elsewhere and subsequently provided SAT-9 scores but no Prior Mathematics Knowledge scores. Similarly, some students who provided a prior mathematics score transferred out of a district before SAT-9 data were collected in 11th grade (information on student transfers was not available). It is also likely that some students simply failed to sit for these tests.

Using available data, we found no evidence that students with missing values differed systematically from those providing data (e.g., that a disproportionate number of missing values were linked to nonnative English speakers or to members of a particular ethnic group). As such discrepancies did not occur, we concluded that there was no clear evidence of bias. However, identifying bias is generally quite difficult and our findings should be interpreted in light of this possibility.

RESULTS

Method 1: Descriptive Summaries of District and Classroom Performance

SAT-9 descriptive statistics for the five districts providing SAT-9 data are presented in Table 3. Perhaps the most striking feature of Table 3 is that 9 out of 10 SAT-9 means were above 50, the national norm for NCE units for the Open Ended and Multiple Choice subtests. This indicates that the performance of *Standards*-based students was generally above average as determined by the test publisher's cutoff. It is also apparent that District E (suburban) had, on average, the highest NCE scores on both the Open Ended and Multiple Choice subtests and District B (urban) the lowest. Four of the five district means for Prior Mathematics Knowledge exceeded 50, with District B falling well below this benchmark. Overall, 16 NCE points separated the highest and lowest performing school districts

Table 3
Descriptive Statistics for the SAT-9 Subtests by District

	Prior Mathematics Knowledge			Open Ended			Multiple Choice		
District	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
A	284	55.2	14.1	380	63.0	20.4	372	54.3	15.1
B	142	41.7	15.3	162	52.1	23.0	171	45.7	17.0
C	44	52.5	14.4	50	62.4	19.0	50	59.9	14.2
D	220	64.0	17.9	163	59.0	24.1	168	57.7	17.2
E	27	57.6	15.7	21	67.7	21.2	21	65.1	17.6
Total	717	55.2	17.5	776	60.0	22.1	782	53.8	16.7

Note. Means (*M*) and standard deviations (*SD*) are reported in NCE units, *n* = sample size.

on Open Ended, and 19 points on Multiple Choice. There was also considerable variability among districts in average Prior Mathematics Knowledge (22 points).

The districts also differed in student characteristics, with District B showing the highest percentage of nonwhite students (just under 60%), which included African American (35.3%) and Asian (20.6%) students who produced similar means on the Open Ended (43.9 and 46.7, respectively) and Multiple Choice subtests (41.7 and 42.1, respectively). The small number of Hispanic students (11) across the five districts led us to exclude them from all analyses. District B also had the highest percentage of students eligible for a free or reduced-price lunch (low SES, just under 46%) and nonnative English speaking students (just over 12%), with the remaining districts showing substantially smaller percentages for these variables ranging between 0 and 10%. The percentage of students in special education in our sample across the districts ranged from 0% to 5.5%.

Table 4 reports average achievement for various student subgroups by district. It is apparent that low SES, African American, nonnative English speakers, and special education students performed below their counterparts, although there were a few exceptions in one suburban district.

The Open Ended, Multiple Choice, and Prior Mathematics Knowledge tests also showed similar patterns across various student subgroups as seen in Table 5. The difference in Prior Mathematics Knowledge means in Districts A and B favored the high SES group over students in the low SES group (9 NCE points and 8 points, respectively), 11 and 13 points for the Open Ended subtest, and 10 and 9 points for Multiple Choice. In most cases, the mean of the low SES group was below 50.

White students in District B scored on average 6–8 points higher than African American and Asian students on Prior Mathematics Knowledge, 8–15 points higher on the Open Ended subtest, and 7–8 points higher on Multiple Choice. White students scored on average 3 and 7 points lower than African American students on Prior Mathematics Knowledge in Districts C and D, respectively. However, White students scored 5–6 points higher than African American students in these districts on the Open Ended and Multiple Choice subtests.

In District B native English speakers scored approximately 9 NCE points higher than nonnative speakers on Prior Mathematics Knowledge, 22 points higher on Open Ended, and 5 points higher for Multiple Choice. The means for nonnative speakers were well below 50.

Additional descriptive statistics for high school students including effect sizes (*d*) appear in Table 5, which were computed across districts. For example, low SES students scored on average .68 standard deviations lower on the Open Ended subtest than high SES students. One obvious finding in Table 5 is the lower average performance for low SES, nonwhite, urban, nonnative English speakers, and special education students. These effect sizes ranged from .11 to 2.00 standard deviation units lower than the comparison group. An examination of test performance by classroom (see Table 6) provided evidence of substantial variation in achievement.

Finally, we examined average performance for the three *Standards*-based curricula in Table 7. On the Open Ended subtest, students in MMOW produced a

Table 4
Average Achievement for Various Subgroups by District

District		SES		Ethnicity			Language	
		High	Low	White	African American	Asian American	English speaker	Nonnative speaker
A	Prior Knowledge	56.0(258)	47.0(26)	55.2(260)	60.8(17)	—	55.1(282)	—
	Open Ended	63.8(351)	52.7(29)	63.6(351)	57.2(20)	—	63.1(378)	—
	Multiple Choice	55.1(343)	44.8(29)	54.8(344)	49.8(19)	—	54.3(370)	—
B	Prior Knowledge	46.1(67)	37.8(75)	45.7(54)	38.4(54)	40.7(34)	43.1(120)	34.3(22)
	Open Ended	57.0(101)	43.9(61)	59.3(80)	43.9(49)	46.8(33)	54.7(143)	32.8(19)
	Multiple Choice	49.2(103)	40.4(68)	49.7(84)	41.7(53)	42.1(34)	46.3(152)	40.7(19)
C	Prior Knowledge	53.8(42)	—	51.7(43)	—	—	52.5(44)	—
	Open Ended	63.1(48)	—	61.7(49)	—	—	62.4(50)	—
	Multiple Choice	60.9(47)	—	59.3(49)	—	—	59.9(50)	—
D	Prior Knowledge	64.3(210)	56.6(10)	64.0(201)	66.7(17)	—	—	—
	Open Ended	59.2(157)	—	59.9(144)	52.5(19)	—	59.3(162)	—
	Multiple Choice	58.0(163)	—	57.3(150)	60.6(18)	—	57.8(167)	—
E	Prior Knowledge	58.6(26)	—	58.1(25)	—	—	57.6(27)	—
	Open Ended	68.3(20)	—	68.9(20)	—	—	67.7(21)	—
	Multiple Choice	65.6(20)	—	65.6(20)	—	—	65.1(21)	—

Note. Sample size is in ( ). —means that $n < 10$ cases were available, which was deemed too small for credible estimation of the mean.

Table 5
SAT-9 Subtest Achievement for Various Subgroups (Across Districts)

		Prior Knowledge				Open Ended				Multiple Choice			
		n	*M*	*SD*	Effect size	*n*	*M*	*SD*	Effect size	*n*	*M*	*SD*	Effect size
SES	High	603	57.8	16.6		677	61.8	21.5		676	55.6	16.0	
	Low	114	41.3	15.0	1.01*	99	47.1	21.8	0.68*	106	42.2	16.4	0.84*
Ethnicity	White	583	57.2	16.3		644	62.2	21.2		647	55.4	16.1	
	African American	90	48.7	20.9	0.42*	90	49.2	25.3	0.52*	92	47.8	18.3	0.46*
	Asian American	44	41.3	15.0	0.98*	42	48.0	18.4	0.68*	43	42.8	15.0	0.78*
Gender	Male	340	56.7	17.0		364	61.3	22.5		370	55.3	16.2	
	Female	377	53.8	17.8	0.16*	412	58.8	21.7	0.11	412	52.5	17.0	0.17*
Location	Suburban	575	58.5	16.3		614	62.0	21.4		611	56.1	15.9	
	Urban	142	41.7	15.3	1.04*	162	52.1	23.0	0.44*	171	45.7	17.0	0.64*
Language status	English speaker	693	55.8	17.1		754	60.8	21.8		760	54.2	16.6	
	Nonnative speaker	24	36.8	16.8	1.11*	22	32.5	13.8	2.00*	22	41.6	14.6	0.76*
Special education	Nonspecial ED	687	55.7	17.4		749	60.5	21.9		757	54.2	16.6	
	Special ED	30	42.7	13.2	0.75*	27	45.9	24.2	0.66*	25	42.4	15.5	0.71*

Note. * $p < 0.05$ (*t* tests). *n* = sample size, *M* = mean, *SD* = standard deviation.

Table 6
Prior Mathematics Knowledge and Classroom Outcomes Expressed in Normal Curve Equivalents (NCEs)

Class	Prior Mathematics Knowledge	Open Ended	Multiple Choice	Location	Curriculum
1	76.3	80.3	70.8	Suburban	CMIC
2	56	68.2	56.9	USB*	CMIC
3	80.8	67.9	67.8	Suburban	CMIC
4	57.6	67.7	65.1	Suburban	CMIC
5	64	65.7	70.3	Suburban	CMIC
6	51.5	60.7	52.9	Suburban	CMIC
7	47.2	60.5	55.1	Suburban	CMIC
8	84.8	55.9	53.9	Suburban	CMIC
9	57.1	54.7	58.4	Suburban	CMIC
10	55.2	52.5	53.4	Suburban	CMIC
11	39.3	61.6	47.6	Urban	IMP
12	40.6	59.7	50.9	Urban	IMP
13	51.9	57.7	50.4	Urban	IMP
14	44.6	55.9	37.5	Urban	IMP
15	51.8	54	54.5	Urban	IMP
16	50.2	46	41.8	Urban	IMP
17	48.4	45.5	42.7	Urban	IMP
18	37.8	44.8	39.2	Urban	IMP
19	35.8	37.6	42.9	Urban	IMP
20	39.8	31.8	49.1	Urban	IMP
21	55.1	68.1	51.6	USB*	MMOW
22	56.8	65.9	54.2	USB*	MMOW
23	53	60.5	53.6	USB*	MMOW
24	54.3	60.2	53.3	USB*	MMOW
25	53.4	54.6	55.8	USB*	MMOW

Note. *USB is Urban-Suburban Boundary.

Table 7
Prior Mathematics Knowledge, Open Ended, and Multiple Choice Achievement for the Standards-based Curricula (Across Districts)

	Prior Mathematics Knowledge			Open Ended			Multiple Choice		
Curriculum	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
IMP	142	41.7	15.3	162	52.1	23.0	171	45.7	17.0
CMIC	291	61.6	17.6	234	60.5	22.9	239	58.7	16.7
MMOW	284	55.2	14.1	380	63.0	20.3	372	54.3	15.4

Note. n = sample size, M = mean, SD = standard deviation.

mean of 63 NCE units, well above the publisher's proficiency cutoff of 50 and higher than those in CMIC (60.5) or IMP (52.1). For the Multiple Choice subtest, CMIC students scored highest on average (58.7), followed by MMOW (54.3) and IMP (45.7). This pattern also emerged for Prior Mathematics Knowledge, with CMIC students scoring on average the highest (61.6), followed by MMOW (55.2) and IMP

(41.7). One-way, fixed-effects ANOVA revealed that the three curriculum groups differed significantly for each of the outcome variables appearing in Table 6. Using Fisher's eta-squared ($\hat{\eta}^2$) as an effect size, .17 or 17% of the variation in Prior Mathematics Knowledge was a function of curriculum, whereas for Open Ended and Multiple Choice the $\hat{\eta}^2$ values were .04 and .08, respectively. Thus, the largest differences between curriculum groups appeared for Prior Mathematics Knowledge.

The fact that the Prior Mathematics Knowledge test was administered in the 8th grade and the Open Ended and Multiple Choice subtests in the 11th grade, and all measures are reported in NCE units, allows a rough comparison of change over time by curriculum. Of course, precise comparisons over time require the use of scale scores and the same students measured repeatedly, and we offer the following only as a potentially interesting facet of the data.

As seen in Table 7, IMP students scored on average the lowest on the Prior Mathematics Knowledge test (41.7) and above the proficiency cutoff of 50 on the Open Ended test administered in the 11th grade (52.1). If these students had shown no change in their mathematics achievement, we might expect the two NCE means to be similar, which was not the case. CMIC students scored about the same on average on these two tests (61.6 and 60.5, respectively), whereas *MMOW* students showed positive change (Prior Mathematics Knowledge = 55.2, Open Ended = 63). For Multiple Choice, IMP students showed on average modest growth between 8th grade and 11th grade, CMIC showed a modest decline, and MMOW students stayed about the same.

The findings above need to be considered in light of the fact that all IMP classrooms were in the urban district, which generally produced lower average test scores than the suburban classrooms. Thus, comparisons of student performances linked to IMP and the other *Standards*-based curricula are confounded with classroom location. We discuss this confounding further when presenting our HLM results.

In sum, there is ample evidence of variability among the school districts and classrooms in achievement as well as background variables. The suburban districts, including the district on the urban-suburban boundary, tended to have far smaller percentages of nonwhite, low SES, and nonnative English speakers. The distribution of special education students showed little relationship with the suburban or urban location of districts. There was also evidence that students in the CMIC and MMOW curricula scored on average well above those in IMP in mathematics achievement.

New Standards Reference Exam in Mathematics

The NSREM was only administered in suburban Districts E and F, and we therefore have NSREM data on only a relatively small number of students: four classrooms ($n = 92$) in District E and one classroom in District F ($n = 22$). These findings are reported as percentages of students who achieved in each of five categories, as denoted in Table 8.

Table 8
Percentage and Number of Students Meeting New Standards Achievement Level—District E and F

	Mathematical skills		Mathematical concepts		Problem solving	
Achievement level	District E &F	National	District E &F	National	District E &F	National
Achieved with honors	79% (90)	11%	49% (56)	4%	18% (20)	2%
Achieved the standard	19% (22)	39%	31% (35)	10%	40% (46)	7%
Nearly achieved	1% (1)	16%	11% (13)	16%	18% (21)	6%
Below the standard	1% (1)	25%	8% (9)	37%	20% (23)	41%
Little achievement	0% (0)	9%	1% (1)	33%	4% (4)	44%

Note. Total sample size $n = 114$.

The percentage of students who achieved the highest category (*Achieved with Honors*) in Mathematical Skills was 79%, which contrasts sharply with the 11% in this category based on the national norm group. Ninety-eight percent of students achieved the standard (top two categories) compared to 50% in the norming group. The pattern was similar for Mathematical Concepts and Problem Solving.

Method 2: HLM Analyses of Student and Classroom Data

The Open Ended and Multiple Choice outcomes were analyzed separately with HLM following the methods described in Raudenbush and Bryk (2002) and using the HLM6 software (Raudenbush, Bryk, & Congdon, 2005). Treating students as clustered within classrooms permitted within-classroom dependency among student mathematics test scores to be modeled, and allowed both student- and classroom-level data to be analyzed simultaneously. HLM also allowed us to assess whether the impact of the three *Standards*-based curricula on achievement was similar with various background variables taken into account.

Student-level regression models containing Prior Mathematics Knowledge, attendance, SES, and gender were fitted to the data for each classroom. These predictors have been shown to frequently impact mathematics achievement (International Association for the Evaluation of Educational Achievement, 2003; Post, Cutler, Davis, Anderson, Maeda, Kahan et al., in press; Reys et al., 2003; Sirin, 2005; Tate, 1997), and largely served as control variables to enhance inferences. For each outcome (Open Ended, Multiple Choice), three models were fitted.

First an unconditional model of the form

$$(1) \qquad \begin{aligned} Y_{ij} &= \beta_{0j} + r_{ij} \\ \beta_{0j} &= \gamma_{00} + u_{0j}, \end{aligned}$$

was fitted, where Y_{ij} is the mathematics score of the ith student in the jth classroom, β_{0j} is the average mathematics score (intercept) for the jth classroom, γ_{00} is the average mathematics performance across classrooms, r_{ij} is a student-level residual,

and u_{0j} represents the unique effect of the jth classroom. The unconditional model results tell us whether average outcomes differ across classrooms.

Next we fitted a student-level model of the form

$$Y_{ij} = \beta_{0j} + \beta_{1j}(SES_{ij} - \bar{X}_1) + \beta_{2j}(Attendance_{ij} - \bar{X}_2) + \beta_{3j}(Gender_{ij} - \bar{X}_3) + \beta_{4j}(Prior_{ij} - \bar{X}_4) + r_{ij}, \quad (2)$$

where β_{1j} is the student-level slope capturing the effect of SES (0 = not eligible for a free or reduced price lunch, 1 = eligible) on mathematics performance with other predictors held constant in the jth classroom, $\bar{X}_1$ is the grand mean for SES, and SES_{ij} is the SES value of the ith student in the jth classroom. We also tested whether slopes for the predictors varied across classrooms. Only attendance slopes varied significantly across classrooms, and we constrained the variances for the SES, gender, and Prior Mathematics Knowledge slopes to zero in the modeling. Grand-mean-centering of student-level predictors ensured that intercepts were adjusted for student-level predictors.

Classroom-level predictive models for intercepts and attendance slopes were developed. That is, we attempted to model variation in intercepts and attendance slopes across classrooms using classroom-level predictors. We again relied on previous findings to guide the selection of key predictors, which included class SES (percentage of students eligible for a free or reduced-price lunch in a classroom) and average Prior Mathematics Knowledge in a classroom. Other classroom predictors that we examined were the effect of different concentrations of various ethnic groups, nonnative English speakers, special education students, female students, and professional development hours of the teacher. The professional development hours linked to a teacher and the percentage of females in the classroom-level model were not statistically significant in any of the analyses, and refitting the model after omitting these variables did not change the findings in an important way. The final classroom-level model for intercepts was

$$\begin{aligned}\beta_{0j} = {} & \gamma_{00} + \gamma_{01}(Class\ SES_j - \bar{W}_1) \\ & + \gamma_{02}(Class\ Nonnative\ Speaker_j - \bar{W}_2) \\ & + \gamma_{03}(Class\ Special\ Education_j - \bar{W}_3) \\ & + \gamma_{04}(Class\ African\ American_j - \bar{W}_4) \\ & + \gamma_{05}(Class\ Prior_j - \bar{W}_5) + \gamma_{06}(IMP - \bar{W}_6) \\ & + \gamma_{07}(CMIC - \bar{W}_7) + u_{0j},\end{aligned} \quad (3)$$

where γ_{01} is the classroom level slope capturing the effect of class SES (percentage of low SES students) on average mathematics performance, $\bar{W}_1$ is class SES averaged across classrooms, and class African American is the percentage of African American students in a classroom. Model (3) was also used to predict variation in attendance slopes.

The predictors IMP (1 = IMP curriculum used in classroom, 0 = not used) and CMIC (1 = CMIC curriculum used in classroom, 0 = not used) captured which of

the three *Standards*-based curricula a student experienced. However, all of the IMP classrooms were in the urban district, whereas CMIC and MMOW classrooms were distributed over the remaining suburban districts. Thus, comparisons of IMP and non-IMP classrooms also potentially reflect differences between urban and suburban classrooms. The confounding of classroom location and which *Standards*-based curriculum was used in a classroom is not easily disentangled, but the fact that HLM statistically controlled for various factors distinguishing urban and suburban classrooms (percentages of low SES students, nonnative English speakers, African American students, etc.) allows us to argue that achievement comparisons between IMP and non-IMP classrooms likely reflect different curricula rather than being solely attributable to classroom location.

The deviance test described in Raudenbush and Bryk (2002, pp. 60–61) was used to test for model fit, allowing us to discriminate among models with more or less explanatory power. Model fitting was followed by extensive model checking to help to ensure validity of inferences. Cases in which normality, homoscedasticity, or linearity appeared to be suspect were examined in detail, and various remedies (e.g., modeling unequal classroom variances) employed. The analyses reported below are based on fitted models in which these assumptions appeared to be at least approximately satisfied. Two classrooms with four or less students were omitted because the student-level model in (2) could not be fitted. The data for these classrooms, although quite sparse, could have been used to estimate fixed effects and variance components using the HLM6 software but could not be used in testing variance components. This led us to omit these two classrooms from the HLM analyses. Missing data resulted in the loss of other cases, leaving 25 classrooms from 5 districts and 560 students for these analyses.

An initial difficulty with several of the classroom-level predictor variables, such as the percentage of nonnative English speakers in a classroom, was their ragged and discontinuous nature. For example, 52% of the classrooms had no nonnative English speakers, 28% of the classrooms had values between 1%–8%, and, at the other end of the distribution, five of the classrooms (20%) had values well above 8%. We explored various transformations of these variables with the goal of representing their variation in a more succinct form, and subsequently polytomized the distributions into quartiles. Thus, each of the classroom predictors in (3), with the exception of Prior Mathematics Achievement, which was not transformed, and percentage nonnative English speakers, which was trichotomous because of its distribution across classrooms, were transformed to a scale in which each was represented with one of four values corresponding to the quartiles. These values are presented in Table 9.

The HLM cross-sectional results are summarized in Table 10. Several findings emerged across the Open Ended and Multiple Choice subtests. First, there was significant, but not overwhelming, between-classroom variation. Approximately 16% of the variation among Open Ended means was accounted for by the classroom that students were in; for Multiple Choice, 22% of the variation in classroom means was accounted for by classroom. Second, at the student level, Prior

Table 9
Definitions of Classroom Level Predictors

Variable	Quartile	*n*	Range (*R*)
Class SES (low)	1	6	$R \leq 5.12\%$
	2	6	$5.12\% < R \leq 8.59\%$
	3	7	$8.59\% < R \leq 40.38\%$
	4	6	$R > 40.38\%$
Class nonnative speaker status	1	13	$R = 0\ \%$
	2	6	$0\% < R \leq 8.01\%$
	3	6	$R > 8.01\%$
Class special education	1	11	$R = 0\ \%$
	2	1	$0\% < R \leq 2.08\%$
	3	7	$2.08\% < R \leq 5.56\%$
	4	6	$R > 5.56\%$
Class African American	1	6	$R \leq 4.21\%$
	2	6	$4.21\% < R \leq 7.81\%$
	3	7	$7.81\% < R \leq 25.14\%$
	4	6	$R > 25.14\%$
Class Asian American	1	9	$R = 0\%$
	2	3	$0\% < R \leq 3.7\%$
	3	7	$3.7\% < R \leq 11.66\%$
	4	6	$R > 11.66\%$
Professional development hours	1	6	$R = <$ 104 hours
	2	7	$104 < R \leq 156$
	3	7	$156 < R \leq 188$
	4	5	$R > 188$
Class Prior Mathematics Knowledge	—	25	Classroom average Prior Mathematics Knowledge

Note. Class Prior Mathematics Knowledge was not polytomized. *n* = number of classrooms.

Mathematics Knowledge was a statistically significant predictor in every model, although its effect expressed in NCE units was quite modest (< 1). Student-level SES was statistically significant in the model for Multiple Choice and demonstrated a moderate effect on mathematics performance (–4.74), meaning that, with the other predictors held constant, low SES students tended to score almost 5 NCE points less than high SES students. Gender and attendance were not statistically significant student-level predictors. However, there was a significant cross-level interaction involving the attendance slopes and the percentage of African American students (expressed in quartiles) in a classroom (1.21), indicating that the effect of attendance on Multiple Choice scores increased as the concentration of African American students in a classroom increased.

Third, the Multiple Choice results showed fewer significant effects than those for the Open Ended subtest. Fourth, statistically significant contextual effects for Prior Mathematics Knowledge on the Open Ended and Multiple Choice subtests suggested that the effect of Prior Mathematics Knowledge on mathematics achievement was stronger at the classroom level than at the student level. Fifth, no differ-

Table 10
HLM Results

Dependent variable	HLM results
Open Ended	1. There was significant between-classroom variation in means (16%). 2. The only significant student-level effect was for Prior Mathematics Knowledge (.68). Open Ended intercepts and attendance slopes varied significantly across classrooms. 3. There was a small but significant contextual effect for Prior Mathematics Knowledge (–.57), meaning that the effect of this factor on Open Ended scores differs at the student and classroom levels. The evidence suggests that the effect is stronger at the classroom level. 4. None of the predictors in equation (3) were significant predictors of classroom intercepts. Although students in the IMP curriculum scored on average 7.7 points lower than those in non-IMP curricula with other predictors held constant, this effect was not significant ($p = .19$). In a similar way, students in the CMIC curriculum scored on average 5.6 points lower than those in non-CMIC curricula; this effect was not significant ($p = .10$). 5. The predictive models for attendance slopes had one significant effect. Increasing concentrations of African American students in a classroom were associated with a stronger effect of attendance on Open Ended scores (1.21).
Multiple Choice	1. There was significant between-classroom variation in means (22%). 2. The largest (significant) student-level effect was for SES (–4.72), meaning that, with other predictors held constant, low SES students tended to score just under 5 NCE points lower than high SES students. Prior Mathematics Knowledge was also a significant student-level predictor (.50). 3. There was a small but significant contextual effect for Prior Mathematics Knowledge (–.37), meaning that the effect of this factor on Open Ended scores differs at the student and classroom levels. The evidence suggests that the effect is stronger at the classroom level.

ence among the three *Standards*-based curricula emerged on the SAT-9 subtests when other variables were taken into account.

DISCUSSION

This was not (and could not be) a true experimental study (Campbell & Stanley, 1966), because the retrospective nature of the study means that students and teachers could not have been assigned at random to urban or suburban classrooms or to a particular *Standards*-based curriculum, implying that strong causal inferences are not warranted. We attempted to describe patterns of student achievement from a number of perspectives, including HLM analyses, which permitted us to view classroom and student data simultaneously and to model within-classroom dependencies.

As noted, the districts were selected for different reasons. One was characterized by enthusiastic district-wide adoption, whereas in another, there were active efforts to undermine the use of *Standards*-based curricula. Our remaining districts

reflected everything in between these extremes. Both affluent suburban districts and one primarily low SES heavily multicultural urban district were represented. Our goal was to provide a view of student mathematics achievement in classrooms that had completed three levels of involvement (three Carnegie units) with *Standards*-based curricula.

The reader will note that we have used the 50th percentile (NCE mean = 50, and scaled scores at the 50th percentile) on the SAT-9 as the benchmark to which students' scores were compared. Why the 50th and not the 65th or 75th percentile? Surely the national mean is not an aspiration for all districts; for example, our high achieving suburban districts would be quite displeased were their students to achieve at this level. On the other hand, the 50th percentile is a realistic goal for our urban district, 73% of whose students (60% in our sample) are minority and which contains a relatively large group of low SES students. In the final analysis, we selected the 50th percentile because we felt it was an appropriate goal for our lower achieving districts, whereas, of course, also permitting us to gauge the degree to which our higher achieving districts exceeded (or did not) the national mean.

No single assessment can accurately assess all that a student knows and understands. This is especially true when asking whether a traditionally oriented standardized test is capable of assessing the efficacy of a *Standards*-based curriculum, given the wide disparity (a validity issue) between content and emphasis. Although there are a number of issues surrounding these tests, attempts such as this to assess traditional topics, and compare results to national norms, is nonetheless an important exercise. Such activity was consistent with, and useful to, our superintendents' perceived need to obtain this kind of information, primarily for parental consumption. The SAT-9 results contained no surprises: Our urban and boundary suburban district achieved at about the national norm, whereas our more affluent suburban districts scored higher. Procedural (Multiple Choice) scores lagged behind Open Ended results in *Standards*-based classrooms.

By 1998, District E, our highest achieving district, had implemented *Standards*-based curricula for *all* its students, adopting Everyday Mathematics (Bell, 1988–1996) at the elementary level, Connected Mathematics (Lappan, Fey, Fitzgerald, Friel, & Phillips, 1991–1997) at the middle grades, and CMIC at the high school level. The criticism from opponents of *Standards*-based curricula that high achieving students would be "held back" was not supported by the results that follow. As part of a district-wide self-assessment, a study distinct from the data included here, District E reported that the number of its high school students taking the BC level of the AP Calculus exam jumped from 50 to 67 between 1999 and 2003. The passing rate with a score of 3 or better during this period increased from 64% to 87%. For AB calculus, the number of students increased from 10 to 16, and there was an increase in the percentage with a grade of 3 or more from 30% to 81%. Similarly, the number of students taking the AP Statistics exam increased from 31 to 71 between 1999 and 2003. The passing rate, with a score of 3 or higher, during this period also increased slightly from 74% to 76%. In 2003, all high school students in District E had been exposed only to the CMP and CMIC programs since grade 6.

Such information is at odds with the negative and cautionary claims of the "mathematically correct" and other groups claiming that *Standards*-based curricula are inappropriate for higher achieving students. In this case, District E teachers, administrators, and most parents are strong in their support for this program and are resolute in their determination to continue with its use. In general, district teachers and administrators (and parents) need to recognize that there is emerging evidence that *Standards*-based curricula do not impede the mathematical performance and development of higher achieving students, and in some cases the curricula appear to have contributed to better student performance. As such evidence continues to mount, school personnel can become increasingly confident in their adoption and implementation of *Standards*-based mathematics curriculum materials with all their students.

Given results such as those obtained from the NSREM, and despite the fact that these were two high achieving districts to begin with, it is difficult to understand the ongoing concern by some that *Standards*-based students are not learning basic mathematical skills and are lacking important mathematical understandings. This issue has been an important motivator for this study, and we are pleased that additional light is being shed on the issue. Coupled with other results from similar studies (e.g., Senk & Thompson, 2003), perhaps it is now time to focus research activity on what mathematics *Standards*-based students are learning that traditional students are not, and vice versa. It will be prudent in future comparative studies to probe the depth of student understanding, their flexibility in accessing various mathematical processes and skills while engaged in problem solving, their realization of the scope of the mathematical connections made, and the ease and accuracy of their applicability to new situations.

Because clear evidence supporting alternatives to *Standards*-based curricula have not been published, one can conclude that opponents would have us "stay the course." Hiebert (2003) suggested that traditional approaches over the past 30 years have not produced acceptable results either in student achievement levels or in the percentages of students who continue the study of mathematics beyond minimum high school requirements. It is unfortunate that the political debates about the efficacy of the reformed mathematics curricula have for the most part unduly influenced research efforts to assess student achievement on traditional topics as measured by traditionally oriented standardized tests. This study was no exception.

Assessing reformed students' achievement on traditional topics is not the most interesting or ultimately the most important question. It is however an important component of an exploratory research agenda. NSF did not fund the development of thirteen complete curricula at the elementary, middle grades, and secondary level in the 1990s to simply increase achievement on existing standardized tests. The NSF initiative occurred because of an increasingly disparate achievement pattern between the "haves" and the "have nots"; because the half-life of mathematics students is 1 year, beginning in 8th grade; and because the United States as the world's technological leader is currently in serious jeopardy (Friedman, 2005). Further, advances in technology have made preoccupation with algorithmic procedures unrealistic. Evolving conditions in the workplace have expanded the need for a deeper under-

standing of probability, statistics, and the problem-solving power of topics in discrete mathematics, traditionally topics not significantly addressed at the secondary level. The traditional separation and isolated treatment of algebra, geometry, and trigonometry have predictably inhibited students' ability to consider problems from multiple perspectives. For example, spreadsheets and statistical regression are two of the most widely used mathematical tools in the business community, but these tools are not represented in traditional approaches.

The preponderance of the research to date has indicated that there are essentially no differences between *Standards*-based and traditional students' achievement on traditional topics, with the probable exception of paper-and-pencil calculation, and speed and accuracy with algorithmic procedures. Given the variety and extent of computer (and calculator) algebra systems, such hand manipulation is a skill of far less importance in the 21st century. This is not to say that such skills are unimportant but there must be additional dimensions to program design for the 21st century for the majority of our student population.

The results obtained here do not suggest that *Standards*-based curricula are the magical solution to ongoing and persistent complaints about students' lack of mathematical preparedness in traditional topical domains. It must be noted that this has been a recurring concern (complaint) for most of the last century. It seems prudent now to refocus research efforts to other and more important areas of investigation that will expand the domain of research to include the learning of probability, statistics, topics in discrete mathematics, and students' ability to solve nonroutine problems and apply mathematics prudently in contextual settings. A second and promising area for investigation will be students' performance in postsecondary mathematics and related coursework. It is time now to devote research energy to the more important question of what mathematics *Standards*-based students are learning that traditional students are not, and vice versa, and what ancillary attributes accompany such learning.

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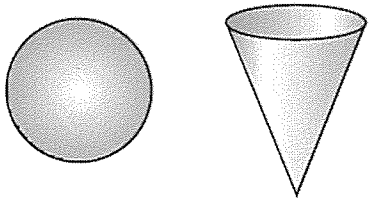
APPENDIX A

Sample Test Items from the
New Standards Reference Examination in Mathematics

High School Short Task—Ice Cream Cone

In this task you are asked to explore if the melted ice cream will fit into the cone or if it will spill over.

The ice cream and cone below are drawn accurately and at full size. If the scoop of ice cream is placed on top of the cone and it melts down into the cone, will all the ice cream fit inside the cone? Show whether the melted ice cream will fit inside the cone. For full credit, you must show your calculations and reasoning.

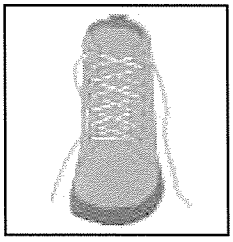


High School Long Task—Shoelaces

You work in a factory that makes shoes. You have to decide what length of shoelace to select when you know the number of lace holes in a shoe. The shoes can have up to ten pairs of lace holes.

1. Estimate the length of shoelace you need for the shoe shown on the next page. (It is shown full size.) Remember to allow enough extra length for tying a bow.
2. Create a rule that will tell you the length of shoelace you need when the number of pairs of lace holes in a shoe is given. Explain how you arrived at your rule.

[A full-page image was used in problem solving.]



3. Now design three signs that can be displayed in the factory to help people select the correct length of shoelaces if they know the number of pairs of lace holes. Your sign should describe your rule as clearly and simply as possible.
   (a) One sign must use a table.
   (b) One sign must use a graph.
   (c) One sign must use a formula.
4. Which of these signs do you think would be the most useful in the factory? Give reasons for your answer.

APPENDIX B

Sample Test Items from the SAT-9

SAT- 9 High School Edited Sample items

1. When Mr. Tillen meets a client, the probability that he will make a sale is 1/4. How many sales can he expect if he meets 144 clients?

   (A) 145 (B) 36 (C) 24 (D) 12

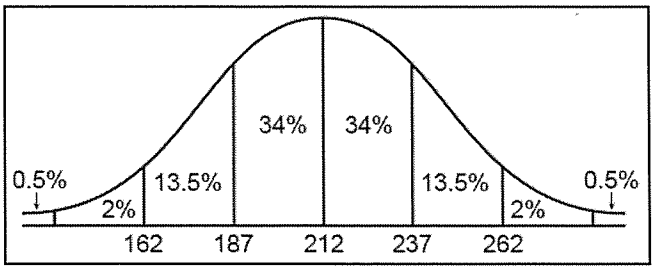
2. After t seconds, the velocity, v, of a basketball thrown upward at 256 feet per second is given by the equation

   $$v = 128 - 32t.$$

   After how many seconds will the velocity equal zero?

   (F) 4 (G) 6 (H) 8 (J) 12

3. The amount of water consumed by single-family homes is distributed as shown.



Water Consumption in Thousands of Gallons per Year

What is the probability that a single-family home will use *more* than 187 thousand gallons per year?

(A) 97% (B) 84% (C) 50% (D) 16%